

cardona-qft-methods — formula report

3760 inline formulas (0 shown: the ones that did not render), 1012 display equations, 8 tables, 40 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08717 [conf 0.003]	275	■0.003 ● C -4	$\begin{aligned} & \hat{T}=\operatorname{tr}_a M_a(u) \\ & \&=\left.\right _L^L \cdots \\ & \cdots \cdots \cdots a. \\ & \cdots 2 \\ & L \end{aligned} a .$	$\hat{T}=\operatorname{tr}_a M_a(u)$ $= \left _L^L \cdots \right _L^L \cdots \left _L^L \right _L^L a .$	
cardona-qft-methods_ E08889 [conf 0.077]	345	■0.077 ● W -2	$\begin{array}{l} \operatorname{Map}(8, M)=L M \\ \times M \rightarrow L M \times M \\ \downarrow \Delta \\ M \times M \end{array}$	$\operatorname{Map}(8, M)=L M \times_M L M \longrightarrow L M \times L M$ $\downarrow \Delta \quad \downarrow \quad \downarrow c M \times M$	$\operatorname{Map}(8, M)=L M \times_M L M \longrightarrow L M \times L M$ $\downarrow \quad \downarrow \quad \downarrow ev \times ev$
cardona-qft-methods_ E08988 [conf 0.080]	350	■0.080 ● U +5	$\begin{array}{l} P_g M: {}_1 \times {}_0 P_h M \\ \downarrow \\ M \\ \downarrow \end{array}$	$P_g M: {}_1 \times {}_0 P_h M \longrightarrow P_g M \times P_h M$ $\downarrow \quad \downarrow$ $M \times M$	$P_g M: {}_1 \times {}_0 P_h M \longrightarrow P_g M \times P_h M$ $\downarrow e_0 \circ \pi_1 \quad \downarrow e_0 \times e_0$ $M \xrightarrow{\Delta_g} M \times M$
cardona-qft-methods_ E08977	046	■0.150 ● C +44	$\left.\begin{array}{l} 1 \\ \operatorname{Inn}(\mathcal{A}) \\ \operatorname{Aut}(\mathcal{A}) \\ \operatorname{Aut}(\mathcal{A}) / \operatorname{Inn}(\mathcal{A}) \end{array}\right\} \rightarrow \mathcal{G} \rightarrow \mathcal{G} \rtimes \operatorname{Diff}(M) \rightarrow \operatorname{Diff}(M) \rightarrow 1 .$	$(not\ rendered)$	$1 \rightarrow \operatorname{Inn}(\mathcal{A}) \rightarrow \operatorname{Aut}(\mathcal{A}) \rightarrow \operatorname{Aut}(\mathcal{A}) / \operatorname{Inn}(\mathcal{A}) \rightarrow 1,$ $1 \rightarrow \mathcal{G} \rightarrow \mathcal{G} \rtimes \operatorname{Diff}(M) \rightarrow \operatorname{Diff}(M) \rightarrow 1 .$
cardona-qft-methods_ E08716	275	■0.155 ● U +2	$\begin{aligned} & M_a(u)=L_{a, L}(u) L_{a, L-1}(u) L_{a, L-1}(u) \\ & \cdots L_{a, 2}(u) L_{a, 1}(u) \&=\left.a\right _L^L \cdots \\ & \cdots \cdots \cdots L \end{aligned}$	$M_a(u)=L_{a, L}(u) L_{a, L-1}(u) \cdots L_{a, 2}(u) L_{a, 1}(u)$ $= a \left _L^L \cdots \right _L^L \left _L^L \right _L^L a .$	$M_a(u)=L_{a, L}(u) L_{a, L-1}(u) \cdots L_{a, 2}(u) L_{a, 1}(u)$ $= a \left _L^L \cdots \right _L^L \left _L^L \right _L^L a .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08654 (4)	261	0.167	$\phi_{\theta}(x)\phi_{\theta}(y)=e^{-\frac{1}{2}\frac{\partial}{\partial x^{\mu}}\theta^{\mu\nu}\frac{\partial}{\partial y^{\nu}}}\phi_0(x)\phi_0(y)e^{\frac{1}{2}\left(\frac{\partial}{\partial x^{\mu}}+\frac{\partial}{\partial y^{\mu}}\right)\theta^{\mu\nu}P_{\nu}}\neq\phi_0(x)\phi_0(y).$	(not rendered)	
cardona-qft-methods_ E08325 (11)	161	0.219 N +0	$\begin{aligned} d\eta_{ab} &= D\eta_{ab} = 0, \\ d\epsilon_{a_1a_2\cdots s_D} &= D\epsilon_{a_1a_2\cdots a_D} = 0. \end{aligned}$	$d\eta_{ab} = D\eta_{ab} = 0, \\ d\epsilon_{a_1a_2\cdots s_D} = D\epsilon_{a_1a_2\cdots a_D} = 0.$	$d\eta_{ab} = D\eta_{ab} = 0, \\ d\epsilon_{a_1a_2\cdots s_D} = D\epsilon_{a_1a_2\cdots a_D} = 0.$
cardona-qft-methods_ E08866 (19)	041	0.286 C +4	$\begin{aligned} & \& a_0(f,P)=(4\pi)^{-d/2}\int_M dvol_g\operatorname{tr}_V(f), \\ & a_2(f,P)=\frac{(4\pi)^{-d/2}}{6}\int_M dvol_g\operatorname{tr}_V[(f(6E+s)], \\ & a_4(f,P)=\frac{(4\pi)^{-d/2}}{360}\int_M dvol_g\operatorname{tr}_V\left[f\left(60E_{;kk}+60Es+180E^2+12R_{;kk} \right. \right. \\ & \qquad \qquad \qquad \left. \left. +5s^2-2R_{ij}R_{ij}+2R_{ijkl}R_{ijkl}+30\Omega_{ij}\Omega_{ij}\right)\right] \end{aligned}$	$a_0(f,P)=(4\pi)^{-d/2}\int_M dvol_g\operatorname{tr}_V(f), \\ a_2(f,P)=\frac{(4\pi)^{-d/2}}{6}\int_M dvol_g\operatorname{tr}_V[(f(6E+s)], \\ a_4(f,P)=\frac{(4\pi)^{-d/2}}{360}\int_M dvol_g\operatorname{tr}_V\left[f\left(60E_{;kk}+60Es+180E^2+12R_{;kk} \right. \right. \\ \qquad \qquad \qquad \left. \left. +5s^2-2R_{ij}R_{ij}+2R_{ijkl}R_{ijkl}+30\Omega_{ij}\Omega_{ij}\right)\right]$	$a_0(f,P)=(4\pi)^{-d/2}\int_M dvol_g\operatorname{tr}_V(f), \\ a_2(f,P)=\frac{(4\pi)^{-d/2}}{6}\int_M dvol_g\operatorname{tr}_V[(f(6E+s)], \\ a_4(f,P)=\frac{(4\pi)^{-d/2}}{360}\int_M dvol_g\operatorname{tr}_V\left[f(60E_{;kk}+60Es+180E^2+12R_{;kk} \right. \\ \qquad \qquad \qquad \left. \left. +5s^2-2R_{ij}R_{ij}+2R_{ijkl}R_{ijkl}+30\Omega_{ij}\Omega_{ij}\right)\right]. \tag{1}$
cardona-qft-methods_ E08824 (3)	313	0.289 N +2	$\begin{gathered} \delta\psi_M=\nabla_M\epsilon+\frac{1}{4}H_M\mathcal{P}\epsilon+\frac{1}{16}e^\phi\sum_n\mathcal{F}_n^{(10)}\Gamma_M\mathcal{P}_n\epsilon, \\ \delta\lambda=\left(\not\partial\phi+\frac{1}{2}\not{H}\mathcal{P}\right)\epsilon+\frac{1}{8}e^\phi\sum_n(-1)^n(5-n)\not{\mathcal{L}}_n^{(10)}\mathcal{P}_n\epsilon. \end{gathered}$	$\delta\psi_M=\nabla_M\epsilon+\frac{1}{4}H_M\mathcal{P}\epsilon+\frac{1}{16}e^\phi\sum_n\mathcal{F}_n^{(10)}\Gamma_M\mathcal{P}_n\epsilon, \\ \delta\lambda=\left(\not\partial\phi+\frac{1}{2}\not{H}\mathcal{P}\right)\epsilon+\frac{1}{8}e^\phi\sum_n(-1)^n(5-n)\not{\mathcal{L}}_n^{(10)}\mathcal{P}_n\epsilon.$	$\delta\psi_M=\nabla_M\epsilon+\frac{1}{4}H_M\mathcal{P}\epsilon+\frac{1}{16}e^\phi\sum_n\mathcal{F}_n^{(10)}\Gamma_M\mathcal{P}_n\epsilon, \\ \delta\lambda=\left(\not\partial\phi+\frac{1}{2}\not{H}\mathcal{P}\right)\epsilon+\frac{1}{8}e^\phi\sum_n(-1)^n(5-n)\mathcal{F}_n^{(10)}\mathcal{P}_n\epsilon.$
cardona-qft-methods_ E08791 (32)	300	0.292 W +0	$\begin{array}{rc} \mathrm{R}-\mathrm{R}: \xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle, & \mathrm{NS}-\mathrm{R}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_0^N 0\rangle, \\ \mathrm{NS}-\mathrm{NS}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_{1/2}^N 0\rangle, & \mathrm{R}-\mathrm{NS}: \xi_{MN}\psi_0^M\tilde{\psi}_{1/2}^N 0\rangle. \end{array}$	$\begin{array}{rc} \mathrm{R}-\mathrm{R}: \xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle, & \mathrm{NS}-\mathrm{R}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_0^N 0\rangle, \\ \mathrm{NS}-\mathrm{NS}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_{1/2}^N 0\rangle, & \mathrm{R}-\mathrm{NS}: \xi_{MN}\psi_0^M\tilde{\psi}_{1/2}^N 0\rangle. \end{array}$	$\begin{array}{rc} \mathrm{R}-\mathrm{R}: \xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle, & \mathrm{NS}-\mathrm{R}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_0^N 0\rangle, \\ \mathrm{NS}-\mathrm{NS}: \xi_{MN}\psi_{1/2}^M\tilde{\psi}_{1/2}^N 0\rangle, & \mathrm{R}-\mathrm{NS}: \xi_{MN}\psi_0^M\tilde{\psi}_{1/2}^N 0\rangle. \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E01012	377	■ 0.300 ● W +0	$\begin{aligned} & \& \left. \omega \right _{x=0} \cong \chi_0(\omega) = \omega(1) = \langle \nu, \nu \rangle = 1, \\ & \& \left. \partial_x \omega \right _{x=0} \cong \chi_0(X\omega) = \langle \nu, X^\dagger \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = 0, \\ & \& \left. \partial_x^2 \omega \right _{x=0} \cong \chi_0(X^2\omega) = \langle \nu, (X^\dagger)^2 \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = -1, \end{aligned}$	$\omega _{x=0} \cong \chi_0(\omega) = \omega(1) = \langle \nu, \nu \rangle = 1,$ $\partial_x \omega _{x=0} \cong \chi_0(X\omega) = \langle \nu, X^\dagger \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = 0,$ $\partial_x^2 \omega _{x=0} \cong \chi_0(X^2\omega) = \langle \nu, (X^\dagger)^2 \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = -1,$	$\omega _{x=0} \cong \chi_0(\omega) = \omega(1) = \langle \nu, \nu \rangle = 1,$ $\partial_x \omega _{x=0} \cong \chi_0(X\omega) = \langle \nu, X^\dagger \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = 0,$ $\partial_x^2 \omega _{x=0} \cong \chi_0(X^2\omega) = \langle \nu, (X^\dagger)^2 \nu \rangle = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = -1,$
cardona-qft-methods_ E00681	269	■ 0.304 ●	$\begin{gathered} \text{" Group" } \approx e^{\{i \text{ "Algebra" } \}} \\ \text{" SU(2) } \approx e^{\{i \mathfrak{su}(2) \}} \end{gathered}$	$\text{" Group" } \approx e^{i \text{ "Algebra" }},$ $SU(2) \approx e^{isu(2)}.$	$\text{"Group"} \approx e^{i \text{"Algebra"}},$ $SU(2) \approx e^{isu(2)}.$
cardona-qft-methods_ E00158 (4)	087	■ 0.311 ● N +0	$\mathrm{t} \text{-} \operatorname{Ind} = i_!^{-1} j_! : K(T^*M) \rightarrow K(\{pt\}) \simeq \mathbb{Z}.$	$\mathrm{t} \text{-} \operatorname{Ind} = i_!^{-1} j_! : K(T^*M) \rightarrow K(\{pt\}) \simeq \mathbb{Z}.$	$\mathrm{t} \text{-} \operatorname{Ind} = i_!^{-1} j_! : K(T^*M) \rightarrow K(\{pt\}) \simeq \mathbb{Z}.$
cardona-qft-methods_ E00501	226	■ 0.333 ● W +0	$\left. \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2) \right _{\Lambda=\Lambda_{unif}} = 8M_W^2 _{\Lambda=\Lambda_{unif}}$	$\sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2) \Big _{\Lambda=\Lambda_{unif}} = 8M_W^2 _{\Lambda=\Lambda_{unif}}$	$\sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2) _{\Lambda=\Lambda_{unif}} = 8M_W^2 _{\Lambda=\Lambda_{unif}}$
cardona-qft-methods_ E00049 (14)	035	■ 0.362 ● N +0	$\left. \mathcal{H} = L^2((M, g), S) \right _{\Gamma^\infty(M, S)} = \left\{ \psi \in \Gamma^\infty(M, S) \mid \int_M \langle \psi, \psi \rangle_x d\operatorname{vol}_g(x) < \infty \right\}$	$\mathcal{H} = L^2((M, g), S) = \left\{ \psi \in \Gamma^\infty(M, S) \mid \int_M \langle \psi, \psi \rangle_x d\operatorname{vol}_g(x) < \infty \right\}$	$\mathcal{H} = L^2((M, g), S) = \{ \psi \in \Gamma^\infty(M, S) \mid \int_M \langle \psi, \psi \rangle_x d\operatorname{vol}_g(x) < \infty \}$
cardona-qft-methods_ E00795 (36)	301	■ 0.396 ● W +0	$\begin{aligned} & \& \operatorname{type} \text{ IIB } \\ & \& \operatorname{NS} \text{-} \operatorname{R} : \mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^1 \oplus \Psi^{1M}, \\ & \& \operatorname{R} \text{-} \operatorname{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M}, \\ & \& \operatorname{type} \text{ IIA } \operatorname{NS} \text{-} \operatorname{R} : \mathbf{8}_v \otimes \mathbf{8}_c = \mathbf{8}_c \oplus \mathbf{56}_c = \lambda^1 \oplus \Psi^{1M}, \\ & \& \operatorname{R} \text{-} \operatorname{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M}. \end{aligned}$	$\text{type IIB NS} \text{-} \text{R} : \mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^1 \oplus \Psi^{1M},$ $\text{R} \text{-} \text{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M},$ $\text{type IIA NS} \text{-} \text{R} : \mathbf{8}_v \otimes \mathbf{8}_c = \mathbf{8}_c \oplus \mathbf{56}_c = \lambda^1 \oplus \Psi^{1M},$ $\text{R} \text{-} \text{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M}.$	$\text{type IIB NS} \text{-} \text{R} : \mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^1 \oplus \Psi^{1M},$ $\text{R} \text{-} \text{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M},$ $\text{type IIA NS} \text{-} \text{R} : \mathbf{8}_v \otimes \mathbf{8}_c = \mathbf{8}_c \oplus \mathbf{56}_c = \lambda^1 \oplus \Psi^{1M},$ $\text{R} \text{-} \text{NS} : \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2M}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00028	027	0.431 W +1	$\begin{aligned} & W R e s \left((1 + \Delta)^{-d / 2} \right) \\ & \& = \int_M d x \int_{\mathbb{S}^{d-1}} \ \xi\ _x^{-d} d \xi \\ & d \, \xi = \int_M d x \sqrt{\det g_x} \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \\ & \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \int_M d \operatorname{vol}_g = \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \end{aligned}$	$W R e s \left((1 + \Delta)^{-d / 2} \right) = \int_M d x \int_{\mathbb{S}^{d-1}} \ \xi\ _x^{-d} d \xi = \int_M d x \sqrt{\det g_x} \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \\ = \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \int_M d \operatorname{vol}_g = \operatorname{Vol} \left(\mathbb{S}^{d-1} \right)$	$W R e s \left((1 + \Delta)^{-d / 2} \right) = \int_M d x \int_{\mathbb{S}^{d-1}} \ \xi\ _x^{-d} d \xi = \int_M d x \sqrt{\det g_x} \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \\ = \operatorname{Vol} \left(\mathbb{S}^{d-1} \right) \int_M d \operatorname{vol}_g = \operatorname{Vol} \left(\mathbb{S}^{d-1} \right). \quad \square$
cardona-qft-methods_ E00148 (1)	083	0.450 N -1	$\left. \mathcal{D} \right\rangle \rightsquigarrow \sigma_L(\mathcal{D}) \rightsquigarrow \text{an element of } K\text{-theory} \rightsquigarrow \text{t-Ind}(\sigma_L(\mathcal{D})) \in \mathbb{Z}.$	$\mathcal{D} \rightsquigarrow \sigma_L(\mathcal{D}) \rightsquigarrow \text{an element of } K\text{-theory} \rightsquigarrow \text{t-Ind}(\sigma_L(\mathcal{D})) \in \mathbb{Z}.$	$\mathcal{D} \rightsquigarrow \sigma_L(\mathcal{D}) \rightsquigarrow \text{an element of } K\text{-theory} \rightsquigarrow \text{t-Ind}(\sigma_L(\mathcal{D})) \in \mathbb{Z}.$
cardona-qft-methods_ E00515 (1)	236	0.478 N -2	$\lambda_C = \frac{\hbar}{Mc} \leq L \text{ or } M \geq \frac{\hbar}{Lc} \simeq 10^{19} \text{GeV (Planck mass)} \simeq 10^{-5} \text{grams}.$	$\lambda_C = \frac{\hbar}{Mc} \leq L \text{ or } M \geq \frac{\hbar}{Lc} \simeq 10^{19} \text{GeV (Planck mass)} \simeq 10^{-5} \text{grams}.$	$\lambda_C = \frac{\hbar}{Mc} \leq L \text{ or } M \geq \frac{\hbar}{Lc} \simeq 10^{19} \text{GeV (Planck mass)} \simeq 10^{-5} \text{grams}.$
cardona-qft-methods_ E00019 (7)	024	0.490 N -1	$W \operatorname{Res} P = \operatorname{Res} \operatorname{Tr} \left(P \mathcal{D} ^{-s} \right).$	$W \operatorname{Res} P = \operatorname{Res} \operatorname{Tr} \left(P \mathcal{D} ^{-s} \right).$	$W \operatorname{Res} P = \operatorname{Res} \operatorname{Tr} \left(P \mathcal{D} ^{-s} \right).$
cardona-qft-methods_ E00324 (10)	161	0.495	$\begin{aligned} & \phi_{ }^a(x) - \phi^a(x) = d\phi^a(x) + \omega_b^a(x)\phi^b(x) = D\phi^a(x) \\ & \omega'^a(x) = \Lambda_c^a(x)\Lambda_b^d(x)\omega_d^c(x) + \Lambda_c^a(x)d\Lambda_b^c(x), \end{aligned}$	$\phi_{ }^a(x) - \phi^a(x) = d\phi^a(x) + \omega_b^a(x)\phi^b(x) = D\phi^a(x) \\ \omega'^a(x) = \Lambda_c^a(x)\Lambda_b^d(x)\omega_d^c(x) + \Lambda_c^a(x)d\Lambda_b^c(x)$	$\phi_{ }^a(x) - \phi^a(x) = d\phi^a(x) + \omega_b^a(x)\phi^b(x) = D\phi^a(x) \\ \omega'^a_b(x) = \Lambda_c^a(x)\Lambda_b^d(x)\omega_d^c(x) + \Lambda_c^a(x)d\Lambda_b^c(x),$
cardona-qft-methods_ E00481	219	0.498 K +0	$P f(\mathfrak{A}) = \int e^{-\frac{1}{2}\mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],$	$P f(\mathfrak{A}) = \int e^{-\frac{1}{2}\mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],$	$P f(\mathfrak{A}) = \int e^{-\frac{1}{2}\mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],$
cardona-qft-methods_ EQ1001	375	0.508 N +0	$\begin{aligned} & X(a \otimes b) \& = (a \otimes b) \circ (X^\dagger(\cdot) \otimes \operatorname{Id} + \operatorname{Id} \otimes X^\dagger(\cdot)) \circ \Delta \\ & \left(X^\dagger(\cdot) \otimes \operatorname{Id} \right) \otimes \operatorname{Id} + \operatorname{Id} \otimes \left(X^\dagger(\cdot) \otimes \operatorname{Id} \right) \\ & \circ \Delta = (X a \otimes b + a \otimes X b) \circ \Delta = (X a) b + a (X b). \end{aligned}$	$X(ab) = (a \otimes b) \circ (X^\dagger(\cdot) \otimes \operatorname{Id} + \operatorname{Id} \otimes X^\dagger(\cdot)) \circ \Delta \\ = (X a \otimes b + a \otimes X b) \circ \Delta = (X a) b + a (X b).$	$X(ab) = (a \otimes b) \circ (X^\dagger(\cdot) \otimes \operatorname{Id} + \operatorname{Id} \otimes X^\dagger(\cdot)) \circ \Delta \\ = (X a \otimes b + a \otimes X b) \circ \Delta = (X a) b + a (X b).$
cardona-qft-methods_ EQ0712	274	0.516 K +0	$R_{\{a, b\}}(u) = u 1_{\{a, b\}} + i P_{\{a, b\}} = \begin{pmatrix} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{pmatrix},$	$R_{a,b}(u) = u 1_{a,b} + i P_{a,b} = \begin{pmatrix} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{pmatrix},$	$R_{a,b}(u) = u 1_{a,b} + i P_{a,b} = \begin{pmatrix} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{pmatrix},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0890	345	0.517 C +5	$\begin{aligned} & H_*(L M) \otimes H_*(L M) \rightarrow H_*(L M \times L M) \\ & \left(\operatorname{Map}(8, M)^{ev^*(TM)} \right) \xrightarrow{\tau_{\Delta*}} H_{*-n}(\operatorname{Map}(8, M)) \\ & \downarrow \\ & H_{*-n}(LM) \end{aligned}$	$H_*(LM) \otimes H_*(LM) \rightarrow H_*(LM \times LM)$ $H_*\left(\operatorname{Map}(8, M)^{ev^*(TM)}\right) \xrightarrow{\tau_{\Delta*}} H_{*-n}(\operatorname{Map}(8, M))$ $\downarrow$ $H_{*-n}(LM)$	$H_*(LM) \otimes H_*(LM) \rightarrow H_*(LM \times LM)$ $\begin{array}{c} \tau_{\Delta*} \downarrow \\ H_*(\operatorname{Map}(8, M)^{ev^*(TM)}) \xrightarrow{Thom} H_{*-n}(\operatorname{Map}(8, M)) \\ \downarrow \\ H_{*-n}(LM) \end{array}$
cardona-qft-methods_ EQ0812	020	0.522 N +0	$\sigma_m^P(x, \xi) = \lim_{t \rightarrow \infty} t^{-m} e^{-i \operatorname{th}(x)} P e^{i \operatorname{th}(x)},$	$\sigma_m^P(x, \xi) = \lim_{t \rightarrow \infty} t^{-m} e^{-i \operatorname{th}(x)} P e^{i \operatorname{th}(x)},$	$\sigma_m^P(x, \xi) = \lim_{t \rightarrow \infty} t^{-m} e^{-i \operatorname{th}(x)} P e^{i \operatorname{th}(x)},$
cardona-qft-methods_ EQ0201	096	0.522 N +0	$\langle d\alpha, \beta \rangle = \langle \alpha, d^*\beta \rangle, \quad \text{for all } \alpha, \beta \in \Omega^*(M, E).$	$\langle d\alpha, \beta \rangle = \langle \alpha, d^*\beta \rangle, \quad \text{for all } \alpha, \beta \in \Omega^*(M, E).$	$\langle d\alpha, \beta \rangle = \langle \alpha, d^*\beta \rangle, \quad \text{for all } \alpha, \beta \in \Omega^*(M, E).$
cardona-qft-methods_ EQ0895	346	0.523 K +0	$LM^{ev^*(-TM)} \wedge LM^{ev^*(-TM)} \rightarrow \operatorname{Map}(8, M)^{ev^*(-TM)}.$	$LM^{ev^*(-TM)} \wedge LM^{ev^*(-TM)} \rightarrow \operatorname{Map}(8, M)^{ev^*(-TM)}.$	$LM^{ev^*(-TM)} \wedge LM^{ev^*(-TM)} \rightarrow \operatorname{Map}(8, M)^{ev^*(-TM)}.$
cardona-qft-methods_ EQ0099	057	0.523	$\begin{aligned} & \operatorname{Tr} \left(f \left(\mathcal{D}^2 / \Lambda^2 \right) \right) \sim \frac{1}{(4\pi)^2} \left[\operatorname{rk}(E) \int_0^\infty x f(x) dx \Lambda^4 + b_2(\mathcal{D}^2) \int_0^\infty f(x) dx \Lambda^2 \right. \\ & \quad \left. + \sum_{m=0}^\infty (-1)^m f^{(m)}(0) b_{2m+4}(\mathcal{D}^2) \Lambda^{-2m} \right], \quad \Lambda \rightarrow \infty \end{aligned}$	$\operatorname{Tr} \left(f \left(\mathcal{D}^2 / \Lambda^2 \right) \right) \sim \frac{1}{(4\pi)^2} \left[\operatorname{rk}(E) \int_0^\infty x f(x) dx \Lambda^4 + b_2(\mathcal{D}^2) \int_0^\infty f(x) dx \Lambda^2 \right. \\ \left. + \sum_{m=0}^\infty (-1)^m f^{(m)}(0) b_{2m+4}(\mathcal{D}^2) \Lambda^{-2m} \right], \quad \Lambda \rightarrow \infty$	$\operatorname{Tr} \left(f \left(\mathcal{D}^2 / \Lambda^2 \right) \right) \sim \frac{1}{(4\pi)^2} \left[\operatorname{rk}(E) \int_0^\infty x f(x) dx \Lambda^4 + b_2(\mathcal{D}^2) \int_0^\infty f(x) dx \Lambda^2 \right. \\ \left. + \sum_{m=0}^\infty (-1)^m f^{(m)}(0) b_{2m+4}(\mathcal{D}^2) \Lambda^{-2m} \right], \quad \Lambda \rightarrow \infty$
cardona-qft-methods_ EQ0154 (3)	085	0.523 K +0	$\sigma_L(\mathcal{D}) : \pi^* E \rightarrow \pi^* F.$	$\sigma_L(\mathcal{D}) : \pi^* E \rightarrow \pi^* F.$	$\sigma_L(\mathcal{D}) : \pi^* E \rightarrow \pi^* F.$
cardona-qft-methods_ EQ0919	352	0.527 K +0	$C_*(P_G M^{-TM}) \xrightarrow{\sim} \mathcal{R} \mathcal{H}om_{C^*(M)^e}(C^*(M), C^*(M) \# G)$	$C_*(P_G M^{-TM}) \xrightarrow{\sim} \mathcal{R} \mathcal{H}om_{C^*(M)^e}(C^*(M), C^*(M) \# G)$	$C_*(P_G M^{-TM}) \xrightarrow{\sim} \mathcal{R} \mathcal{H}om_{C^*(M)^e}(C^*(M), C^*(M) \# G)$
cardona-qft-methods_ EQ0356 (25)	171	0.527 S +0	$\begin{aligned} & \mathbf{P}_8 = R_{a_2}^{a_1} R_{a_3}^{a_2} \dots R_{a_1}^{a_4}, \\ & (\mathbf{P}_4)^2 = (R_b^a R_a^b)^2, \end{aligned}$	$\begin{aligned} & \mathbf{P}_8 = R_{a_2}^{a_1} R_{a_3}^{a_2} \dots R_{a_1}^{a_4}, \\ & (\mathbf{P}_4)^2 = (R_b^a R_a^b)^2, \end{aligned}$	$\begin{aligned} & \mathbf{P}_8 = R_{a_2}^{a_1} R_{a_3}^{a_2} \dots R_{a_1}^{a_4}, \\ & (\mathbf{P}_4)^2 = (R_b^a R_a^b)^2, \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00424	202	■ 0.529 ● N +0	$\begin{aligned} C_{\lambda\mu\nu\kappa} &= R_{\lambda\mu\nu\kappa} - \frac{1}{2} (g_{\lambda\nu} R_{\mu\kappa} - g_{\lambda\kappa} R_{\mu\nu} - g_{\mu\nu} R_{\lambda\kappa} + g_{\mu\kappa} R_{\lambda\nu}) \\ &\quad + \frac{1}{6} R_{\alpha}^{\alpha} (g_{\lambda\nu} g_{\mu\kappa} - g_{\lambda\kappa} g_{\mu\nu}). \end{aligned}$	$C_{\lambda\mu\nu\kappa} = R_{\lambda\mu\nu\kappa} - \frac{1}{2} (g_{\lambda\nu} R_{\mu\kappa} - g_{\lambda\kappa} R_{\mu\nu} - g_{\mu\nu} R_{\lambda\kappa} + g_{\mu\kappa} R_{\lambda\nu}) + \frac{1}{6} R_{\alpha}^{\alpha} (g_{\lambda\nu} g_{\mu\kappa} - g_{\lambda\kappa} g_{\mu\nu}).$	$C_{\lambda\mu\nu\kappa} = R_{\lambda\mu\nu\kappa} - \frac{1}{2} (g_{\lambda\nu} R_{\mu\kappa} - g_{\lambda\kappa} R_{\mu\nu} - g_{\mu\nu} R_{\lambda\kappa} + g_{\mu\kappa} R_{\lambda\nu}) + \frac{1}{6} R_{\alpha}^{\alpha} (g_{\lambda\nu} g_{\mu\kappa} - g_{\lambda\kappa} g_{\mu\nu}).$
cardona-qft-methods_ E00026	027	■ 0.531 ● N +1	$T(P)=\mu\left(c_{\{P\}}(x)\right)=c\int_{\{M\}}c_{\{P\}}(x) d\,x =c\,W\,R\,e\,s(P)$	$T(P)=\mu\left(c_P(x)\right)=c\int_Mc_P(x) dx =cWRes(P)$	$T(P)=\mu\left(c_P(x)\right)=c\int_Mc_P(x) dx =c\,WRes(P).$
cardona-qft-methods_ E00747	283	■ 0.535 ● N +0	$\mathcal{D}=\mathcal{D}_0+\frac{g_{\mathrm{YM}}^2}{16\pi^2}\hat{H}_{XXX}+O\left(g_{\mathrm{YM}}^4\right).$	$\mathcal{D}=\mathcal{D}_0+\frac{g_{\mathrm{YM}}^2}{16\pi^2}\hat{H}_{XXX}+O\left(g_{\mathrm{YM}}^4\right).$	$\mathcal{D}=\mathcal{D}_0+\frac{g_{\mathrm{YM}}^2}{16\pi^2}\hat{H}_{XXX}+O(g_{\mathrm{YM}}^4).$
cardona-qft-methods_ E00831 (10)	314	■ 0.539 ● C +3	$\begin{aligned} &\epsilon_{\mathrm{IIA}}^1=\xi_{+}^1\otimes\eta_{+}^1+\xi_{-}^1\otimes\eta_{-}^1 \\ &\epsilon_{\mathrm{IIA}}^2=\xi_{+}^2\otimes\eta_{-}^2+\xi_{-}^2\otimes\eta_{+}^2 \end{aligned}$	$\begin{aligned} \epsilon_{\mathrm{IIA}}^1 &= \xi_+^1 \otimes \eta_+^1 + \xi_-^1 \otimes \eta_-^1 \\ \epsilon_{\mathrm{IIA}}^2 &= \xi_+^2 \otimes \eta_-^2 + \xi_-^2 \otimes \eta_+^2 \end{aligned}$	$\begin{aligned} \epsilon_{\mathrm{IIA}}^1 &= \xi_+^1 \otimes \eta_+^1 + \xi_-^1 \otimes \eta_-^1, \\ \epsilon_{\mathrm{IIA}}^2 &= \xi_+^2 \otimes \eta_-^2 + \xi_-^2 \otimes \eta_+^2, \end{aligned}$
cardona-qft-methods_ E00834 (13)	317	■ 0.540 ● N +1	$\begin{aligned} &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_+)=0 \\ &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_-)=\mathrm{d}A\wedge\bar{\Phi}_-+\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIA}}+ \end{aligned}$	$\begin{aligned} &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_+)=0 \\ &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_-)=\mathrm{d}A\wedge\bar{\Phi}_-+\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIA}}+ \end{aligned}$	$\begin{aligned} &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_+)=0, \\ &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_-)=\mathrm{d}A\wedge\bar{\Phi}_-+\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIA}}+ \end{aligned}$
cardona-qft-methods_ E00031	029	■ 0.541 ●	$\begin{aligned} &\sigma_N(T_1+T_2)\leq\sigma_N(T_1)+\sigma_N(T_2) \\ &\sigma_{N_1}(T_1)+\sigma_{N_2}(T_2)\leq\sigma_{N_1+N_2}(T_1+T_2) \text{ when } T_1,T_2\geq 0. \end{aligned}$	$\begin{aligned} &\sigma_N(T_1+T_2)\leq\sigma_N(T_1)+\sigma_N(T_2) \\ &\sigma_{N_1}(T_1)+\sigma_{N_2}(T_2)\leq\sigma_{N_1+N_2}(T_1+T_2) \text{ when } T_1,T_2\geq 0. \end{aligned}$	$\begin{aligned} &\sigma_N(T_1+T_2)\leq\sigma_N(T_1)+\sigma_N(T_2), \\ &\sigma_{N_1}(T_1)+\sigma_{N_2}(T_2)\leq\sigma_{N_1+N_2}(T_1+T_2) \text{ when } T_1,T_2\geq 0. \end{aligned}$
cardona-qft-methods_ E00748	284	■ 0.547 ● C -27	$\begin{aligned} &AdS_5:\quad -R^2=-Z_0^2+Z_1^2+Z_2^2+Z_3^2+Z_4^2-Z_5^2 \\ &S^5:\quad R^2=Y_1^2+Y_2^2+Y_3^2+Y_4^2+Y_5^2+Y_6^2 \end{aligned}$	$\begin{aligned} AdS_5:\quad &-R^2=-Z_0^2+Z_1^2+Z_2^2+Z_3^2+Z_4^2-Z_5^2 \\ S^5:\quad &R^2=Y_1^2+Y_2^2+Y_3^2+Y_4^2+Y_5^2+Y_6^2 \end{aligned}$	$\begin{aligned} AdS_5:\quad &-R^2=-Z_0^2+Z_1^2+Z_2^2+Z_3^2+Z_4^2-Z_5^2, \\ S^5:\quad &R^2=Y_1^2+Y_2^2+Y_3^2+Y_4^2+Y_5^2+Y_6^2. \end{aligned}$
cardona-qft-methods_ E00936	354	■ 0.552 ● N +1	$HH^*(\mathbb{Z}G,\mathbb{Z}G)\cong H_{\ast}^{\mathrm{pro}}(LBG^{-TBG}).$	$HH^*(\mathbb{Z}G,\mathbb{Z}G)\cong H_{\ast}^{\mathrm{pro}}(LBG^{-TBG}).$	$HH^*(\mathbb{Z}G,\mathbb{Z}G)\cong H_{\ast}^{\mathrm{pro}}(LBG^{-TBG}).$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00778 (19)	297	■ 0.557 ● W +0	$\text{\texttt{T}_{\alpha}^{\alpha}\{\alpha\}\text{\texttt{}}_{\alpha}=-\frac{1}{2}\alpha^{\prime}\beta_{MN}^Gg^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{i}{2\alpha'}\beta_{MN}^B\epsilon^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{1}{2}\beta^{\Phi}R.$	$T^{\alpha}_{\alpha}=-\frac{1}{2\alpha'}\beta_{MN}^Gg^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{i}{2\alpha'}\beta_{MN}^B\epsilon^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{1}{2}\beta^{\Phi}R.$	$T^{\alpha}_{\alpha}=-\frac{1}{2\alpha'}\beta_{MN}^Gg^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{i}{2\alpha'}\beta_{MN}^B\epsilon^{\alpha\beta}\partial_{\alpha}X^M\partial_{\beta}X^N-\frac{1}{2}\beta^{\Phi}R.$
cardona-qft-methods_ E00149	083	■ 0.565 ● K +0	$\text{\texttt{\text{elliptic operators}}}\rightarrow\mathbb{Z},\quad\mathcal{D}\mapsto\text{\texttt{t-Ind}}(\sigma_L(\mathcal{D}))$	elliptic operators $\rightarrow \mathbb{Z}$, $\mathcal{D} \mapsto \text{t-Ind}(\sigma_L(\mathcal{D}))$	elliptic operators $\rightarrow \mathbb{Z}$, $\mathcal{D} \mapsto \text{t-Ind}(\sigma_L(\mathcal{D}))$
cardona-qft-methods_ E00978 (18)	364	■ 0.570 ● N +0	$\left[\mathcal{S}_n\right]=\left[\Sigma_n\right]-n\mathbb{T}^{n-2}=\sum_{i=2}^{n-1}\binom{n}{i}\mathbb{T}^{n-1-i}.$	$[\mathcal{S}_n] = [\Sigma_n] - n\mathbb{T}^{n-2} = \sum_{i=2}^{n-1} \binom{n}{i} \mathbb{T}^{n-1-i}.$	$[\mathcal{S}_n] = [\Sigma_n] - n\mathbb{T}^{n-2} = \sum_{i=2}^{n-1} \binom{n}{i} \mathbb{T}^{n-1-i}.$
cardona-qft-methods_ E00558	246	■ 0.574 ● N +0	$\begin{aligned} \Delta(\eta) &= \sum_{\alpha} \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \equiv \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \\ &\equiv \eta^{(1)} \otimes \eta^{(2)}. \end{aligned}$	$\Delta(\eta) = \sum_{\alpha} \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \equiv \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \equiv \eta^{(1)} \otimes \eta^{(2)}.$	$\Delta(\eta) = \sum_{\alpha} \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \equiv \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \equiv \eta^{(1)} \otimes \eta^{(2)}.$
cardona-qft-methods_ E00548	244	■ 0.577 ● W +1	$\begin{aligned} &\text{\texttt{m}_{r}}(\xi\otimes\eta\otimes\rho)=\eta\xi\otimes\rho \\ &\text{\texttt{m}_{r}}'(\xi\otimes\eta\otimes\rho)=\rho\otimes\eta\xi \end{aligned}$	$\begin{aligned} m_r(\xi \otimes \eta \otimes \rho) &= \eta \xi \otimes \rho \\ m'_r(\xi \otimes \eta \otimes \rho) &= \rho \otimes \eta \xi \end{aligned}$	$\begin{aligned} m_r(\xi \otimes \eta \otimes \rho) &= \eta \xi \otimes \rho \\ m'_r(\xi \otimes \eta \otimes \rho) &= \rho \otimes \eta \xi. \end{aligned}$
cardona-qft-methods_ E00035 (10)	029	■ 0.578 ● K +0	$\mathcal{L}^1(\mathcal{H})\subset\mathcal{L}^{1,\infty}\subset\mathcal{L}^p(\mathcal{H})\quad\text{for }p>1,\quad\ T\ \leq\ T\ _{1,\infty}\leq\ T\ _1.$	$\mathcal{L}^1(\mathcal{H})\subset\mathcal{L}^{1,\infty}\subset\mathcal{L}^p(\mathcal{H})\quad\text{for }p>1,\quad\ T\ \leq\ T\ _{1,\infty}\leq\ T\ _1.$	$\mathcal{L}^1(\mathcal{H})\subset\mathcal{L}^{1,\infty}\subset\mathcal{L}^p(\mathcal{H})\quad\text{for }p>1,\quad\ T\ \leq\ T\ _{1,\infty}\leq\ T\ _1.$
cardona-qft-methods_ E00674	267	■ 0.578 ● N +0	$Z_{\text{\texttt{s t r}}}\big[\phi\big _{\partial AdS}=J\big]=Z_{CFT}[J]$	$Z_{str}[\phi _{\partial AdS}=J]=Z_{CFT}[J]$	$Z_{str}[\phi _{\partial AdS}=J]=Z_{CFT}[J]$
cardona-qft-methods_ E00735 (3)	280	■ 0.578 ● W +0	$\begin{aligned} &\left\{\left[M_{\mu\nu},M_{\rho\sigma}\right]=i\left(\eta_{\mu\rho}M_{\nu\sigma}+\eta_{\nu\sigma}M_{\mu\rho}-\eta_{\mu\sigma}M_{\nu\rho}-\eta_{\nu\rho}M_{\mu\sigma}\right),\right. \\ &\left.\left[M_{\mu\nu},P_{\rho}\right]=i\left(\eta_{\rho\nu}P_{\mu}-\eta_{\mu\rho}P_{\nu}\right),\right. \\ &\left.\left[P_{\mu},P_{\nu}\right]=0.\right. \end{aligned}$	$\begin{aligned} [M_{\mu\nu}, M_{\rho\sigma}] &= i(\eta_{\mu\rho}M_{\nu\sigma} + \eta_{\nu\sigma}M_{\mu\rho} - \eta_{\mu\sigma}M_{\nu\rho} - \eta_{\nu\rho}M_{\mu\sigma}), \\ [M_{\mu\nu}, P_{\rho}] &= i(\eta_{\rho\nu}P_{\mu} - \eta_{\mu\rho}P_{\nu}), \\ [P_{\mu}, P_{\nu}] &= 0. \end{aligned}$	$\begin{aligned} [M_{\mu\nu}, M_{\rho\sigma}] &= i(\eta_{\mu\rho}M_{\nu\sigma} + \eta_{\nu\sigma}M_{\mu\rho} - \eta_{\mu\sigma}M_{\nu\rho} - \eta_{\nu\rho}M_{\mu\sigma}), \\ [M_{\mu\nu}, P_{\rho}] &= i(\eta_{\rho\nu}P_{\mu} - \eta_{\mu\rho}P_{\nu}), \\ [P_{\mu}, P_{\nu}] &= 0. \end{aligned}$
cardona-qft-methods_ E00773 (14)	295	■ 0.590 ●	$\begin{aligned} &\xi_t\equiv\frac{1}{D}\eta^{MN}\xi_{MN} \\ &\xi^s_{MN}=\frac{1}{2}(\xi_{MN}+\xi_{NM}-2\xi_t\eta_{MN}), \\ &\xi^a_{MN}=\frac{1}{2}(\xi_{MN}-\xi_{NM}). \end{aligned}$	$\begin{aligned} \xi_t &\equiv \frac{1}{D} \eta^{MN} \xi_{MN} \\ \xi^s_{MN} &= \frac{1}{2} (\xi_{MN} + \xi_{NM} - 2\xi_t \eta_{MN}), \\ \xi^a_{MN} &= \frac{1}{2} (\xi_{MN} - \xi_{NM}). \end{aligned}$	$\begin{aligned} \xi_t &\equiv \frac{1}{D} \eta^{MN} \xi_{MN}, \\ \xi^s_{MN} &= \frac{1}{2} (\xi_{MN} + \xi_{NM} - 2\xi_t \eta_{MN}), \\ \xi^a_{MN} &= \frac{1}{2} (\xi_{MN} - \xi_{NM}). \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ8399	182	■ 0.599 ● S +1	$\begin{aligned} & \& \mathbf{N}_{\{4\}} \mathbf{P}_{\{4\}}^{\wedge \text{\texttt{Lor}}} \\ & \& \left(T^a T_a - e^a e^b R_{ab} \right) R^c{}_d R^d{}_c \\ & \& \right) R^{\{c\}\{ \}_d} R^{\{d\}\{ \}_c} \end{aligned}$	$\mathbf{N}_4 \mathbf{P}_4^{\text{\texttt{Lor}}} =$ $(T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} =$ $(T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
cardona-qft-methods_ EQ8611 (19)	254	■ 0.600 ● S +1	$\begin{array}{l} P_{\mu} c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} + \\ c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle (-\sum_i P_{i\mu}) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = 0 \end{array}$	$P_{\mu} c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} +$ $c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle (-\sum_i P_{i\mu}) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = 0$	$P_{\mu} c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} +$ $c_{p_1}^{\dagger} c_{p_2}^{\dagger} \dots c_{p_N}^{\dagger} 0\rangle (-\sum_i P_{i\mu}) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = 0,$
cardona-qft-methods_ EQ8478	219	■ 0.600 ● N +0	$S=\operatorname{Tr}\left(f\left(D_A/\Lambda\right) / \Lambda\right) \frac{1}{2}\left\langle J \tilde{\xi}, D_A \tilde{\xi}\right\rangle, \quad \tilde{\xi} \in \mathcal{H}_{cl}^{+},$	$S=\operatorname{Tr}\left(f\left(D_A/\Lambda\right)\right)+\frac{1}{2}\left\langle J \tilde{\xi}, D_A \tilde{\xi}\right\rangle, \quad \tilde{\xi} \in \mathcal{H}_{cl}^{+},$	$S=\operatorname{Tr}\left(f\left(D_A/\Lambda\right)\right)+\frac{1}{2}\left\langle J \tilde{\xi}, D_A \tilde{\xi}\right\rangle, \quad \tilde{\xi} \in \mathcal{H}_{cl}^{+},$
cardona-qft-methods_ EQ8889 (9)	308	■ 0.604 ● N +0	$\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .$	$\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .$	$\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .$
cardona-qft-methods_ EQ8465	217	■ 0.612 ●	$\begin{gathered} A_{\{q\}^{\{(0,1)\}}}=\left(\begin{array}{cc} \theta & X \\ X^{\prime} & \theta \end{array}\right) \otimes 1_3 \\ \otimes X \otimes X^{\prime} \otimes \theta \\ \otimes 1_{\{3\}} \quad A_{\{1\}^{\{(0,1)\}}}=\left(\begin{array}{cc} \theta & Y \\ Y^{\prime} & \theta \end{array}\right) \otimes 1_3 \\ \otimes Y \otimes Y^{\prime} \otimes \theta \\ \otimes X=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1 & \Upsilon_{\nu}^* \varphi_2 \\ -\Upsilon_{\nu}^* \bar{\varphi}_2 & \Upsilon_{\nu}^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad X^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1^{\prime} & \Upsilon_{\nu}^* \varphi_2^{\prime} \\ -\Upsilon_{\nu}^* \bar{\varphi}_2^{\prime} & \Upsilon_{\nu}^* \bar{\varphi}_1^{\prime} \end{array}\right) \\ Y=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1 & \Upsilon_{\nu}^* \varphi_2 \\ -\Upsilon_{\nu}^* \bar{\varphi}_2 & \Upsilon_{\nu}^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1^{\prime} & \Upsilon_{\nu}^* \varphi_2^{\prime} \\ -\Upsilon_{\nu}^* \bar{\varphi}_2^{\prime} & \Upsilon_{\nu}^* \bar{\varphi}_1^{\prime} \end{array}\right) \end{gathered}$	$A_q^{(0,1)}=\left(\begin{array}{cc} 0 & X \\ X^{\prime} & 0 \end{array}\right) \otimes 1_3 \quad A_1^{(0,1)}=\left(\begin{array}{cc} 0 & Y \\ Y^{\prime} & 0 \end{array}\right)$ $X=\left(\begin{array}{cc} \Upsilon_u^* \varphi_1 & \Upsilon_u^* \varphi_2 \\ -\Upsilon_d^* \bar{\varphi}_2 & \Upsilon_d^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad X^{\prime}=\left(\begin{array}{cc} \Upsilon_u \varphi_1^{\prime} & \Upsilon_d \varphi_2^{\prime} \\ -\Upsilon_u \bar{\varphi}_2^{\prime} & \Upsilon_d \bar{\varphi}_1^{\prime} \end{array}\right)$ $Y=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1 & \Upsilon_{\nu}^* \varphi_2 \\ -\Upsilon_e^* \bar{\varphi}_2 & \Upsilon_e^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu} \varphi_1^{\prime} & \Upsilon_e \varphi_2^{\prime} \\ -\Upsilon_{\nu} \bar{\varphi}_2^{\prime} & \Upsilon_e \bar{\varphi}_1^{\prime} \end{array}\right)$	$A_q^{(0,1)}=\left(\begin{array}{cc} 0 & X \\ X^{\prime} & 0 \end{array}\right) \otimes 1_3 \quad A_1^{(0,1)}=\left(\begin{array}{cc} 0 & Y \\ Y^{\prime} & 0 \end{array}\right)$ $X=\left(\begin{array}{cc} \Upsilon_u^* \varphi_1 & \Upsilon_u^* \varphi_2 \\ -\Upsilon_d^* \bar{\varphi}_2 & \Upsilon_d^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad X^{\prime}=\left(\begin{array}{cc} \Upsilon_u \varphi_1^{\prime} & \Upsilon_d \varphi_2^{\prime} \\ -\Upsilon_u \bar{\varphi}_2^{\prime} & \Upsilon_d \bar{\varphi}_1^{\prime} \end{array}\right)$ $Y=\left(\begin{array}{cc} \Upsilon_{\nu}^* \varphi_1 & \Upsilon_{\nu}^* \varphi_2 \\ -\Upsilon_e^* \bar{\varphi}_2 & \Upsilon_e^* \bar{\varphi}_1 \end{array}\right) \quad \text{and} \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu} \varphi_1^{\prime} & \Upsilon_e \varphi_2^{\prime} \\ -\Upsilon_{\nu} \bar{\varphi}_2^{\prime} & \Upsilon_e \bar{\varphi}_1^{\prime} \end{array}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0326 (13)	161	0.613 N +1	$\begin{gathered} \eta_{ac}\omega^c{}_b = -\eta_{bc}\omega^c{}_a \\ \omega^c{}_b = \epsilon_{b_1 a_2 \dots a_D} \omega^{b_1}{}_{a_1} + \epsilon_{a_1 b_2 \dots a_D} \omega^{b_2}{}_{a_2} + \dots + \epsilon_{a_1 a_2 \dots b_D} \omega^{b_D}{}_{a_D} = 0. \end{gathered}$	$\eta_{ac}\omega^c{}_b = -\eta_{bc}\omega^c{}_a$ $\epsilon_{b_1 a_2 \dots a_D} \omega^{b_1}{}_{a_1} + \epsilon_{a_1 b_2 \dots a_D} \omega^{b_2}{}_{a_2} + \dots + \epsilon_{a_1 a_2 \dots b_D} \omega^{b_D}{}_{a_D} = 0.$	$\eta_{ac}\omega^c{}_b = -\eta_{bc}\omega^c{}_a,$ $\epsilon_{b_1 a_2 \dots a_D} \omega^{b_1}{}_{a_1} + \epsilon_{a_1 b_2 \dots a_D} \omega^{b_2}{}_{a_2} + \dots + \epsilon_{a_1 a_2 \dots b_D} \omega^{b_D}{}_{a_D} = 0.$
cardona-qft-methods_ EQ1000	374	0.621 W +0	$\begin{aligned} \Delta \left(X^\dagger E \right) &= \Delta \left(X^\dagger \right) \Delta(E) = \left(X^\dagger \otimes 1 + 1 \otimes X^\dagger \right) \Delta(E) \\ &= \left(\left(X^\dagger(\cdot) \otimes \text{Id} + \text{Id} \otimes X^\dagger(\cdot) \right) \circ \Delta \right) (E). \end{aligned}$	$\Delta \left(X^\dagger E \right) = \Delta \left(X^\dagger \right) \Delta(E) = \left(X^\dagger \otimes 1 + 1 \otimes X^\dagger \right) \Delta(E)$ $= \left(\left(X^\dagger(\cdot) \otimes \text{Id} + \text{Id} \otimes X^\dagger(\cdot) \right) \circ \Delta \right) (E).$	$\Delta(X^\dagger E) = \Delta(X^\dagger)\Delta(E) = (X^\dagger \otimes 1 + 1 \otimes X^\dagger)\Delta(E)$ $= ((X^\dagger(\cdot) \otimes \text{Id} + \text{Id} \otimes X^\dagger(\cdot)) \circ \Delta)(E).$
cardona-qft-methods_ EQ0041 (11)	032	0.623 K +0	$c\left(\omega_1\right)c\left(\omega_2\right)+c\left(\omega_2\right)c\left(\omega_1\right)=2g\left(\omega_1,\omega_2\right)\text{id}_E.$	$c\left(\omega_1\right)c\left(\omega_2\right)+c\left(\omega_2\right)c\left(\omega_1\right)=2g\left(\omega_1,\omega_2\right)\text{id}_E.$	$c(\omega_1)c(\omega_2)+c(\omega_2)c(\omega_1)=2g(\omega_1,\omega_2)\text{id}_E.$
cardona-qft-methods_ EQ0916	351	0.638 N +0	$C^*(X) \stackrel{L}{\otimes}_{C^*(X)^e} C^*(X) \# G \stackrel{\sim}{\rightarrow} C^*(P_G X)$	$C^*(X) \stackrel{L}{\otimes}_{C^*(X)^e} C^*(X) \# G \stackrel{\sim}{\rightarrow} C^*(P_G X)$	$C^*(X) \stackrel{L}{\otimes}_{C^*(X)^e} C^*(X) \# G \stackrel{\sim}{\rightarrow} C^*(P_G X)$
cardona-qft-methods_ EQ0217 (4)	099	0.641 K +0	$c(\xi+v(x)):\pi^*E_{x,\xi}^+\rightarrow\pi^*E_{x,\xi}^-.$	$c(\xi+v(x)):\pi^*E_{x,\xi}^+\rightarrow\pi^*E_{x,\xi}^-.$	$c(\xi+v(x)):\pi^*E_{x,\xi}^+\rightarrow\pi^*E_{x,\xi}^-.$
cardona-qft-methods_ EQ0214 (2)	099	0.644 N +0	$v(x):=\left.\frac{d}{dt}\right _{t=0}\exp\left(t\mathbf{v}(x)\right)\cdot x.$	$v(x):=\left.\frac{d}{dt}\right _{t=0}\exp\left(t\mathbf{v}(x)\right)\cdot x.$	$v(x):=\left.\frac{d}{dt}\right _{t=0}\exp\left(t\mathbf{v}(x)\right)\cdot x.$
cardona-qft-methods_ EQ0339 (7)	166	0.644	$\Omega_n\int_{M^{2n}}\mathbf{E}_{2n}\in\mathbb{Z},\quad\tilde{\Omega}_k\int_{M^{4k}}\mathbf{P}_{4k}\in\mathbb{Z},$	$\Omega_n\int_{M^{2n}}\mathbf{E}_{2n}\in\mathbb{Z},\quad\tilde{\Omega}_k\int_{M^{4k}}\mathbf{P}_{4k}\in\mathbb{Z},$	$\Omega_n\int_{M^{2n}}\mathbf{E}_{2n}\in\mathbb{Z},\quad\tilde{\Omega}_k\int_{M^{4k}}\mathbf{P}_{4k}\in\mathbb{Z},$
cardona-qft-methods_ EQ0846 (2)	329	0.651 N +1	$\begin{aligned} &\frac{\delta\mathcal{L}}{\delta X}=dX+(\Pi^\#\circ X)\eta=0 \\ &\frac{\delta\mathcal{L}}{\delta\eta}=d\eta+\frac{1}{2}\langle(\partial\Pi^\#\circ X)\eta,\eta\rangle=0 \end{aligned}$	$\frac{\delta\mathcal{L}}{\delta X}=dX+(\Pi^\#\circ X)\eta=0$ $\frac{\delta\mathcal{L}}{\delta\eta}=d\eta+\frac{1}{2}\langle(\partial\Pi^\#\circ X)\eta,\eta\rangle=0$	$\frac{\delta\mathcal{L}}{\delta X}=dX+(\Pi^\#\circ X)\eta=0$ $\frac{\delta\mathcal{L}}{\delta\eta}=d\eta+\frac{1}{2}\langle(\partial\Pi^\#\circ X)\eta,\eta\rangle=0.$
cardona-qft-methods_ EQ0104	062	0.660 N +2	$\begin{aligned} &\gamma^{\alpha_1\alpha_2}=\frac{1}{2}(\gamma^{\alpha_1}\gamma^{\alpha_2}-\gamma^{\alpha_2}\gamma^{\alpha_1}) \\ &\Omega_{\alpha_1\alpha_2}=\left[\delta_{\alpha_1}+\tilde{A}_{\alpha_1},\delta_{\alpha_2}+\tilde{A}_{\alpha_2}\right]=L(F_{\alpha_1\alpha_2})-R(F_{\alpha_1\alpha_2}) \\ &F_{\alpha_1\alpha_2}=\delta_{\alpha_1}(A_{\alpha_2})-\delta_{\alpha_2}(A_{\alpha_1})+[A_{\alpha_1},A_{\alpha_2}] \end{aligned}$	$\gamma^{\alpha_1\alpha_2}=\frac{1}{2}(\gamma^{\alpha_1}\gamma^{\alpha_2}-\gamma^{\alpha_2}\gamma^{\alpha_1})$ $\Omega_{\alpha_1\alpha_2}=\left[\delta_{\alpha_1}+\tilde{A}_{\alpha_1},\delta_{\alpha_2}+\tilde{A}_{\alpha_2}\right]=L(F_{\alpha_1\alpha_2})-R(F_{\alpha_1\alpha_2})$ $F_{\alpha_1\alpha_2}=\delta_{\alpha_1}(A_{\alpha_2})-\delta_{\alpha_2}(A_{\alpha_1})+[A_{\alpha_1},A_{\alpha_2}]$	$\gamma^{\alpha_1\alpha_2}=\frac{1}{2}(\gamma^{\alpha_1}\gamma^{\alpha_2}-\gamma^{\alpha_2}\gamma^{\alpha_1}),$ $\Omega_{\alpha_1\alpha_2}=[\delta_{\alpha_1}+\tilde{A}_{\alpha_1},\delta_{\alpha_2}+\tilde{A}_{\alpha_2}]=L(F_{\alpha_1\alpha_2})-R(F_{\alpha_1\alpha_2})$ $F_{\alpha_1\alpha_2}=\delta_{\alpha_1}(A_{\alpha_2})-\delta_{\alpha_2}(A_{\alpha_1})+[A_{\alpha_1},A_{\alpha_2}].$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0659 (9)	262	0.662 K +0	$S=T\exp\left(-i\int d^4xH_I(x)\right)$.	$S=T\exp\left(-i\int d^4xH_I(x)\right)$.	$S=T\exp\left(-i\int d^4xH_I(x)\right)$.
cardona-qft-methods_ EQ0896	346	0.664 N +1	$\begin{aligned} &L^M\{e^{v^*}(-TM)\}\wedge L^M\{e^{v^*}(-TM)\}= & (LM\times LM)^{ev*}(-2TM) \\ &\downarrow \\ &Map(8,M)^{ev*}(-TM)\rightarrow LM^{ev*}(-TM) \end{aligned}$	$LM^{ev*}(-TM)\wedge LM^{ev*}(-TM)=(LM\times LM)^{ev*}(-2TM)$ $\downarrow$ $Map(8,M)^{ev*}(-TM)\rightarrow LM^{ev*}(-TM)$	$LM^{ev*}(-TM)\wedge LM^{ev*}(-TM)=(LM\times LM)^{ev*}(-2TM)$ $\downarrow$ $Map(8,M)^{ev*}(-TM)\rightarrow LM^{ev*}(-TM)$
cardona-qft-methods_ EQ0467	217	0.689 W +1	$\begin{gathered} D_{\{A\}^2}=\left(D^{\{1,0\}}\right)^2+1_{\{4\}}\otimes\left(D^{\{0,1\}}\right)^2-\gamma_5\left[D^{\{1,0\}},1_{\{4\}}\right]\otimes D^{\{0,1\}}\right]\backslash\left[D^{\{1,0\}},1_{\{4\}}\right]\otimes D^{\{0,1\}}\right]=\sqrt{-1}\gamma^{\mu}\left[\left(\nabla_{\mu}^s+\mathbb{A}_{\mu}\right),1_{\{4\}}\right]\otimes D^{\{0,1\}}\right]\backslash\left[D^{\{1,0\}},1_{\{4\}}\right]\otimes D^{\{0,1\}}\right]\end{gathered}$	$D_A^2=\left(D^{1,0}\right)^2+1_4\otimes\left(D^{0,1}\right)^2-\gamma_5\left[D^{1,0},1_4\otimes D^{0,1}\right]\left[D^{1,0},1_4\otimes D^{0,1}\right]=\sqrt{-1}\gamma^{\mu}\left[\left(\nabla_{\mu}^s+\mathbb{A}_{\mu}\right),1_4\otimes D^{0,1}\right]$	$D_A^2=\left(D^{1,0}\right)^2+1_4\otimes\left(D^{0,1}\right)^2-\gamma_5\left[D^{1,0},1_4\otimes D^{0,1}\right]$ $\left[D^{1,0},1_4\otimes D^{0,1}\right]=\sqrt{-1}\gamma^{\mu}\left[\left(\nabla_{\mu}^s+\mathbb{A}_{\mu}\right),1_4\otimes D^{0,1}\right]$
cardona-qft-methods_ EQ0808 (8)	307	0.696 N +0	$[X+\zeta,Y+\eta]_C=[X,Y]+\mathcal{L}_X\eta-\mathcal{L}_Y\zeta-\frac{1}{2}\,\mathrm{d}\left(\iota_X\eta-\iota_Y\zeta\right)$.	$[X+\zeta,Y+\eta]_C=[X,Y]+\mathcal{L}_X\eta-\mathcal{L}_Y\zeta-\frac{1}{2}\,\mathrm{d}\left(\iota_X\eta-\iota_Y\zeta\right)$.	$[X+\zeta,Y+\eta]_C=[X,Y]+\mathcal{L}_X\eta-\mathcal{L}_Y\zeta-\frac{1}{2}\mathrm{d}(\iota_X\eta-\iota_Y\zeta)$.
cardona-qft-methods_ EQ0330 (17)	162	0.706 W +0	$R^{\{a\ b\}}=\frac{1}{2}e_{\alpha}^ae_{\beta}^bR_{\mu\nu}^{\alpha\beta}dx^{\mu}\wedge dx^{\nu}$.	$R^{ab}=\frac{1}{2}e_{\alpha}^ae_{\beta}^bR_{\mu\nu}^{\alpha\beta}dx^{\mu}\wedge dx^{\nu}$.	$R^{ab}=\frac{1}{2}e_{\alpha}^ae_{\beta}^bR_{\mu\nu}^{\alpha\beta}dx^{\mu}\wedge dx^{\nu}$.
cardona-qft-methods_ EQ0070 (21)	042	0.716 W -1	$a_{\{2\}}\left(\not{D}^{\{2\}}\right)=-\frac{1}{12(4\pi)^{d/2}}\int_Ms(x)dvol_g(x)$.	$a_2\left(\not{D}^2\right)=-\frac{1}{12(4\pi)^{d/2}}\int_Ms(x)dvol_g(x)$.	$a_2(\not{D}^2)=-\frac{1}{12(4\pi)^{d/2}}\int_Ms(x)dvol_g(x)$.
cardona-qft-methods_ EQ0233	109	0.718 K +0	$\chi(Y\cup Z)=\chi(Y)+\chi(Z)-\chi(Y\cap Z)$.	$\chi(Y\cup Z)=\chi(Y)+\chi(Z)-\chi(Y\cap Z)$.	$\chi(Y\cup Z)=\chi(Y)+\chi(Z)-\chi(Y\cap Z)$.
cardona-qft-methods_ EQ0017 (5)	022	0.722	$c_{\{P\}}(x)=\frac{1}{(2\pi)^d}\int_{\mathbb{S}^{d-1}}\mathrm{tr}\left(\sigma_{-d}^P(x,\xi)\right)d\xi$.	$c_P(x)=\frac{1}{(2\pi)^d}\int_{\mathbb{S}^{d-1}}\mathrm{tr}\left(\sigma_{-d}^P(x,\xi)\right)d\xi$.	$c_P(x)=\frac{1}{(2\pi)^d}\int_{\mathbb{S}^{d-1}}tr\left(\sigma_{-d}^P(x,\xi)\right)d\xi$.
cardona-qft-methods_ EQ0043	032	0.725 N +0	$\sigma_1^d(x,\xi)=\lim_{t\rightarrow\infty}\frac{1}{t}\left(e^{-ith(x)}de^{ith(x)}\right)(x)=\lim_{t\rightarrow\infty}\frac{1}{t}itd_xh=id_xh=i\xi,$	$\sigma_1^d(x,\xi)=\lim_{t\rightarrow\infty}\frac{1}{t}\left(e^{-ith(x)}de^{ith(x)}\right)(x)=\lim_{t\rightarrow\infty}\frac{1}{t}itd_xh=id_xh=i\xi,$	$\sigma_1^d(x,\xi)=\lim_{t\rightarrow\infty}\frac{1}{t}\left(e^{-ith(x)}de^{ith(x)}\right)(x)=\lim_{t\rightarrow\infty}\frac{1}{t}itd_xh=id_xh=i\xi,$
cardona-qft-methods_ EQ0227	102	0.728 N +0	$e^{\{i\ t\}}\cdot(z,\nu):=\left(e^{\{i\ t\}}z,\nu\right),\quad z\in M,\,t,\,\nu\in\mathbb{R}.$	$e^{it}\cdot(z,\nu):=\left(e^{it}z,\nu\right),\quad z\in M,\,t,\,\nu\in\mathbb{R}.$	$e^{it}\cdot(z,\nu)\, :=\, (e^{it}z,\nu),\qquad z\in M,\,t,\,\nu\in\mathbb{R}.$
cardona-qft-methods_ EQ0855	331	0.737 W +1	$G\times_{(s,t)}G\overset{\mu}{\rightarrow}G\overset{i}{\rightarrow}G\overset{s}{\overset{\leftarrow}{\rightleftarrows}}_tM$	$G\times_{(s,t)}G\overset{\mu}{\rightarrow}G\overset{i}{\rightarrow}G\overset{s}{\overset{\leftarrow}{\rightleftarrows}}_tM$	$G\times_{(s,t)}G\overset{\mu}{\longrightarrow}G\overset{i}{\longrightarrow}G\overset{s}{\overset{\leftarrow}{\rightleftarrows}}_tM$
cardona-qft-methods_ EQ0794 (35)	301	0.738 N +0	$\begin{aligned} &\text{ type IIB R - R : } & \mathbf{8}_s\otimes\mathbf{8}_s=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_+=C_0\oplus C_2\oplus C_{4+} \\ &\text{ type IIA R - R : } & \mathbf{8}_s\otimes\mathbf{8}_c=\mathbf{8}_v\oplus\mathbf{56}_t=C_1\oplus C_3. \end{aligned}$	$\begin{aligned} &\text{ type IIB R - R : } & \mathbf{8}_s\otimes\mathbf{8}_s=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_+=C_0\oplus C_2\oplus C_{4+} \\ &\text{ type IIA R - R : } & \mathbf{8}_s\otimes\mathbf{8}_c=\mathbf{8}_v\oplus\mathbf{56}_t=C_1\oplus C_3. \end{aligned}$	$\begin{aligned} &\text{ type IIB R - R : } & \mathbf{8}_s\otimes\mathbf{8}_s=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_+=C_0\oplus C_2\oplus C_{4+} \\ &\text{ type IIA R - R : } & \mathbf{8}_s\otimes\mathbf{8}_c=\mathbf{8}_v\oplus\mathbf{56}_t=C_1\oplus C_3. \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0069	042	■0.751 ● N +1	$\operatorname{Tr}_{\operatorname{Dix}}\left(D^{-d}\right)=\operatorname{Tr}_{\omega}\left(D^{-d}\right)=\frac{a_0\left(D^2\right)}{\Gamma(d / 2+1)}=\frac{1}{d} W \operatorname{Res}\left(D^{-d}\right) .$	$\operatorname{Tr}_{D i x}\left(D^{-d}\right)=\operatorname{Tr}_{\omega}\left(D^{-d}\right)=\frac{a_0\left(D^2\right)}{\Gamma(d / 2+1)}=\frac{1}{d} W \operatorname{Res}\left(D^{-d}\right) .$	
cardona-qft-methods_ EQ0087	269	■0.753 ● W +0	$\begin{array}{ll} S^{+} \uparrow\rangle=\uparrow \downarrow, & S^{-} \uparrow\rangle= \downarrow\rangle, \\ S^{+} \downarrow\rangle= \uparrow\rangle, & S^{-} \downarrow\rangle=0 . \end{array}$	$\begin{array}{ll} S^{+} \uparrow\rangle=0, & S^{-} \uparrow\rangle= \downarrow\rangle, \\ S^{+} \downarrow\rangle= \uparrow\rangle, & S^{-} \downarrow\rangle=0 . \end{array}$	
cardona-qft-methods_ EQ0282	137	■0.755 ● K +0	$\underline{t} \in Y_G \Longleftrightarrow \Psi_G(\underline{t}) \neq 0 \Longleftrightarrow \Psi_{G / e}(\underline{t}') \neq-t_e \Psi_{G \setminus e}(\underline{t}') ,$	$\underline{t} \in Y_G \Longleftrightarrow \Psi_G(\underline{t}) \neq 0 \Longleftrightarrow \Psi_{G / e}(\underline{t}') \neq-t_e \Psi_{G \setminus e}(\underline{t}') \quad ,$	
cardona-qft-methods_ EQ0652 (2)	261	■0.755 ● N +0	$\begin{gathered} \exp \left(-\beta V_{S T A T}\left(x_1, x_2\right)\right)=\left\langle x_1, x_2\left e^{-\beta H}\right x_1, x_2\right\rangle, \\ H=\frac{1}{2 m}\left(p_1^2+p_2^2\right) . \end{gathered}$	$\begin{array}{l} \exp \left(-\beta V_{S T A T}\left(x_1, x_2\right)\right)=\left\langle x_1, x_2\left e^{-\beta H}\right x_1, x_2\right\rangle , \\ H=\frac{1}{2 m}\left(p_1^2+p_2^2\right) . \end{array}$	
cardona-qft-methods_ EQ0236 (2)	113	■0.763 ● N +0	$[X]=a_0+a_1 \mathbb{L}+\cdots+a_r \mathbb{L}^r,$	$[X]=a_0+a_1 \mathbb{L}+\cdots+a_r \mathbb{L}^r \quad ,$	
cardona-qft-methods_ EQ0883	343	■0.769 ● N +0	$\begin{gathered} H^{\{n-p\}}(M) \times H^{\{p\}}(M) \rightarrow \mathbf{k} \\ (\alpha, \beta) \rightarrow\langle\alpha \cup \beta,[M]\rangle \end{gathered}$	$\begin{array}{l} H^{n-p}(M) \times H^p(M) \rightarrow \mathbf{k} \\ (\alpha, \beta) \rightarrow\langle\alpha \cup \beta,[M]\rangle \end{array}$	
cardona-qft-methods_ EQ0181 (7)	092	■0.772 ● K +0	$\mathrm{t}-\operatorname{Ind}_{\mathcal{G}}\left(T_{\mathcal{G}}^* M\right) \longrightarrow \widehat{R}(G)$	$\mathrm{t}\text{-}\operatorname{Ind}_G: K_G\left(T_G^* M\right) \longrightarrow \widehat{R}(G)$	
cardona-qft-methods_ EQ0408	186	■0.772 ●	$\Delta B\left(\phi^r\right _{\partial M})+\int_{\partial M} \Theta(\phi, \delta \phi)=0 .$	$\delta B\left(\phi^r\right _{\partial M})+\int_{\partial M} \Theta(\phi, \delta \phi)=0 .$	
cardona-qft-methods_ EQ0022	025	■0.776 ● N -2	$W \operatorname{Res}\left(P_1 P_2\right)=\operatorname{Res} \operatorname{Tr}\left(P_2\left \mathcal{D}\right ^{-s} P_1\right)=\operatorname{Res} \operatorname{Tr}\left(P_2 P_1\left \mathcal{D}\right ^{-s}\right)=W \operatorname{Res}\left(P_2 P_1\right)$	$W \operatorname{Res}\left(P_1 P_2\right)=\operatorname{Res}_{s=0} \operatorname{Tr}\left(P_2\left \mathcal{D}\right ^{-s} P_1\right)=\operatorname{Res}_{s=0} \operatorname{Tr}\left(P_2 P_1\left \mathcal{D}\right ^{-s}\right)=W \operatorname{Res}\left(P_2 P_1\right)$	
cardona-qft-methods_ EQ0928	353	■0.782 ● N +1	$H H^*\left(C^*(M) \# G, C^*(M) \# G\right) \cong H_*^{\text {pro }}\left(\mathbb{L}\left(M \times_G E G\right)^{-T\left(M \times_G E G\right)}\right) .$	$H H^*\left(C^*(M) \# G, C^*(M) \# G\right) \cong H_*^{\text {pro }}\left(\mathbb{L}\left(M \times_G E G\right)^{-T\left(M \times_G E G\right)}\right) .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00435	208	0.790 S +1	$\begin{array}{r} \mathbf{2}_L \otimes \mathbf{1}^0 \otimes \mathbf{2}_R \otimes \mathbf{1}^0 \otimes \mathbf{2}_L \otimes \mathbf{3}^0 \otimes \mathbf{2}_R \otimes \mathbf{3}^0 \\ \oplus \mathbf{1} \otimes \mathbf{2}_L^0 \oplus \mathbf{1} \otimes \mathbf{2}_R^0 \oplus \mathbf{3} \otimes \mathbf{2}_L^0 \oplus \mathbf{3} \otimes \mathbf{2}_R^0 \end{array}$	$\mathbf{2}_L \otimes \mathbf{1}^0 \oplus \mathbf{2}_R \otimes \mathbf{1}^0 \oplus \mathbf{2}_L \otimes \mathbf{3}^0 \oplus \mathbf{2}_R \otimes \mathbf{3}^0$	$\mathbf{2}_L \otimes \mathbf{1}^0 \oplus \mathbf{2}_R \otimes \mathbf{1}^0 \oplus \mathbf{2}_L \otimes \mathbf{3}^0 \oplus \mathbf{2}_R \otimes \mathbf{3}^0$
cardona-qft-methods_ E00838 (17)	317	0.790 N +0	$F^{(10)} = F + \operatorname{vol}_4 \wedge \lambda(*F),$	$F^{(10)} = F + \operatorname{vol}_4 \wedge \lambda(*F),$	$F^{(10)} = F + \operatorname{vol}_4 \wedge \lambda(*F) ,$
cardona-qft-methods_ E00959 (11)	361	0.793 K +0	$\mathcal{C}: X_{\Gamma} \backslash \Sigma_n \rightarrow X_{\Gamma^{\vee}} \backslash \Sigma_n,$	$\mathcal{C}: X_{\Gamma} \backslash \Sigma_n \rightarrow X_{\Gamma^{\vee}} \backslash \Sigma_n,$	$\mathcal{C}: X_{\Gamma} \backslash \Sigma_n \rightarrow X_{\Gamma^{\vee}} \backslash \Sigma_n ,$
cardona-qft-methods_ E00359 (30)	171	0.794 C +3	$\begin{aligned} & \mathbf{d} \, \mathbf{C}_{\{3\}}^{\{L \, o \, r\}} = \mathbf{R}^{\{a \, b\}} \, \mathbf{R}_{\{a \, b\}} = \\ & \mathbf{P}_{\{4\}} \, \, \& \, \mathbf{d} \, \mathbf{C}_{\{3\}}^{\{T \, o \, r\}} = \mathbf{T}^{\{a\}} \, \mathbf{T}_{\{a\}} - \mathbf{e}^{\{a\}} \, \mathbf{e}^{\{b\}} \, \mathbf{R}_{\{a \, b\}} = \mathbf{N}_{\{4\}} \end{aligned}$	$\begin{aligned} dC_3^{Lor} &= R^{ab} R_{ab} = \mathbf{P}_4 \\ dC_3^{Tor} &= T^a T_a - e^a e^b R_{ab} = \mathbf{N}_4 \end{aligned}$	$\begin{aligned} dC_3^{Lor} &= R^{ab} R_{ab} = \mathbf{P}_4, \\ dC_3^{Tor} &= T^a T_a - e^a e^b R_{ab} = \mathbf{N}_4. \end{aligned}$
cardona-qft-methods_ E00818 (18)	309	0.794 S +2	$\begin{aligned} & 6 \rightarrow 3 + \overline{3} \, \, \& \, 15 \rightarrow 8 + 3 + \overline{3} + 1 \, \, \& \, 10 + 10^* \rightarrow 6 + \overline{6} + 3 + \overline{3} + 1 + 1 . \end{aligned}$	$\begin{aligned} & 6 \rightarrow 3 + \overline{3} \\ & 15 \rightarrow 8 + 3 + \overline{3} + 1 \\ & 10 + 10^* \rightarrow 6 + \overline{6} + 3 + \overline{3} + 1 + 1. \end{aligned}$	$\begin{aligned} & \mathbf{6} \rightarrow \mathbf{3} + \overline{\mathbf{3}} , \\ & \mathbf{15} \rightarrow \mathbf{8} + \mathbf{3} + \overline{\mathbf{3}} + \mathbf{1} , \\ & \mathbf{10} + \mathbf{10}^* \rightarrow \mathbf{6} + \overline{\mathbf{6}} + \mathbf{3} + \overline{\mathbf{3}} + \mathbf{1} + \mathbf{1} . \end{aligned}$
cardona-qft-methods_ E00906	349	0.798 N +0	$\begin{aligned} & G \times P_G X \rightarrow P_G X \\ & ((f,k),g) \mapsto (fg,g^{-1}kg) . \end{aligned}$	$\begin{aligned} G \times P_G X &\rightarrow P_G X \\ ((f,k),g) &\mapsto (fg,g^{-1}kg) . \end{aligned}$	$\begin{aligned} G \times P_G X &\rightarrow P_G X \\ ((f,k),g) &\mapsto (fg,g^{-1}kg). \end{aligned}$
cardona-qft-methods_ E00362 (2)	174	0.805 W +0	$\begin{aligned} & \Delta \, \omega_b^a = d \lambda_b^a + \omega_c^a \lambda_b^c - \omega_b^c \lambda_c^a \\ & = D_\omega \lambda_b^a, \end{aligned}$	$\begin{aligned} \delta \omega_b^a &= d \lambda_b^a + \omega_c^a \lambda_b^c - \omega_b^c \lambda_c^a \\ &= D_\omega \lambda_b^a, \end{aligned}$	$\begin{aligned} \delta \omega_b^a &= d \lambda_b^a + \omega_c^a \lambda_b^c - \omega_b^c \lambda_c^a \\ &= D_\omega \lambda_b^a, \end{aligned}$
cardona-qft-methods_ E00146	082	0.808 W +0	$\sigma_L(\Delta)(\xi_1,\xi_2,,\dots,\xi_n) = \xi_1^2 + \xi_2^2 + \dots + \xi_n^2,$	$\sigma_L(\Delta)(\xi_1,\xi_2,,\dots,\xi_n) = \xi_1^2 + \xi_2^2 + \dots + \xi_n^2,$	$\sigma_L(\Delta)(\xi_1,\xi_2,,\dots,\xi_n) = \xi_1^2 + \xi_2^2 + \dots + \xi_n^2,$
cardona-qft-methods_ E00177 (4)	090	0.808 S +1	$T_{\mathcal{G}}^*M := \{ \xi \in T^*M \mid \xi \text{ is perpendicular to the orbits of } \mathcal{G} \} .$	$T_G^*M := \{ \xi \in T^*M \mid \xi \text{ is perpendicular to the orbits of } G \} .$	$T_G^*M := \{ \xi \in T^*M \mid \xi \text{ is perpendicular to the orbits of } G \} .$
cardona-qft-methods_ E00059	039	0.810	$\begin{aligned} & k_t(x,y) \, \& \, = \left\langle x, e^{t\Delta} y \right\rangle = \sum_{n,m=1}^{\infty} \left\langle x, \psi_m \right\rangle \left\langle \psi_m, e^{t\Delta} \psi_n \right\rangle \left\langle \psi_n, y \right\rangle \\ & = \sum_{n,m=1}^{\infty} \overline{\psi_n(x)} \psi_n(y) e^{t\lambda_n} . \end{aligned}$	$\begin{aligned} k_t(x,y) &= \langle x, e^{t\Delta} y \rangle = \sum_{n,m=1}^{\infty} \langle x, \psi_m \rangle \langle \psi_m, e^{t\Delta} \psi_n \rangle \langle \psi_n, y \rangle \\ &= \sum_{n,m=1}^{\infty} \overline{\psi_n(x)} \psi_n(y) e^{t\lambda_n} . \end{aligned}$	$\begin{aligned} k_t(x,y) &= \langle x, e^{t\Delta} y \rangle = \sum_{n,m=1}^{\infty} \langle x, \psi_m \rangle \langle \psi_m, e^{t\Delta} \psi_n \rangle \langle \psi_n, y \rangle \\ &= \sum_{n,m=1}^{\infty} \overline{\psi_n(x)} \psi_n(y) e^{t\lambda_n} . \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0254	119	0.811 K +0	$P_{\{G\}}(\underline{t}, p) = \sum_{\{C\}} s_{\{C\}} \prod_{e \in C} t_e,$	$P_G(\underline{t}, p) = \sum_C s_C \prod_{e \in C} t_e,$	$P_G(\underline{t}, p) = \sum_C s_C \prod_{e \in C} t_e \quad ,$
cardona-qft-methods_ EQ0758	286	0.813 S +0	$\begin{aligned} &\text{string energy} \quad \& \\ &\longleftrightarrow \text{scaling dimension} \quad \& \\ &E(\lambda) = \Delta(\lambda). \end{aligned}$	string energy $\longleftrightarrow$ scaling dimension $E(\lambda) = \Delta(\lambda).$	string energy $\longleftrightarrow$ scaling dimension $E(\lambda) = \Delta(\lambda).$
cardona-qft-methods_ EQ0185 (1)	093	0.816 K +0	$c(v)^2 = - v ^2 \text{ Id. }$	$c(v)^2 = - v ^2 \text{ Id.}$	$c(v)^2 = - v ^2 \text{ Id.}$
cardona-qft-methods_ EQ0615 (23)	255	0.817 K +0	$\begin{gathered} S_{\{N\}} = \langle \tau_{i,i+1} i \in \{1, 2, \dots, N-1\}, \tau_{i,i+1}^2 = \mathbb{I}, \\ \tau_{i,i+1} \tau_{i+1,i+2} \tau_{i,i+1} = \tau_{i+1,i+2} \tau_{i,i+1} \tau_{i+1,i+2} \rangle \end{gathered}$	$S_N = \langle \tau_{i,i+1} i \in \{1, 2, \dots, N-1\}, \tau_{i,i+1}^2 = \mathbb{I}, \\ \tau_{i,i+1} \tau_{i+1,i+2} \tau_{i,i+1} = \tau_{i+1,i+2} \tau_{i,i+1} \tau_{i+1,i+2} \rangle$	$S_N = \langle \tau_{i,i+1} i \in \{1, 2, \dots, N-1\}, \tau_{i,i+1}^2 = \mathbb{I}, \\ \tau_{i,i+1} \tau_{i+1,i+2} \tau_{i,i+1} = \tau_{i+1,i+2} \tau_{i,i+1} \tau_{i+1,i+2} \rangle$
cardona-qft-methods_ EQ0320 (6)	160	0.820 W +0	$\begin{aligned} &\phi_{\parallel}(x) \equiv \phi^a(x+dx) + dx^\mu \omega_{b\mu}^a(x) \phi^b(x) \\ &= \phi^a(x) + dx^\mu [\partial_\mu \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x)]. \end{aligned}$	$\begin{aligned} \phi_{\parallel}^a(x) &\equiv \phi^a(x+dx) + dx^\mu \omega_{b\mu}^a(x) \phi^b(x) \\ &= \phi^a(x) + dx^\mu [\partial_\mu \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x)]. \end{aligned}$	$\begin{aligned} \phi_{\parallel}^a(x) &\equiv \phi^a(x+dx) + dx^\mu \omega_{b\mu}^a(x) \phi^b(x) \\ &= \phi^a(x) + dx^\mu [\partial_\mu \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x)]. \end{aligned}$
cardona-qft-methods_ EQ0650 (57)	260	0.831 N +0	$\varphi_{\theta}^* = \exp \frac{i}{2} \partial \wedge P \varphi_0,$	$\varphi_{\theta}^* = \exp \frac{i}{2} \partial \wedge P \varphi_0,$	$\varphi_{\theta}^* = \exp \frac{i}{2} \partial \wedge P \varphi_0,$
cardona-qft-methods_ EQ0029	028	0.834 N +0	$\operatorname{Tr}_{Dix}(P) = \frac{1}{d} WRes(P) = \frac{1}{d} \int_M \int_{S^*M} \sigma_{-d}^P(x, \xi) d\xi dx $	$\operatorname{Tr}_{Dix}(P) = \frac{1}{d} WRes(P) = \frac{1}{d} \int_M \int_{S^*M} \sigma_{-d}^P(x, \xi) d\xi dx $	$\operatorname{Tr}_{Dix}(P) = \frac{1}{d} WRes(P) = \frac{1}{d} \int_M \int_{S^*M} \sigma_{-d}^P(x, \xi) d\xi dx $
cardona-qft-methods_ EQ0519 (5)	237	0.835 N +1	$f * g(x) = f e^{i/2 \overleftarrow{\delta}_\mu \theta^{\mu\nu} \overrightarrow{\partial}_\nu} g$	$f * g(x) = f e^{i/2 \overleftarrow{\delta}_\mu \theta^{\mu\nu} \overrightarrow{\partial}_\nu} g$	$f * g(x) = f e^{i/2 \overleftarrow{\delta}_\mu \theta^{\mu\nu} \overrightarrow{\partial}_\nu} g.$
cardona-qft-methods_ EQ0613 (21)	254	0.835 N +1	$\begin{gathered} \int \prod_i d\mu(p_i) c_{\Delta p_1}^\dagger c_{\Delta p_2}^\dagger \dots c_{\Delta p_N}^\dagger 0\rangle e_{\Delta p_1} \otimes e_{\Delta p_1} \otimes \dots \otimes e_{\Delta p_N} = \\ \int \prod_i d\mu(p_i) c_{p_1}^\dagger c_{p_2}^\dagger \dots c_{p_N}^\dagger 0\rangle e_{p_1} \otimes e_{p_1} \otimes \dots \otimes e_{p_N}, \end{gathered}$	$\int \prod_i d\mu(p_i) c_{\Delta p_1}^\dagger c_{\Delta p_2}^\dagger \dots c_{\Delta p_N}^\dagger 0\rangle e_{\Delta p_1} \otimes e_{\Delta p_1} \otimes \dots \otimes e_{\Delta p_N} = \\ \int \prod_i d\mu(p_i) c_{p_1}^\dagger c_{p_2}^\dagger \dots c_{p_N}^\dagger 0\rangle e_{p_1} \otimes e_{p_1} \otimes \dots \otimes e_{p_N},$	$\int \prod_i d\mu(p_i) c_{\Delta p_1}^\dagger c_{\Delta p_2}^\dagger \dots c_{\Delta p_N}^\dagger 0\rangle e_{\Delta p_1} \otimes e_{\Delta p_1} \otimes \dots \otimes e_{\Delta p_N} = \\ \int \prod_i d\mu(p_i) c_{p_1}^\dagger c_{p_2}^\dagger \dots c_{p_N}^\dagger 0\rangle e_{p_1} \otimes e_{p_1} \otimes \dots \otimes e_{p_N},$
cardona-qft-methods_ EQ0512 (5)	230	0.840 N +0	$\partial_t x_j = \beta_j^{\text{SM}} + \beta_j^{\text{grav}},$	$\partial_t x_j = \beta_j^{\text{SM}} + \beta_j^{\text{grav}},$	$\partial_t x_j = \beta_j^{\text{SM}} + \beta_j^{\text{grav}},$
cardona-qft-methods_ EQ0046	035	0.840	$\mathcal{C} \ell_x M = \left\{ \begin{array}{l} M_{2^m}(\mathbb{C}) \text{ when } d=2m \text{ is even,} \\ M_{2^m}(\mathbb{C}) \oplus M_{2^m}(\mathbb{C}) \text{ when } d=2m+1. \end{array} \right.$	$\mathcal{C} \ell_x M = \left\{ \begin{array}{l} M_{2^m}(\mathbb{C}) \text{ when } d=2m \text{ is even,} \\ M_{2^m}(\mathbb{C}) \oplus M_{2^m}(\mathbb{C}) \text{ when } d=2m+1. \end{array} \right.$	$\mathcal{C} \ell_x M = \left\{ \begin{array}{l} M_{2^m}(\mathbb{C}) \text{ when } d=2m \text{ is even,} \\ M_{2^m}(\mathbb{C}) \oplus M_{2^m}(\mathbb{C}) \text{ when } d=2m+1. \end{array} \right.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0749	284	■0.842 ●N +1	$\text{Super}\left[A d S_5 \times S^5\right]:=\frac{\widetilde{\text{PSU}}(2,2\mid 4)}{\text{SO}(1,4)\times \text{SO}(5)}$	$\text{Super}\left[AdS_5\times S^5\right]:=\frac{\widetilde{\text{PSU}}(2,2\mid 4)}{\text{SO}(1,4)\times \text{SO}(5)}$	$\text{Super}\Big[AdS_5\times S^5\Big]:=\frac{\widetilde{\text{PSU}}(2,2 4)}{\text{SO}(1,4)\times \text{SO}(5)},$
cardona-qft-methods_ EQ0335 (21)	164	■0.848 ●N +0	$D\left(R^{\{a\}}\}_{\{b\}}\wedge\phi^b\right)=R^{\{a\}}\}_{\{b\}}\wedge D\phi^b.$	$D\left(R^a_b\phi^b\right)=R^a_b\wedge D\phi^b.$	$D(R^a_b\phi^b)=R^a_b\wedge D\phi^b.$
cardona-qft-methods_ EQ0898	347	■0.850 ●W +0	$\begin{gathered} A\stackrel{\text{L}}{\stackrel{\text{L}}{\stackrel{\text{L}}{\times}}A^{\{e\}}}M=B(A)\stackrel{\text{L}}{\stackrel{\text{L}}{\stackrel{\text{L}}{\times}}M} \\ \mathcal{RH}\mathbb{I}_{A^e}(A,M)=\text{Hom}_{A^e}(B(A),M) \end{gathered}$	$A\stackrel{\text{L}}{\otimes}_{A^e}M=B(A)\stackrel{\text{L}}{\otimes}_{A^e}M$ $\mathcal{RH}\mathbb{I}_{A^e}(A,M)=\text{Hom}_{A^e}(B(A),M)$	$A\stackrel{\text{L}}{\otimes}_{A^e}M=B(A)\stackrel{\text{L}}{\otimes}_{A^e}M$ $\mathcal{RH}om_{A^e}(A,M)=Hom_{A^e}(B(A),M)$
cardona-qft-methods_ EQ0100	058	■0.850 ●W -2	$\text{Tr}\left(e^{-tD^2}\right)=\frac{1}{t^2}\left(\frac{2}{3}+\frac{2}{3}t+\sum_{k=0}^na_kt^{k+2}+\mathcal{O}\left(t^{n+3}\right)\right),$	$\text{Tr}\left(e^{-tD^2}\right)=\frac{1}{t^2}\left(\frac{2}{3}+\frac{2}{3}t+\sum_{k=0}^na_kt^{k+2}+\mathcal{O}\left(t^{n+3}\right)\right),$	$\text{Tr}(e^{-tD^2})=\frac{1}{t^2}\left(\frac{2}{3}+\frac{2}{3}t+\sum_{k=0}^na_kt^{k+2}+\mathcal{O}(t^{n+3})\right),$
cardona-qft-methods_ EQ0957 (9)	361	■0.850 ●N +0	$\mathcal{C}\left(X_\Gamma\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)\right)=X_{\Gamma^\vee}\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)$	$\mathcal{C}\left(X_\Gamma\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)\right)=X_{\Gamma^\vee}\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)$	$\mathcal{C}\left(X_\Gamma\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)\right)=X_{\Gamma^\vee}\cap\left(\mathbb{P}_k^{n-1}\backslash\Sigma_n\right)$
cardona-qft-methods_ EQ0284	137	■0.856 ●N +1	$\Psi_{G_{2e}}=t_e t_{e'}\Psi_{G\backslash e}+(t_e+t_{e'})\Psi_{G/e}.$	$\Psi_{G_{2e}}=t_e t_{e'}\Psi_{G\backslash e}+(t_e+t_{e'})\Psi_{G/e}.$	$\Psi_{G_{2e}}=t_e t_{e'}\Psi_{G\backslash e}+(t_e+t_{e'})\Psi_{G/e}.$
cardona-qft-methods_ EQ0337 (23)	164	■0.856 ●N +0	$e^a\equiv e_\mu^a(x)dx^\mu\quad\text{and}\quad\omega_b^a\equiv\omega_{b\mu}^a(x)dx^\mu.$	$e^a\equiv e_\mu^a(x)dx^\mu\quad\text{and}\quad\omega_b^a\equiv\omega_{b\mu}^a(x)dx^\mu.$	$e^a\equiv e_\mu^a(x)dx^\mu\quad\text{and}\quad\omega_b^a\equiv\omega_{b\mu}^a(x)dx^\mu.$
cardona-qft-methods_ EQ0037	029	■0.857 ●N +0	$\tau_\lambda(T_1+T_2)-\tau_\lambda(T_1)-\tau_\lambda(T_2)\underset{\lambda\rightarrow\infty}{\simeq}\mathcal{O}\left(\frac{\log(\log\lambda)}{\log\lambda}\right),\text{ when }0\leq T_1,T_2\in\mathcal{L}^{1,\infty}.$	$\tau_\lambda(T_1+T_2)-\tau_\lambda(T_1)-\tau_\lambda(T_2)\underset{\lambda\rightarrow\infty}{\simeq}\mathcal{O}\left(\frac{\log(\log\lambda)}{\log\lambda}\right),\text{ when }0\leq T_1,T_2\in\mathcal{L}^{1,\infty}.$	$\tau_\lambda(T_1+T_2)-\tau_\lambda(T_1)-\tau_\lambda(T_2)\underset{\lambda\rightarrow\infty}{\simeq}\mathcal{O}\left(\frac{\log(\log\lambda)}{\log\lambda}\right),\text{ when }0\leq T_1,T_2\in\mathcal{L}^{1,\infty}.$
cardona-qft-methods_ EQ0068	042	■0.858 ●W +1	$\begin{aligned} &\operatorname{WRes}\left(D^{\{-p\}}\right) \\ &=\frac{2}{\Gamma(p/2)}a_{d-p}\left(D^2\right) \\ &=\frac{2}{(4\pi)^{d/2}\Gamma(p/2)}\int_M\operatorname{tr}\left(k_{(d-p)/2}(x,x)\right)d\operatorname{vol}_g(x) \end{aligned}$	$\operatorname{WRes}\left(D^{-p}\right)=\frac{2}{\Gamma(p/2)}a_{d-p}\left(D^2\right)$ $=\frac{2}{(4\pi)^{d/2}\Gamma(p/2)}\int_M\operatorname{tr}\left(k_{(d-p)/2}(x,x)\right)d\operatorname{vol}_g(x)$	$\operatorname{WRes}\left(D^{-p}\right)=\frac{2}{\Gamma(p/2)}a_{d-p}\left(D^2\right)$ $=\frac{2}{(4\pi)^{d/2}\Gamma(p/2)}\int_M\operatorname{tr}\left(k_{(d-p)/2}(x,x)\right)d\operatorname{vol}_g(x).$
cardona-qft-methods_ EQ0265	125	■0.863 ●N +0	$\langle G\rangle=\text{‘propagator’}\cdot\langle G_1\rangle\cdot\langle G_2\rangle,$	$\langle G\rangle=\text{‘propagator’}\cdot\langle G_1\rangle\cdot\langle G_2\rangle,$	$\langle G\rangle=\text{‘propagator’}\cdot\langle G_1\rangle\cdot\langle G_2\rangle,$
cardona-qft-methods_ EQ0261	123	■0.872 ●K +0	$t_{\{1\}}t_{\{2\}}+t_{\{1\}}t_{\{3\}}+t_{\{1\}}t_{\{5\}}+t_{\{2\}}t_{\{4\}}+t_{\{2\}}t_{\{5\}}+t_{\{3\}}t_{\{4\}}+t_{\{3\}}t_{\{5\}}+t_{\{4\}}t_{\{5\}},$	$t_1t_2+t_1t_3+t_1t_5+t_2t_4+t_2t_5+t_3t_4+t_3t_5+t_4t_5,$	$t_1t_2+t_1t_3+t_1t_5+t_2t_4+t_2t_5+t_3t_4+t_3t_5+t_4t_5,$
cardona-qft-methods_ EQ0967 (7)	362	■0.873 ●W +0	$\left[X_{\Gamma^\vee}\right]=\left[\Sigma_n\right]=\frac{(1+\mathbb{T})^n-1-\mathbb{T}^n}{\mathbb{T}}=\sum_{i=1}^{n-1}\binom{n}{i}\mathbb{T}^{n-1-i}.$	$\left[X_{\Gamma^\vee}\right]=\left[\Sigma_n\right]=\frac{(1+\mathbb{T})^n-1-\mathbb{T}^n}{\mathbb{T}}=\sum_{i=1}^{n-1}\binom{n}{i}\mathbb{T}^{n-1-i}.$	$\left[X_{\Gamma^\vee}\right]=\left[\Sigma_n\right]=\frac{(1+\mathbb{T})^n-1-\mathbb{T}^n}{\mathbb{T}}=\sum_{i=1}^{n-1}\binom{n}{i}\mathbb{T}^{n-1-i}.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08338 (4)	165	0.874	$\begin{aligned} \mathbf{P}_{2k} &=: R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_{a_1} \\ v_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b e^b, \text{ odd } k \\ \tau_k &=: T_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b, \text{ even } k \\ \zeta_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b \\ \mathbf{E}_D &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{D-1} a_D}, \text{ even } D \\ L_p &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{2p-1} a_{2p}} e^{a_{2p+1}} \cdots e^{a_D}. \end{aligned}$	$\begin{aligned} \mathbf{P}_{2k} &=: R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_{a_1} \\ v_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b e^b, \text{ odd } k \\ \tau_k &=: T_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b, \text{ even } k \\ \zeta_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b \\ \mathbf{E}_D &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{D-1} a_D}, \text{ even } D \\ L_p &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{2p-1} a_{2p}} e^{a_{2p+1}} \cdots e^{a_D}. \end{aligned}$	$\begin{aligned} \mathbf{P}_{2k} &=: R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_{a_1} \\ v_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b e^b, \text{ odd } k \\ \tau_k &=: T_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b, \text{ even } k \\ \zeta_k &=: e_{a_1} R^{a_1}_{a_2} R^{a_2}_{a_3} \cdots R^{a_k}_b T^b \\ \mathbf{E}_D &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{D-1} a_D}, \text{ even } D \\ L_p &=: \epsilon_{a_1 a_2 \cdots a_D} R^{a_1 a_2} R^{a_3 a_4} \cdots R^{a_{2p-1} a_{2p}} e^{a_{2p+1}} \cdots e^{a_D}. \end{aligned}$
cardona-qft-methods_ E09915	351	0.874 N +0	$H \, H_* \left((C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Tor}_{\mathbb{Z}G}^* \left(\mathbb{Z}, C^*(X) \otimes_{C^*(X)^e}^L C^*(X) \# G \right) \right)$	$HH_* \left((C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Tor}_{\mathbb{Z}G}^* \left(\mathbb{Z}, C^*(X) \otimes_{C^*(X)^e}^L C^*(X) \# G \right) \right)$	$HH_* \left((C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Tor}_{\mathbb{Z}G}^* \left(\mathbb{Z}, C^*(X) \otimes_{C^*(X)^e}^L C^*(X) \# G \right) \right)$
cardona-qft-methods_ E08870	337	0.875 N +1	$\begin{aligned} \mathcal{G} &= G \\ L &= \{(g_1, g_2, g_3) \mid (g_1, g_2) \in G \times_{(s,t)} G, g_3 = \mu(g_1, g_2)\} \\ I &= g \mapsto g^{-1}, g \in G. \end{aligned}$	$\begin{aligned} \mathcal{G} &= G \\ L &= \{(g_1, g_2, g_3) \mid (g_1, g_2) \in G \times_{(s,t)} G, g_3 = \mu(g_1, g_2)\} \\ I &= g \mapsto g^{-1}, g \in G. \end{aligned}$	$\begin{aligned} \mathcal{G} &= G. \\ L &= \{(g_1, g_2, g_3) \mid (g_1, g_2) \in G \times_{(s,t)} G, g_3 = \mu(g_1, g_2)\}. \\ I &= g \mapsto g^{-1}, g \in G. \end{aligned}$
cardona-qft-methods_ E08670	264	0.875 N +1	$\begin{aligned} &\left\langle a_{lm} a_{l'm'}^* \right\rangle_{\theta} \\ &= * \pi^2 \int dk \sum_{l''=0, l'': \text{even}}^{\infty} i^{l+l'} (-1)^{l+m} (2l''+1) k^2 \Delta_l(k) \Delta_{l'}'(k) P_{\Phi}(k) i_{l''}(\theta k H) \\ &\quad \times \sqrt{(2l+1)(2l'+1)} \begin{bmatrix} l & l' & l'' \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} l & l' & l'' \\ -m & m' & 0 \end{bmatrix}, \end{aligned}$	$\begin{aligned} &\left\langle a_{lm} a_{l'm'}^* \right\rangle_{\theta} \\ &= * \pi^2 \int dk \sum_{l''=0, l'': \text{even}}^{\infty} i^{l+l'} (-1)^{l+m} (2l''+1) k^2 \Delta_l(k) \Delta_{l'}'(k) P_{\Phi}(k) i_{l''}(\theta k H) \\ &\quad \times \sqrt{(2l+1)(2l'+1)} \begin{bmatrix} l & l' & l'' \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} l & l' & l'' \\ -m & m' & 0 \end{bmatrix} \end{aligned}$	$\begin{aligned} &\left\langle a_{lm} a_{l'm'}^* \right\rangle_{\theta} \\ &= * \pi^2 \int dk \sum_{l''=0, l'': \text{even}}^{\infty} i^{l+l'} (-1)^{l+m} (2l''+1) k^2 \Delta_l(k) \Delta_{l'}'(k) P_{\Phi}(k) i_{l''}(\theta k H) \\ &\quad \times \sqrt{(2l+1)(2l'+1)} \begin{bmatrix} l & l' & l'' \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} l & l' & l'' \\ -m & m' & 0 \end{bmatrix}, \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08357 (26)	171	■ 0.877 ● N +2	$\begin{aligned} & \left(\mathbf{N}_{\{4\}}\right)^2=\left(T^aT_a-e^ae^bR_{ab}\right)^2\quad\&\\ & \mathbf{N}_4\mathbf{P}_4=\left(T^aT_a-e^ae^bR_{ab}\right)\left(R^c_dR^d_c\right) \end{aligned}$	$\left(\mathbf{N}_4\right)^2=\left(T^aT_a-e^ae^bR_{ab}\right)^2$ $\mathbf{N}_4\mathbf{P}_4=\left(T^aT_a-e^ae^bR_{ab}\right)\left(R^c_dR^d_c\right)$	$\left(\mathbf{N}_4\right)^2=\left(T^aT_a-e^ae^bR_{ab}\right)^2,$ $\mathbf{N}_4\mathbf{P}_4=\left(T^aT_a-e^ae^bR_{ab}\right)\left(R^c_dR^d_c\right),$
cardona-qft-methods_ E08267	127	■ 0.880 ● K +0	$\Psi_{G_1}\left(t_1,\ldots,t_{n_1}\right)\Psi_{G_2}\left(u_1,\ldots,u_{n_2}\right)\neq 0,$	$\Psi_{G_1}\left(t_1,\ldots,t_{n_1}\right)\Psi_{G_2}\left(u_1,\ldots,u_{n_2}\right)\neq 0,$	$\Psi_{G_1}\left(t_1,\ldots,t_{n_1}\right)\Psi_{G_2}\left(u_1,\ldots,u_{n_2}\right)\neq 0,$
cardona-qft-methods_ E08314 (18)	156	■ 0.882 ●	$\begin{aligned} & \left[\mathcal{H}_{\perp}(x),\mathcal{H}_{\perp}(y)\right]=g^{ij}(x)\delta(x,y)_{,i}\mathcal{H}_j(y)-g^{ij}(y)\delta(y,x)_{,i}\mathcal{H}_j(x)\\ & \left[\mathcal{H}_i(x),\mathcal{H}_j(y)\right]=\delta(x,y)_{,i}\mathcal{H}_j(y)-\delta(x,y)_{,j}\mathcal{H}_i(y)\\ & \left[\mathcal{H}_{\perp}(x),\mathcal{H}_i(y)\right]=\delta(x,y)_{,i}\mathcal{H}_{\perp}(y) \end{aligned}$	$\left[\mathcal{H}_{\perp}(x),\mathcal{H}_{\perp}(y)\right]=g^{ij}(x)\delta(x,y)_{,i}\mathcal{H}_j(y)-g^{ij}(y)\delta(y,x)_{,i}\mathcal{H}_j(x)$ $\left[\mathcal{H}_i(x),\mathcal{H}_j(y)\right]=\delta(x,y)_{,i}\mathcal{H}_j(y)-\delta(x,y)_{,j}\mathcal{H}_i(y)$ $\left[\mathcal{H}_{\perp}(x),\mathcal{H}_i(y)\right]=\delta(x,y)_{,i}\mathcal{H}_{\perp}(y)$	$\left[\mathcal{H}_{\perp}(x),\mathcal{H}_{\perp}(y)\right]=g^{ij}(x)\delta(x,y)_{,i}\mathcal{H}_j(y)-g^{ij}(y)\delta(y,x)_{,i}\mathcal{H}_j(x)$ $\left[\mathcal{H}_i(x),\mathcal{H}_j(y)\right]=\delta(x,y)_{,i}\mathcal{H}_j(y)-\delta(x,y)_{,j}\mathcal{H}_i(y)$ $\left[\mathcal{H}_{\perp}(x),\mathcal{H}_i(y)\right]=\delta(x,y)_{,i}\mathcal{H}_{\perp}(y)$
cardona-qft-methods_ E08175 (3)	090	■ 0.890 ● N +0	$\operatorname{Ind}_{\{G\}}(\tilde{\operatorname{mathcal{D}}}):=\operatorname{bigoplus}_{V\in\operatorname{Irr}(G)}(\operatorname{dim}V)(\operatorname{dim}\operatorname{Ker}(\mathcal{D})-\operatorname{dim}\operatorname{Ker}(\tilde{\mathcal{D}}))V.$	$\operatorname{Ind}_G(\tilde{\mathcal{D}}):=\bigoplus_{V\in\operatorname{Irr}(G)}(\operatorname{dim}V)(\operatorname{dim}\operatorname{Ker}(\mathcal{D})-\operatorname{dim}\operatorname{Ker}(\tilde{\mathcal{D}}))V.$	$\operatorname{Ind}_G(\tilde{\mathcal{D}}):=\bigoplus_{V\in\operatorname{Irr}(G)}(\operatorname{dim}V)(\operatorname{dim}\operatorname{Ker}(\mathcal{D})-\operatorname{dim}\operatorname{Ker}(\tilde{\mathcal{D}}))V.$
cardona-qft-methods_ E08820 (20)	310	■ 0.890 ● N +0	$\begin{aligned} & \Phi_{+}=\eta_{+}^1\otimes\eta_{+}^{2\dagger}=-\frac{1}{8}\omega\wedge e^{-\mathrm{i}v\wedge v'}\,,\\ & \Phi_{-}=\eta_{+}^1\otimes\eta_{-}^{2\dagger}=-\frac{1}{8}e^{-\mathrm{i}j}\wedge(v+\mathrm{i}v')\,. \end{aligned}$	$\Phi_{+}=\eta_{+}^1\otimes\eta_{+}^{2\dagger}=-\frac{1}{8}\omega\wedge e^{-\mathrm{i}v\wedge v'}\,,$ $\Phi_{-}=\eta_{+}^1\otimes\eta_{-}^{2\dagger}=-\frac{1}{8}e^{-\mathrm{i}j}\wedge(v+\mathrm{i}v')\,.$	$\Phi_{+}=\eta_{+}^1\otimes\eta_{+}^{2\dagger}=-\frac{1}{8}\omega\wedge e^{-\mathrm{i}v\wedge v'}\,,$ $\Phi_{-}=\eta_{+}^1\otimes\eta_{-}^{2\dagger}=-\frac{1}{8}e^{-\mathrm{i}j}\wedge(v+\mathrm{i}v')\,.$
cardona-qft-methods_ E08449	211	■ 0.896 ● K +0	$Y_{\{\downarrow 1\}}^{\prime}=\mathcal{V}_1Y_{\{\downarrow 1\}}V_3^*,\quad Y_{\{\uparrow 1\}}^{\prime}=V_2Y_{\{\uparrow 1\}}V_3^*,\quad Y_R^{\prime}=V_2Y_R\bar{V}_2^*,$	$Y_{\{\downarrow 1\}}^{\prime}=\mathcal{V}_1Y_{\{\downarrow 1\}}V_3^*,\quad Y_{\{\uparrow 1\}}^{\prime}=V_2Y_{\{\uparrow 1\}}V_3^*,\quad Y_R^{\prime}=V_2Y_R\bar{V}_2^*,$	$Y_{\{\downarrow 1\}}^{\prime}=\mathcal{V}_1Y_{\{\downarrow 1\}}V_3^*,\quad Y_{\{\uparrow 1\}}^{\prime}=V_2Y_{\{\uparrow 1\}}V_3^*,\quad Y_R^{\prime}=V_2Y_R\bar{V}_2^*,$
cardona-qft-methods_ E08427	205	■ 0.896 ● K +0	$J^2=\varepsilon,\quad JD=\varepsilon'DJ,\quad\text{and}\quad J\gamma=\varepsilon''\gamma J,$	$J^2=\varepsilon,\quad JD=\varepsilon'DJ,\quad\text{and}\quad J\gamma=\varepsilon''\gamma J,$	$J^2=\varepsilon,\quad JD=\varepsilon'DJ,\quad\text{and}\quad J\gamma=\varepsilon''\gamma J,$
cardona-qft-methods_ E08728	277	■ 0.905 ● W +0	$\mathbf{M}_{\{a\}}(u)=\left(\begin{array}{ll}\hat{A}(u)&\hat{B}(u)\\\hat{C}(u)&\hat{D}(u)\end{array}\right),$	$M_a(u)=\left(\begin{array}{cc}\hat{A}(u)&\hat{B}(u)\\\hat{C}(u)&\hat{D}(u)\end{array}\right),$	$M_a(u)=\left(\begin{array}{cc}\hat{A}(u)&\hat{B}(u)\\\hat{C}(u)&\hat{D}(u)\end{array}\right),$
cardona-qft-methods_ E08980	347	■ 0.907 ● N +0	$C_{\{*\}}\left(L\,M^{\{-T\,M\}}\right)\stackrel{\sim}{\rightarrow}\mathcal{R}Hom_{C^*(M)^e}\left(C^*(M),C^*(M)\right)$	$C_{\ast}\left(LM^{-TM}\right)\stackrel{\sim}{\rightarrow}\mathcal{R}Hom_{C^*(M)^e}\left(C^*(M),C^*(M)\right)$	$C_{\ast}(LM^{-TM})\stackrel{\sim}{\rightarrow}\mathcal{R}Hom_{C^*(M)^e}(C^*(M),C^*(M))$
cardona-qft-methods_ E08979 (19)	364	■ 0.907 ● S +0	$\left[X_{\{\Gamma^{\vee}\}}\right]=\frac{(\mathbb{T}+1)^n-1}{\mathbb{T}}-\frac{\mathbb{T}^n-(-1)^n}{\mathbb{T}}-n\mathbb{T}^{n-2}.$	$\left[X_{\Gamma^{\vee}}\right]=\frac{(\mathbb{T}+1)^n-1}{\mathbb{T}}-\frac{\mathbb{T}^n-(-1)^n}{\mathbb{T}}-n\mathbb{T}^{n-2}.$	$\left[X_{\Gamma^{\vee}}\right]=\frac{(\mathbb{T}+1)^n-1}{\mathbb{T}}-\frac{\mathbb{T}^n-(-1)^n}{\mathbb{T}}-n\mathbb{T}^{n-2}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E09549 (18)	244	■0.911 ●K +0	$\begin{aligned} m_r[(S \otimes \mathrm{Id} \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= e \otimes \alpha \\ m'_r[(\mathrm{Id} \otimes S \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= \alpha \otimes e, \quad \alpha \in H. \end{aligned}$	$\begin{aligned} m_r[(S \otimes \mathrm{Id} \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= e \otimes \alpha \\ m'_r[(\mathrm{Id} \otimes S \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= \alpha \otimes e, \quad \alpha \in H. \end{aligned}$	$\begin{aligned} m_r[(S \otimes \mathrm{Id} \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= e \otimes \alpha \\ m'_r[(\mathrm{Id} \otimes S \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta)\Delta(\alpha)] &= \alpha \otimes e, \quad \alpha \in H. \end{aligned}$
cardona-qft-methods_ E0492	222	■0.913 ●	$\begin{array}{l} \frac{1}{2} \kappa_0^2 = \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \gamma_0 = \frac{1}{\pi^2} (48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d}) \\ \alpha_0 = -\frac{3 f_0}{10 \pi^2} \quad \tau_0 = \frac{11 f_0}{60 \pi^2} \\ \mu_0^2 = 2 \frac{f_2 \Lambda^2}{f_0} - \frac{\mathfrak{c}}{\mathfrak{a}} \quad \xi_0 = \frac{1}{12} \\ \lambda_0 = \frac{\pi^2 \mathfrak{b}}{2 f_0 \mathfrak{a}^2} \end{array}$	$\begin{array}{l} \frac{1}{2} \kappa_0^2 = \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \gamma_0 = \frac{1}{\pi^2} (48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d}) \\ \alpha_0 = -\frac{3 f_0}{10 \pi^2} \quad \tau_0 = \frac{11 f_0}{60 \pi^2} \\ \mu_0^2 = 2 \frac{f_2 \Lambda^2}{f_0} - \frac{\mathfrak{c}}{\mathfrak{a}} \quad \xi_0 = \frac{1}{12} \\ \lambda_0 = \frac{\pi^2 \mathfrak{b}}{2 f_0 \mathfrak{a}^2} \end{array}$	$\begin{array}{l} \frac{1}{2} \kappa_0^2 = \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \gamma_0 = \frac{1}{\pi^2} (48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d}) \\ \alpha_0 = -\frac{3 f_0}{10 \pi^2} \quad \tau_0 = \frac{11 f_0}{60 \pi^2} \\ \mu_0^2 = 2 \frac{f_2 \Lambda^2}{f_0} - \frac{\mathfrak{c}}{\mathfrak{a}} \quad \xi_0 = \frac{1}{12} \\ \lambda_0 = \frac{\pi^2 \mathfrak{b}}{2 f_0 \mathfrak{a}^2} \end{array}$
cardona-qft-methods_ E0105	063	■0.913 ●N +0	$\begin{aligned} & \left D_A \right ^{-n} = f \left D \right ^{-n} = 2^{m+1} \pi^{n/2} \Gamma \left(\frac{n}{2} \right)^{-1}, \\ & \left D_A \right ^{-n+k} = 0 \text{ for } k \text{ odd}, \\ & \left D_A \right ^{-n+2} = 0. \end{aligned}$	$\begin{aligned} f \left D_A \right ^{-n} &= f \left D \right ^{-n} = 2^{m+1} \pi^{n/2} \Gamma \left(\frac{n}{2} \right)^{-1}, \\ f \left D_A \right ^{-n+k} &= 0 \text{ for } k \text{ odd}, \\ f \left D_A \right ^{-n+2} &= 0. \end{aligned}$	$\begin{aligned} f \left D_A \right ^{-n} &= f \left D \right ^{-n} = 2^{m+1} \pi^{n/2} \Gamma \left(\frac{n}{2} \right)^{-1}, \\ f \left D_A \right ^{-n+k} &= 0 \text{ for } k \text{ odd}, \\ f \left D_A \right ^{-n+2} &= 0. \end{aligned}$
cardona-qft-methods_ E0096	056	■0.914 ●N +1	$\operatorname{Tr} \left(g \left(t \mathcal{D}^2 \right) \right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^-} \left[\frac{1}{2} \operatorname{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s) \int_0^\infty g(y) y^{-\alpha-1} dy \right] t^\alpha$	$\operatorname{Tr} \left(g \left(t \mathcal{D}^2 \right) \right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^-} \left[\frac{1}{2} \operatorname{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s) \int_0^\infty g(y) y^{-\alpha-1} dy \right] t^\alpha.$	$\operatorname{Tr} \left(g \left(t \mathcal{D}^2 \right) \right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^-} \left[\frac{1}{2} \operatorname{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s) \int_0^\infty g(y) y^{-\alpha-1} dy \right] t^\alpha.$
cardona-qft-methods_ E0386 (26)	179	■0.918 ●K +0	$\mathbf{E}_4 = d L_3^{AdS}.$	$\mathbf{E}_4 = d L_3^{AdS}.$	$\mathbf{E}_4 = d L_3^{AdS}.$
cardona-qft-methods_ E09397 (32)	181	■0.919 ●N +0	$L_3^{\mathrm{Exotic}} = L_3^{\mathrm{Lor}} \pm \frac{2}{l^2} L_3^{\mathrm{Tor}}.$	$L_3^{\mathrm{Exotic}} = L_3^{\mathrm{Lor}} \pm \frac{2}{l^2} L_3^{\mathrm{Tor}}.$	$L_3^{\mathrm{Exotic}} = L_3^{\mathrm{Lor}} \pm \frac{2}{l^2} L_3^{\mathrm{Tor}}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0779 (20)	297	■0.922 ●C +4	$\begin{aligned} \beta_{MN}^G &= \alpha' \left(R_{MN} + 2 \nabla_M \nabla_N \Phi - \frac{1}{4} H_{MPQ} H_N^{PQ} \right) + O(\alpha'^2), \\ \beta_{MN}^B &= \alpha' \left(-\frac{1}{2} \nabla^P H_{PMN} + \nabla^P \Phi H_{PMN} \right) + O(\alpha'^2), \\ \beta^\Phi &= \alpha' \left(\frac{D-26}{6\alpha'} - \frac{1}{2} \nabla^2 \Phi + \nabla_M \Phi \nabla^M \Phi - \frac{1}{24} H_{MNP} H^{MNP} \right) + O(\alpha'^2), \end{aligned}$	$\beta_{MN}^G = \alpha' \left(R_{MN} + 2 \nabla_M \nabla_N \Phi - \frac{1}{4} H_{MPQ} H_N^{PQ} \right) + O(\alpha'^2),$ $\beta_{MN}^B = \alpha' \left(-\frac{1}{2} \nabla^P H_{PMN} + \nabla^P \Phi H_{PMN} \right) + O(\alpha'^2),$ $\beta^\Phi = \alpha' \left(\frac{D-26}{6\alpha'} - \frac{1}{2} \nabla^2 \Phi + \nabla_M \Phi \nabla^M \Phi - \frac{1}{24} H_{MNP} H^{MNP} \right) + O(\alpha'^2),$	$\beta_{MN}^G = \alpha' \left(R_{MN} + 2 \nabla_M \nabla_N \Phi - \frac{1}{4} H_{MPQ} H_N^{PQ} \right) + O(\alpha'^2),$ $\beta_{MN}^B = \alpha' \left(-\frac{1}{2} \nabla^P H_{PMN} + \nabla^P \Phi H_{PMN} \right) + O(\alpha'^2),$ $\beta^\Phi = \alpha' \left(\frac{D-26}{6\alpha'} - \frac{1}{2} \nabla^2 \Phi + \nabla_M \Phi \nabla^M \Phi - \frac{1}{24} H_{MNP} H^{MNP} \right) + O(\alpha'^2),$
cardona-qft-methods_ EQ0238	114	■0.922 ●	$c(E) = 1 + c_1(E) + c_2(E) + \cdots + c_{\mathrm{rk}}(E).$	$c(E) = 1 + c_1(E) + c_2(E) + \cdots + c_{\mathrm{rk}}(E).$	$c(E) = 1 + c_1(E) + c_2(E) + \cdots + c_{\mathrm{rk}}(E).$
cardona-qft-methods_ EQ0264	125	■0.926 ●N +0	$\sum_{\text{graphs } G} \langle G \rangle = \exp \left(\sum_{\text{connected graphs } G} \langle G \rangle \right).$	$\sum_{\text{graphs } G} \langle G \rangle = \exp \left(\sum_{\text{connected graphs } G} \langle G \rangle \right).$	$\sum_{\text{graphs } G} \langle G \rangle = \exp \left(\sum_{\text{connected graphs } G} \langle G \rangle \right).$
cardona-qft-methods_ EQ0768 (9)	294	■0.927 ●K +0	$\left[\alpha_n^P, \alpha_m^Q \right] = \left[\tilde{\alpha}_n^P, \tilde{\alpha}_m^Q \right] = n \delta_{n+m} \eta^{PQ}, \quad [x^P, p^Q] = i \eta^{PQ}.$	$[\alpha_n^P, \alpha_m^Q] = [\tilde{\alpha}_n^P, \tilde{\alpha}_m^Q] = n \delta_{n+m} \eta^{PQ}, \quad [x^P, p^Q] = i \eta^{PQ}.$	$[\alpha_n^P, \alpha_m^Q] = [\tilde{\alpha}_n^P, \tilde{\alpha}_m^Q] = n \delta_{n+m} \eta^{PQ}, \quad [x^P, p^Q] = i \eta^{PQ}.$
cardona-qft-methods_ EQ0098 (35)	056	■0.928 ●N +1	$\begin{aligned} & \mathcal{A}_k \left(\mathcal{D}^2 \right) = \frac{1}{2} \Gamma \left(\frac{d-k}{2} \right) f \mathcal{D} ^{-d+k} \quad \text{for } k = 0, \dots, d-1, \\ & \mathcal{A}_d \left(\mathcal{D}^2 \right) = \dim \mathcal{D} + \zeta_{\mathcal{D}^2}(0). \end{aligned}$	$a_k(\mathcal{D}^2) = \frac{1}{2} \Gamma \left(\frac{d-k}{2} \right) f \mathcal{D} ^{-d+k} \quad \text{for } k = 0, \dots, d-1,$ $a_d(\mathcal{D}^2) = \dim \mathcal{D} + \zeta_{\mathcal{D}^2}(0).$	$a_k(\mathcal{D}^2) = \frac{1}{2} \Gamma \left(\frac{d-k}{2} \right) \int \mathcal{D} ^{-d+k} \quad \text{for } k = 0, \dots, d-1,$ $a_d(\mathcal{D}^2) = \dim \mathcal{D} + \zeta_{\mathcal{D}^2}(0).$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C ●W ●S ●N ●K ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0734 (1)	279	■ 0.929 ● C +4	$\begin{aligned} S=&\frac{N}{\lambda}\int\frac{d^4x}{(2\pi)^4}\mathrm{Tr}\left(\frac{1}{4}F^{\mu\nu}F_{\mu\nu}+\frac{1}{2}(D^\mu)^*\varphi_mD_\mu\varphi_m+\bar\psi_\alpha^a(\sigma_\mu)^{\dot\alpha\beta}D^\mu\psi_{a\beta}\right.\\&\left.-\frac{1}{4}[\varphi^m,\varphi^n][\varphi_m,\varphi_n]\right.\\&\left.-\frac{i}{2}\psi_{a\alpha}(\Sigma_m)^{ab}\varepsilon^{\alpha\beta}[\varphi^m\psi_{a\beta}]-\frac{i}{2}\bar\psi_\alpha^a(\Sigma_m)^{ab}\varepsilon^{\dot\alpha\dot\beta}[\varphi^m\bar\psi_{a\dot\beta}]\right)\\&+\text{ghosts}+\text{gauge fixing}+\text{topological term}\end{aligned}$	$S=\frac{N}{\lambda}\int\frac{d^4x}{(2\pi)^4}\mathrm{Tr}\left(\frac{1}{4}F^{\mu\nu}F_{\mu\nu}+\frac{1}{2}(D^\mu)^*\varphi_mD_\mu\varphi_m+\bar\psi_\alpha^a(\sigma_\mu)^{\dot\alpha\beta}D^\mu\psi_{a\beta}\right.\\-\frac{1}{4}[\varphi^m,\varphi^n][\varphi_m,\varphi_n]\\-\frac{i}{2}\psi_{a\alpha}(\Sigma_m)^{ab}\varepsilon^{\alpha\beta}[\varphi^m\psi_{a\beta}]-\frac{i}{2}\bar\psi_\alpha^a(\Sigma_m)^{ab}\varepsilon^{\dot\alpha\dot\beta}[\varphi^m\bar\psi_{a\dot\beta}]\Bigg)\\+\text{ghosts}+\text{gauge fixing}+\text{topological term}$	$S=\frac{N}{\lambda}\int\frac{d^4x}{(2\pi)^4}\mathrm{Tr}\left(\frac{1}{4}F^{\mu\nu}F_{\mu\nu}+\frac{1}{2}(D^\mu)^*\varphi_mD_\mu\varphi_m+\bar\psi_\alpha^a(\sigma_\mu)^{\dot\alpha\beta}D^\mu\psi_{a\beta}\right.\\-\frac{1}{4}[\varphi^m,\varphi^n][\varphi_m,\varphi_n]\\-\frac{i}{2}\psi_{a\alpha}(\Sigma_m)^{ab}\varepsilon^{\alpha\beta}[\varphi^m\psi_{a\beta}]-\frac{i}{2}\bar\psi_\alpha^a(\Sigma_m)^{ab}\varepsilon^{\dot\alpha\dot\beta}[\varphi^m\bar\psi_{a\dot\beta}]\Bigg)\\+\text{ghosts}+\text{gauge fixing}+\text{topological term}.\quad(1)$
cardona-qft-methods_ EQ0972 (12)	363	■ 0.929 ● K +0	$\left[X_\Gamma\backslash\Sigma_n\right]=\left[X_\Gamma\vee\backslash\Sigma_n\right].$	$[X_\Gamma\backslash\Sigma_n]=[X_{\Gamma\vee}\backslash\Sigma_n].$	$[X_\Gamma\backslash\Sigma_n]=[X_{\Gamma\vee}\backslash\Sigma_n].$
cardona-qft-methods_ EQ0398	181	■ 0.929 ● S +1	$\begin{aligned} d\,L_3^{\text{Exotic}}&=R_b^aR_a^b\pm\frac{2}{l^2}\left(T^aT_a-e^ae^bR_{ab}\right)\\&=F_B^AF_A^B\end{aligned}$	$dL_3^{\text{Exotic}}=R_b^aR_a^b\pm\frac{2}{l^2}\left(T^aT_a-e^ae^bR_{ab}\right)=F_B^AF_A^B$	$dL_3^{Exotic}=R_b^aR_a^b\pm\frac{2}{l^2}\left(T^aT_a-e^ae^bR_{ab}\right)=F_B^AF_A^B.$
cardona-qft-methods_ EQ0044	033	■ 0.933 ● N +0	$i[D,f]=i\left[d+d^*,f\right]=i\left(i\sigma_1^d(df)-i\sigma_1^{d^*}(df)\right)=-i(\epsilon+\iota)(df).$	$i[D,f]=i\left[d+d^*,f\right]=i\left(i\sigma_1^d(df)-i\sigma_1^{d^*}(df)\right)=-i(\epsilon+\iota)(df).$	$i[D,f]=i[d+d^*,f]=i\left(i\sigma_1^d(df)-i\sigma_1^{d^*}(df)\right)=-i(\epsilon+\iota)(df).$
cardona-qft-methods_ EQ0112	073	■ 0.933 ● K +0	$\operatorname{Ind}(A)=-\operatorname{Ind}\left(A^*\right)\operatorname{left}(A^{[*]}\operatorname{right})=\operatorname{operatorname{dimKer}}(A)-\operatorname{operatorname{dimKer}}\left(\operatorname{left}(A^{[*]}\operatorname{right})\right).$	$\operatorname{Ind}(A)=-\operatorname{Ind}\left(A^*\right)=\operatorname{dimKer}(A)-\operatorname{dimKer}\left(A^*\right).$	$\operatorname{Ind}(A)\;=\;-\;\operatorname{Ind}(A^*)\;=\;\operatorname{dim}\operatorname{Ker}(A)\;-\;\operatorname{dim}\operatorname{Ker}(A^*).$
cardona-qft-methods_ EQ0845 (3)	329	■ 0.933 ●	$\delta S=\int_\Sigma\frac{\delta\mathcal{L}}{\delta X}\delta X+\frac{\delta\mathcal{L}}{\delta\eta}\delta\eta+\text{boundary terms}$	$\delta S=\int_\Sigma\frac{\delta\mathcal{L}}{\delta X}\delta X+\frac{\delta\mathcal{L}}{\delta\eta}\delta\eta+\text{boundary terms}$	$\delta S=\int_\Sigma\frac{\delta\mathcal{L}}{\delta X}\delta X+\frac{\delta\mathcal{L}}{\delta\eta}\delta\eta+\text{boundary terms}.$
cardona-qft-methods_ EQ0040	032	■ 0.936 ● K +0	$\begin{gathered} \epsilon\left(\omega_1\right)\omega_2=\omega_1\wedge\omega_2,\\ \omega_{\{2\}}=\omega_{\{1\}}\wedge\omega_{\{2\}},\quad\forall\,\iota(\omega)\\ \left(\omega_{\{1\}}\wedge\cdots\wedge\omega_{\{m\}}\right)=\\ \sum_{j=1}^m(-1)^{j-1}g\left(\omega,\omega_{\{j\}}\right)\omega_1\wedge\cdots\wedge\widehat{\omega_{\{j\}}}\wedge\cdots\wedge\omega_m\\ \omega_{\{1\}}\wedge\cdots\wedge\widehat{\omega_{\{j\}}}\wedge\cdots\wedge\omega_m\end{gathered}$	$\iota(\omega)\left(\omega_1\wedge\cdots\wedge\omega_m\right)=\sum_{j=1}^m(-1)^{j-1}g\left(\omega,\omega_{\{j\}}\right)\omega_1\wedge\cdots\wedge\widehat{\omega_{\{j\}}}\wedge\cdots\wedge\omega_m$	$\epsilon(\omega_1)\omega_2=\omega_1\wedge\omega_2,$ $\iota(\omega)\left(\omega_1\wedge\cdots\wedge\omega_m\right)=\sum_{j=1}^m(-1)^{j-1}g(\omega,\omega_j)\omega_1\wedge\cdots\wedge\widehat{\omega_j}\wedge\cdots\wedge\omega_m$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0745	283	■ 0.941 ● N +0	$\backslash \langle \mathrm{mathcal{0}}(x), \, \overline{\mathrm{mathcal{0}}}(y) \rangle = \frac{1}{(x-y)^{2\Delta}} \sim \frac{1}{(x-y)^{2\Delta_0}} \left(1 - g_{\mathrm{YM}}^2 \gamma_1 \log \left[(x-y)^2 \Lambda^2 \right] \right),$	$\langle \mathcal{O}(x), \overline{\mathcal{O}}(y) \rangle = \frac{1}{(x-y)^{2\Delta}} \sim \frac{1}{(x-y)^{2\Delta_0}} \left(1 - g_{\mathrm{YM}}^2 \gamma_1 \log \left[(x-y)^2 \Lambda^2 \right] \right),$	$\langle \mathcal{O}(x), \overline{\mathcal{O}}(y) \rangle = \frac{1}{(x-y)^{2\Delta}} \sim \frac{1}{(x-y)^{2\Delta_0}} \left(1 - g_{\mathrm{YM}}^2 \gamma_1 \log \left[(x-y)^2 \Lambda^2 \right] \right),$
cardona-qft-methods_ EQ0661 (11)	262	■ 0.943 ● W +2	$\begin{aligned} S^{\{2\}} &= \frac{-i^2}{2!} \int d^4x \, d^4y \, T(H_I(x), H_I(y)) \\ &= \frac{-i^2}{2!} \int d^4x \, d^4y \, (\theta(x_0 - y_0) [H_I(x), H_I(y)] + H_I(y) H_I(x)), \end{aligned}$	$S^{(2)} = \frac{-i^2}{2!} \int d^4x d^4y T(H_I(x), H_I(y))$ $= \frac{-i^2}{2!} \int d^4x d^4y (\theta(x_0 - y_0) [H_I(x), H_I(y)] + H_I(y) H_I(x))$	$S^{(2)} = \frac{-i^2}{2!} \int d^4_x d^4_y T(H_I(x), H_I(y))$ $= \frac{-i^2}{2!} \int d^4_x d^4_y (\theta(x_0 - y_0) [H_I(x), H_I(y)] + H_I(y) H_I(x)),$
cardona-qft-methods_ EQ0308 (12)	154	■ 0.944 ● N +0	$d\left(\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}=0\right).$	$d\left(\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}=0\right).$	$d(\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}=0).$
cardona-qft-methods_ EQ0333 (19)	163	■ 0.947 ● W +0	$R_{\mathrm{b}}^{\mathrm{a}}=\bar{R}_{\mathrm{b}}^{\mathrm{a}}+\bar{D}\kappa_{\mathrm{b}}^{\mathrm{a}}+\kappa_{\mathrm{c}}^{\mathrm{a}}\kappa_{\mathrm{b}}^{\mathrm{c}},$	$R_b^a = \bar{R}_b^a + \bar{D}\kappa_b^a + \kappa_c^a \kappa_b^c,$	$R_b^a = \bar{R}_b^a + \bar{D}\kappa_b^a + \kappa_c^a \kappa_b^c,$
cardona-qft-methods_ EQ0432	208	■ 0.947 ● N +0	$\mathrm{mathcal{B}}\!=\!\mathrm{oplus}^{4-\mathrm{times}}\mathrm{M}_2(\mathrm{mathbb{C}})\!\mathrm{oplus}\!\mathrm{M}_6(\mathrm{mathbb{C}})~.$	$\mathcal{B} = \oplus^{4-\mathrm{times}} M_2(\mathbb{C}) \oplus M_6(\mathbb{C}).$	$\mathcal{B} = \oplus^{4-\mathrm{times}} M_2(\mathbb{C}) \oplus M_6(\mathbb{C}).$
cardona-qft-methods_ EQ0278	135	■ 0.947 ● S +0	$(\mathrm{mathbb{T}}-1)\,\mathrm{mathbb{T}}(\mathrm{mathbb{T}}+1)^2+\mathrm{mathbb{T}}(\mathrm{mathbb{T}}+1)^2+(\mathrm{T}+1)\mathrm{T}(\mathrm{T}+1)=\mathrm{T}(\mathrm{T}+1)^3~.$	$(\mathbb{T}-1)\mathbb{T}(\mathbb{T}+1)^2+\mathbb{T}(\mathbb{T}+1)^2+(\mathbb{T}+1)\mathbb{T}(\mathbb{T}+1)=\mathbb{T}(\mathbb{T}+1)^3.$	$(\mathbb{T}-1)\mathbb{T}(\mathbb{T}+1)^2+\mathbb{T}(\mathbb{T}+1)^2+(\mathbb{T}+1)\mathbb{T}(\mathbb{T}+1)=\mathbb{T}(\mathbb{T}+1)^3~.$
cardona-qft-methods_ EQ0132	078	■ 0.948 ● N +0	$f\!=\!\left(f_{\mathrm{1}},\,\ldots,\,f_{\mathrm{N}}\right)\!:\,\mathrm{mathbb{R}}^{\mathrm{n}}\!\rightarrow\!\mathrm{mathbb{R}}^{\mathrm{N}},\,\,\,\mathrm{quad}\,\,f_{\mathrm{j}}\!\in\!\mathcal{C}^{\infty}\!\left(\mathrm{mathbb{R}}^{\mathrm{n}}\right).$	$f = (f_1, \dots, f_N) : \mathbb{R}^n \longrightarrow \mathbb{R}^N, \quad f_j \in C^\infty(\mathbb{R}^n).$	$f = (f_1, \dots, f_N) : \mathbb{R}^n \longrightarrow \mathbb{R}^N, \quad f_j \in C^\infty(\mathbb{R}^n).$
cardona-qft-methods_ EQ0237 (3)	113	■ 0.948 ● N +0	$N_{\mathrm{q}}(X)\!=\!a_{\mathrm{0}}\!+\!a_{\mathrm{1}}q\!+\!\cdots\!+\!a_rq^r,$	$N_q(X) = a_0 + a_1q + \cdots + a_rq^r,$	$N_q(X) = a_0 + a_1q + \cdots + a_rq^r \quad ,$
cardona-qft-methods_ EQ0316 (2)	159	■ 0.949 ● N +0	$\begin{aligned} d\,s^{\{2\}} &= \eta_{\mathrm{a}\,\mathrm{b}}\,d\,z^{\mathrm{a}}\,d\,z^{\mathrm{b}}\,\backslash \\ \backslash &= \eta_{\mathrm{a}\,\mathrm{b}}\,e_{\mathrm{\mu}}^{\mathrm{a}}(x)\,d\,x^{\mathrm{\mu}}\,e_{\mathrm{\nu}}^{\mathrm{b}}(x)\,d\,x^{\mathrm{\nu}}\,\backslash\backslash \\ &= g_{\mathrm{\mu}\,\mathrm{\nu}}\,d\,x^{\mathrm{\mu}}\,d\,x^{\mathrm{\nu}}, \end{aligned}$	$ds^2 = \eta_{ab} dz^a dz^b$ $= \eta_{ab} e_\mu^a(x) dx^\mu e_\nu^b(x) dx^\nu$ $= g_{\mu\nu}(x) dx^\mu dx^\nu,$	$ds^2 = \eta_{ab} dz^a dz^b$ $= \eta_{ab} e_\mu^a(x) dx^\mu e_\nu^b(x) dx^\nu$ $= g_{\mu\nu}(x) dx^\mu dx^\nu,$
cardona-qft-methods_ EQ0363	174	■ 0.949 ●	$\begin{aligned} \delta e^{\mathrm{a}} &= -\lambda^{\mathrm{a}}_{\mathrm{c}}e^{\mathrm{c}} \\ \backslash\backslash\delta R^{\mathrm{a}}_{\mathrm{b}} &= -\left(\lambda^{\mathrm{a}}_{\mathrm{c}}R^{\mathrm{cb}}+\lambda^{\mathrm{b}}_{\mathrm{c}}R^{\mathrm{ac}}\right) \\ \backslash\backslash\delta\epsilon_{\mathrm{abc}} &= -\left(\lambda^{\mathrm{d}}_{\mathrm{a}}\epsilon_{\mathrm{dbc}}+\lambda^{\mathrm{d}}_{\mathrm{b}}\epsilon_{\mathrm{adc}}+\lambda^{\mathrm{d}}_{\mathrm{c}}\epsilon_{\mathrm{abd}}\right)\equiv 0. \end{aligned}$	$\delta e^a = -\lambda^a_{\,c}e^c$ $\delta R^{ab} = -\left(\lambda^a_{\,c}R^{cb} + \lambda^b_{\,c}R^{ac}\right)$ $\delta \epsilon_{abc} = -\left(\lambda^d_{\,a}\epsilon_{dbc} + \lambda^d_{\,b}\epsilon_{adc} + \lambda^d_{\,c}\epsilon_{abd}\right) \equiv 0.$	$\delta e^a = -\lambda^a_{\,c}e^c$ $\delta R^{ab} = -(\lambda^a_{\,c}R^{cb} + \lambda^b_{\,c}R^{ac}),$ $\delta \epsilon_{abc} = -(\lambda^d_{\,a}\epsilon_{dbc} + \lambda^d_{\,b}\epsilon_{adc} + \lambda^d_{\,c}\epsilon_{abd}) \equiv 0.$
cardona-qft-methods_ EQ0837 (16)	317	■ 0.950 ● K +0	$F_{\mathrm{IIA}\pm}=F_{\mathrm{0}}\pm F_{\mathrm{2}}+F_{\mathrm{4}}\pm F_{\mathrm{6}},\quad\mathrm{quad}\quad F_{\mathrm{IIB}\pm}=F_{\mathrm{1}}\pm F_{\mathrm{3}}+F_{\mathrm{5}}~.$	$F_{\mathrm{IIA}\pm} = F_0 \pm F_2 + F_4 \pm F_6, \quad F_{\mathrm{IIB}\pm} = F_1 \pm F_3 + F_5.$	$F_{\mathrm{IIA}\pm} = F_0 \pm F_2 + F_4 \pm F_6 \quad , \quad F_{\mathrm{IIB}\pm} = F_1 \pm F_3 + F_5~.$
cardona-qft-methods_ EQ0186	093	■ 0.950 ● K +0	$c(v)\,c(u)+c(u)\,c(v)=-2\langle v,u\rangle\mathrm{Id},\quad\mathrm{quad}\quad\text{for all }u,v\in V~.$	$c(v)c(u)+c(u)c(v)=-2\langle v,u\rangle\mathrm{Id}, \quad \text{for all }u,v\in V.$	$c(v)\,c(u)+c(u)\,c(v) \;=\; -2\langle v,u\rangle\,\mathrm{Id}, \quad \text{for all }u,v\in V.$
cardona-qft-methods_ EQ0931	353	■ 0.950 ● N +0	$E\,\mathrm{G}_{\mathrm{n}}\,/\,\mathrm{C}(g)\,\times\,\mathrm{E}\,\mathrm{G}_{\mathrm{n}}\,/\,\mathrm{C}(h)\,\stackrel{\Delta}{\stackrel{\mathrm{rel}}{\longrightarrow}}\,\mathrm{E}\,\mathrm{G}_{\mathrm{n}}\,/\,(\mathrm{C}(g)\,\cap\,\mathrm{C}(h))\,\longrightarrow\,\mathrm{E}\,\mathrm{G}_{\mathrm{n}}\,/\,\mathrm{C}(g\,h)$	$EG_n/C(g) \times EG_n/C(h) \stackrel{\Delta}{\leftarrow} EG_n/(C(g) \cap C(h)) \longrightarrow EG_n/C(gh)$	$EG_n/C(g) \times EG_n/C(h) \stackrel{\Delta}{\leftarrow} EG_n/(C(g) \cap C(h)) \longrightarrow EG_n/C(gh)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0360 (31)	171	■0.951 ● N +2	$\mathcal{C}_{\{7\}}^{\{L\ o\ r\}}=\operatorname{Tr}\left[\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\frac{4}{5}\omega^2(d\omega)\omega(d\omega)+2\omega^5(d\omega)+\frac{4}{7}\omega^7\right]$	$C_7^{Lor}=\mathrm{Tr}\left[\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\frac{4}{5}\omega^2(d\omega)\omega(d\omega)+2\omega^5(d\omega)+\frac{4}{7}\omega^7\right]$	$C_7^{Lor}=\mathrm{Tr}[\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\frac{4}{5}\omega^2(d\omega)\omega(d\omega)+2\omega^5(d\omega)+\frac{4}{7}\omega^7],$
cardona-qft-methods_ EQ0315 (1)	159	■0.953 ● K +0	$dz^a=e_{\mu}^a(x)\,d x^{\mu},$	$dz^a=e_{\mu}^a(x)dx^{\mu},$	$dz^a=e_{\mu}^a(x)dx^{\mu},$
cardona-qft-methods_ EQ0765 (6)	293	■0.953 ● W +0	$\begin{aligned} X^M(\tau,\sigma)&=X_R^M(\sigma^-)+X_L^M(\sigma^+)\\ X_R^M(\sigma^-)&=\frac{1}{2}x^M+\alpha'p^M\sigma^-+i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-2in\sigma^-}\\ X_L^M(\sigma^+)&=\frac{1}{2}x^M+\alpha'p^M\sigma^++i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\tilde{\alpha}_n^Me^{-2in\sigma^+}, \end{aligned}$	$\begin{aligned} X^M(\tau,\sigma)&=X_R^M(\sigma^-)+X_L^M(\sigma^+)\\ X_R^M(\sigma^-)&=\frac{1}{2}x^M+\alpha'p^M\sigma^-+i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-2in\sigma^-}\\ X_L^M(\sigma^+)&=\frac{1}{2}x^M+\alpha'p^M\sigma^++i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\tilde{\alpha}_n^Me^{-2in\sigma^+}, \end{aligned}$	$\begin{aligned} X^M(\tau,\sigma)&=X_R^M(\sigma^-)+X_L^M(\sigma^+)\\ X_R^M(\sigma^-)&=\frac{1}{2}x^M+\alpha'p^M\sigma^-+i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-2in\sigma^-}\\ X_L^M(\sigma^+)&=\frac{1}{2}x^M+\alpha'p^M\sigma^++i\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\tilde{\alpha}_n^Me^{-2in\sigma^+}, \end{aligned}$
cardona-qft-methods_ EQ0736 (6)	280	■0.954 ● N +1	$\begin{aligned} &\left\{\left[M_{\mu\nu},K_{\rho}\right]\right\} &=i\left(\eta_{\rho\nu}K_{\mu}-\eta_{\mu\rho}K_{\nu}\right),\\ &\left[P_{\mu},K_{\nu}\right] &=2i\left(\eta_{\mu\nu}\mathcal{D}-M_{\mu\nu}\right). \end{aligned}$	$\begin{aligned} [M_{\mu\nu},K_{\rho}]&=i\left(\eta_{\rho\nu}K_{\mu}-\eta_{\mu\rho}K_{\nu}\right),\\ [P_{\mu},K_{\nu}]&=2i\left(\eta_{\mu\nu}\mathcal{D}-M_{\mu\nu}\right). \end{aligned}$	$\begin{aligned} [M_{\mu\nu},K_{\rho}]&=i(\eta_{\rho\nu}K_{\mu}-\eta_{\mu\rho}K_{\nu}),\\ [P_{\mu},K_{\nu}]&=2i(\eta_{\mu\nu}\mathcal{D}-M_{\mu\nu}). \end{aligned}$
cardona-qft-methods_ EQ0323 (9)	161	■0.954 ● N +0	$\varphi_{\parallel}^{\mu}(x)=\varphi^{\mu}(x)+dx^{\lambda}\left[\partial_{\lambda}\varphi^{\mu}(x)+\Gamma_{\lambda\rho}^{\mu}(x)\varphi^{\rho}(x)\right].$	$\varphi_{\parallel}^{\mu}(x)=\varphi^{\mu}(x)+dx^{\lambda}\left[\partial_{\lambda}\varphi^{\mu}(x)+\Gamma_{\lambda\rho}^{\mu}(x)\varphi^{\rho}(x)\right].$	$\varphi_{\parallel}^{\mu}(x)=\varphi^{\mu}(x)+dx^{\lambda}[\partial_{\lambda}\varphi^{\mu}(x)+\Gamma_{\lambda\rho}^{\mu}(x)\varphi^{\rho}(x)].$
cardona-qft-methods_ EQ0147	083	■0.957 ● N +0	$\mathcal{D} \coloneqq -i\,\frac{d}{dt} + \sin t \colon C^{\infty}(M) \rightarrow C^{\infty}(M).$	$\mathcal{D} \coloneqq -i\,\frac{d}{dt} + \sin t \colon C^{\infty}(M) \rightarrow C^{\infty}(M).$	$\mathcal{D} \coloneqq -i\,\frac{d}{dt} + \sin t \colon C^{\infty}(M) \rightarrow C^{\infty}(M).$
cardona-qft-methods_ EQ0065	040	■0.957 ●	$\begin{aligned} \omega_{\nu}&=\frac{1}{2}g_{\nu\mu}\left(\mathbb{A}^{\mu}+g^{\sigma\varepsilon}\Gamma_{\sigma\varepsilon}^{\mu}\operatorname{id}_V\right),\\ E&=\mathbb{B}-g^{\nu\mu}\left(\partial_{\nu}\omega_{\mu}+\omega_{\nu}\omega_{\mu}-\omega_{\sigma}\Gamma_{\nu\mu}^{\sigma}\right). \end{aligned}$	$\begin{aligned} \omega_{\nu}&=\frac{1}{2}g_{\nu\mu}\left(\mathbb{A}^{\mu}+g^{\sigma\varepsilon}\Gamma_{\sigma\varepsilon}^{\mu}\operatorname{id}_V\right)\\ E&=\mathbb{B}-g^{\nu\mu}\left(\partial_{\nu}\omega_{\mu}+\omega_{\nu}\omega_{\mu}-\omega_{\sigma}\Gamma_{\nu\mu}^{\sigma}\right). \end{aligned}$	$\begin{aligned} \omega_{\nu}&=\tfrac{1}{2}g_{\nu\mu}\left(\mathbb{A}^{\mu}+g^{\sigma\varepsilon}\Gamma_{\sigma\varepsilon}^{\mu}\operatorname{id}_V\right),\\ E&=\mathbb{B}-g^{\nu\mu}\left(\partial_{\nu}\omega_{\mu}+\omega_{\nu}\omega_{\mu}-\omega_{\sigma}\Gamma_{\nu\mu}^{\sigma}\right). \end{aligned}$
cardona-qft-methods_ EQ0618 (26)	256	■0.957 ● N +0	$\left(f_1f_2\right)(p)=f_1(p)f_2(p).$	$(f_1f_2)(p)=f_1(p)f_2(p).$	$(f_1f_2)(p)=f_1(p)f_2(p).$
cardona-qft-methods_ EQ0751	284	■0.957 ● N +0	$\left.\frac{30\mid 32}{10\mid 0\oplus 10\mid 0}\right _{\oplus 10\mid 0}=10\mid 32,$	$\frac{30\mid 32}{10\mid 0\oplus 10\mid 0}=10\mid 32,$	$\frac{30\mid 32}{10\mid 0\oplus 10\mid 0}=10\mid 32,$
cardona-qft-methods_ EQ0816 (16)	309	■0.958 ● N +0	$\mathcal{J}_{\{m\ n\}}=\mp 2\,\mathrm{i}\,\eta_{\pm}^{\dagger}\,\eta_{mn}\,\eta_{\pm}\quad\Omega_{mnp}=-2\mathrm{i}\,\eta_{\pm}^{\dagger}\,\gamma_{mnp}\,\eta_{\pm}.$	$J_{mn}=\mp 2\mathrm{i}\,\eta_{\pm}^{\dagger}\gamma_{mn}\eta_{\pm}\quad\Omega_{mnp}=-2\mathrm{i}\,\eta_{\pm}^{\dagger}\gamma_{mnp}\eta_{\pm}.$	$J_{mn}=\mp 2\mathrm{i}\,\eta_{\pm}^{\dagger}\gamma_{mn}\eta_{\pm}\quad\Omega_{mnp}=-2\mathrm{i}\,\eta_{\pm}^{\dagger}\gamma_{mnp}\eta_{\pm}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0488	221	■ 0.958 ● W -1	$\begin{aligned} a_0(x,P) &= (4\pi)^{-m/2} \mathrm{Tr}(\mathrm{id}) \\ a_2(x,P) &= (4\pi)^{-m/2} \mathrm{Tr}\left(-\frac{R}{6}\mathrm{id} + E\right) \\ a_4(x,P) &= (4\pi)^{-m/2} \frac{1}{360} \mathrm{Tr}\left(-12R_{;\mu}{}^\mu + 5R^2 - 2R_{\mu\nu}R^{\mu\nu} \right. \\ &\quad \left. + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 60RE + 180E^2 + 60E_{;\mu}{}^\mu \right. \\ &\quad \left. + 30\Omega_{\mu\nu}\Omega^{\mu\nu}\right). \end{aligned}$	$a_0(x,P) = (4\pi)^{-m/2} \mathrm{Tr}(\mathrm{id})$ $a_2(x,P) = (4\pi)^{-m/2} \mathrm{Tr}\left(-\frac{R}{6}\mathrm{id} + E\right)$ $a_4(x,P) = (4\pi)^{-m/2} \frac{1}{360} \mathrm{Tr}\left(-12R_{;\mu}{}^\mu + 5R^2 - 2R_{\mu\nu}R^{\mu\nu} \right. \\ \left. + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 60RE + 180E^2 + 60E_{;\mu}{}^\mu \right. \\ \left. + 30\Omega_{\mu\nu}\Omega^{\mu\nu}\right).$	$a_0(x,P) = (4\pi)^{-m/2} \mathrm{Tr}(\mathrm{id})$ $a_2(x,P) = (4\pi)^{-m/2} \mathrm{Tr}\left(-\frac{R}{6}\mathrm{id} + E\right)$ $a_4(x,P) = (4\pi)^{-m/2} \frac{1}{360} \mathrm{Tr}\left(-12R_{;\mu}{}^\mu + 5R^2 - 2R_{\mu\nu}R^{\mu\nu} \right. \\ \left. + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 60RE + 180E^2 + 60E_{;\mu}{}^\mu \right. \\ \left. + 30\Omega_{\mu\nu}\Omega^{\mu\nu}\right).$
cardona-qft-methods_ EQ0500	226	■ 0.959 ● N +0	$\frac{2f_2\Lambda_{unif}^2}{f_0} \leq \mathfrak{c}(\Lambda_{unif}) \leq \frac{6f_2\Lambda_{unif}^2}{f_0}$	$\frac{2f_2\Lambda_{unif}^2}{f_0} \leq \mathfrak{c}(\Lambda_{unif}) \leq \frac{6f_2\Lambda_{unif}^2}{f_0}$	$\frac{2f_2\Lambda_{unif}^2}{f_0} \leq \mathfrak{c}(\Lambda_{unif}) \leq \frac{6f_2\Lambda_{unif}^2}{f_0}$
cardona-qft-methods_ EQ0234 (1)	112	■ 0.961 ● K +0	$[\widetilde{X}] - [E] = [X] - [Z]$	$[\tilde{X}] - [E] = [X] - [Z]$	$[\tilde{X}] - [E] = [X] - [Z]$
cardona-qft-methods_ EQ0262	123	■ 0.961 ● N -1	$\hat{Y}_{\{G\}} \text{ \textit{mixed-Tate} } \Longrightarrow \left[\hat{Y}_G\right] \in \mathbb{Z}[\mathbb{L}] \Longrightarrow N_q(G) \text{ is a polynomial in } q$	$\hat{Y}_G \text{ mixed-Tate} \implies \left[\hat{Y}_G\right] \in \mathbb{Z}[\mathbb{L}] \implies N_q(G) \text{ is a polynomial in } q.$	$\hat{Y}_G \text{ mixed-Tate} \implies \left[\hat{Y}_G\right] \in \mathbb{Z}[\mathbb{L}] \implies N_q(G) \text{ is a polynomial in } q.$
cardona-qft-methods_ EQ0332	163	■ 0.963 ● N +0	$d\,e^{\{a\}+\overline{\omega}_b\}_{b\wedge\{a\}}\wedge e^{\{b\}}\equiv 0,\,\,\,\,\,\text{and}\,\,\,\,\,T^a=\kappa_b^a\wedge e^b.$	$de^a+\bar{\omega}_b^a\wedge e^b\equiv 0,\quad\text{and}\quad T^a=\kappa_b^a\wedge e^b.$	$de^a+\bar{\omega}_b^a\wedge e^b\equiv 0,\quad\text{and}\quad T^a=\kappa_b^a\wedge e^b.$
cardona-qft-methods_ EQ0142 (6)	081	■ 0.963 ● N +0	$\sigma_L(\mathcal{D})(x_0,\xi)=\sum_{ \alpha =k}a_\alpha(x_0)\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$	$\sigma_L(\mathcal{D})(x_0,\xi)=\sum_{ \alpha =k}a_\alpha(x_0)\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$	$\sigma_L(\mathcal{D})(x_0,\xi)=\sum_{ \alpha =k}a_\alpha(x_0)\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$
cardona-qft-methods_ EQ0454	214	■ 0.964 ● N +0	$\left(C^\infty(M,\mathcal{E}),L^2(M,\mathcal{E}\otimes S),\mathcal{D}_\mathcal{E}\right),$	$(C^\infty(M,\mathcal{E}),L^2(M,\mathcal{E}\otimes S),\mathcal{D}_\mathcal{E}),$	$(C^\infty(M,\mathcal{E}),L^2(M,\mathcal{E}\otimes S),\mathcal{D}_\mathcal{E}),$
cardona-qft-methods_ EQ0086	050	■ 0.965 ●	$\begin{gathered} \operatorname{ Tad}_{\{ \mathcal{D} \} + A}(d-k) = -(d-k) \int A \mathcal{D} \mathcal{D} ^{-(d-k)-2}, \quad \forall k \neq d \\ \operatorname{ Tad}_{\mathcal{D} + A}(0) = - \int A \mathcal{D}^{-1}. \end{gathered}$	$\operatorname{ Tad}_{\mathcal{D} + A}(d-k) = -(d-k) \int A \mathcal{D} \mathcal{D} ^{-(d-k)-2}, \quad \forall k \neq d$ $\operatorname{ Tad}_{\mathcal{D} + A}(0) = - \int A \mathcal{D}^{-1}.$	$\mathrm{Tad}_{\mathcal{D}+A}(d-k) = -(d-k) \int A \mathcal{D} \mathcal{D} ^{-(d-k)-2}, \quad \forall k \neq d,$ $\mathrm{Tad}_{\mathcal{D}+A}(0) = - \int A \mathcal{D}^{-1}.$
cardona-qft-methods_ EQ0106 (1)	072	■ 0.965 ● N +0	$\operatorname{Ind}(A) := \operatorname{dimKer}(A) - \operatorname{dimCoker}(A) \in \mathbb{Z},$	$\operatorname{Ind}(A) := \operatorname{dimKer}(A) - \operatorname{dimCoker}(A) \in \mathbb{Z},$	$\operatorname{Ind}(A) := \operatorname{dimKer}(A) - \operatorname{dimCoker}(A) \in \mathbb{Z},$
cardona-qft-methods_ EQ0272	131	■ 0.966 ● K +0	$C_{G_1\amalg G_2}(T)=\chi(Y_{G_1})\chi(Y_{G_2})T^2+\text{h.o.t}:$	$C_{G_1\amalg G_2}(T)=\chi(Y_{G_1})\chi(Y_{G_2})T^2+\text{h.o.t}:$	$C_{G_1\amalg G_2}(T)=\chi(Y_{G_1})\chi(Y_{G_2})T^2+\text{h.o.t}:$
cardona-qft-methods_ EQ0055	038	■ 0.970 ● C +7	$\sigma_d^{\mathcal{P}_{g'}}(x,\xi)=e^{-h(x)/2}U_{g',g}^{-1}(x)\sigma_d^{\mathcal{P}_g}(x,\xi)U_{g',g}(x)e^{-h(x)/2},\quad \xi\in T_x^*M.$	$\sigma_d^{\mathcal{P}_{g'}}(x,\xi)=e^{-h(x)/2}U_{g',g}^{-1}(x)\sigma_d^{\mathcal{P}_g}(x,\xi)U_{g',g}(x)e^{-h(x)/2},\quad \xi\in T_x^*M.$	$\sigma_d^{\mathcal{P}_{g'}}(x,\xi)=e^{-h(x)/2}U_{g',g}^{-1}(x)\sigma_d^{\mathcal{P}_g}(x,\xi)U_{g',g}(x)e^{-h(x)/2},\quad \xi\in T_x^*M.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0321 (7)	161	0.971 ● N +0	$\begin{aligned} & \mathrm{d} \, x^{\mu} \left[\partial_{\mu} \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x) \right] \\ & = \mathrm{d} \, x^{\mu} D_{\mu} \phi^a(x) \end{aligned}$	$dx^{\mu} \left[\partial_{\mu} \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x) \right] = dx^{\mu} D_{\mu} \phi^a(x) = D\phi^a(x).$	$dx^{\mu} [\partial_{\mu} \phi^a(x) + \omega_{b\mu}^a(x) \phi^b(x)] = dx^{\mu} D_{\mu} \phi^a(x) = D\phi^a(x).$
cardona-qft-methods_ EQ0969 (9)	363	0.971 ● K +0	$X_{\{\Gamma\}} = \left\{ t \in \mathbb{P}_k^{n-1} \mid t_1 + t_2 + \cdots + t_n = 0 \right\} =: \mathcal{L}.$	$X_{\Gamma} = \left\{ t \in \mathbb{P}_k^{n-1} \mid t_1 + t_2 + \cdots + t_n = 0 \right\} =: \mathcal{L}.$	$X_{\Gamma} = \left\{ t \in \mathbb{P}_k^{n-1} \mid t_1 + t_2 + \cdots + t_n = 0 \right\} =: \mathcal{L}.$
cardona-qft-methods_ EQ0452	213	0.972 ● W +1	$\begin{gathered} \mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2 \quad \mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \\ D = D_1 \otimes 1 + \gamma_1 \otimes D_2 \\ \gamma = \gamma_1 \otimes \gamma_2 \quad J = J_1 \otimes J_2. \end{gathered}$	$\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2 \quad \mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \\ D = D_1 \otimes 1 + \gamma_1 \otimes D_2 \\ \gamma = \gamma_1 \otimes \gamma_2 \quad J = J_1 \otimes J_2$	$\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2 \quad \mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \\ D = D_1 \otimes 1 + \gamma_1 \otimes D_2 \\ \gamma = \gamma_1 \otimes \gamma_2 \quad J = J_1 \otimes J_2.$
cardona-qft-methods_ EQ0971 (11)	363	0.974 ● K +0	$\Psi_{\Gamma^{\vee}}(t) = t_2 t_3 \cdots t_n + t_1 t_3 \cdots t_n + \cdots + t_1 t_2 \cdots \hat{t}_i \cdots t_n + \cdots + t_1 t_2 \cdots t_{n-1},$	$\Psi_{\Gamma^{\vee}}(t) = t_2 t_3 \cdots t_n + t_1 t_3 \cdots t_n + \cdots + t_1 t_2 \cdots \hat{t}_i \cdots t_n + \cdots + t_1 t_2 \cdots t_{n-1},$	$\Psi_{\Gamma^{\vee}}(t) = t_2 t_3 \cdots t_n + t_1 t_3 \cdots t_n + \cdots + t_1 t_2 \cdots \hat{t}_i \cdots t_n + \cdots + t_1 t_2 \cdots t_{n-1},$
cardona-qft-methods_ EQ0591 (25)	251	0.974 ● N +1	$\begin{aligned} & \phi \otimes_{S_{\theta}} \chi \equiv \left(\frac{1 + \tau_0}{2} \right) (\phi \otimes \chi), \\ & \phi \otimes_{A_{\theta}} \chi \equiv \left(\frac{1 - \tau_0}{2} \right) (\phi \otimes \chi), \end{aligned}$	$\phi \otimes_{S_{\theta}} \chi \equiv \left(\frac{1 + \tau_0}{2} \right) (\phi \otimes \chi), \\ \phi \otimes_{A_{\theta}} \chi \equiv \left(\frac{1 - \tau_0}{2} \right) (\phi \otimes \chi),$	$\phi \otimes_{S_{\theta}} \chi \equiv \left(\frac{1 + \tau_0}{2} \right) (\phi \otimes \chi), \\ \phi \otimes_{A_{\theta}} \chi \equiv \left(\frac{1 - \tau_0}{2} \right) (\phi \otimes \chi),$
cardona-qft-methods_ EQ0960 (12)	361	0.974 ● K +0	$\pi_2 : \pi_1^{-1}(X_{\Gamma} \setminus \Sigma_n) \rightarrow X_{\Gamma^{\vee}} \setminus \Sigma_n.$	$\pi_2 : \pi_1^{-1}(X_{\Gamma} \setminus \Sigma_n) \rightarrow X_{\Gamma^{\vee}} \setminus \Sigma_n.$	$\pi_2 : \pi_1^{-1}(X_{\Gamma} \setminus \Sigma_n) \rightarrow X_{\Gamma^{\vee}} \setminus \Sigma_n.$
cardona-qft-methods_ EQ0302 (6)	151	0.975 ● K +0	$\delta I = \int_M \delta \mathcal{C} \wedge \star j = \int_M (d\Omega) \wedge \star j = \int_M d(\Omega \wedge \star j) - \int_M \Omega \wedge d \star j,$	$\delta I = \int_M \delta \mathcal{C} \wedge \star j = \int_M (d\Omega) \wedge \star j = \int_M d(\Omega \wedge \star j) - \int_M \Omega \wedge d \star j,$	$\delta I = \int_M \delta \mathcal{C} \wedge \star j = \int_M (d\Omega) \wedge \star j = \int_M d(\Omega \wedge \star j) - \int_M \Omega \wedge d \star j,$
cardona-qft-methods_ EQ0358 (28)	171	0.976 ●	$\begin{aligned} & C_3^{\text{Lor}} = \omega_b^a d\omega_a^b + \frac{2}{3} \omega_b^a \omega_c^b \omega_a^c \\ & C_3^{\text{Tor}} = e^a T_a. \end{aligned}$	$C_3^{\text{Lor}} = \omega_b^a d\omega_a^b + \frac{2}{3} \omega_b^a \omega_c^b \omega_a^c \\ C_3^{\text{Tor}} = e^a T_a.$	$C_3^{\text{Lor}} = \omega_b^a d\omega_a^b + \frac{2}{3} \omega_b^a \omega_c^b \omega_a^c \\ C_3^{\text{Tor}} = e^a T_a.$
cardona-qft-methods_ EQ0027 (8)	027	0.976 ● N +1	$\operatorname{WRes}\left((1+\Delta)^{-d/2}\right) = \operatorname{Vol}\left(\mathbb{S}^{d-1}\right) = \frac{2\pi^{d/2}}{\Gamma(d/2)}.$	$\operatorname{WRes}\left((1+\Delta)^{-d/2}\right) = \operatorname{Vol}\left(\mathbb{S}^{d-1}\right) = \frac{2\pi^{d/2}}{\Gamma(d/2)}.$	$\operatorname{WRes}\left((1+\Delta)^{-d/2}\right) = \operatorname{Vol}\left(\mathbb{S}^{d-1}\right) = \frac{2\pi^{d/2}}{\Gamma(d/2)}.$
cardona-qft-methods_ EQ0336 (22)	164	0.978 ● N +0	$D \, T^{\{a\}} = R_{\{b\}}^{\{a\}} \wedge e^{\{b\}}.$	$DT^a = R_b^a \wedge e^b.$	$DT^a = R_b^a \wedge e^b.$
cardona-qft-methods_ EQ0925	352	0.978 ● K +0	$\left(P_G M \times_G EG_n \right)^{-e_0^* TM_n}.$	$(P_G M \times_G EG_n)^{-e_0^* TM_n}.$	$(P_G M \times_G EG_n)^{-e_0^* TM_n}.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0303 (7)	152	■ 0.979 ● N +1	$\begin{aligned} \mathbf{A} &= \mathbf{A}_{\mu} dx^{\mu} \\ &= A_{\mu}^a \mathbf{K}_a dx^{\mu} \end{aligned}$	$\mathbf{A} = \mathbf{A}_{\mu} dx^{\mu} = A_{\mu}^a \mathbf{K}_a dx^{\mu},$	$\mathbf{A} = \mathbf{A}_{\mu} dx^{\mu} = A_{\mu}^a \mathbf{K}_a dx^{\mu},$
cardona-qft-methods_ EQ0489	221	■ 0.980 ● K +0	$\int \frac{1}{2} D_{\mu} \mathbf{H} ^2 \sqrt{g} d^4 x;$	$\int \frac{1}{2} D_{\mu} \mathbf{H} ^2 \sqrt{g} d^4 x;$	$\int \frac{1}{2} D_{\mu} \mathbf{H} ^2 \sqrt{g} d^4 x;$
cardona-qft-methods_ EQ0268	127	■ 0.981 ● K +0	$\Psi_{G_1}(t_1, \dots, t_{n_1}) \neq 0 \quad \text{and} \quad \Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0,$	$\Psi_{G_1}(t_1, \dots, t_{n_1}) \neq 0 \quad \text{and} \quad \Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0,$	$\Psi_{G_1}(t_1, \dots, t_{n_1}) \neq 0 \quad \text{and} \quad \Psi_{G_2}(u_1, \dots, u_{n_2}) \neq 0,$
cardona-qft-methods_ EQ0095	056	■ 0.983 ● W +2	$g\left(t\mathcal{D}^2\right)=\int_0^\infty e^{-st\mathcal{D}^2}\phi(s)ds,\mathrm{Tr}\left(g\left(t\mathcal{D}^2\right)\right)\underset{t\downarrow 0}{\sim}\sum_{\alpha\in\mathrm{Sp}^+}a_\alpha t^\alpha\int_0^\infty s^\alpha g(s)ds$	$g\left(t\mathcal{D}^2\right)=\int_0^\infty e^{-st\mathcal{D}^2}\phi(s)ds,\mathrm{Tr}\left(g\left(t\mathcal{D}^2\right)\right)\underset{t\downarrow 0}{\sim}\sum_{\alpha\in\mathrm{Sp}^+}a_\alpha t^\alpha\int_0^\infty s^\alpha g(s)ds.$	$g\left(t\mathcal{D}^2\right)=\int_0^\infty e^{-st\mathcal{D}^2}\phi(s)ds,\mathrm{Tr}\left(g\left(t\mathcal{D}^2\right)\right)\underset{t\downarrow 0}{\sim}\sum_{\alpha\in\mathrm{Sp}^+}a_\alpha t^\alpha\int_0^\infty s^\alpha g(s)ds.$
cardona-qft-methods_ EQ0379 (20)	177	■ 0.983 ● K +0	$L_4=\epsilon_{abcd}R^{ab}e^ce^d.$	$L_4=\epsilon_{abcd}R^{ab}e^ce^d.$	$L_4=\epsilon_{abcd}R^{ab}e^ce^d.$
cardona-qft-methods_ EQ0490	221	■ 0.983 ● N +1	$\frac{1}{4}G_{\mu\nu}^i\bar{G}^{\mu\nu i}+\frac{1}{4}F_{\mu\nu}^\alpha\bar{F}^{\mu\nu\alpha}+\frac{1}{4}B_{\mu\nu}\bar{B}^{\mu\nu}.$	$\frac{1}{4}G_{\mu\nu}^i\bar{G}^{\mu\nu i}+\frac{1}{4}F_{\mu\nu}^\alpha\bar{F}^{\mu\nu\alpha}+\frac{1}{4}B_{\mu\nu}\bar{B}^{\mu\nu}.$	$\frac{1}{4}G_{\mu\nu}^i\bar{G}^{\mu\nu i}+\frac{1}{4}F_{\mu\nu}^\alpha\bar{F}^{\mu\nu\alpha}+\frac{1}{4}B_{\mu\nu}\bar{B}^{\mu\nu}.$
cardona-qft-methods_ EQ0698	271	■ 0.983 ● N +0	$\hat{H}=\sum_{l=1}^L\mathcal{H}_{l,l+1}.$	$\hat{H}=\sum_{l=1}^L\mathcal{H}_{l,l+1}.$	$\hat{H}=\sum_{l=1}^L\mathcal{H}_{l,l+1}.$
cardona-qft-methods_ EQ0368 (7)	175	■ 0.984 ● N +0	$\begin{aligned} \delta\omega^{ab} &= d\lambda^{ab} + \omega_c^a \lambda^{cb} + \omega_c^b \lambda^{ac} \pm [e^a \lambda^b - \lambda^a e^b] l^{-2} \\ \delta e^a &= d\lambda^a + \omega_b^a \lambda^b - \lambda_b^a e^b. \end{aligned}$	$\delta\omega^{ab} = d\lambda^{ab} + \omega_c^a \lambda^{cb} + \omega_c^b \lambda^{ac} \pm [e^a \lambda^b - \lambda^a e^b] l^{-2}$ $\delta e^a = d\lambda^a + \omega_b^a \lambda^b - \lambda_b^a e^b.$	$\delta\omega^{ab} = d\lambda^{ab} + \omega_c^a \lambda^{cb} + \omega_c^b \lambda^{ac} \pm [e^a \lambda^b - \lambda^a e^b] l^{-2}$ $\delta e^a = d\lambda^a + \omega_b^a \lambda^b - \lambda_b^a e^b.$
cardona-qft-methods_ EQ0730	277	■ 0.984 ● N +0	$\left(\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\right)^L=\prod_{\substack{j=1\\j\neq k}}^Mu_k-u_j+i.$	$\left(\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\right)^L=\prod_{\substack{j=1\\j\neq k}}^Mu_k-u_j+i.$	$\left(\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\right)^L=\prod_{\substack{j=1\\j\neq k}}^Mu_k-u_j+i.$
cardona-qft-methods_ EQ0781 (22)	298	■ 0.984 ●	$X^{\{M\}}(\tau,\sigma)=x^{\{M\}}+2\,\alpha^{\{\prime\}}p^{\{M\}}\,\tau+\mathrm{i}\left(2\,\alpha^{\{\prime\}}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-in\tau}\cos(n\sigma).$	$X^M(\tau,\sigma)=x^M+2\alpha'p^M\tau+\mathrm{i}(2\alpha')^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-in\tau}\cos(n\sigma).$	$X^M(\tau,\sigma)=x^M+2\alpha'p^M\tau+\mathrm{i}(2\alpha')^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha_n^Me^{-in\tau}\cos(n\sigma).$
cardona-qft-methods_ EQ0434	208	■ 0.985 ● K +0	$\mathcal{E}=\mathbf{2}_L\otimes\mathbf{1}^0\oplus\mathbf{2}_R\otimes\mathbf{1}^0\oplus\mathbf{2}_L\otimes\mathbf{3}^0\oplus\mathbf{2}_R\otimes\mathbf{3}^0,$	$\mathcal{E}=\mathbf{2}_L\otimes\mathbf{1}^0\oplus\mathbf{2}_R\otimes\mathbf{1}^0\oplus\mathbf{2}_L\otimes\mathbf{3}^0\oplus\mathbf{2}_R\otimes\mathbf{3}^0,$	$\mathcal{E}=\mathbf{2}_L\otimes\mathbf{1}^0\oplus\mathbf{2}_R\otimes\mathbf{1}^0\oplus\mathbf{2}_L\otimes\mathbf{3}^0\oplus\mathbf{2}_R\otimes\mathbf{3}^0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0485	221	0.985 C -4	$\begin{aligned} & \& \mathfrak{a}=\operatorname{Tr} \left(Y_{\nu}^{\dagger} Y_{\nu}+Y_e^{\dagger} Y_e+3\left(Y_u^{\dagger} Y_u+Y_d^{\dagger} Y_d\right)\right. \\ & \left. \left(Y_{\nu}^{\dagger} Y_{\nu}\right)^2+\left(Y_e^{\dagger} Y_e\right)^2+3\left(Y_u^{\dagger} Y_u\right)^2+3\left(Y_d^{\dagger} Y_d\right)^2\right) \\ & \& \mathfrak{b}=\operatorname{Tr} \left(M M^{\dagger}\right) \\ & \& \mathfrak{d}=\operatorname{Tr} \left(\left(M M^{\dagger}\right)^2\right) \\ & \& \mathfrak{e}=\operatorname{Tr} \left(M M^{\dagger} Y_{\nu}^{\dagger} Y_{\nu}\right) . \end{aligned}$	$\begin{aligned} \mathfrak{a} &= \operatorname{Tr}\left(Y_{\nu}^{\dagger} Y_{\nu}+Y_e^{\dagger} Y_e+3\left(Y_u^{\dagger} Y_u+Y_d^{\dagger} Y_d\right)\right) \\ \mathfrak{b} &= \operatorname{Tr}\left(\left(Y_{\nu}^{\dagger} Y_{\nu}\right)^2+\left(Y_e^{\dagger} Y_e\right)^2+3\left(Y_u^{\dagger} Y_u\right)^2+3\left(Y_d^{\dagger} Y_d\right)^2\right) \\ \mathfrak{c} &= \operatorname{Tr}\left(M M^{\dagger}\right) \\ \mathfrak{d} &= \operatorname{Tr}\left(\left(M M^{\dagger}\right)^2\right) \\ \mathfrak{e} &= \operatorname{Tr}\left(M M^{\dagger} Y_{\nu}^{\dagger} Y_{\nu}\right) . \end{aligned}$	
cardona-qft-methods_ EQ0085 (30)	049	0.985 C +4	$\begin{aligned} & \& f\left D_A\right ^{\left(-d\right)}=f\left D\right ^{\left(-d\right)}, \\ & \& f\left D_A\right ^{\left(-\left(d-1\right)\right)}=f\left D\right ^{\left(-\left(d-1\right)\right)}-\left(\frac{d-1}{2}\right) f X\left D\right ^{\left(-d-1\right)}, \\ & \& f\left D_A\right ^{\left(-\left(d-2\right)\right)}=f\left D\right ^{\left(-\left(d-2\right)\right)}+\frac{d-2}{2}\left(-f X\left D\right ^{\left(-d\right)}+\frac{d}{4} f X^2\left D\right ^{\left(-2-d\right)}\right) . \end{aligned}$	$\begin{aligned} f\left D_A\right ^{\left(-d\right)} &=f\left D\right ^{\left(-d\right)}, \\ f\left D_A\right ^{\left(-\left(d-1\right)\right)} &=f\left D\right ^{\left(-\left(d-1\right)\right)}-\left(\frac{d-1}{2}\right) f X\left D\right ^{\left(-d-1\right)}, \\ f\left D_A\right ^{\left(-\left(d-2\right)\right)} &=f\left D\right ^{\left(-\left(d-2\right)\right)}+\frac{d-2}{2}\left(-f X\left D\right ^{\left(-d\right)}+\frac{d}{4} f X^2\left D\right ^{\left(-2-d\right)}\right) . \end{aligned}$	
cardona-qft-methods_ EQ0250 (1)	118	0.986 N +0	$S_{\left\{\mathrm{eff}\right\}}\left(\phi\right)=\sum _G\left\langle G\right\rangle$	$S_{\mathrm{eff}}(\phi)=\sum_G\langle G\rangle$	$S_{\mathrm{eff}}(\phi)=\sum_G\langle G\rangle$
cardona-qft-methods_ EQ0102	061	0.986 N +0	$U_{\left\{k\right\}}U_{\left\{q\right\}}=e^{\left\{-\frac{i}{2}k\cdot\Theta q\right\}}U_{k+q},\text{ and }U_{\left\{k\right\}}U_{\left\{q\right\}}=e^{\left\{-ik\cdot\Theta q\right\}}U_qU_k$	$U_kU_q=e^{-\frac{i}{2}k\cdot\Theta q}U_{k+q},\text{ and }U_kU_q=e^{-ik\cdot\Theta q}U_qU_k$	$U_kU_q=e^{-\frac{i}{2}k\cdot\Theta q}U_{k+q},\text{ and }U_kU_q=e^{-ik\cdot\Theta q}U_qU_k$
cardona-qft-methods_ EQ0445	211	0.986 N +0	$S_{\left\{3\right\}}=\left(\begin{array}{cccc} 0&0&Y_{\left(\uparrow 3\right)}^*&0\\ 0&0&0&Y_{\left(\downarrow 3\right)}^*\\ Y_{\left(\uparrow 3\right)}&0&0&0\\ 0&Y_{\left(\downarrow 3\right)}&0&0 \end{array}\right)$	$S_3=\left(\begin{array}{cccc} 0&0&Y_{\left(\uparrow 3\right)}^*&0\\ 0&0&0&Y_{\left(\downarrow 3\right)}^*\\ Y_{\left(\uparrow 3\right)}&0&0&0\\ 0&Y_{\left(\downarrow 3\right)}&0&0 \end{array}\right)$	$S_3=\left(\begin{array}{cccc} 0&0&Y_{\left(\uparrow 3\right)}^*&0\\ 0&0&0&Y_{\left(\downarrow 3\right)}^*\\ Y_{\left(\uparrow 3\right)}&0&0&0\\ 0&Y_{\left(\downarrow 3\right)}&0&0 \end{array}\right)$
cardona-qft-methods_ EQ0441	210	0.987	$\begin{array}{ccccccc} \uparrow \otimes \mathbf{1}^0 & \downarrow \otimes \mathbf{1}^0 & \uparrow \otimes \mathbf{3}^0 & \downarrow \otimes \mathbf{3}^0 & & & \\ \mathbf{2}_L & & -1 & & -1 & \frac{1}{3} & \frac{1}{3} \\ \mathbf{2}_R & & 0 & & -2 & \frac{4}{3} & -\frac{2}{3} \end{array}$	$\begin{array}{ccccccc} \uparrow \otimes \mathbf{1}^0 & \downarrow \otimes \mathbf{1}^0 & \uparrow \otimes \mathbf{3}^0 & \downarrow \otimes \mathbf{3}^0 & & & \\ \mathbf{2}_L & & -1 & & -1 & \frac{1}{3} & \frac{1}{3} \\ \mathbf{2}_R & & 0 & & -2 & \frac{4}{3} & -\frac{2}{3} \end{array}$	
Conf. MathPix confidence: 0.985					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0078 (26)	046	0.987 ● N +0	$\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A},$ $\quad \widetilde{A}=A+\epsilon JAJ^{-1}, \quad A=A^*$.	$\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A}, \quad \widetilde{A}=A+\epsilon JAJ^{-1}, \quad A=A^*.$	$\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A}, \quad \widetilde{A}=A+\epsilon JAJ^{-1}, \quad A=A^*.$
cardona-qft-methods_ EQ0164	088	0.988 ● N +0	$\mathrm{t}\text{-}\operatorname{Ind}_{\{G\}}:K_{\{G\}}(M)\rightarrow K_{\{G\}}(\{pt\})\simeq R(G).$ $K_{\{G\}}(\backslash\{p\text{ t}\})\backslash\operatorname{simeq} R(G) \text{ .}$	$\mathrm{t}\text{-}\operatorname{Ind}_G:K_G(M)\rightarrow K_G(\{pt\})\simeq R(G).$	$\mathrm{t}\text{-}\operatorname{Ind}_G:K_G(M)\rightarrow K_G(\{pt\})\simeq R(G).$
cardona-qft-methods_ EQ0956 (8)	361	0.988 ● K +0	$\Psi_{\Gamma}(\mathrm{t})=\left(\prod_{i=1}^n\mathrm{t}_i\right)\Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$ $\Psi_{\Gamma}(\mathrm{t})=\left(\prod_{i=1}^n\mathrm{t}_i\right)\Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$	$\Psi_{\Gamma}(t)=\left(\prod_{i=1}^nt_i\right)\Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$	$\Psi_{\Gamma}(t)=\left(\prod_{i=1}^nt_i\right)\Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$
cardona-qft-methods_ EQ0655 (5)	262	0.988 ● C -7	$\begin{aligned} \langle \theta \rho_{\theta}(x) \eta_{\theta}(y) 0 \rangle &= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} \langle 0 \rho_0(x) \eta_0(y) 0 \rangle \\ &= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y) = F(x-y) \\ &= \langle 0 \rho_0(x) \eta_0(y) 0 \rangle \end{aligned}$ $\begin{aligned} \langle \theta \rho_{\theta}(x) \eta_{\theta}(y) 0 \rangle &= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} \langle 0 \rho_0(x) \eta_0(y) 0 \rangle \\ &= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y) = F(x-y) \\ &= \langle 0 \rho_0(x) \eta_0(y) 0 \rangle \end{aligned}$	$\langle 0 \rho_{\theta}(x) \eta_{\theta}(y) 0 \rangle = e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} \langle 0 \rho_0(x) \eta_0(y) 0 \rangle$ $= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y) = F(x-y)$ $= \langle 0 \rho_0(x) \eta_0(y) 0 \rangle$	$\langle 0 \rho_{\theta}(x) \eta_{\theta}(y) 0 \rangle = e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} \langle 0 \rho_0(x) \eta_0(y) 0 \rangle$ $= e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y) = F(x-y)$ $= \langle 0 \rho_0(x) \eta_0(y) 0 \rangle$
cardona-qft-methods_ EQ0375 (16)	176	0.989 ● N +0	$\Upsilon_{\{0\}^A B}=\left[\begin{array}{cc} : \eta^a{}_b & 0 \\ 0 & 0 \end{array}\right],$ $\Upsilon_{\{0\}^A B}=\left[\begin{array}{cc} : \eta^a{}_b & 0 \\ 0 & 0 \end{array}\right],$	$\Upsilon_0^{AB}=\left[\begin{array}{cc} : \eta^{ab} & 0 \\ 0 & 0 \end{array}\right],$	$\Upsilon_0^{AB}=\left[\begin{array}{cc} : \eta^{ab} & 0 \\ 0 & 0 \end{array}\right],$
cardona-qft-methods_ EQ0884	343	0.989 ● N +0	$P\text{ . }D: H_{\{p\}}(M)\cong\left(H^{\{p\}}(M)\right)^*\cong H^{\{n-p\}}(M) \text{ .}$ $H^{\{n-p\}}(M) \text{ .}$	$P.D: H_p(M)\cong (H^p(M))^*\cong H^{n-p}(M).$	$P.D: H_p(M)\cong (H^p(M))^*\cong H^{n-p}(M).$
cardona-qft-methods_ EQ0937	354	0.989 ● N +0	$H_{\{*\}^{\{p\text{ r o}\}}}\left(L\text{ B }G^{-T\text{ B }G}\right)\cong H^{\{*\}}(L\text{ B }G) \text{ .}$ $H_{\{*\}^{\{p\text{ r o}\}}}\left(L\text{ B }G^{-T\text{ B }G}\right)\cong H^{\{*\}}(L\text{ B }G) \text{ .}$	$H_*^{pro}\left(LBG^{-TBG}\right)\cong H^*(LBG).$	$H_*^{pro}(LBG^{-TBG})\cong H^*(LBG).$
cardona-qft-methods_ EQ0897	347	0.989 ● W +0	$\begin{aligned} &\& H\,H_{\{*\}}(A,M)\cong \\ &\operatorname{Tor}_{\{A\otimes A^{\circ p}\}^{\{*\}}(A,M)}\& \\ &H\,H^{\{*\}}(A,M)\cong \operatorname{Ext}_{\{A\otimes A^{\circ p}\}^{\{*\}}(A,M)}\end{aligned}$ $\begin{aligned} &\& H\,H_{\{*\}}(A,M)\cong \\ &\operatorname{Tor}_{\{A\otimes A^{\circ p}\}^{\{*\}}(A,M)}\& \\ &H\,H^{\{*\}}(A,M)\cong \operatorname{Ext}_{\{A\otimes A^{\circ p}\}^{\{*\}}(A,M)}\end{aligned}$	$HH_*(A,M)\cong \operatorname{Tor}_{A\otimes A^{op}}^*(A,M)$ $HH^*(A,M)\cong \operatorname{Ext}_{A\otimes A^{op}}^*(A,M)$	$HH_*(A,M)\cong \operatorname{Tor}_{A\otimes A^{op}}^*(A,M)$ $HH^*(A,M)\cong \operatorname{Ext}_{A\otimes A^{op}}^*(A,M)$
cardona-qft-methods_ EQ0513 (6)	230	0.990 ● N +0	$\beta_{\{j\}^{\{\text{grav}\}}}=\frac{a_{\{j\}}}{8\pi}\frac{\Lambda^2}{M_P^2(\Lambda)}x_j,$ $\beta_{\{j\}^{\{\text{grav}\}}}=\frac{a_{\{j\}}}{8\pi}\frac{\Lambda^2}{M_P^2(\Lambda)}x_j,$	$\beta_j^{\text{grav}}=\frac{a_j}{8\pi}\frac{\Lambda^2}{M_P^2(\Lambda)}x_j,$	$\beta_j^{\text{grav}}=\frac{a_j}{8\pi}\frac{\Lambda^2}{M_P^2(\Lambda)}x_j,$
cardona-qft-methods_ EQ0002	018	0.990 ● W +1	$\left \partial_x^\alpha\partial_y^\gamma\partial_\xi^\beta a(x,y,\xi)\right \leq C_{K\alpha\beta\gamma}(1+ \xi)^{\Re(m)- \beta }, \quad \forall x,y\in K,\xi\in\mathbb{R}^d$ $\left \partial_x^\alpha\partial_y^\gamma\partial_\xi^\beta a(x,y,\xi)\right \leq C_{K\alpha\beta\gamma}(1+ \xi)^{\Re(m)- \beta }, \quad \forall x,y\in K,\xi\in\mathbb{R}^d.$	$\left \partial_x^\alpha\partial_y^\gamma\partial_\xi^\beta a(x,y,\xi)\right \leq C_{K\alpha\beta\gamma}(1+ \xi)^{\Re(m)- \beta }, \quad \forall x,y\in K,\xi\in\mathbb{R}^d$	$ \partial_x^\alpha\partial_y^\gamma\partial_\xi^\beta a(x,y,\xi) \leq C_{K\alpha\beta\gamma}(1+ \xi)^{\Re(m)- \beta }, \quad \forall x,y\in K,\xi\in\mathbb{R}^d.$
cardona-qft-methods_ EQ0199 (9)	095	0.990 ●	$\mathcal{D}=\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_1}+\left(\begin{array}{ll} 0&-i\\ i&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_2}+\left(\begin{array}{ll} 1&0\\ 0&-1 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_3}.$ $\mathcal{D}=\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_1}+\left(\begin{array}{ll} 0&-i\\ i&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_2}+\left(\begin{array}{ll} 1&0\\ 0&-1 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_3}.$	$\mathcal{D}=\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_1}+\left(\begin{array}{ll} 0&-i\\ i&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_2}+\left(\begin{array}{ll} 1&0\\ 0&-1 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_3}.$	$\mathcal{D}=\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_1}+\left(\begin{array}{ll} 0&-i\\ i&0 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_2}+\left(\begin{array}{ll} 1&0\\ 0&-1 \end{array}\right)\frac{1}{i}\frac{\partial}{\partial x_3}.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0301 (5)	151	0.990 K +0	$I=\int \mathcal{C} \wedge \star j,$	$I = \int \mathcal{C} \wedge \star j,$	$I = \int \mathcal{C} \wedge \star j,$
cardona-qft-methods_ EQ0409 (39)	187	0.991 S +1	$B_{2n}=-\kappa n \int_0^1 dt \int_0^t ds \epsilon \theta e \left(\widetilde{R} + t^2 \theta^2 + s^2 e^2 \right)^{n-1}$	$B_{2n} = -\kappa n \int_0^1 dt \int_0^t ds \epsilon \theta e \left(\widetilde{R} + t^2 \theta^2 + s^2 e^2 \right)^{n-1}$	$B_{2n} = -\kappa n \int_0^1 dt \int_0^t ds \epsilon \theta e \left(\widetilde{R} + t^2 \theta^2 + s^2 e^2 \right)^{n-1},$
cardona-qft-methods_ EQ0787 (28)	299	0.991 S +0	$\begin{array}{rc} \mathrm{R}: & \Psi^M(\tau,0)=\Psi^M(\tau,2\pi) \\ \mathrm{NS}: & \Psi^M(\tau,0)=-\Psi^M(\tau,2\pi) \end{array}$	$\mathrm{R}:$ $\Psi^M(\tau,0)=\Psi^M(\tau,2\pi)$ $\mathrm{NS}:$ $\Psi^M(\tau,0)=-\Psi^M(\tau,2\pi)$	$\mathrm{R}:$ $\Psi^M(\tau,0)=\Psi^M(\tau,2\pi)$ $\mathrm{NS}:$ $\Psi^M(\tau,0)=-\Psi^M(\tau,2\pi)$
cardona-qft-methods_ EQ0995	374	0.991 N +1	$\Delta X^{\alpha}=\sum_{\beta\leq\alpha}\binom{\alpha}{\beta}X^{\beta}\otimes X^{\alpha-\beta},$	$\Delta X^{\alpha} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}$	$\Delta X^{\alpha} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta},$
cardona-qft-methods_ EQ1003	375	0.991 C +3	$\begin{aligned} f_{ab}(x) &= \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} ab(X^{\alpha}) = \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) (\Delta X^{\alpha}) \\ &= \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) \left(\sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta} \right) \\ &= \sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta! \gamma!} a(X^{\beta}) b(X^{\gamma}) = f_a(x) f_b(x). \end{aligned}$	$f_{ab}(x) = \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} ab(X^{\alpha}) = \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) (\Delta X^{\alpha})$ $= \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) \left(\sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta} \right)$ $= \sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta! \gamma!} a(X^{\beta}) b(X^{\gamma}) = f_a(x) f_b(x).$	$f_{ab}(x) = \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} ab(X^{\alpha}) = \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) (\Delta X^{\alpha})$ $= \sum_{\alpha} \frac{x^{\alpha}}{\alpha!} (a \otimes b) \left(\sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta} \right)$ $= \sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta! \gamma!} a(X^{\beta}) b(X^{\gamma}) = f_a(x) f_b(x).$
cardona-qft-methods_ EQ0118 (6)	076	0.991 N +1	$\left\{\bigoplus_{V\in\mathrm{Irr}(G)}m_VV\mid m_V\in\mathbb{Z},\text{ only finitely many numbers }m_V\neq 0\right\}.$	$\left\{\bigoplus_{V\in\mathrm{Irr}(G)}m_VV\mid m_V\in\mathbb{Z},\text{ only finitely many numbers }m_V\neq 0\right\}.$	$\left\{\bigoplus_{V\in\mathrm{Irr}(G)}m_VV\mid m_V\in\mathbb{Z},\text{ only finitely many numbers }m_V\neq 0\right\}.$
cardona-qft-methods_ EQ0835 (14)	317	0.991 N +1	$\begin{aligned} &e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_+)=\mathrm{d}A\wedge\bar{\Phi}_+-\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIB}}-\\ &e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_-)=0, \end{aligned}$	$e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_+)=\mathrm{d}A\wedge\bar{\Phi}_+-\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIB}}-$ $e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_-)=0$	$e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_+)=\mathrm{d}A\wedge\bar{\Phi}_+-\frac{\mathrm{i}}{16}e^{\phi+A}\ast F_{\mathrm{IIB}}-$ $e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_-)=0,$
cardona-qft-methods_ EQ0088	050	0.992	$\operatorname{Tr}\left(P\,e^{-t\mathcal{D}^2}\right)_{t\downarrow 0^+}\sim\sum_{k=0}^{\infty}a_kt^{(k-\operatorname{ord}(P)-d)/2}+\sum_{k=0}^{\infty}(-a'_k\log t+b_k)t^k,$	$\operatorname{Tr}\left(Pe^{-t\mathcal{D}^2}\right)_{t\downarrow 0^+}\sim\sum_{k=0}^{\infty}a_kt^{(k-\operatorname{ord}(P)-d)/2}+\sum_{k=0}^{\infty}(-a'_k\log t+b_k)t^k$	$\operatorname{Tr}(Pe^{-t\mathcal{D}^2})_{t\downarrow 0^+}\sim\sum_{k=0}^{\infty}a_kt^{(k-\operatorname{ord}(P)-d)/2}+\sum_{k=0}^{\infty}(-a'_k\log t+b_k)t^k,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08582	226	■ 0.992 ● N +0	$\begin{gathered} \zeta=\exp\left(2\pi i/3\right)\\ U_{PMNS}(\Lambda_{unif})=\frac{1}{3}\begin{pmatrix} 1 & \zeta & \zeta^2 \\ \zeta & 1 & \zeta \\ \zeta^2 & \zeta & 1 \end{pmatrix} \end{gathered}$	$\zeta=\exp(2\pi i/3)$ $U_{PMNS}(\Lambda_{unif})=\frac{1}{3}\begin{pmatrix} 1 & \zeta & \zeta^2 \\ \zeta & 1 & \zeta \\ \zeta^2 & \zeta & 1 \end{pmatrix}$	$\zeta=\exp(2\pi i/3)$ $U_{PMNS}(\Lambda_{unif})=\frac{1}{3}\begin{pmatrix} 1 & \zeta & \zeta^2 \\ \zeta & 1 & \zeta \\ \zeta^2 & \zeta & 1 \end{pmatrix}$
cardona-qft-methods_ E08141 (5)	081	■ 0.992 ● N +0	$\sigma_L(\mathcal{D})(x_0,\xi)e:=\lim_{t\rightarrow\infty}t^{-k}\mathcal{D}\left(e^{itf(x)}s(x)\right)\Big _{x=x_0}.$	$\sigma_L(\mathcal{D})(x_0,\xi)e:=\lim_{t\rightarrow\infty}t^{-k}\mathcal{D}\left(e^{itf(x)}s(x)\right)\Big _{x=x_0}.$	$\sigma_L(\mathcal{D})(x_0,\xi)e:=\lim_{t\rightarrow\infty}t^{-k}\mathcal{D}\left(e^{itf(x)}s(x)\right)\Big _{x=x_0}.$
cardona-qft-methods_ E08461	216	■ 0.992 ● K +0	$V^{\prime}=-V-\frac{1}{3}\begin{pmatrix} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{pmatrix}=-V-\frac{1}{3}\Lambda 1_3.$	$V'=-V-\frac{1}{3}\begin{pmatrix} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{pmatrix}=-V-\frac{1}{3}\Lambda 1_3.$	$V'=-V-\frac{1}{3}\begin{pmatrix} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{pmatrix}=-V-\frac{1}{3}\Lambda 1_3.$
cardona-qft-methods_ E08372 (12)	176	■ 0.993 ● N +0	$\Upsilon^{AB}=\begin{bmatrix}\eta^{ab}&0\\0&\mp 1\end{bmatrix},$	$\Upsilon^{AB}=\begin{bmatrix}\eta^{ab}&0\\0&\mp 1\end{bmatrix},$	$\Upsilon^{AB}=\begin{bmatrix}\eta^{ab}&0\\0&\mp 1\end{bmatrix},$
cardona-qft-methods_ E08374 (14)	176	■ 0.993 ● N +1	$\begin{aligned} \delta\omega^{ab} &= d\lambda^{ab} + \omega_c^a\lambda^{cb} + \omega_c^b\lambda^{ac} = D_\omega\lambda^{ab} \\ \delta e^a &= d\lambda^a + \omega_b^a\lambda^b - \lambda_b^ae^b = D_\omega\lambda^a \end{aligned}$	$\delta\omega^{ab} = d\lambda^{ab} + \omega_c^a\lambda^{cb} + \omega_c^b\lambda^{ac} = D_\omega\lambda^{ab}$ $\delta e^a = d\lambda^a + \omega_b^a\lambda^b - \lambda_b^ae^b = D_\omega\lambda^a$	$\delta\omega^{ab} = d\lambda^{ab} + \omega_c^a\lambda^{cb} + \omega_c^b\lambda^{ac} = D_\omega\lambda^{ab}$ $\delta e^a = d\lambda^a + \omega_b^a\lambda^b - \lambda_b^ae^b = D_\omega\lambda^a.$
cardona-qft-methods_ E08685 (14)	253	■ 0.993 ● K +0	$e_p(x)=\exp{-ip\cdot x},\quad p_0 =(\mathbf{p}^2+m^2)^{\frac{1}{2}},\quad d\mu(p)=\frac{d^dp}{2 p_0 },$	$e_p(x)=\exp{-ip\cdot x},\quad p_0 =(\mathbf{p}^2+m^2)^{\frac{1}{2}},\quad d\mu(p)=\frac{d^dp}{2 p_0 },$	$e_p(x)=\exp{-ip\cdot x},\quad p_0 =(\mathbf{p}^2+m^2)^{\frac{1}{2}},\quad d\mu(p)=\frac{d^dp}{2 p_0 },$
cardona-qft-methods_ E08748 (8)	282	■ 0.993 ● N +0	$\begin{aligned} Q\cdot\varphi&\sim\psi, & \bar{Q}\cdot\varphi&\sim\bar{\psi} \\ Q\cdot\psi&\sim F+g_{YM}[\varphi,\varphi], & \bar{Q}\cdot\psi&\sim\mathcal{D}\varphi \end{aligned}$	$Q\cdot\varphi\sim\psi,\quad\bar{Q}\cdot\varphi\sim\bar{\psi}$ $Q\cdot\psi\sim F+g_{YM}[\varphi,\varphi],\quad\bar{Q}\cdot\psi\sim\mathcal{D}\varphi$	$Q\cdot\varphi\sim\psi\quad,\quad\bar{Q}\cdot\varphi\sim\bar{\psi}$ $Q\cdot\psi\sim F+g_{YM}[\varphi,\varphi]\quad,\quad\bar{Q}\cdot\psi\sim\mathcal{D}\varphi$
cardona-qft-methods_ E09075	045	■ 0.993 ● K +0	$\Psi(\mathcal{A})=\{T\, \,\forall N\in\mathbb{N}\exists P\in\mathcal{D}(\mathcal{A}),R\in OP^{-N},p\in\mathbb{N}\text{ s.t. }T=P \mathcal{D} ^{-p}+R\}.$	$\Psi(\mathcal{A})=\{T\, \,\forall N\in\mathbb{N}\exists P\in\mathcal{D}(\mathcal{A}),R\in OP^{-N},p\in\mathbb{N}\text{ s.t. }T=P \mathcal{D} ^{-p}+R\}.$	$\Psi(\mathcal{A})=\{T\, \,\forall N\in\mathbb{N}\exists P\in\mathcal{D}(\mathcal{A}),R\in OP^{-N},p\in\mathbb{N}\text{ s.t. }T=P \mathcal{D} ^{-p}+R\}.$
cardona-qft-methods_ E09927	352	■ 0.993 ● N +1	$H_{*}^{pro}\left(\mathbb{L}(M\times_G EG)^{-T(M\times_G EG)}\right):=\lim_{\leftarrow n}H_{*}\left((P_G M\times_G EG_n)^{-e_0^*T(M\times_G EG_n)}\right)$	$H_{*}^{pro}\left(\mathbb{L}(M\times_G EG)^{-T(M\times_G EG)}\right):=\lim_{\leftarrow n}H_{*}\left((P_G M\times_G EG_n)^{-e_0^*T(M\times_G EG_n)}\right)$	$H_{*}^{pro}(\mathbb{L}(M\times_G EG)^{-T(M\times_G EG)}):=\lim_{\leftarrow n}H_{*}((P_G M\times_G EG_n)^{-e_0^*T(M\times_G EG_n)})$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0899	347	■ 0.993	$\begin{gathered} H\,H_*(A,M)=H^*\left(A\stackrel{L}{\otimes}_{A^e}M\right)\\ H\,H^*(A,M)=H_*\left(\mathcal{R}Hom_{A^e}(A,M)\right) \end{gathered}$	$HH_*(A,M)=H^*\left(A\stackrel{L}{\otimes}_{A^e}M\right)$ $HH^*(A,M)=H_*\left(\mathcal{R}Hom_{A^e}(A,M)\right)$	$HH_*(A,M)=H^*(A\stackrel{L}{\otimes}_{A^e}M)$ $HH^*(A,M)=H_*(\mathcal{R}Hom_{A^e}(A,M))$
cardona-qft-methods_ EQ0138	080	■ 0.994 ● N +0	$\phi_i:U_i\stackrel{\sim}{\rightarrow}V_i\subset\mathbb{R}^n\quad(i=1,\ldots,m)$	$\phi_i:U_i\stackrel{\sim}{\rightarrow}V_i\subset\mathbb{R}^n\quad(i=1,\ldots,m)$	$\phi_i:U_i\stackrel{\sim}{\rightarrow}V_i\subset\mathbb{R}^n\quad(i=1,\ldots,m)$
cardona-qft-methods_ EQ0292	142	■ 0.994 ● K +0	$\left[B\,\ell_{B}\,X\right]_{SB}-[E]_{SB}=[X]_{SB}-[B]_{SB},$	$[B\ell_B X]_{SB}-[E]_{SB}=[X]_{SB}-[B]_{SB},$	$[B\ell_B X]_{SB}-[E]_{SB}=[X]_{SB}-[B]_{SB}\quad,$
cardona-qft-methods_ EQ0917	351	■ 0.994 ● N +0	$\operatorname{Tor}_{C^*(X)^e}(C^*(X),C^*(X)\#G)\cong H^*(P_GX).$	$\operatorname{Tor}_{C^*(X)^e}(C^*(X),C^*(X)\#G)\cong H^*(P_GX).$	$\operatorname{Tor}_{C^*(X)^e}(C^*(X),C^*(X)\#G)\cong H^*(P_GX).$
cardona-qft-methods_ EQ0209	097	■ 0.994 ● N +0	$E^{+}:=\bigoplus_{j\text{ even}}\Lambda^j(T^*M\otimes\mathcal{C}),\quad E^{-}:=\bigoplus_{j\text{ odd}}\Lambda^j(T^*M\otimes\mathcal{C}).$	$E^{+}:=\bigoplus_{j\text{ even}}\Lambda^j(T^*M\otimes\mathcal{C}),\quad E^{-}:=\bigoplus_{j\text{ odd}}\Lambda^j(T^*M\otimes\mathcal{C}).$	$E^{+}:=\bigoplus_{j\text{ even}}\Lambda^j(T^*M\otimes\mathcal{C}),\quad E^{-}:=\bigoplus_{j\text{ odd}}\Lambda^j(T^*M\otimes\mathcal{C}).$
cardona-qft-methods_ EQ0263	125	■ 0.994 ● N +1	$\langle G\rangle=\prod_i\langle G_i\rangle/\text{extra symmetries}$	$\langle G\rangle=\prod_i\langle G_i\rangle/\text{extra symmetries}$	$\langle G\rangle=\prod_i\langle G_i\rangle/\text{extra symmetries}\quad.$
cardona-qft-methods_ EQ0422 (7)	193	■ 0.994 ● N +0	$\int_M\Theta\operatorname{Tr}[F\wedge F]=\int_{\partial\bar{M}}j\wedge\operatorname{Tr}[A\wedge dA],$	$\int_M\Theta\operatorname{Tr}[F\wedge F]=\int_{\partial\bar{M}}j\wedge\operatorname{Tr}[A\wedge dA],$	$\int_M\Theta Tr[F\wedge F]=\int_{\partial\bar{M}}j\wedge Tr[A\wedge dA],$
cardona-qft-methods_ EQ0888	345	■ 0.994 ● N +0	$\circ:H_p(LM)\otimes H_q(LM)\rightarrow H_{p+q-n}(LM).$	$\circ:H_p(LM)\otimes H_q(LM)\rightarrow H_{p+q-n}(LM).$	$\circ:H_p(LM)\otimes H_q(LM)\rightarrow H_{p+q-n}(LM).$
cardona-qft-methods_ EQ0364 (3)	174	■ 0.994 ● N +1	$\begin{aligned} \delta e^a &= d\lambda^a + \omega_b^a \lambda^b \\ &= D_\omega \lambda^a \end{aligned}$	$\begin{aligned} \delta e^a &= d\lambda^a + \omega_b^a \lambda^b \\ &= D_\omega \lambda^a \end{aligned}$	$\begin{aligned} \delta e^a &= d\lambda^a + \omega_b^a \lambda^b, \\ &= D_\omega \lambda^a \end{aligned}$
cardona-qft-methods_ EQ0412 (42)	188	■ 0.995 ● K +0	$\mathcal{T}_{2n+1}(A,\bar{A})=\mathcal{C}_{2n+1}(A)-\mathcal{C}_{2n+1}(\bar{A})+d\mathcal{B}_{2n}(A,\bar{A}).$	$\mathcal{T}_{2n+1}(A,\bar{A})=\mathcal{C}_{2n+1}(A)-\mathcal{C}_{2n+1}(\bar{A})+d\mathcal{B}_{2n}(A,\bar{A}).$	$\mathcal{T}_{2n+1}(A,\bar{A})=\mathcal{C}_{2n+1}(A)-\mathcal{C}_{2n+1}(\bar{A})+d\mathcal{B}_{2n}(A,\bar{A}).$
cardona-qft-methods_ EQ0942 (2)	358	■ 0.995 ● K +0	$[\emptyset]=0.$	$[\emptyset]=0.$	$[\emptyset]=0.$
cardona-qft-methods_ EQ0349 (17)	168	■ 0.995 ● N +0	$\begin{aligned} &\&\mathcal{E}_{a_1\cdots a_{2p}}^{b_1\cdots b_{2p}}:=\epsilon_{a_1\cdots a_{2p}}\epsilon^{b_1\cdots b_{2p}}\\ &\&\mathcal{E}_{a_1\cdots a_{2p}}^{b_1\cdots b_{2p}}:=\epsilon_{a_1\cdots a_{2p}}\epsilon^{b_1\cdots b_{2p}} \end{aligned}$	$\begin{aligned} \mathcal{E}_a^{(p)} &:= \epsilon_{ab_2\cdots b_D}R^{b_2b_3}\cdots R^{b_{2p}b_{2p+1}}e^{b_{2p+2}}\cdots e^{b_D}, \\ \mathcal{E}_{ab}^{(p)} &:= \epsilon_{aba_3\cdots a_D}R^{a_3a_4}\cdots R^{a_{2p-1}a_{2p}}T^{a_{2p+1}}e^{a_{2p+2}}\cdots e^{a_D}. \end{aligned}$	$\begin{aligned} \mathcal{E}_a^{(p)} &:= \epsilon_{ab_2\cdots b_D}R^{b_2b_3}\cdots R^{b_{2p}b_{2p+1}}e^{b_{2p+2}}\cdots e^{b_D}, \\ \mathcal{E}_{ab}^{(p)} &:= \epsilon_{aba_3\cdots a_D}R^{a_3a_4}\cdots R^{a_{2p-1}a_{2p}}T^{a_{2p+1}}e^{a_{2p+2}}\cdots e^{a_D}. \end{aligned}$
cardona-qft-methods_ EQ0796 (37)	302	■ 0.995 ● K +0	$NS-N\mathcal{S}:\text{ type IIA and IIB } \mathbf{8_v}\otimes\mathbf{8_v}=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}=\Phi\oplus\mathbf{B}_2\oplus\mathbf{G_{MN}}$	$NS-NS:\text{ type IIA and IIB } \mathbf{8_v}\otimes\mathbf{8_v}=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}=\Phi\oplus\mathbf{B}_2\oplus\mathbf{G_{MN}}$	$NS-NS:\text{ type IIA and IIB } \mathbf{8_v}\otimes\mathbf{8_v}=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}=\Phi\oplus\mathbf{B}_2\oplus\mathbf{G_{MN}}$
Conf. MathPix confidence: ■ ■ ■ — ≥0.90.5-0.9<0.5no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● ● ● ● ● ● C componentW weakS stableN noiseK cleannot measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0471	218	0.995 ● N -1	$\operatorname{Tr}(f(D/\Lambda))\sim\sum_kf_k\Lambda^kf D ^{-k}+f(0)\zeta_D(0)+o(1),$	$\operatorname{Tr}(f(D/\Lambda))\sim\sum_kf_k\Lambda^kf D ^{-k}+f(0)\zeta_D(0)+o(1),$	$\operatorname{Tr}(f(D/\Lambda))\sim\sum_kf_k\Lambda^k\int D ^{-k}+f(0)\zeta_D(0)+o(1),$
cardona-qft-methods_ EQ0574 (9)	249	0.995 ●	$\widehat{x}_{\mu}\star\widehat{x}_{\nu}-\widehat{x}_{\nu}\star\widehat{x}_{\mu}=\left[\widehat{x}_{\mu},\widehat{x}_{\nu}\right]_{\star}=i\theta_{\mu\nu}\quad\mu,\nu=0,1,\ldots d.$	$\widehat{x}_{\mu}\star\widehat{x}_{\nu}-\widehat{x}_{\nu}\star\widehat{x}_{\mu}=\left[\widehat{x}_{\mu},\widehat{x}_{\nu}\right]_{\star}=i\theta_{\mu\nu}\quad\mu,\nu=0,1,\ldots d.$	$\widehat{x}_{\mu}\star\widehat{x}_{\nu}-\widehat{x}_{\nu}\star\widehat{x}_{\mu}=\left[\widehat{x}_{\mu},\widehat{x}_{\nu}\right]_{\star}=i\theta_{\mu\nu}\quad\mu,\nu=0,1,\ldots d.$
cardona-qft-methods_ EQ0111	073	0.996 ● K +0	$\operatorname{Ind}(B\ A)=\operatorname{Ind}(A)+\operatorname{Ind}(B).$	$\operatorname{Ind}(BA)=\operatorname{Ind}(A)+\operatorname{Ind}(B).$	$\operatorname{Ind}(BA)=\operatorname{Ind}(A)+\operatorname{Ind}(B).$
cardona-qft-methods_ EQ0429	206	0.996 ● N +0	$\mathcal{A}=\oplus_{i=1}^N M_{n_i}(\mathbb{K}_i)$	$\mathcal{A}=\oplus_{i=1}^N M_{n_i}(\mathbb{K}_i)$	$\mathcal{A}=\oplus_{i=1}^N M_{n_i}(\mathbb{K}_i)$
cardona-qft-methods_ EQ0494	223	0.996 ● C +3	$\begin{gathered} 16\,\pi^{\wedge\{2\}}\,\beta_{g_i}=b_i\,g_i^{\wedge\{3\}}\\ \quad\quad\quad\text{with}\,\,\left(b_{SU(3)},b_{SU(2)},b_{U(1)}\right)=\left(-7,-\frac{19}{6},\frac{41}{10}\right)\\ \quad\quad\quad 16\pi^2\,\beta_{Y_u}=Y_u\left(\frac{3}{2}Y_u^{\dagger}Y_u-\frac{3}{2}Y_d^{\dagger}Y_d+\mathfrak{a}-\frac{17}{20}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ \quad\quad\quad 16\pi^2\,\beta_{Y_d}=Y_d\left(\frac{3}{2}Y_d^{\dagger}Y_d-\frac{3}{2}Y_u^{\dagger}Y_u+\mathfrak{a}-\frac{1}{4}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ \quad\quad\quad 16\pi^2\,\beta_{Y_{\nu}}=Y_{\nu}\left(\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}-\frac{3}{2}Y_e^{\dagger}Y_e+\mathfrak{a}-\frac{9}{20}g_1^2-\frac{9}{4}g_2^2\right)\\ \quad\quad\quad 16\pi^2\,\beta_{Y_e}=Y_e\left(\frac{3}{2}Y_e^{\dagger}Y_e-\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}+\mathfrak{a}-\frac{9}{4}g_1^2-\frac{9}{4}g_2^2\right)\\ \quad\quad\quad 16\pi^2\,\beta_M=Y_{\nu}Y_{\nu}^{\dagger}M+M\left(Y_{\nu}Y_{\nu}^{\dagger}\right)^T\\ 16\pi^2\,\beta_{\lambda}=6\lambda^2-3\lambda\left(3g_2^2+\frac{3}{5}g_1^2\right)+3g_2^4+\frac{3}{2}\left(\frac{3}{5}g_1^2+g_2^2\right)^2+4\lambda\mathfrak{a}-8\mathfrak{b}. \end{gathered}$	$\begin{gathered} 16\pi^2\beta_{g_i}=b_i g_i^3\quad\text{with}\,\,\left(b_{SU(3)},b_{SU(2)},b_{U(1)}\right)=\left(-7,-\frac{19}{6},\frac{41}{10}\right)\\ 16\pi^2\beta_{Y_u}=Y_u\left(\frac{3}{2}Y_u^{\dagger}Y_u-\frac{3}{2}Y_d^{\dagger}Y_d+\mathfrak{a}-\frac{17}{20}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ 16\pi^2\beta_{Y_d}=Y_d\left(\frac{3}{2}Y_d^{\dagger}Y_d-\frac{3}{2}Y_u^{\dagger}Y_u+\mathfrak{a}-\frac{1}{4}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ 16\pi^2\beta_{Y_{\nu}}=Y_{\nu}\left(\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}-\frac{3}{2}Y_e^{\dagger}Y_e+\mathfrak{a}-\frac{9}{20}g_1^2-\frac{9}{4}g_2^2\right)\\ 16\pi^2\beta_{Y_e}=Y_e\left(\frac{3}{2}Y_e^{\dagger}Y_e-\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}+\mathfrak{a}-\frac{9}{4}g_1^2-\frac{9}{4}g_2^2\right)\\ 16\pi^2\beta_M=Y_{\nu}Y_{\nu}^{\dagger}M+M\left(Y_{\nu}Y_{\nu}^{\dagger}\right)^T\\ 16\pi^2\beta_{\lambda}=6\lambda^2-3\lambda\left(3g_2^2+\frac{3}{5}g_1^2\right)+3g_2^4+\frac{3}{2}\left(\frac{3}{5}g_1^2+g_2^2\right)^2+4\lambda\mathfrak{a}-8\mathfrak{b} \end{gathered}$	$\begin{gathered} 16\pi^2\,\beta_{g_i}=b_i\,g_i^3\quad\text{with}\,\,\left(b_{SU(3)},b_{SU(2)},b_{U(1)}\right)=\left(-7,-\frac{19}{6},\frac{41}{10}\right)\\ 16\pi^2\,\beta_{Y_u}=Y_u\left(\frac{3}{2}Y_u^{\dagger}Y_u-\frac{3}{2}Y_d^{\dagger}Y_d+\mathfrak{a}-\frac{17}{20}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ 16\pi^2\,\beta_{Y_d}=Y_d\left(\frac{3}{2}Y_d^{\dagger}Y_d-\frac{3}{2}Y_u^{\dagger}Y_u+\mathfrak{a}-\frac{1}{4}g_1^2-\frac{9}{4}g_2^2-8g_3^2\right)\\ 16\pi^2\,\beta_{Y_{\nu}}=Y_{\nu}\left(\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}-\frac{3}{2}Y_e^{\dagger}Y_e+\mathfrak{a}-\frac{9}{20}g_1^2-\frac{9}{4}g_2^2\right)\\ 16\pi^2\,\beta_{Y_e}=Y_e\left(\frac{3}{2}Y_e^{\dagger}Y_e-\frac{3}{2}Y_{\nu}^{\dagger}Y_{\nu}+\mathfrak{a}-\frac{9}{4}g_1^2-\frac{9}{4}g_2^2\right)\\ 16\pi^2\,\beta_M=Y_{\nu}Y_{\nu}^{\dagger}M+M\left(Y_{\nu}Y_{\nu}^{\dagger}\right)^T\\ 16\pi^2\,\beta_{\lambda}=6\lambda^2-3\lambda\left(3g_2^2+\frac{3}{5}g_1^2\right)+3g_2^4+\frac{3}{2}\left(\frac{3}{5}g_1^2+g_2^2\right)^2+4\lambda\mathfrak{a}-8\mathfrak{b}. \end{gathered}$
cardona-qft-methods_ EQ0241	115	0.996 ● N +0	$\kappa_*\colon A_*X\rightarrow A_*\{\mathrm{pt}\}\backslash\{\mathrm{mathrm{pt}}\}\!=\mathbb{Z},$	$\kappa_*\colon A_*X\rightarrow A_*\{\mathrm{pt}\}=\mathbb{Z},$	$\kappa_*\colon A_*X\rightarrow A_*\{\mathrm{pt}\}=\mathbb{Z}\quad,$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0279	136	■ 0.996 ●	$\begin{aligned} \sum_{m \geq 0} \mathbb{U}(G_{me}) \frac{s^m}{m!} = & \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G) + \frac{e^{\mathbb{T}s} + \mathbb{T}e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G \setminus e) \\ & + \left(s e^{\mathbb{T}s} - \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \right) \mathbb{U}(G/e) \end{aligned}$	$\sum_{m \geq 0} \mathbb{U}(G_{me}) \frac{s^m}{m!} = \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G) + \frac{e^{\mathbb{T}s} + \mathbb{T}e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G \setminus e) + \left(s e^{\mathbb{T}s} - \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \right) \mathbb{U}(G/e)$	$\sum_{m \geq 0} \mathbb{U}(G_{me}) \frac{s^m}{m!} = \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G) + \frac{e^{\mathbb{T}s} + \mathbb{T}e^{-s}}{\mathbb{T} + 1} \mathbb{U}(G \setminus e) + \left(s e^{\mathbb{T}s} - \frac{e^{\mathbb{T}s} - e^{-s}}{\mathbb{T} + 1} \right) \mathbb{U}(G/e)$
cardona-qft-methods_ EQ0588 (21)	251	■ 0.996 ● K +0	$\tau_0 \Delta_\theta(\Lambda) \neq \Delta_\theta(\Lambda) \tau_0,$	$\tau_0 \Delta_\theta(\Lambda) \neq \Delta_\theta(\Lambda) \tau_0,$	$\tau_0 \Delta_\theta(\Lambda) \neq \Delta_\theta(\Lambda) \tau_0,$
cardona-qft-methods_ EQ0709	273	■ 0.996 ● K +0	$e^{ip_k L} = \prod_{j \neq k}^M S(p_k, p_j) \quad \text{for } k = 1, \dots, M,$	$e^{ip_k L} = \prod_{j \neq k}^M S(p_k, p_j) \quad \text{for } k = 1, \dots, M,$	$e^{ip_k L} = \prod_{j \neq k}^M S(p_k, p_j) \quad \text{for } k = 1, \dots, M,$
cardona-qft-methods_ EQ0803 (3)	306	■ 0.996 ● K +0	$0 \longrightarrow T^*M \longrightarrow E \xrightarrow{\pi} TM \longrightarrow 0.$	$0 \longrightarrow T^*M \longrightarrow E \xrightarrow{\pi} TM \longrightarrow 0.$	$0 \longrightarrow T^*M \longrightarrow E \xrightarrow{\pi} TM \longrightarrow 0.$
cardona-qft-methods_ EQ0202 (10)	096	■ 0.997 ● K +0	$\mathcal{D} = d + d^*.$	$\mathcal{D} = d + d^*.$	$\mathcal{D} = d + d^*.$
cardona-qft-methods_ EQ0290	139	■ 0.997 ● W +1	$\begin{aligned} \sum_{m \geq 0} C_{G_{(m+1)e}}(t) \frac{s^m}{m!} = & \left(e^{ts} - e^{(t-1)s} \right) C_{G_{2e}}(t) \\ & - \left((t-1)e^{ts} - te^{(t-1)s} \right) C_G(t) + t \left((s-1)e^{ts} + e^{(t-1)s} \right) C_{G/e}(t). \end{aligned}$	$\sum_{m \geq 0} C_{G_{(m+1)e}}(t) \frac{s^m}{m!} = \left(e^{ts} - e^{(t-1)s} \right) C_{G_{2e}}(t) - \left((t-1)e^{ts} - te^{(t-1)s} \right) C_G(t) + t \left((s-1)e^{ts} + e^{(t-1)s} \right) C_{G/e}(t)$	$\sum_{m \geq 0} C_{G_{(m+1)e}}(t) \frac{s^m}{m!} = \left(e^{ts} - e^{(t-1)s} \right) C_{G_{2e}}(t) - \left((t-1)e^{ts} - te^{(t-1)s} \right) C_G(t) + t \left((s-1)e^{ts} + e^{(t-1)s} \right) C_{G/e}(t).$
cardona-qft-methods_ EQ0439	210	■ 0.997 ● W +0	$\begin{gathered} \mathcal{A}_F = \left\{ \left(\lambda, q_L, \lambda, m \right) \mid \lambda \in \mathbb{C}, q_L \in \mathbb{H}, m \in M_3(\mathbb{C}) \right\} \\ \sim \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \end{gathered}$	$\mathcal{A}_F = \{(\lambda, q_L, \lambda, m) \mid \lambda \in \mathbb{C}, q_L \in \mathbb{H}, m \in M_3(\mathbb{C})\} \sim \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$	$\mathcal{A}_F = \{(\lambda, q_L, \lambda, m) \mid \lambda \in \mathbb{C}, q_L \in \mathbb{H}, m \in M_3(\mathbb{C})\} \sim \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$
cardona-qft-methods_ EQ0958 (10)	361	■ 0.997 ● K +0	$X_{\Gamma^\vee} = \pi_2 \left(\pi_1^{-1}(X_\Gamma) \right),$	$X_{\Gamma^\vee} = \pi_2 \left(\pi_1^{-1}(X_\Gamma) \right),$	$X_{\Gamma^\vee} = \pi_2 \left(\pi_1^{-1}(X_\Gamma) \right),$
cardona-qft-methods_ EQ0804 (4)	306	■ 0.997 ● N +0	$x_{(\alpha)} + \xi_{(\alpha)} = a_{(\alpha\beta)} \cdot x_{(\beta)} + \left[a_{(\alpha\beta)}^{-T} \xi_{(\beta)} - \iota_{a_{(\alpha\beta)}x_{(\beta)}} b_{(\alpha\beta)} \right],$	$x_{(\alpha)} + \xi_{(\alpha)} = a_{(\alpha\beta)} \cdot x_{(\beta)} + \left[a_{(\alpha\beta)}^{-T} \xi_{(\beta)} - \iota_{a_{(\alpha\beta)}x_{(\beta)}} b_{(\alpha\beta)} \right],$	$x_{(\alpha)} + \xi_{(\alpha)} = a_{(\alpha\beta)} \cdot x_{(\beta)} + \left[a_{(\alpha\beta)}^{-T} \xi_{(\beta)} - \iota_{a_{(\alpha\beta)}x_{(\beta)}} b_{(\alpha\beta)} \right],$
cardona-qft-methods_ EQ0157	086	■ 0.997 ● N +0	$j: T^*M \hookrightarrow \mathbb{R}^N \oplus \mathbb{R}^N \simeq \mathcal{C}^N.$	$j: T^*M \hookrightarrow \mathbb{R}^N \oplus \mathbb{R}^N \simeq \mathcal{C}^N.$	$j: T^*M \hookrightarrow \mathbb{R}^N \oplus \mathbb{R}^N \simeq \mathcal{C}^N.$
cardona-qft-methods_ EQ0286	138	■ 0.997 ● N +0	$C_{G_{2e}}(T) = (2T-1)C_G(T) - T(T-1)C_{G \setminus e}(T) + C_{G/e}(T).$	$C_{G_{2e}}(T) = (2T-1)C_G(T) - T(T-1)C_{G \setminus e}(T) + C_{G/e}(T).$	$C_{G_{2e}}(T) = (2T-1)C_G(T) - T(T-1)C_{G \setminus e}(T) + C_{G/e}(T).$
cardona-qft-methods_ EQ0543	243	■ 0.997 ● N +1	$\Delta \circ m(a \otimes b) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(a \otimes b).$	$\Delta \circ m(a \otimes b) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(a \otimes b).$	$\Delta \circ m(a \otimes b) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(a \otimes b).$
cardona-qft-methods_ EQ0811 (11)	308	■ 0.997 ● K +0	$\Phi_\pm = \eta_+^1 \otimes \eta_\pm^{2\dagger}.$	$\Phi_\pm = \eta_+^1 \otimes \eta_\pm^{2\dagger}.$	$\Phi_\pm = \eta_+^1 \otimes \eta_\pm^{2\dagger}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0612 (20)	254	0.999 ● K +0	$\left(\Lambda e_p\right)(x)=e_p\left(\Lambda^{-1} x\right. \\ \left.\right)=e_{\Lambda p(x)},$	$(\Lambda e_p)(x)=e_p\left(\Lambda^{-1} x\right)=e_{\Lambda p(x)},$	$(\Lambda e_p)(x)=e_p\left(\Lambda^{-1} x\right)=e_{\Lambda p(x)},$
cardona-qft-methods_ EQ0624 (32)	257	0.999 ● N +1	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta} ; \quad F_{\theta}=\exp ^{\frac{i}{2} \hat{P}_{\mu} \otimes \theta_{\mu \nu} \hat{P}_{\nu}},$	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta} ; \quad F_{\theta}=\exp ^{\frac{i}{2} \hat{P}_{\mu} \otimes \theta_{\mu \nu} \hat{P}_{\nu}},$	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta} ; \quad F_{\theta}=\exp ^{\frac{i}{2} \hat{P}_{\mu} \otimes \theta_{\mu \nu} \hat{P}_{\nu}},$
cardona-qft-methods_ EQ0744	283	0.999 ●	$[K, \mathcal{O}]=0 \quad, \quad [S, \mathcal{O}]=0 \quad, \quad [\bar{S}, \mathcal{O}]=0.$	$[K, \mathcal{O}]=0 \quad, \quad [S, \mathcal{O}]=0 \quad, \quad [\bar{S}, \mathcal{O}]=0.$	$[K, \mathcal{O}]=0 \quad, \quad [S, \mathcal{O}]=0 \quad, \quad [\bar{S}, \mathcal{O}]=0.$
cardona-qft-methods_ EQ0767 (8)	293	0.999 ● K +0	$\eta_{M N} \partial_{\sigma}+X_L^M \partial_{\sigma}+X_L^N=\eta_{M N} \partial_{\sigma}-X_R^M \partial_{\sigma}-X_R^N=0 .$	$\eta_{M N} \partial_{\sigma}+X_L^M \partial_{\sigma}+X_L^N=\eta_{M N} \partial_{\sigma}-X_R^M \partial_{\sigma}-X_R^N=0 .$	$\eta_{M N} \partial_{\sigma}+X_L^M \partial_{\sigma}+X_L^N=\eta_{M N} \partial_{\sigma}-X_R^M \partial_{\sigma}-X_R^N=0 .$
cardona-qft-methods_ EQ0943 (3)	358	0.999 ● K +0	$[X] \cdot[Y]=[X \times Y] \quad .$	$[X] \cdot[Y]=[X \times Y].$	$[X] \cdot[Y]=[X \times Y].$
cardona-qft-methods_ EQ0989	371	0.999 ● K +0	$A \, B=\text { the closure of } \left. A \, B\right _{\mathcal{D}}, \quad \mathcal{D}=B^{-1} \operatorname{dom} A .$	$AB = \text{ the closure of } AB _{\mathcal{D}}, \quad \mathcal{D} = B^{-1} \operatorname{dom} A.$	$AB = \text{ the closure of } AB _{\mathcal{D}}, \quad \mathcal{D} = B^{-1} \operatorname{dom} A.$
cardona-qft-methods_ EQ0289	138	0.999 ● W +1	$\begin{aligned} C_{G_{(m+3) e}}(T)= & (3 T-1) C_{G_{(m+2) e}}(T)-\left(3 T^2-2 T\right) C_{G_{(m+1) e}}(T) \\ & +\left(T^3-T^2\right) C_{G_{m e}}(T) \end{aligned}$	$C_{G_{(m+3) e}}(T)=\left(3 T-1\right) C_{G_{(m+2) e}}(T)-\left(3 T^2-2 T\right) C_{G_{(m+1) e}}(T) \\ +\left(T^3-T^2\right) C_{G_{m e}}(T)$	$C_{G_{(m+3) e}}(T)=\left(3 T-1\right) C_{G_{(m+2) e}}(T)-\left(3 T^2-2 T\right) C_{G_{(m+1) e}}(T) \\ +\left(T^3-T^2\right) C_{G_{m e}}(T)$
cardona-qft-methods_ EQ0389 (27)	179	0.999 ● K +0	$\mathbf{E}_{2 n}=\epsilon_{A_1 \cdots A_{2 n}} F^{A_1 A_2} \ldots F^{A_{2 n-1} A_{2 n}} .$	$\mathbf{E}_{2 n}=\epsilon_{A_1 \cdots A_{2 n}} F^{A_1 A_2} \ldots F^{A_{2 n-1} A_{2 n}} .$	$\mathbf{E}_{2 n}=\epsilon_{A_1 \cdots A_{2 n}} F^{A_1 A_2} \ldots F^{A_{2 n-1} A_{2 n}} .$
cardona-qft-methods_ EQ0392 (30)	180	0.999 ● N +1	$I_{2 \, n-1}=\frac{\kappa}{l} \int_M \int_0^1 d t \epsilon_{a_1 \cdots a_{2 n-1}} R_t^{a_1 a_2} \ldots R_t^{a_{2 n-3} a_{2 n-2}} e^{a_{2 n-1}},$	$I_{2 n-1}=\frac{\kappa}{l} \int_M \int_0^1 d t \epsilon_{a_1 \cdots a_{2 n-1}} R_t^{a_1 a_2} \ldots R_t^{a_{2 n-3} a_{2 n-2}} e^{a_{2 n-1}},$	$I_{2 n-1}=\frac{\kappa}{l} \int_M \int_0^1 d t \epsilon_{a_1 \cdots a_{2 n-1}} R_t^{a_1 a_2} \ldots R_t^{a_{2 n-3} a_{2 n-2}} e^{a_{2 n-1}},$
cardona-qft-methods_ EQ0405 (37)	184	0.999 ● N +0	$\begin{aligned} L_{2 n}^{B I} & =\epsilon_{a_1 b_1 a_2 b_2 \cdots a_n b_n}\left[R^{a_1 b_1}+\frac{1}{l^2} e^{a_1} e^{b_1}\right] \ldots\left[R^{a_n b_n}+\frac{1}{l^2} e^{a_n} e^{b_n}\right] \\ & =\operatorname{Pfaff}\left[R^{a b}+\frac{1}{l^2} e^a e^b\right] . \end{aligned}$	$L_{2 n}^{B I}=\epsilon_{a_1 b_1 a_2 b_2 \cdots a_n b_n}\left[R^{a_1 b_1}+\frac{1}{l^2} e^{a_1} e^{b_1}\right] \ldots\left[R^{a_n b_n}+\frac{1}{l^2} e^{a_n} e^{b_n}\right] \\ =\operatorname{Pfaff}\left[R^{a b}+\frac{1}{l^2} e^a e^b\right] .$	$L_{2 n}^{B I}=\epsilon_{a_1 b_1 a_2 b_2 \cdots a_n b_n}\left[R^{a_1 b_1}+\frac{1}{l^2} e^{a_1} e^{b_1}\right] \ldots\left[R^{a_n b_n}+\frac{1}{l^2} e^{a_n} e^{b_n}\right] \\ =\operatorname{Pfaff}\left[R^{a b}+\frac{1}{l^2} e^a e^b\right] .$
cardona-qft-methods_ EQ0436	209	0.999 ● N +0	$J_{\tilde{F}}^2=1, \quad \xi b=J_F b^* J_F \xi \quad \xi \in \mathcal{M}_F, b \in \mathcal{A}_{L R} .$	$J_{\tilde{F}}^2=1, \quad \xi b=J_F b^* J_F \xi \quad \xi \in \mathcal{M}_F, b \in \mathcal{A}_{L R} .$	$J_F^2=1, \quad \xi b=J_F b^* J_F \xi \quad \xi \in \mathcal{M}_F, b \in \mathcal{A}_{L R} .$
cardona-qft-methods_ EQ0498	226	0.999 ● W +0	$\lambda\left(\Lambda_{\text {unif }}\right)=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}\left(\Lambda_{\text {unif }}\right)}{\mathfrak{a}\left(\Lambda_{\text {unif }}\right)^2}$	$\lambda\left(\Lambda_{\text {unif }}\right)=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}\left(\Lambda_{\text {unif }}\right)}{\mathfrak{a}\left(\Lambda_{\text {unif }}\right)^2}$	$\lambda\left(\Lambda_{\text {unif }}\right)=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}\left(\Lambda_{\text {unif }}\right)}{\mathfrak{a}\left(\Lambda_{\text {unif }}\right)^2}$
cardona-qft-methods_ EQ0520 (6)	237	0.999 ● N +0	$\hat{x}_{\mu} \star \hat{x}_{\nu}-\hat{x}_{\nu} \star \hat{x}_{\mu}=\left[\hat{x}_{\mu}, \hat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu}, \quad \mu, \nu=0,1, \ldots, d .$	$\hat{x}_{\mu} \star \hat{x}_{\nu}-\hat{x}_{\nu} \star \hat{x}_{\mu}=\left[\hat{x}_{\mu}, \hat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu}, \quad \mu, \nu=0,1, \ldots, d .$	$\hat{x}_{\mu} \star \hat{x}_{\nu}-\hat{x}_{\nu} \star \hat{x}_{\mu}=\left[\hat{x}_{\mu}, \hat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu}, \quad \mu, \nu=0,1, \ldots, d .$
cardona-qft-methods_ EQ0592 (1)	252	0.999 ● N +0	$(a, \Lambda) \in \mathcal{P}: (a, \Lambda) x=\Lambda x+a \quad .$	$(a, \Lambda) \in \mathcal{P}: (a, \Lambda) x=\Lambda x+a .$	$(a, \Lambda) \in \mathcal{P}: (a, \Lambda) x=\Lambda x+a .$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0598 (7)	252	0.999 K +0	$(\operatorname{Ad} U \otimes V) \Delta_0((a, \Lambda)) \varphi = \varphi$ $\varphi=\varphi$	$(\operatorname{Ad} U \otimes V) \Delta_0((a, \Lambda)) \varphi = \varphi$	$(\operatorname{Ad} U \otimes V) \Delta_0((a, \Lambda)) \varphi = \varphi$
cardona-qft-methods_ EQ0623 (31)	257	0.999 N +1	$\mathcal{P} \ni (a, \Delta) : \alpha \rightarrow (a, \Delta) \alpha, \quad ((a, \Delta) \alpha)(x) = \alpha \left(((a, \Delta)^{-1} x) \right).$ $\mathcal{P} \ni (a, \Delta) : \alpha \rightarrow (a, \Delta) \alpha, \quad ((a, \Delta) \alpha)(x) = \alpha \left(((a, \Delta)^{-1} x) \right).$	$\mathcal{P} \ni (a, \Delta) : \alpha \rightarrow (a, \Delta) \alpha, \quad ((a, \Delta) \alpha)(x) = \alpha \left(((a, \Delta)^{-1} x) \right).$	$\mathcal{P} \ni (a, \Delta) : \alpha \rightarrow (a, \Delta) \alpha, \quad ((a, \Delta) \alpha)(x) = \alpha \left(((a, \Delta)^{-1} x) \right).$
cardona-qft-methods_ EQ0678 (3)	269	0.999 N +0	$\left[S^{\mathrm{i}}, S^{\mathrm{j}}\right]=\mathrm{i} \, \epsilon^{\mathrm{ijk}} S^{\mathrm{k}} \, .$	$\left[S^i, S^j\right] = i \epsilon^{ijk} S^k.$	$\left[S^i, S^j\right] = i \epsilon^{ijk} S^k.$
cardona-qft-methods_ EQ0697	271	0.999	$\mathcal{H}_{\{l, \, l+1\}}=2\left(\mathbb{I}_{\{l, \, l+1\}}-\mathbb{P}_{\{l, \, l+1\}}\right)=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & +2 & -2 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right),$	$\mathcal{H}_{l,l+1} = 2 \left(\mathbb{I}_{l,l+1} - \mathbb{P}_{l,l+1} \right) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & +2 & -2 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$	$\mathcal{H}_{l,l+1} = 2 \left(\mathbb{I}_{l,l+1} - \mathbb{P}_{l,l+1} \right) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & +2 & -2 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$
cardona-qft-methods_ EQ0699	271	0.999 K +0	$\left[\vec{S}_{\mathrm{tot}}, \hat{H}\right]=0,$	$\left[\vec{S}_{\mathrm{tot}}, \hat{H}\right] = 0,$	$\left[\vec{S}_{\mathrm{tot}}, \hat{H}\right] = 0,$
cardona-qft-methods_ EQ0823 (1)	312	0.999 N +0	$\mathrm{d} \, s^2 = e^{2 \, A(y)} \eta_{\mu \nu} \mathrm{d} x^{\mu} \mathrm{d} x^{\nu} + g_{mn} \, \mathrm{d} y^m \, \mathrm{d} y^n, \quad \mu = 0, 1, 2, 3 \quad m = 1, \ldots, 6$	$\mathrm{d} s^2 = e^{2 A(y)} \eta_{\mu \nu} \mathrm{d} x^{\mu} \mathrm{d} x^{\nu} + g_{mn} \, \mathrm{d} y^m \, \mathrm{d} y^n, \quad \mu = 0, 1, 2, 3 \quad m = 1, \ldots, 6$	$\mathrm{d} s^2 = e^{2 A(y)} \eta_{\mu \nu} \mathrm{d} x^{\mu} \mathrm{d} x^{\nu} + g_{mn} \mathrm{d} y^m \mathrm{d} y^n, \quad \mu = 0, 1, 2, 3 \quad m = 1, ..., 6$
cardona-qft-methods_ EQ0842 (1)	329	0.999 N +0	$\Pi^{\mathrm{sr}}(x) (\partial_r) \Pi^{\mathrm{lk}}(x) + \Pi^{\mathrm{kr}}(x) (\partial_r) \Pi^{\mathrm{sl}}(x) + \Pi^{\mathrm{lr}}(x) (\partial_r) \Pi^{\mathrm{ks}}(x) = 0,$	$\Pi^{sr}(x) (\partial_r) \Pi^{lk}(x) + \Pi^{kr}(x) (\partial_r) \Pi^{sl}(x) + \Pi^{lr}(x) (\partial_r) \Pi^{ks}(x) = 0,$	$\Pi^{sr}(x) (\partial_r) \Pi^{lk}(x) + \Pi^{kr}(x) (\partial_r) \Pi^{sl}(x) + \Pi^{lr}(x) (\partial_r) \Pi^{ks}(x) = 0,$
cardona-qft-methods_ EQ0887	344	0.999 N +0	$H_{\{p\}}(M) \otimes H_{\{q\}}(M) \longrightarrow H_{\{p+q\}}(M \times M) \xrightarrow{\tau_{\Delta^*}} H_{p+q} \left(M^{TM} \right) \xrightarrow{Thom} H_{p+q-n}(M)$	$H_p(M) \otimes H_q(M) \longrightarrow H_{p+q}(M \times M) \xrightarrow{\tau_{\Delta^*}} H_{p+q} \left(M^{TM} \right) \xrightarrow{Thom} H_{p+q-n}(M)$	$H_p(M) \otimes H_q(M) \longrightarrow H_{p+q}(M \times M) \xrightarrow{\tau_{\Delta^*}} H_{p+q}(M^{TM}) \xrightarrow{Thom} H_{p+q-n}(M)$
cardona-qft-methods_ EQ0988	371	0.999 K +0	$A+B=\text{the closure of } A\left _{\mathcal{D}}\right +\text{the closure of } B\left _{\mathcal{D}}\right , \quad \mathcal{D}=\operatorname{dom} A \cap \operatorname{dom} B.$	$A+B=\text{the closure of } A _{\mathcal{D}}+B _{\mathcal{D}}, \quad \mathcal{D}=\operatorname{dom} A \cap \operatorname{dom} B.$	$A+B=\text{the closure of } A _{\mathcal{D}}+B _{\mathcal{D}}, \quad \mathcal{D}=\operatorname{dom} A \cap \operatorname{dom} B.$
cardona-qft-methods_ EQ0759	287	0.999 W +1	$\begin{aligned} &\uparrow = Z = \text{nothing}, \\ &\downarrow = M_{AA} = 8 + 8 \text{ magnons transforming under } \mathfrak{su}(2 \mid 2) \otimes \overline{\mathfrak{su}}(2 \mid 2), \end{aligned}$	$\uparrow = Z = \text{nothing},$ $\downarrow = M_{AA} = 8 + 8 \text{ magnons transforming under } \mathfrak{su}(2 \mid 2) \otimes \overline{\mathfrak{su}}(2 \mid 2),$	$\uparrow = Z = \text{nothing},$ $\downarrow = M_{A\dot{A}} = 8 + 8 \text{ magnons transforming under } \mathfrak{su}(2 2) \otimes \overline{\mathfrak{su}}(2 2),$
cardona-qft-methods_ EQ0039	031	1.000 S -1	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n \left((1 + \Delta)^{-d/2} \right) = \frac{\alpha}{\Gamma(d/2 + 1)} = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}.$	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n \left((1 + \Delta)^{-d/2} \right) = \frac{\alpha}{\Gamma(d/2 + 1)} = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}.$	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n \left((1 + \Delta)^{-d/2} \right) = \frac{\alpha}{\Gamma(d/2+1)} = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$
cardona-qft-methods_ EQ0047 (13)	035	1.000 K +0	$\not{D}=-i\,c\,\circ\nabla^S\,.$	$\not{D}=-i c \circ \nabla^S.$	$\not{D}=-i\,c\,\circ\nabla^S\,.$
cardona-qft-methods_ EQ0087	050	1.000 N +0	$f\,A\,\mathcal{D}^{-1}=0, \forall A\in\Omega^1_{\mathcal{D}}(\mathcal{A})\Longleftrightarrow f\,a\,b=f\,a\,\alpha(b), \forall a,b\in\mathcal{A},$	$fAD^{-1} = 0, \forall A \in \Omega^1_{\mathcal{D}}(\mathcal{A}) \Longleftrightarrow fab = fa\alpha(b), \forall a, b \in \mathcal{A},$	$f\,AD^{-1}=0, \forall A\in\Omega^1_{\mathcal{D}}(\mathcal{A})\Longleftrightarrow f\,ab=f\,a\alpha(b), \forall a,b\in\mathcal{A},$
cardona-qft-methods_ EQ0125	077	1.000 N +0	$D^{\alpha}:=D_1^{\alpha_1}D_2^{\alpha_1}\cdots D_n^{\alpha_n}:C^{\infty}(\mathbb{R}^n)\longrightarrow C^{\infty}(\mathbb{R}^n).$	$D^{\alpha} := D_1^{\alpha_1} D_2^{\alpha_1} \cdots D_n^{\alpha_n} : C^{\infty}(\mathbb{R}^n) \longrightarrow C^{\infty}(\mathbb{R}^n).$	$D^{\alpha} := D_1^{\alpha_1} D_2^{\alpha_1} \cdots D_n^{\alpha_n} : C^{\infty}(\mathbb{R}^n) \longrightarrow C^{\infty}(\mathbb{R}^n).$
cardona-qft-methods_ EQ0135	079	1.000 N +0	$\mathcal{H}_{\{K\}^m}^{\{L\}^n} \left(\mathbb{R}^{\{N\}^n}, \mathbb{R}^{\{N\}^n} \right) \longrightarrow \mathcal{H}_{\{K\}^m}^{\{L\}^n} \left(\mathbb{R}^{\{N\}^n}, \mathbb{R}^{\{N\}^n} \right), \quad \text{for } m > k.$	$\mathcal{D} : H_K^m \left(\mathbb{R}^n, \mathbb{R}^N \right) \longrightarrow L^2 \left(\mathbb{R}^n, \mathbb{R}^N \right), \quad \text{for } m > k.$	$\mathcal{D} : H_K^m(\mathbb{R}^n, \mathbb{R}^N) \longrightarrow L^2(\mathbb{R}^n, \mathbb{R}^N), \quad \text{for } m > k.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0162	087	■1.000 ●N +0	$\backslash\mathrm{mathcal{D}}\colon C^{\infty}(M,E)\rightarrow C^{\infty}(M,E),$	$\mathcal{D}:C^{\infty}(M,E)\longrightarrow C^{\infty}(M,E),$	$\mathcal{D}:C^{\infty}(M,E)\longrightarrow C^{\infty}(M,E),$
cardona-qft-methods_ EQ0187 (2)	093	■1.000 ●K +0	$\backslash\mathrm{sigma}_{\mathrm{1}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ll}\}\ 0\ \&\ 1\ \backslash\backslash\ 1\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ \backslash\mathrm{sigma}_{\mathrm{2}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ 0\ \&\ -\mathrm{i}\ \backslash\backslash\ \mathrm{i}\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ \backslash\mathrm{sigma}_{\mathrm{3}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ 1\ \&\ 0\ \backslash\backslash\ 0\ \&\ -1\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})\ .$	$\sigma_1=\left(\begin{array}{cc}0&1\\1&0\end{array}\right),\quad\sigma_2=\left(\begin{array}{cc}0&-i\\i&0\end{array}\right),\quad\sigma_3=\left(\begin{array}{cc}1&0\\0&-1\end{array}\right).$	$\sigma_1\ =\ \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right),\quad\sigma_2\ =\ \left(\begin{smallmatrix}0&-i\\i&0\end{smallmatrix}\right),\quad\sigma_3\ =\ \left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right).$
cardona-qft-methods_ EQ0283	137	■1.000 ●N +1	$\backslash\mathrm{mathbb{L}}\cdot\backslash\mathrm{left}[\backslash\mathrm{left}(Y_{G\backslash e}\cup Y_{G/e})\backslash Y_{G\backslash e}\ / \ e]\backslash\mathrm{backslash}\ Y_{G\backslash e}\backslash\mathrm{backslash}\ e]\backslash\mathrm{right}],$	$\mathbb{L}\cdot\left[\left(Y_{G\backslash e}\cup Y_{G/e}\right)\backslash Y_{G\backslash e}\right],$	$\mathbb{L}\cdot\left[\left(Y_{G\backslash e}\cup Y_{G/e}\right)\backslash Y_{G\backslash e}\right]\quad,$
cardona-qft-methods_ EQ0287	138	■1.000 ●	$T(1+T)^{\mathrm{2}}\quad\backslash\mathrm{quad},\quad\backslash\mathrm{quad}(1+T)^{\mathrm{2}}\quad\backslash\mathrm{quad},\quad\backslash\mathrm{quad}T(1+T)$	$T(1+T)^2\quad,\quad(1+T)^2\quad,\quad T(1+T)$	$T(1+T)^2\quad,\quad(1+T)^2\quad,\quad T(1+T)$
cardona-qft-methods_ EQ0296	143	■1.000 ●W +0	$\backslash\mathrm{mathbb{U}}(G)=\backslash\mathrm{left}\backslash\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{lll}\}\ 0\ \&\ \backslash\mathrm{bmod}\ (\backslash\mathrm{mathbb{L}})\ \&\ \backslash\mathrm{text}\ \{\mathrm{if}\ \}\ G\ \backslash\mathrm{text}\ \{\mathrm{has}\ \mathrm{edges}\ \mathrm{that}\ \mathrm{are}\ \mathrm{not}\ \mathrm{looping}\ \mathrm{edges}\ \}\ \backslash\backslash\ (-1)^{\mathrm{n}}\ \&\ \backslash\mathrm{bmod}\ (\backslash\mathrm{mathbb{L}})\ \&\ \backslash\mathrm{text}\ \{\mathrm{if}\ \}\ G\ \backslash\mathrm{text}\ \{\mathrm{has}\ \}\ n\ \backslash\mathrm{text}\ \{\mathrm{looping}\ \mathrm{edges}\ \mathrm{and}\ \mathrm{no}\ \mathrm{other}\ \mathrm{edge}.\ \}\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}.$	$\mathbb{U}(G)=\left\{\begin{array}{cc}0&\mathrm{mod}(\mathbb{L})\\(-1)^n&\mathrm{mod}(\mathbb{L})\end{array}\right.\begin{array}{l}\text{if }G\text{ has edges that are not looping edges}\\\text{if }G\text{ has }n\text{ looping edges and no other edge.}\end{array}$	$\mathbb{U}(G)=\left\{\begin{array}{cc}0&\mathrm{mod}\left(\mathbb{L}\right)\text{ if }G\text{ has edges that are not looping edges}\\(-1)^n&\mathrm{mod}\left(\mathbb{L}\right)\text{ if }G\text{ has }n\text{ looping edges and no other edge.}\end{array}\right.$
cardona-qft-methods_ EQ0329 (16)	162	■1.000 ●N +0	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\ R_{\mathrm{b}}^{\mathrm{a}}\ \&\ =\mathrm{d}\ \backslash\mathrm{omega}_{\mathrm{b}}^{\mathrm{a}}+\backslash\mathrm{omega}_{\mathrm{c}}^{\mathrm{a}}\ \backslash\mathrm{wedge}\ \backslash\mathrm{omega}_{\mathrm{b}}^{\mathrm{c}}\ \backslash\backslash\ \&\ =\backslash\mathrm{frac}\{1\}\{2\}\ R_{\mathrm{b}\ \backslash\mathrm{mu}\ \backslash\mathrm{nu}}^{\mathrm{a}}\ \mathrm{d}\ x^{\backslash\mathrm{mu}}\ \backslash\mathrm{wedge}\ \mathrm{d}\ x^{\backslash\mathrm{nu}}\ \backslash\mathrm{end}\{\mathrm{aligned}\}$	$R_b^a=d\omega_b^a+\omega_c^a\wedge\omega_b^c\\=\frac{1}{2}R_{b\mu\nu}^adx^\mu\wedge dx^\nu$	$R_b^a=d\omega_b^a+\omega_c^a\wedge\omega_b^c\\=\frac{1}{2}R_{b\mu\nu}^adx^\mu\wedge dx^\nu.$
cardona-qft-methods_ EQ0334 (20)	163	■1.000 ●W +0	$\mathrm{D}\ R_{\mathrm{b}}^{\mathrm{a}}=\mathrm{d}\ R_{\mathrm{b}}^{\mathrm{a}}+\backslash\mathrm{omega}_{\mathrm{c}}^{\mathrm{a}}\ \backslash\mathrm{wedge}\ R_{\mathrm{b}}^{\mathrm{c}}-\backslash\mathrm{omega}_{\mathrm{b}}^{\mathrm{c}}\ \backslash\mathrm{wedge}\ R_{\mathrm{c}}^{\mathrm{a}}\ \backslash\mathrm{equiv}\ 0\ .$	$DR_b^a=dR_b^a+\omega_c^a\wedge R_b^c-\omega_b^c\wedge R_c^a\equiv 0.$	$DR_b^a=dR_b^a+\omega_c^a\wedge R_b^c-\omega_b^c\wedge R_c^a\equiv 0\ .$
cardona-qft-methods_ EQ0343 (11)	167	■1.000 ●S +2	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\ I_{\mathrm{2}}\ \&\ =\kappa\ \backslash\mathrm{int}_{\mathrm{M}}\ \backslash\mathrm{epsilon}_{\mathrm{a}\ \mathrm{b}}\ \backslash\mathrm{left}[\backslash\mathrm{alpha}_{\mathrm{1}}\ R^{\mathrm{a}}\ \mathrm{b}+\backslash\mathrm{alpha}_{\mathrm{0}}\ e^{\mathrm{a}}\ e^{\mathrm{b}}\backslash\mathrm{right}]\ \backslash\backslash\ \&\ =\kappa\ \backslash\mathrm{int}_{\mathrm{M}}\ \backslash\mathrm{sqrt}\{ g \}\backslash\mathrm{left}(\backslash\mathrm{alpha}_{\mathrm{1}}\ R+\backslash\mathrm{alpha}_{\mathrm{0}}\backslash\mathrm{right})\ \mathrm{d}^{\mathrm{2}}\ x\ \backslash\backslash\ \&\ =\kappa\ \backslash\mathrm{alpha}_{\mathrm{1}}\ \backslash\mathrm{cdot}\ \backslash\mathrm{mathbf{E}}_{\mathrm{2}}+2\ \kappa\ \backslash\mathrm{alpha}_{\mathrm{0}}\ \backslash\mathrm{cdot}\ V_{\mathrm{2}}\ \backslash\mathrm{end}\{\mathrm{aligned}\}$	$I_2=\kappa\int_M\epsilon_{ab}\left[\alpha_1R^{ab}+\alpha_0e^ae^b\right]\\=\kappa\int_M\sqrt{ g }\left(\alpha_1R+2\alpha_0\right)d^2x\\=\kappa\alpha_1\cdot\mathbf{E}_2+2\kappa\alpha_0\cdot V_2$	$I_2=\kappa\int_M\epsilon_{ab}[\alpha_1R^{ab}+\alpha_0e^ae^b]\\=\kappa\int_M\sqrt{ g }\left(\alpha_1R+2\alpha_0\right)d^2x\\=\kappa\alpha_1\cdot\mathbf{E}_2+2\kappa\alpha_0\cdot V_2.$
cardona-qft-methods_ EQ0351 (20)	168	■1.000 ●N +0	$\backslash\mathrm{omega}^{\mathrm{a}}_{\mathrm{b}\ \mathrm{mu}}=-E_{\mathrm{b}}^{\nu}\left(\partial_{\mathrm{mu}}e_{\nu}^{\mathrm{a}}-\Gamma_{\mathrm{mu}\ \mathrm{nu}}^{\lambda}\ e_{\lambda}^{\mathrm{a}}\right),$	$\omega^a_{\ b\mu}=-E_b^\nu\left(\partial_\mu e_\nu^a-\Gamma_{\mu\nu}^\lambda e_\lambda^a\right),$	$\omega^a_{\ b\mu}=-E_b^\nu(\partial_\mu e_\nu^a-\Gamma_{\mu\nu}^\lambda e_\lambda^a),$
cardona-qft-methods_ EQ0354 (22)	170	■1.000 ●K +0	$\mathrm{v}_{\mathrm{1}}=\mathrm{e}^{\mathrm{a}}\ \mathrm{e}^{\mathrm{b}}\ R_{\mathrm{a}\ \mathrm{b}},\quad\backslash\mathrm{quad}\ \backslash\mathrm{tau}_{\mathrm{0}}=\mathrm{T}^{\mathrm{a}}\ T_{\mathrm{a}},\quad\backslash\mathrm{quad}\ \backslash\mathrm{mathbf{P}}_{\mathrm{4}}=\mathrm{R}^{\mathrm{a}}\ \mathrm{b}\ R_{\mathrm{a}\ \mathrm{b}}\ .$	$v_1=e^ae^bR_{ab},\quad\tau_0=T^aT_a,\quad\mathbf{P}_4=R^{ab}R_{ab}.$	$v_1=e^ae^bR_{ab},\quad\tau_0=T^aT_a,\quad\mathbf{P}_4=R^{ab}R_{ab}.$
cardona-qft-methods_ EQ0355 (23)	170	■1.000 ●K +0	$\backslash\mathrm{mathbf{N}}_{\mathrm{4}}=\mathrm{T}^{\mathrm{a}}\ T_{\mathrm{a}}-\mathrm{e}^{\mathrm{a}}\ \mathrm{e}^{\mathrm{b}}\ R_{\mathrm{a}\ \mathrm{b}},$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab},$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab},$
cardona-qft-methods_ EQ0378 (19)	177	■1.000 ●K +0	$S\ 0(\mathrm{D}-1,1)\ \backslash\mathrm{hookrightarrow}\ I\ S\ 0(\mathrm{D}-1,1)\ .$	$SO(D-1,1)\hookrightarrow ISO(D-1,1).$	$SO(D-1,1)\hookrightarrow ISO(D-1,1).$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0643 (51)	259	■ 1.000 ● N +0	$\begin{aligned} & e_{\{p_1\}} \otimes e_{\{p_2\}} + \frac{1}{2} \left[e_{\{p_1\}} \otimes e_{\{p_2\}} + \mathcal{F}_{\theta}^{-2} e_{\{p_2\}} \otimes e_{\{p_1\}} \right] \\ & = \frac{1}{2} \left[e_{\{p_1\}} \otimes e_{\{p_2\}} + e^{ip_2 \wedge p_1} e_{\{p_2\}} \otimes e_{\{p_1\}} \right] \\ & = e^{ip_2 \wedge p_1} e_{\{p_2\}} \otimes e_{\{p_1\}}. \end{aligned}$	$e_{p_1} \otimes_{S_{\theta}} e_{p_2} = \frac{1}{2} \left[e_{p_1} \otimes e_{p_2} + \mathcal{F}_{\theta}^{-2} e_{p_2} \otimes e_{p_1} \right]$ $= \frac{1}{2} \left[e_{p_1} \otimes e_{p_2} + e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1} \right]$ $= e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1}.$	$e_{p_1} \otimes_{S_{\theta}} e_{p_2} = \frac{1}{2} [e_{p_1} \otimes e_{p_2} + \mathcal{F}_{\theta}^{-2} e_{p_2} \otimes e_{p_1}]$ $= \frac{1}{2} [e_{p_1} \otimes e_{p_2} + e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1}]$ $= e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1}.$
cardona-qft-methods_ EQ0646 (53)	260	■ 1.000 ● K +0	$a_{\{p_1\}}^{\dagger} a_{\{p_2\}}^{\dagger} = e^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger}.$	$a_{p_1}^{\dagger} a_{p_2}^{\dagger} = e^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger}.$	$a_{p_1}^{\dagger} a_{p_2}^{\dagger} = e^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger}.$
cardona-qft-methods_ EQ0660 (10)	262	■ 1.000 ● N +0	$\left[H_I(x), H_I(y) \right] = 0,$	$[H_I(x), H_I(y)] = 0,$	$[H_I(x), H_I(y)] = 0,$
cardona-qft-methods_ EQ0684	269	■ 1.000 ● N +0	$S^{+} = \frac{1}{2} \sigma^{+} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S^{-} = \frac{1}{2} \sigma^{-} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$	$S^{+} = \frac{1}{2} \sigma^{+} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S^{-} = \frac{1}{2} \sigma^{-} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$	$S^{+} = \frac{1}{2} \sigma^{+} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S^{-} = \frac{1}{2} \sigma^{-} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$
cardona-qft-methods_ EQ0695 (10)	271	■ 1.000 ●	$\begin{aligned} & \hat{H} = \sum_{l=1}^L (1 - \vec{\sigma}_l \cdot \vec{\sigma}_{l+1}) \\ & = \sum_{l=1}^L \left(1 - \sigma_l^3 \sigma_{l+1}^3 - \frac{1}{2} \sigma_l^{+} \sigma_{l+1}^{-} - \frac{1}{2} \sigma_l^{-} \sigma_{l+1}^{+} \right) \\ & = 2 \sum_{l=1}^L (\mathbb{I}_{l,l+1} - \mathbb{P}_{l,l+1}). \end{aligned}$	$\hat{H} = \sum_{l=1}^L (1 - \vec{\sigma}_l \cdot \vec{\sigma}_{l+1})$ $= \sum_{l=1}^L \left(1 - \sigma_l^3 \sigma_{l+1}^3 - \frac{1}{2} \sigma_l^{+} \sigma_{l+1}^{-} - \frac{1}{2} \sigma_l^{-} \sigma_{l+1}^{+} \right)$ $= 2 \sum_{l=1}^L (\mathbb{I}_{l,l+1} - \mathbb{P}_{l,l+1}).$	$\hat{H} = \sum_{l=1}^L (1 - \vec{\sigma}_l \cdot \vec{\sigma}_{l+1})$ $= \sum_{l=1}^L \left(1 - \sigma_l^3 \sigma_{l+1}^3 - \frac{1}{2} \sigma_l^{+} \sigma_{l+1}^{-} - \frac{1}{2} \sigma_l^{-} \sigma_{l+1}^{+} \right)$ $= 2 \sum_{l=1}^L (\mathbb{I}_{l,l+1} - \mathbb{P}_{l,l+1}).$
cardona-qft-methods_ EQ0718	275	■ 1.000 ● N +0	$R_{\{a, b\}}(u - u') M_{\{a\}}(u) M_{\{b\}}(u') = M_{\{b\}}(u') M_{\{a\}}(u) R_{\{a, b\}}(u - u').$	$R_{a,b}(u - u') M_a(u) M_b(u') = M_b(u') M_a(u) R_{a,b}(u - u').$	$R_{a,b}(u - u') M_a(u) M_b(u') = M_b(u') M_a(u) R_{a,b}(u - u').$
cardona-qft-methods_ EQ0766 (7)	293	■ 1.000 ● W +1	$T^{\alpha\beta} \equiv -\frac{2\pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha\beta}} = -\frac{1}{\alpha'} \left(\partial^{\alpha} X^M \partial^{\beta} X_M - \frac{1}{2} \gamma^{\alpha\beta} \gamma_{\lambda\rho} \partial^{\lambda} X^M \partial^{\rho} X_M \right) = 0.$	$T^{\alpha\beta} \equiv -\frac{2\pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha\beta}} = -\frac{1}{\alpha'} \left(\partial^{\alpha} X^M \partial^{\beta} X_M - \frac{1}{2} \gamma^{\alpha\beta} \gamma_{\lambda\rho} \partial^{\lambda} X^M \partial^{\rho} X_M \right) = 0.$	$T^{\alpha\beta} \equiv -\frac{2\pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha\beta}} = -\frac{1}{\alpha'} \left(\partial^{\alpha} X^M \partial^{\beta} X_M - \frac{1}{2} \gamma^{\alpha\beta} \gamma_{\lambda\rho} \partial^{\lambda} X^M \partial^{\rho} X_M \right) = 0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08775 (16)	295	1.000 C +4	$\begin{aligned} Z &= \int \mathcal{D}X \mathcal{D}\gamma e^{-S} \\ &= \int \mathcal{D}X \mathcal{D}\gamma e^{-S_0} \left(1 + \frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} h_{MN} \partial_\alpha X^M \partial_\beta X^N + \dots \right) \end{aligned}$	$Z = \int \mathcal{D}X \mathcal{D}\gamma e^{-S} = \int \mathcal{D}X \mathcal{D}\gamma e^{-S_0} \left(1 + \frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} h_{MN} \partial_\alpha X^M \partial_\beta X^N + \dots \right)$	$Z = \int \mathcal{D}X \mathcal{D}\gamma e^{-S} = \int \mathcal{D}X \mathcal{D}\gamma e^{-S_0} \left(1 + \frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} h_{MN} \partial_\alpha X^M \partial_\beta X^N + \dots \right), \quad (16)$
cardona-qft-methods_ E08780 (21)	297	1.000 C +3	$\begin{aligned} \mathrm{S} &= \frac{1}{2\kappa_0^2} \int d^D X (-G)^{1/2} e^{-2\Phi} \left[R + 4\nabla_M \Phi \nabla^M \Phi - \frac{1}{12} H_{MNP} H^{MNP} - \frac{2(D-26)}{3\alpha'} + O(\alpha') \right] \\ &\quad + O(\alpha') \end{aligned}$	$S = \frac{1}{2\kappa_0^2} \int d^D X (-G)^{1/2} e^{-2\Phi} \left[R + 4\nabla_M \Phi \nabla^M \Phi - \frac{1}{12} H_{MNP} H^{MNP} - \frac{2(D-26)}{3\alpha'} + O(\alpha') \right]$	$S = \frac{1}{2\kappa_0^2} \int d^D X (-G)^{1/2} e^{-2\Phi} \left[R + 4\nabla_M \Phi \nabla^M \Phi - \frac{1}{12} H_{MNP} H^{MNP} - \frac{2(D-26)}{3\alpha'} + O(\alpha') \right]. \quad ($
cardona-qft-methods_ E08819 (19)	310	1.000 N +0	$\begin{aligned} \Phi_- &= -\frac{i}{8} \Omega \leftrightarrow \mathcal{J}_1 \\ \Phi_+ &= \frac{1}{8} e^{-iJ} \leftrightarrow \mathcal{J}_2 \end{aligned}$	$\Phi_- = -\frac{i}{8} \Omega \leftrightarrow \mathcal{J}_1$ $\Phi_+ = \frac{1}{8} e^{-iJ} \leftrightarrow \mathcal{J}_2$	$\Phi_- = -\frac{i}{8} \Omega \leftrightarrow \mathcal{J}_1$ $\Phi_+ = \frac{1}{8} e^{-iJ} \leftrightarrow \mathcal{J}_2$
cardona-qft-methods_ E08851	330	1.000 K +0	$[df, dg] := d\{f, g\}, \forall f, g \in \mathcal{C}^\infty(M),$	$[df, dg] := d\{f, g\}, \forall f, g \in \mathcal{C}^\infty(M),$	$[df, dg] := d\{f, g\}, \forall f, g \in \mathcal{C}^\infty(M),$
cardona-qft-methods_ E08863	334	1.000 K +0	$L_{\{3\}} \circ \left(L_{\{1\}} \times L_{\{1\}} \right) = L_{\{1\}}.$	$L_3 \circ (L_1 \times L_1) = L_1.$	$L_3 \circ (L_1 \times L_1) = L_1.$
cardona-qft-methods_ E08947 (7)	359	1.000 K +0	$[X] - [X \cap Y] = [X \setminus (X \cap Y)] = [(X \cup Y) \setminus Y] = [X \cup Y] - [Y],$	$[X] - [X \cap Y] = [X \setminus (X \cap Y)] = [(X \cup Y) \setminus Y] = [X \cup Y] - [Y],$	$[X] - [X \cap Y] = [X \setminus (X \cap Y)] = [(X \cup Y) \setminus Y] = [X \cup Y] - [Y],$
cardona-qft-methods_ E08992	372	1.000	$\mathcal{S} = \text{the multiplicative set generated by } \{1 + E^\dagger E \mid E \in \mathcal{E}\}$	$\mathcal{S} = \text{the multiplicative set generated by } \{1 + E^\dagger E \mid E \in \mathcal{E}\}$	$\mathcal{S} = \text{the multiplicative set generated by } \{1 + E^\dagger E \mid E \in \mathcal{E}\}$
cardona-qft-methods_ E09001	018	1.000 N +1	$\left \partial_x^\alpha \partial_\xi^\beta \left(\sigma - \sum_{j < N} \sigma_{m-j} \right) (x, \xi) \right \leq C_{N\alpha\beta}(x) \xi ^{\Re(m) - N - \beta }$	$\left \partial_x^\alpha \partial_\xi^\beta \left(\sigma - \sum_{j < N} \sigma_{m-j} \right) (x, \xi) \right \leq C_{N\alpha\beta}(x) \xi ^{\Re(m) - N - \beta }$	$ \partial_x^\alpha \partial_\xi^\beta (\sigma - \sum_{j < N} \sigma_{m-j})(x, \xi) \leq C_{N\alpha\beta}(x) \xi ^{\Re(m) - N - \beta }$
cardona-qft-methods_ E09003 (1)	018	1.000 N +0	$\sigma(\cdot, D)(u)(x) = \sigma(x, D)(u) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) \widehat{u}(\xi) e^{ix \cdot \xi} d\xi$	$\sigma(\cdot, D)(u)(x) = \sigma(x, D)(u) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) \widehat{u}(\xi) e^{ix \cdot \xi} d\xi$	$\sigma(\cdot, D)(u)(x) = \sigma(x, D)(u) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) \widehat{u}(\xi) e^{ix \cdot \xi} d\xi$
cardona-qft-methods_ E09004	018	1.000 N +0	$k^{\sigma(x, D)}(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) e^{i(x-y) \cdot \xi} d\xi = \check{\sigma}_{\xi \rightarrow y}(x, x-y)$	$k^{\sigma(x, D)}(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) e^{i(x-y) \cdot \xi} d\xi = \check{\sigma}_{\xi \rightarrow y}(x, x-y)$	$k^{\sigma(x, D)}(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(x, \xi) e^{i(x-y) \cdot \xi} d\xi = \check{\sigma}_{\xi \rightarrow y}(x, x-y)$
cardona-qft-methods_ E09005 (2)	018	1.000 N +2	$k^a(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} a(x, y, \xi) e^{i(x-y) \cdot \xi} d\xi$	$k^a(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} a(x, y, \xi) e^{i(x-y) \cdot \xi} d\xi$	$k^a(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} a(x, y, \xi) e^{i(x-y) \cdot \xi} d\xi.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0007	019	■ 1.000 ● N +0	$\sigma^A \simeq \sum_{\alpha} \frac{(-i)^{\alpha}}{\alpha!} \partial_{\xi}^{\alpha} \partial_y^{\alpha} k_{\sigma}^A(x, y, \xi) _{y=x}$		
cardona-qft-methods_ EQ0008	019	■ 1.000 ● W +1	$k_{\sigma}^A(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i(x-y) \cdot \xi} k^A(x, y, \xi) d\xi$		$k_{\sigma}^A(x, y) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i(x-y) \cdot \xi} k^A(x, y, \xi) d\xi,$
cardona-qft-methods_ EQ0009	019	■ 1.000 ● N +0	$\sigma^{P_1 P_2}(x, \xi) \simeq \sum_{\alpha \in \mathbb{N}^d} \frac{(-i)^{\alpha}}{\alpha!} \partial_{\xi}^{\alpha} \sigma^{P_1}(x, \xi) \partial_x^{\alpha} \sigma^{P_2}(x, \xi).$		$\sigma^{P_1 P_2}(x, \xi) \simeq \sum_{\alpha \in \mathbb{N}^d} \frac{(-i)^{\alpha}}{\alpha!} \partial_{\xi}^{\alpha} \sigma^{P_1}(x, \xi) \partial_x^{\alpha} \sigma^{P_2}(x, \xi).$
cardona-qft-methods_ EQ0011	020	■ 1.000 ● N +0	$\sigma_m^{\phi_* P}(x, \xi) = \sigma_m^P(\phi^{-1}(x), (d\phi)^t \xi).$		$\sigma_m^{\phi_* P}(x, \xi) = \sigma_m^P(\phi^{-1}(x), (d\phi)^t \xi).$
cardona-qft-methods_ EQ0013 (3)	021	■ 1.000 ● N +0	$\sigma_{-m}^P(x, \xi) = \lim_{t \rightarrow \infty} t^{-m} (e^{-ith} \cdot P \cdot e^{ith})(x)$ for $x \in M, \xi \in T_x^* M$		$\sigma_{-m}^P(x, \xi) = \lim_{t \rightarrow \infty} t^{-m} (e^{-ith} \cdot P \cdot e^{ith})(x)$ for $x \in M, \xi \in T_x^* M$
cardona-qft-methods_ EQ0014	022	■ 1.000 ● N +0	$k^P(x, y) = \sum_{-(m+d) \leq j \leq 0} a_j(x, x-y) - c_P(x) \log x-y + \mathcal{O}(1)$		$k^P(x, y) = \sum_{-(m+d) \leq j \leq 0} a_j(x, x-y) - c_P(x) \log x-y + \mathcal{O}(1)$
cardona-qft-methods_ EQ0015 (4)	022	■ 1.000 ● N +1	$c_P(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d\xi$		$c_P(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d\xi.$
cardona-qft-methods_ EQ0016	022	■ 1.000 ● N +0	$\text{tr}(k^P(x, y)) = \sum_{j=-(m+d)}^0 a_j(x, x-y) - c_P(x) \log x-y + \mathcal{O}(1),$		$\text{tr}(k^P(x, y)) = \sum_{j=-(m+d)}^0 a_j(x, x-y) - c_P(x) \log x-y + \mathcal{O}(1),$
cardona-qft-methods_ EQ0018 (6)	022	■ 1.000 ● N +0	$c_{\phi_* P}(x) = \phi_*(c_P(x)).$		$c_{\phi_* P}(x) = \phi_*(c_P(x)).$
cardona-qft-methods_ EQ0020 (8)	024	■ 1.000 ● K +0	$\text{WRes } P = - \int_M c_P(x) dx $		$\text{WRes } P = - \int_M c_P(x) dx $
cardona-qft-methods_ EQ0021	025	■ 1.000 ●	$\text{Tr}(\phi P) = \int_U \phi(x) k^P(x, x) dx = \frac{1}{(2\pi)^d} \int_U \phi(x) \sigma(x, \xi) d\xi dx $ $= \int_U \phi(x) L(\sigma(x, \cdot)) dx $		$\text{Tr}(\phi P) = \int_U \phi(x) k^P(x, x) dx = \frac{1}{(2\pi)^d} \int_U \phi(x) \sigma(x, \xi) d\xi dx $ $= \int_U \phi(x) L(\sigma(x, \cdot)) dx ,$
cardona-qft-methods_ EQ0023 (9)	026	■ 1.000 ● N +0	$\text{if } \phi \in \text{Diff}(M), \text{WRes}(P) = \text{WRes}(\phi_* P)$		$\text{if } \phi \in \text{Diff}(M), \text{WRes}(P) = \text{WRes}(\phi_* P)$
cardona-qft-methods_ EQ0024	026	■ 1.000 ● N +0	$[x^j, \sigma] = i \partial_{\xi_j} \sigma, \quad [\xi_j, \sigma] = -i \partial_{x^j} \sigma.$		$[x^j, \sigma] = i \partial_{\xi_j} \sigma, \quad [\xi_j, \sigma] = -i \partial_{x^j} \sigma.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0025	026	■ 1.000 ● N +0	$\sigma_{-d}^P = \sum_{k=1}^d \partial_{\xi_k}^2 \tilde{h} = -i \sum_{k=1}^d [\partial_{\xi_k} \tilde{h}, x^k]$		$\sigma_{-d}^P = \sum_{k=1}^d \partial_{\xi_k}^2 \tilde{h} = -i \sum_{k=1}^d [\partial_{\xi_k} \tilde{h}, x^k]$
cardona-qft-methods_ EQ0030	028	■ 1.000 ● W +1	$\begin{aligned} & \mu_n(T) = \mu_n(T^*) = \mu_n(T). \\ & T \in \mathcal{L}^1(\mathcal{H}) \text{ (meaning } \ T\ _1 = \text{Tr}(T) < \infty) \iff \sum_{n \in \mathbb{N}} \mu_n(T) < \infty. \\ & \mu_n(ATB) \leq \ A\ \mu_n(T) \ B\ \text{ when } A, B \in \mathcal{B}(\mathcal{H}). \\ & \mu_N(UTU^*) = \mu_N(T) \text{ when } U \text{ is a unitary.} \end{aligned}$		$\mu_n(T) = \mu_n(T^*) = \mu_n(T). \\ T \in \mathcal{L}^1(\mathcal{H}) \text{ (meaning } \ T\ _1 = \text{Tr}(T) < \infty) \iff \sum_{n \in \mathbb{N}} \mu_n(T) < \infty. \\ \mu_n(ATB) \leq \ A\ \mu_n(T) \ B\ \text{ when } A, B \in \mathcal{B}(\mathcal{H}). \\ \mu_N(UTU^*) = \mu_N(T) \text{ when } U \text{ is a unitary.}$
cardona-qft-methods_ EQ0032	029	■ 1.000 ● K +0	$\sigma_N(T) = \inf \left\{ \ x\ _1 + N\ y\ \mid T = x + y \text{ with } x \in \mathcal{L}^1(\mathcal{H}), y \in \mathcal{K}(\mathcal{H}) \right\}.$		$\sigma_N(T) = \inf \left\{ \ x\ _1 + N\ y\ \mid T = x + y \text{ with } x \in \mathcal{L}^1(\mathcal{H}), y \in \mathcal{K}(\mathcal{H}) \right\}.$
cardona-qft-methods_ EQ0033	029	■ 1.000 ● K +0	$\sigma_\lambda(T) = \inf \left\{ \ x\ _1 + \lambda\ y\ \mid T = x + y \text{ with } x \in \mathcal{L}^1(\mathcal{H}), y \in \mathcal{K}(\mathcal{H}) \right\}.$		$\sigma_\lambda(T) = \inf \left\{ \ x\ _1 + \lambda\ y\ \mid T = x + y \text{ with } x \in \mathcal{L}^1(\mathcal{H}), y \in \mathcal{K}(\mathcal{H}) \right\}.$
cardona-qft-methods_ EQ0034	029	■ 1.000 ● N +0	$\mathcal{L}^{1,\infty} = \left\{ T \in \mathcal{K}(\mathcal{H}) \mid \ T\ _{1,\infty} = \sup_{\lambda \geq e} \frac{\sigma_\lambda(T)}{\log \lambda} < \infty \right\}.$		$\mathcal{L}^{1,\infty} = \left\{ T \in \mathcal{K}(\mathcal{H}) \mid \ T\ _{1,\infty} = \sup_{\lambda \geq e} \frac{\sigma_\lambda(T)}{\log \lambda} < \infty \right\}.$
cardona-qft-methods_ EQ0036	029	■ 1.000 ● N +0	$\mathcal{L}^{1,+} = \left\{ T \in \mathcal{K}(\mathcal{H}) \mid \ T\ _{1,+} = \sup_{N \geq 2} \frac{\sigma_N(T)}{\log(N)} < \infty \right\}.$		$\mathcal{L}^{1,+} = \left\{ T \in \mathcal{K}(\mathcal{H}) \mid \ T\ _{1,+} = \sup_{N \geq 2} \frac{\sigma_N(T)}{\log(N)} < \infty \right\}.$
cardona-qft-methods_ EQ0038	029	■ 1.000 ● N +1	$ \tau_\lambda(T_1 + T_2) - \tau_\lambda(T_1) - \tau_\lambda(T_2) \leq \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right) \ T_1 + T_2\ _{1,\infty},$		$ \tau_\lambda(T_1 + T_2) - \tau_\lambda(T_1) - \tau_\lambda(T_2) \leq \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right) \ T_1 + T_2\ _{1,\infty},$
cardona-qft-methods_ EQ0042 (12)	032	■ 1.000 ● N +0	$\sigma_1^d(\omega) = i\epsilon(\omega), \quad \sigma_1^{d*}(\omega) = -i\iota(\omega).$		$\sigma_1^d(\omega) = i\epsilon(\omega), \quad \sigma_1^{d*}(\omega) = -i\iota(\omega).$
cardona-qft-methods_ EQ0045	033	■ 1.000 ● N +0	$D_\nabla = -ic \circ \nabla, \quad \Gamma(M, E) \xrightarrow{\nabla} \Gamma(M, T^*M \otimes E) \xrightarrow{c \otimes 1} \Gamma(M, E),$		$D_\nabla = -ic \circ \nabla, \quad \Gamma(M, E) \xrightarrow{\nabla} \Gamma(M, T^*M \otimes E) \xrightarrow{c \otimes 1} \Gamma(M, E),$
cardona-qft-methods_ EQ0048	035	■ 1.000 ●	$c(dx^j)(x_0) = \gamma^j, \quad \sigma_1^D(x_0, \xi) = c(\xi) = \gamma^j \xi_j$		$c(dx^j)(x_0) = \gamma^j, \quad \sigma_1^D(x_0, \xi) = c(\xi) = \gamma^j \xi_j$
cardona-qft-methods_ EQ0050	036	■ 1.000 ● N -1	$\Delta^S = -\text{Tr}_g(\nabla^S \circ \nabla^S) : \Gamma^\infty(M, S) \rightarrow \Gamma^\infty(M, S).$		$\Delta^S = -\text{Tr}_g(\nabla^S \circ \nabla^S) : \Gamma^\infty(M, S) \rightarrow \Gamma^\infty(M, S).$
cardona-qft-methods_ EQ0051	037	■ 1.000 ● K +0	$U_{g',g} := \sqrt{f_{g,g'}} I_{g',g} : \mathcal{H}_{g'} \rightarrow \mathcal{H}_g.$		$U_{g',g} := \sqrt{f_{g,g'}} I_{g',g} : \mathcal{H}_{g'} \rightarrow \mathcal{H}_g.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0053 (15)	037	■1.000 ●N +0	$\mathcal{D}_{g^{\prime}}: \mathcal{H}_g \rightarrow \mathcal{H}_g, \quad \mathcal{D}_{g^{\prime}} := U_{g,g^{\prime}}^{-1} \mathcal{D}_{g^{\prime}} U_{g,g^{\prime}}.$	$D_{g^{\prime}}: \mathcal{H}_g \rightarrow \mathcal{H}_g, \quad D_{g^{\prime}} := U_{g,g^{\prime}}^{-1} \mathcal{D}_{g^{\prime}} U_{g,g^{\prime}}.$	$D_{g^{\prime}}: \mathcal{H}_g \rightarrow \mathcal{H}_g, \quad D_{g^{\prime}} := U_{g,g^{\prime}}^{-1} \mathcal{D}_{g^{\prime}} U_{g,g^{\prime}}.$
cardona-qft-methods_ EQ0054	037	■1.000 ●W +0	$\begin{aligned} & \not D_{g^{\prime}} I_{g,g^{\prime}} \psi = e^{-h} I_{g,g^{\prime}} \left(\mathcal{D}_g \psi - i \frac{d-1}{2} c_g(\text{grad } h) \psi \right), \\ & \mathcal{D}_{g^{\prime}} = e^{-\frac{d+1}{2} h} I_{g,g^{\prime}} \mathcal{D}_g I_{g,g^{\prime}}^{-1} e^{\frac{d-1}{2} h}, \\ & D_{g^{\prime}} = e^{-h/2} \mathcal{D}_g e^{-h/2}. \end{aligned}$	$\begin{aligned} \mathcal{D}_{g^{\prime}} I_{g,g^{\prime}} \psi &= e^{-h} I_{g,g^{\prime}} \left(\mathcal{D}_g \psi - i \frac{d-1}{2} c_g(\text{grad } h) \psi \right), \\ \mathcal{D}_{g^{\prime}} &= e^{-\frac{d+1}{2} h} I_{g,g^{\prime}} \mathcal{D}_g I_{g,g^{\prime}}^{-1} e^{\frac{d-1}{2} h}, \\ D_{g^{\prime}} &= e^{-h/2} \mathcal{D}_g e^{-h/2}. \end{aligned}$	$\begin{aligned} \mathcal{D}_{g^{\prime}} I_{g,g^{\prime}} \psi &= e^{-h} I_{g,g^{\prime}} \left(\mathcal{D}_g \psi - i \frac{d-1}{2} c_g(\text{grad } h) \psi \right), \\ \mathcal{D}_{g^{\prime}} &= e^{-\frac{d+1}{2} h} I_{g,g^{\prime}} \mathcal{D}_g I_{g,g^{\prime}}^{-1} e^{\frac{d-1}{2} h}, \\ D_{g^{\prime}} &= e^{-h/2} \mathcal{D}_g e^{-h/2}. \end{aligned}$
cardona-qft-methods_ EQ0056 (16)	038	■1.000 ●N +0	$k_t(x,y)=\frac{1}{(4\pi t)^{d/2}}e^{- x-y ^2/4t} \text{ for } x \neq y$	$k_t(x,y) = \frac{1}{(4\pi t)^{d/2}}e^{- x-y ^2/4t} \text{ for } x \neq y$	$k_t(x,y) = \frac{1}{(4\pi t)^{d/2}}e^{- x-y ^2/4t} \text{ for } x \neq y$
cardona-qft-methods_ EQ0057	038	■1.000 ●N +1	$\partial_t k_t(x,y) = \Delta_x k_t(x,y), \quad \forall t > 0, x,y \in \mathbb{R}^d$	$\partial_t k_t(x,y) = \Delta_x k_t(x,y), \quad \forall t > 0, x,y \in \mathbb{R}^d$	$\partial_t k_t(x,y) = \Delta_x k_t(x,y), \quad \forall t > 0, x,y \in \mathbb{R}^d,$
cardona-qft-methods_ EQ0058	039	■1.000 ●N +0	$-\Delta \phi = \lambda \psi \text{ in } U, \quad \psi = 0 \text{ on } \partial U.$	$-\Delta \phi = \lambda \psi \text{ in } U, \quad \psi = 0 \text{ on } \partial U.$	$-\Delta \phi = \lambda \psi \text{ in } U, \quad \psi = 0 \text{ on } \partial U.$
cardona-qft-methods_ EQ0060	039	■1.000 ●N +0	$k_t(x,x) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(x) t^{(-d+k)/m}$	$k_t(x,x) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(x) t^{(-d+k)/m}$	$k_t(x,x) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(x) t^{(-d+k)/m}$
cardona-qft-methods_ EQ0061 (17)	039	■1.000 ●N +0	$k(t,f,P) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(f,P) t^{(-d+k)/m}.$	$k(t,f,P) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(f,P) t^{(-d+k)/m}.$	$k(t,f,P) \underset{t \downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k(f,P) t^{(-d+k)/m}.$
cardona-qft-methods_ EQ0062 (18)	040	■1.000 ●N +0	$P = -(g^{\mu\nu} \partial_\mu \partial_\nu + \mathbb{A}^\mu \partial_\mu + \mathbb{B})$	$P = -(g^{\mu\nu} \partial_\mu \partial_\nu + \mathbb{A}^\mu \partial_\mu + \mathbb{B})$	$P = -(g^{\mu\nu} \partial_\mu \partial_\nu + \mathbb{A}^\mu \partial_\mu + \mathbb{B})$
cardona-qft-methods_ EQ0063	040	■1.000 ●N -1	$P = -(\text{Tr}_g \nabla^2 + E), \quad \nabla^2(X,Y) := [\nabla_X, \nabla_Y] - \nabla_{\nabla_X^L C Y},$	$P = -(\text{Tr}_g \nabla^2 + E), \quad \nabla^2(X,Y) := [\nabla_X, \nabla_Y] - \nabla_{\nabla_X^L C Y},$	$P = -(\text{Tr}_g \nabla^2 + E), \quad \nabla^2(X,Y) := [\nabla_X, \nabla_Y] - \nabla_{\nabla_X^{L^C} Y},$
cardona-qft-methods_ EQ0064	040	■1.000 ●C +3	$\text{Tr}_g \nabla^2 := g^{\mu\nu} (\nabla_\mu \nabla_\nu - \Gamma_{\mu\nu}^\rho \nabla_\rho)$	$\text{Tr}_g \nabla^2 := g^{\mu\nu} (\nabla_\mu \nabla_\nu - \Gamma_{\mu\nu}^\rho \nabla_\rho)$	$\text{Tr}_g \nabla^2 := g^{\mu\nu} (\nabla_\mu \nabla_\nu - \Gamma_{\mu\nu}^\rho \nabla_\rho)$
cardona-qft-methods_ EQ0067 (20)	041	■1.000 ●N -1	$\text{Tr} \left(e^{-tD^2} \right) \underset{t \downarrow 0+}{\sim} \sum_{j \geq 0} t^{(j-d)/2} a_j(D^2)$	$\text{Tr} \left(e^{-tD^2} \right) \underset{t \downarrow 0+}{\sim} \sum_{j \geq 0} t^{(j-d)/2} a_j(D^2)$	$\text{Tr} \left(e^{-tD^2} \right) \underset{t \downarrow 0+}{\sim} \sum_{j \geq 0} t^{(j-d)/2} a_j(D^2)$
cardona-qft-methods_ EQ0071 (23)	043	■1.000 ●K +0	$[\pi(a), \pi(b)^\circ] = 0, \quad [[\mathcal{D}, \pi(a)], \pi(b)^\circ] = 0, \forall a, b \in \mathcal{A}$	$[\pi(a), \pi(b)^\circ] = 0, \quad [[\mathcal{D}, \pi(a)], \pi(b)^\circ] = 0, \forall a, b \in \mathcal{A}$	$[\pi(a), \pi(b)^\circ] = 0, \quad [[\mathcal{D}, \pi(a)], \pi(b)^\circ] = 0, \forall a, b \in \mathcal{A}$
cardona-qft-methods_ EQ0072 (24)	044	■1.000 ●	$d(\phi_1, \phi_2) = \sup \{ \phi_1(a) - \phi_2(a) \mid \ [\mathcal{D}, \pi(a)]\ \leq 1, a \in \mathcal{A} \}$	$d(\phi_1, \phi_2) = \sup \{ \phi_1(a) - \phi_2(a) \mid \ [\mathcal{D}, \pi(a)]\ \leq 1, a \in \mathcal{A} \}$	$d(\phi_1, \phi_2) = \sup \{ \phi_1(a) - \phi_2(a) \mid \ [\mathcal{D}, \pi(a)]\ \leq 1, a \in \mathcal{A} \}$
cardona-qft-methods_ EQ0073	045	■1.000 ●N +0	$OP^\alpha OP^\beta \subset OP^{\alpha+\beta}, \quad OP^\alpha \subset OP^\beta \text{ if } \alpha \leq \beta, \quad \delta(OP^\alpha) \subset OP^\alpha$	$OP^\alpha OP^\beta \subset OP^{\alpha+\beta}, \quad OP^\alpha \subset OP^\beta \text{ if } \alpha \leq \beta, \quad \delta(OP^\alpha) \subset OP^\alpha$	$OP^\alpha OP^\beta \subset OP^{\alpha+\beta}, \quad OP^\alpha \subset OP^\beta \text{ if } \alpha \leq \beta, \quad \delta(OP^\alpha) \subset OP^\alpha$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00076 (25)	045	1.000 N ·1	$\zeta_{\mathcal{D}}\{s\in\mathbb{C}\}:\mathbb{C}\rightarrow\mathrm{Tr}\left(P \mathcal{D} ^{-s}\right)$	$\zeta_{\mathcal{D}}^P:s\in\mathbb{C}\rightarrow\mathrm{Tr}\left(P \mathcal{D} ^{-s}\right)$	$\zeta_{\mathcal{D}}^P:s\in\mathbb{C}\rightarrow\mathrm{Tr}\left(P \mathcal{D} ^{-s}\right)$
cardona-qft-methods_ E00079	047	1.000 N +1	$\sigma_z(T)= D ^zT D ^{-z},z\in\mathbb{C},\quad\epsilon(T)=\nabla(T)D^{-2}$	$\sigma_z(T)= D ^zT D ^{-z},z\in\mathbb{C},\quad\epsilon(T)=\nabla(T)D^{-2}$	$\sigma_z(T)= D ^zT D ^{-z},z\in\mathbb{C},\quad\epsilon(T)=\nabla(T)D^{-2},$
cardona-qft-methods_ E00080 (27)	047	1.000 N +0	$\sigma_z(T)\sim\sum_{r=0}^Ng(z,r)\varepsilon^r(T)\mod OP^{-N-1+q}$	$\sigma_z(T)\sim\sum_{r=0}^Ng(z,r)\varepsilon^r(T)\mod OP^{-N-1+q}$	$\sigma_z(T)\sim\sum_{r=0}^Ng(z,r)\varepsilon^r(T)\mod OP^{-N-1+q}$
cardona-qft-methods_ E00082	048	1.000 W +0	$Y\sim\sum_{p=1}^N\sum_{k_1,\dots,k_p=0}^{N-p}\frac{(-1)^{ k _1+p+1}}{ k _1+p}\nabla^{k_p}\left(X\nabla^{k_{p-1}}\left(\dots X\nabla^{k_1}(X)\dots\right)\right)D^{-2(k _1+p)},$	$Y\sim\sum_{p=1}^N\sum_{k_1,\dots,k_p=0}^{N-p}\frac{(-1)^{ k _1+p+1}}{ k _1+p}\nabla^{k_p}\left(X\nabla^{k_{p-1}}\left(\dots X\nabla^{k_1}(X)\dots\right)\right)D^{-2(k _1+p)},$	$Y\sim\sum_{p=1}^N\sum_{k_1,\dots,k_p=0}^{N-p}\frac{(-1)^{ k _1+p+1}}{ k _1+p}\nabla^{k_p}\left(X\nabla^{k_{p-1}}\left(\dots X\nabla^{k_1}(X)\dots\right)\right)D^{-2(k _1+p)},$
cardona-qft-methods_ E00084	049	1.000 W +0	$\sum_{p=1}^k\sum_{r_1,\dots,r_p=0}^{k-p}\mathrm{Res}_{s=d-k}h(s,r,p)\mathrm{Tr}\left(\varepsilon^{r_1}(Y)\dots\varepsilon^{r_p}(Y) D ^{-s}\right),$	$\sum_{p=1}^k\sum_{r_1,\dots,r_p=0}^{k-p}\mathrm{Res}_{s=d-k}h(s,r,p)\mathrm{Tr}\left(\varepsilon^{r_1}(Y)\dots\varepsilon^{r_p}(Y) D ^{-s}\right),$	$\sum_{p=1}^k\sum_{r_1,\dots,r_p=0}^{k-p}\mathrm{Res}_{s=d-k}h(s,r,p)\mathrm{Tr}\left(\varepsilon^{r_1}(Y)\dots\varepsilon^{r_p}(Y) D ^{-s}\right),$
cardona-qft-methods_ E00089	051	1.000 N +1	$\zeta_{\mathcal{D}_A}(0)=c_1\int_M\left(5R^2-8R\mu\nu r^{\mu\nu}-7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}dvol+c_2\int_M\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right)dvol,\right.$	$\zeta_{\mathcal{D}_A}(0)=c_1\int_M\left(5R^2-8R\mu\nu r^{\mu\nu}-7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}dvol+c_2\int_M\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right)dvol\right.$	$\zeta_{\mathcal{D}_A}(0)=c_1\int_M\left(5R^2-8R\mu\nu r^{\mu\nu}-7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}dvol+c_2\int_M\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right)dvol,\right.$
cardona-qft-methods_ E00090	052	1.000 N +1	$\mathcal{A}:=\mathcal{A}_1\otimes\mathcal{A}_2,\quad\mathcal{D}:=\mathcal{D}_1\otimes 1+\chi_1\otimes\mathcal{D}_2,\quad\mathcal{H}:=\mathcal{H}_1\otimes\mathcal{H}_2.$	$\mathcal{A}:=\mathcal{A}_1\otimes\mathcal{A}_2,\quad\mathcal{D}:=\mathcal{D}_1\otimes 1+\chi_1\otimes\mathcal{D}_2,\quad\mathcal{H}:=\mathcal{H}_1\otimes\mathcal{H}_2.$	$\mathcal{A}:=\mathcal{A}_1\otimes\mathcal{A}_2,\quad\mathcal{D}:=\mathcal{D}_1\otimes 1+\chi_1\otimes\mathcal{D}_2,\quad\mathcal{H}:=\mathcal{H}_1\otimes\mathcal{H}_2.$
cardona-qft-methods_ E00091	052	1.000 N +0	$\mathrm{Res}_{s=d_1+d_2}(\zeta_{\mathcal{D}}(s))=\frac{1}{2}\frac{\Gamma(d_1/2)\Gamma(d_2/2)}{\Gamma(d/2)}\mathrm{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s))\mathrm{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s)).$	$\mathrm{Res}_{s=d_1+d_2}(\zeta_{\mathcal{D}}(s))=\frac{1}{2}\frac{\Gamma(d_1/2)\Gamma(d_2/2)}{\Gamma(d/2)}\mathrm{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s))\mathrm{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s)).$	$\mathrm{Res}_{s=d_1+d_2}(\zeta_{\mathcal{D}}(s))=\frac{1}{2}\frac{\Gamma(d_1/2)\Gamma(d_2/2)}{\Gamma(d/2)}\mathrm{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s))\mathrm{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s)).$
cardona-qft-methods_ E00092 (31)	055	1.000 N +0	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{\alpha\in Sd}a_{\alpha}t^{\alpha}\quad\text{with }a_{\alpha}\neq 0.$	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{\alpha\in Sd}a_{\alpha}t^{\alpha}\quad\text{with }a_{\alpha}\neq 0.$	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{\alpha\in Sd}a_{\alpha}t^{\alpha}\quad\text{with }a_{\alpha}\neq 0.$
cardona-qft-methods_ E00093 (32)	055	1.000 N +0	$a_{\alpha}=\frac{1}{2}\mathrm{Res}_{s=-2\alpha}\left(\Gamma(s/2)\zeta_{\mathcal{D}}(s)\right).$	$a_{\alpha}=\frac{1}{2}\mathrm{Res}_{s=-2\alpha}\left(\Gamma(s/2)\zeta_{\mathcal{D}}(s)\right).$	$a_{\alpha}=\frac{1}{2}\mathrm{Res}_{s=-2\alpha}\left(\Gamma(s/2)\zeta_{\mathcal{D}}(s)\right).$
cardona-qft-methods_ E00094 (33)	055	1.000 N +0	$\mathrm{Tr}(f(\mathcal{D}/\Lambda))_{\Lambda\rightarrow+\infty}\sim\sum_{\beta\in Sd^+}f_{\beta}\Lambda^{\beta}f \mathcal{D} ^{\beta}+f(0)\zeta_{\mathcal{D}}(0)+\dots$	$\mathrm{Tr}(f(\mathcal{D}/\Lambda))_{\Lambda\rightarrow+\infty}\sim\sum_{\beta\in Sd^+}f_{\beta}\Lambda^{\beta}f \mathcal{D} ^{\beta}+f(0)\zeta_{\mathcal{D}}(0)+\dots$	$\mathrm{Tr}(f(\mathcal{D}/\Lambda))_{\Lambda\rightarrow+\infty}\sim\sum_{\beta\in Sd^+}f_{\beta}\Lambda^{\beta}f \mathcal{D} ^{\beta}+f(0)\zeta_{\mathcal{D}}(0)+\dots$
cardona-qft-methods_ E00097 (34)	056	1.000	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{k\in\{0,\dots,d\}}t^{(k-d)/2}a_k\left(\mathcal{D}^2\right)+\dots,$	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{k\in\{0,\dots,d\}}t^{(k-d)/2}a_k\left(\mathcal{D}^2\right)+\dots,$	$\mathrm{Tr}\left(e^{-t\mathcal{D}^2}\right)_{t\downarrow 0}\sim\sum_{k\in\{0,\dots,d\}}t^{(k-d)/2}a_k\left(\mathcal{D}^2\right)+\dots,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0128 (2)	077	1.000 ● N +0	$\langle f, g \rangle_k := \sum_{ \alpha =k} \langle D^\alpha f, D^\alpha g \rangle.$	$\langle f, g \rangle_k := \sum_{ \alpha =k} \langle D^\alpha f, D^\alpha g \rangle.$	$\langle f, g \rangle_k := \sum_{ \alpha =k} \langle D^\alpha f, D^\alpha g \rangle.$
cardona-qft-methods_ EQ0129 (3)	077	1.000 ● N +0	$\ f\ _k := \sqrt{\langle f, f \rangle_k}$	$\ f\ _k := \sqrt{\langle f, f \rangle_k}$	$\ f\ _k := \sqrt{\langle f, f \rangle_k}$
cardona-qft-methods_ EQ0130	078	1.000 ● N +0	$H^{\{k\}}(\mathbb{R}^n) = \{f : \mathbb{R}^n \rightarrow \mathcal{C} \mid D^\alpha f \in L^2(\mathbb{R}^n) \text{ for all } \alpha \in \mathbb{Z}_+^n \text{ with } \alpha \leq k\}.$	$H^k(\mathbb{R}^n) = \{f : \mathbb{R}^n \rightarrow \mathcal{C} \mid D^\alpha f \in L^2(\mathbb{R}^n) \text{ for all } \alpha \in \mathbb{Z}_+^n \text{ with } \alpha \leq k\}.$	$H^k(\mathbb{R}^n) = \{f : \mathbb{R}^n \rightarrow \mathcal{C} \mid D^\alpha f \in L^2(\mathbb{R}^n) \text{ for all } \alpha \in \mathbb{Z}_+^n \text{ with } \alpha \leq k\}.$
cardona-qft-methods_ EQ0131	078	1.000 ● N +0	$H_{\{K\}}(\mathbb{R}^n) = \{f \in H^k(\mathbb{R}^n) \mid \text{supp}(f) \subset K\}.$	$H_K^k(\mathbb{R}^n) = \{f \in H^k(\mathbb{R}^n) \mid \text{supp}(f) \subset K\}.$	$H_K^k(\mathbb{R}^n) = \{f \in H^k(\mathbb{R}^n) \mid \text{supp}(f) \subset K\}.$
cardona-qft-methods_ EQ0133	079	1.000 ● N +0	$a_{\alpha}(x) = \left\{ a_{\alpha, ij}(x) \right\}_{\substack{1 \leq i \leq N_2 \\ 1 \leq j \leq N_1}} \in \text{Mat}_{N_1 \times N_2}, \quad a_{\alpha, ij}(x) \in C^\infty(\mathbb{R}^n).$	$a_{\alpha}(x) = \{a_{\alpha, ij}(x)\}_{\substack{1 \leq i \leq N_2 \\ 1 \leq j \leq N_1}} \in \text{Mat}_{N_1 \times N_2}, \quad a_{\alpha, ij}(x) \in C^\infty(\mathbb{R}^n).$	$a_{\alpha}(x) = \{a_{\alpha, ij}(x)\}_{\substack{1 \leq i \leq N_2 \\ 1 \leq j \leq N_1}} \in \text{Mat}_{N_1 \times N_2}, \quad a_{\alpha, ij}(x) \in C^\infty(\mathbb{R}^n).$
cardona-qft-methods_ EQ0134	079	1.000 ● N +0	$\mathcal{D}: H^{\{k\}}(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n, \mathbb{R}^N)$	$\mathcal{D}: H^k(\mathbb{R}^n, \mathbb{R}^N) \rightarrow L^2(\mathbb{R}^n, \mathbb{R}^N)$	$\mathcal{D}: H^k(\mathbb{R}^n, \mathbb{R}^N) \rightarrow L^2(\mathbb{R}^n, \mathbb{R}^N)$
cardona-qft-methods_ EQ0136	079	1.000 ● N +0	$\pi(s(x)) = x, \quad \text{for all } x \in M.$	$\pi(s(x)) = x, \quad \text{for all } x \in M.$	$\pi(s(x)) = x, \quad \text{for all } x \in M.$
cardona-qft-methods_ EQ0137 (4)	080	1.000 ● N +1	$\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$	$\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$	$\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$
cardona-qft-methods_ EQ0139	080	1.000 ● N +0	$\mathcal{D}_i: C^\infty(V_i, \mathbb{R}^{N_1}) \rightarrow C^\infty(V_i, \mathbb{R}^{N_2}).$	$\mathcal{D}_i: C^\infty(V_i, \mathbb{R}^{N_1}) \rightarrow C^\infty(V_i, \mathbb{R}^{N_2}).$	$\mathcal{D}_i: C^\infty(V_i, \mathbb{R}^{N_1}) \rightarrow C^\infty(V_i, \mathbb{R}^{N_2}).$
cardona-qft-methods_ EQ0140	081	1.000 ● N +0	$\mathcal{D}: H_K^m(M, E) \rightarrow L^2(M, F).$	$\mathcal{D}: H_K^m(M, E) \rightarrow L^2(M, F).$	$\mathcal{D}: H_K^m(M, E) \rightarrow L^2(M, F).$
cardona-qft-methods_ EQ0143	081	1.000 ● N +0	$\sigma(\mathcal{D})(x_0, \xi) = \sum_{ \alpha \leq k} a_\alpha(x_0) \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n}.$	$\sigma(\mathcal{D})(x_0, \xi) = \sum_{ \alpha \leq k} a_\alpha(x_0) \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n}.$	$\sigma(\mathcal{D})(x_0, \xi) = \sum_{ \alpha \leq k} a_\alpha(x_0) \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n}.$
cardona-qft-methods_ EQ0144	082	1.000 ● N +0	$\sigma_L(\mathcal{D})(x, \xi) : \pi^* E_{(x, \xi)} \rightarrow \pi^* F_{(x, \xi)}.$	$\sigma_L(\mathcal{D})(x, \xi) : \pi^* E_{(x, \xi)} \rightarrow \pi^* F_{(x, \xi)}.$	$\sigma_L(\mathcal{D})(x, \xi) : \pi^* E_{(x, \xi)} \rightarrow \pi^* F_{(x, \xi)}.$
cardona-qft-methods_ EQ0145	082	1.000 ● N +0	$\Delta = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} - \cdots - \frac{\partial^2}{\partial x_n^2}$	$\Delta = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} - \cdots - \frac{\partial^2}{\partial x_n^2}$	$\Delta = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} - \cdots - \frac{\partial^2}{\partial x_n^2}$
cardona-qft-methods_ EQ0150 (2)	085	1.000 ● N +0	$\sigma_1 \oplus \text{Id} : E_1 \oplus \mathcal{C}^{k_1} \rightarrow F_1 \oplus \mathcal{C}^{k_1}, \quad \text{and} \quad \sigma_2 \oplus \text{Id} : E_2 \oplus \mathcal{C}^{k_2} \rightarrow F_2 \oplus \mathcal{C}^{k_2},$	$\sigma_1 \oplus \text{Id} : E_1 \oplus \mathcal{C}^{k_1} \rightarrow F_1 \oplus \mathcal{C}^{k_1}, \quad \text{and} \quad \sigma_2 \oplus \text{Id} : E_2 \oplus \mathcal{C}^{k_2} \rightarrow F_2 \oplus \mathcal{C}^{k_2},$	$\sigma_1 \oplus \text{Id} : E_1 \oplus \mathcal{C}^{k_1} \rightarrow F_1 \oplus \mathcal{C}^{k_1}, \quad \text{and} \quad \sigma_2 \oplus \text{Id} : E_2 \oplus \mathcal{C}^{k_2} \rightarrow F_2 \oplus \mathcal{C}^{k_2},$
cardona-qft-methods_ EQ0151	085	1.000 ● K +0	$K(X) = \{\sigma : E \rightarrow F \mid \sigma(x) \text{ is invertible outside of a compact set}\} / \sim$	$K(X) = \{\sigma : E \rightarrow F \mid \sigma(x) \text{ is invertible outside of a compact set}\} / \sim$	$K(X) = \{\sigma : E \rightarrow F \mid \sigma(x) \text{ is invertible outside of a compact set}\} / \sim$
cardona-qft-methods_ EQ0152	085	1.000 ● K +0	$\dim E - \dim F \in \mathbb{Z}.$	$\dim E - \dim F \in \mathbb{Z}.$	$\dim E - \dim F \in \mathbb{Z}.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0153	085	■1.000	$K(\backslash\{p\ t\})\ \backslash\mathrm{cong}\ \backslash\mathrm{mathbb{Z}}$	$K(\{pt\})\cong\mathbb{Z}$	$K(\{pt\})\cong\mathbb{Z}.$
cardona-qft-methods_ EQ0155	086	■1.000 ●N +0	$j_{!}: K(Y)\rightarrow K(X),$	$j_{!}: K(Y)\rightarrow K(X),$	$j_{!}: K(Y)\rightarrow K(X),$
cardona-qft-methods_ EQ0156	086	■1.000 ●N +1	$i_{!}: K(p\ t)\rightarrow\mathcal{K}\backslash\mathrm{left}(\backslash\mathrm{mathcal{C}}^{\wedge\{N\}}\backslash\mathrm{right})$	$i_{!}: K(pt)\longrightarrow K\left(\mathcal{C}^N\right)$	$i_{!}: K(pt)\longrightarrow K(\mathcal{C}^N)$
cardona-qft-methods_ EQ0159	087	■1.000 ●K +0	$g\mapsto\psi(g): E\rightarrow E,\quad g\in G,$	$g\mapsto\psi(g): E\rightarrow E,\quad g\in G,$	$g\mapsto\psi(g): E\rightarrow E,\quad g\in G,$
cardona-qft-methods_ EQ0160	087	■1.000 ●N +0	$\pi\circ\psi(g)=\phi\circ\pi(g)\ .$	$\pi\circ\psi(g)=\phi\circ\pi(g).$	$\pi\circ\psi(g)\,=\,\phi\circ\pi(g).$
cardona-qft-methods_ EQ0161 (5)	087	■1.000 ●N +0	$g\mapsto l_g: \mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{E})\rightarrow\mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{E}),\quad\backslash\mathrm{quad}\ l_g(f)(x):=g\cdot\mathrm{f}\backslash\mathrm{left}(g^{\wedge[-1]}\backslash\mathrm{cdot}\ x\backslash\mathrm{right})\ .$	$g\mapsto l_g: C^\infty(M,\mathcal{E})\longrightarrow C^\infty(M,\mathcal{E}),\quad l_g(f)(x):=g\cdot f\left(g^{-1}\cdot x\right).$	$g\mapsto l_g: C^\infty(M,\mathcal{E})\longrightarrow C^\infty(M,\mathcal{E}),\quad l_g(f)(x):=g\cdot f(g^{-1}\cdot x).$
cardona-qft-methods_ EQ0163 (6)	088	■1.000 ●K +0	$K_{\{G\}}(\backslash\{p\ t\})=R(G),$	$K_G(\{pt\})=R(G),$	$K_G(\{pt\})\,=\,R(G),$
cardona-qft-methods_ EQ0165	088	■1.000 ●K +0	$\backslash\mathrm{mathcal{D}}: \mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{E})\rightarrow\mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{F})$	$\mathcal{D}: C^\infty(M,\mathcal{E})\rightarrow C^\infty(M,\mathcal{F})$	$\mathcal{D}: C^\infty(M,\mathcal{E})\rightarrow C^\infty(M,\mathcal{F})$
cardona-qft-methods_ EQ0166	088	■1.000 ●N +0	$\backslash\mathrm{sigma}_{\{L\}}(\backslash\mathrm{mathcal{D}}): \backslash\mathrm{pi}^{\wedge\{*\}}\mathcal{E}\rightarrow\backslash\mathrm{pi}^{\wedge\{*\}}\mathcal{F},$	$\sigma_L(\mathcal{D}): \pi^*\mathcal{E}\longrightarrow\pi^*\mathcal{F},$	$\sigma_L(\mathcal{D}): \pi^*\mathcal{E}\longrightarrow\pi^*\mathcal{F},$
cardona-qft-methods_ EQ0167	089	■1.000 ●K +0	$\backslash\mathrm{operatorname{Ind}}_{\{G\}}(\backslash\mathrm{mathcal{D}})=\mathrm{t}-\backslash\mathrm{operatorname{Ind}}_{\{G\}}(\backslash\mathrm{sigma})\ .$	$\mathrm{Ind}_G(\mathcal{D})=\mathrm{t}-\mathrm{Ind}_G(\sigma).$	$\mathrm{Ind}_G(\mathcal{D})\,=\,\mathrm{t}\text{-}\mathrm{Ind}_G(\sigma).$
cardona-qft-methods_ EQ0168	089	■1.000 ●K +0	$M:=N\times G$	$M:=N\times G$	$M\,:=\,N\times G$
cardona-qft-methods_ EQ0170	089	■1.000 ●N +0	$\backslash\mathrm{tilde}(\backslash\mathrm{mathcal{D}}): \mathcal{C}^{\wedge\{\infty\}}(M)\rightarrow\mathcal{C}^{\wedge\{\infty\}}(M)$	$\tilde{\mathcal{D}}: C^\infty(M)\longrightarrow C^\infty(M)$	$\tilde{\mathcal{D}}: C^\infty(M)\longrightarrow C^\infty(M)$
cardona-qft-methods_ EQ0171	089	■1.000 ●N +0	$\backslash\mathrm{mathcal{D}}=\sum_{ \backslash\mathrm{alpha} \leq k}a_{\backslash\mathrm{alpha}}(x)\backslash\mathrm{frac}\{\backslash\mathrm{partial}^{\wedge\backslash\mathrm{alpha}}_{\{1\}}\}\{\backslash\mathrm{partial}\ x_{\{1\}}^{\wedge\backslash\mathrm{alpha}}_{\{1\}}\}\backslash\mathrm{cdots}\backslash\mathrm{frac}\{\backslash\mathrm{partial}^{\wedge\backslash\mathrm{alpha}}_{\{n\}}\}\{\backslash\mathrm{partial}\ x_{\{n\}}^{\wedge\backslash\mathrm{alpha}}_{\{n\}}\}\ .$	$\mathcal{D}=\sum_{ \alpha \leq k}a_\alpha(x)\frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}}\cdots\frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}.$	$\mathcal{D}\,=\,\sum_{ \alpha \leq k}a_\alpha(x)\frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}}\cdots\frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}.$
cardona-qft-methods_ EQ0172	089	■1.000 ●N +0	$\backslash\mathrm{tilde}(\backslash\mathrm{mathcal{D}}):=\sum_{ \backslash\mathrm{alpha} \leq k}a_{\backslash\mathrm{alpha}}(x)\backslash\mathrm{frac}\{\backslash\mathrm{partial}^{\wedge\backslash\mathrm{alpha}}_{\{1\}}\}\{\backslash\mathrm{partial}\ x_{\{1\}}^{\wedge\backslash\mathrm{alpha}}_{\{1\}}\}\backslash\mathrm{cdots}\backslash\mathrm{frac}\{\backslash\mathrm{partial}^{\wedge\backslash\mathrm{alpha}}_{\{n\}}\}\{\backslash\mathrm{partial}\ x_{\{n\}}^{\wedge\backslash\mathrm{alpha}}_{\{n\}}\}\ .$	$\tilde{\mathcal{D}}:=\sum_{ \alpha \leq k}a_\alpha(x)\frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}}\cdots\frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}.$	$\tilde{\mathcal{D}}\,:=\,\sum_{ \alpha \leq k}a_\alpha(x)\frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}}\cdots\frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}.$
cardona-qft-methods_ EQ0173 (1)	090	■1.000 ●K +0	$\backslash\mathrm{operatorname{Ker}}(\backslash\mathrm{tilde}(\backslash\mathrm{mathcal{D}}))=\backslash\mathrm{operatorname{Ker}}(\backslash\mathrm{mathcal{D}})\backslash\mathrm{otimes}\mathcal{L}^{\wedge\{2\}}(G),\quad\backslash\mathrm{quad}\backslash\mathrm{operatorname{Coker}}(\backslash\mathrm{tilde}(\backslash\mathrm{mathcal{D}}))=\backslash\mathrm{operatorname{Coker}}(\backslash\mathrm{mathcal{D}})\backslash\mathrm{otimes}\mathcal{L}^{\wedge\{2\}}(G)\ .$	$\mathrm{Ker}(\tilde{\mathcal{D}})=\mathrm{Ker}(\mathcal{D})\otimes L^2(G),\quad\mathrm{Coker}(\tilde{\mathcal{D}})=\mathrm{Coker}(\mathcal{D})\otimes L^2(G).$	$\mathrm{Ker}(\tilde{\mathcal{D}})\,=\,\mathrm{Ker}(\mathcal{D})\otimes L^2(G),\quad\mathrm{Coker}(\tilde{\mathcal{D}})\,=\,\mathrm{Coker}(\mathcal{D})\otimes L^2(G).$
cardona-qft-methods_ EQ0174 (2)	090	■1.000 ●N +1	$\mathcal{L}^{\wedge\{2\}}(G)=\bigoplus_{V\in\backslash\mathrm{operatorname{Irr}}(G)}(\backslash\mathrm{operatorname{dim}}\ V)\ V\ .$	$L^2(G)=\bigoplus_{V\in\mathrm{Irr}(G)}(\dim V)V.$	$L^2(G)\,=\,\bigoplus_{V\in\mathrm{Irr}(G)}(\dim V)V.$
cardona-qft-methods_ EQ0176	090	■1.000 ●K +0	$\backslash\mathrm{mathcal{O}}(x):=\{g\cdot x\mid g\in G\}\ .$	$\mathcal{O}(x):=\{g\cdot x\mid g\in G\}.$	$\mathcal{O}(x)\,:=\,\big\{g\cdot x\big g\in G\big\}.$
cardona-qft-methods_ EQ0178	091	■1.000 ●N +0	$\backslash\mathrm{mathcal{D}}: \mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{E})\rightarrow\mathcal{C}^{\wedge\{\infty\}}(M,\mathcal{F})$	$\mathcal{D}: C^\infty(M,\mathcal{E})\longrightarrow C^\infty(M,\mathcal{F})$	$\mathcal{D}: C^\infty(M,\mathcal{E})\longrightarrow C^\infty(M,\mathcal{F})$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0179 (5)	091	■ 1.000 ● N +0	$\operatorname{Ker}(\mathcal{D})=\bigoplus_V \operatorname{Irr}(G) m_V^{+} V, \quad \operatorname{Coker}(\mathcal{D})=\bigoplus_V \operatorname{Irr}(G) m_V^{-} V$.	$\operatorname{Ker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_V^{+} V, \quad \operatorname{Coker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_V^{-} V.$	$\operatorname{Ker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_V^{+} V, \quad \operatorname{Coker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_V^{-} V.$
cardona-qft-methods_ EQ0180 (6)	091	■ 1.000 ●	$\operatorname{Ind}_G(\mathcal{D}) := \bigoplus_V \operatorname{Irr}(G) \left(m_V^{+} - m_V^{-} \right) V \in \widehat{R}(G),$	$\operatorname{Ind}_G(\mathcal{D}) := \bigoplus_{V \in \operatorname{Irr}(G)} (m_V^{+} - m_V^{-}) V \in \widehat{R}(G),$	$\operatorname{Ind}_G(\mathcal{D}) := \bigoplus_{V \in \operatorname{Irr}(G)} (m_V^{+} - m_V^{-}) V \in \widehat{R}(G),$
cardona-qft-methods_ EQ0182 (8)	092	■ 1.000 ● K +0	$\operatorname{t-Ind}_G(\sigma) := \operatorname{Ind}_G(\mathcal{D})$.	$\operatorname{t-Ind}_G(\sigma) := \operatorname{Ind}_G(\mathcal{D}).$	$\operatorname{t-Ind}_G(\sigma) := \operatorname{Ind}_G(\mathcal{D}).$
cardona-qft-methods_ EQ0183	092	■ 1.000 ● K +0	$\operatorname{Ind}_G(\mathcal{D}) = \operatorname{t-Ind}_G(\sigma) \in \widehat{R}(G)$.	$\operatorname{Ind}_G(\mathcal{D}) = \operatorname{t-Ind}_G(\sigma) \in \widehat{R}(G).$	$\operatorname{Ind}_G(\mathcal{D}) = \operatorname{t-Ind}_G(\sigma) \in \widehat{R}(G).$
cardona-qft-methods_ EQ0184	093	■ 1.000 ● K +0	$c: V \rightarrow \operatorname{End}(W)$	$c: V \rightarrow \operatorname{End}(W)$	$c: V \rightarrow \operatorname{End}(W)$
cardona-qft-methods_ EQ0188 (3)	093	■ 1.000 ● N +1	$c: \mathbb{R}^3 \rightarrow \operatorname{End}(\mathcal{C}^2), \quad c(x_1, x_2, x_3) := \frac{1}{i} (x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3)$	$c: \mathbb{R}^3 \rightarrow \operatorname{End}(\mathcal{C}^2), \quad c(x_1, x_2, x_3) := \frac{1}{i} (x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3)$	$c: \mathbb{R}^3 \rightarrow \operatorname{End}(\mathcal{C}^2), \quad c(x_1, x_2, x_3) := \frac{1}{i} (x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3)$
cardona-qft-methods_ EQ0189	093	■ 1.000 ● N +0	$W = \Lambda^*(V^{\mathcal{C}})$	$W = \Lambda^*(V^{\mathcal{C}})$	$W = \Lambda^*(V^{\mathcal{C}})$
cardona-qft-methods_ EQ0190	093	■ 1.000 ● K +0	$\varepsilon(v) \alpha := v \wedge \alpha, \quad \alpha \in W.$	$\varepsilon(v) \alpha := v \wedge \alpha, \quad \alpha \in W.$	$\varepsilon(v) \alpha := v \wedge \alpha, \quad \alpha \in W.$
cardona-qft-methods_ EQ0191 (4)	093	■ 1.000 ● K +0	$c(v) := \varepsilon_v - \iota_v: W \rightarrow W.$	$c(v) := \varepsilon_v - \iota_v: W \rightarrow W.$	$c(v) := \varepsilon_v - \iota_v: W \rightarrow W.$
cardona-qft-methods_ EQ0192 (5)	094	■ 1.000 ● N +0	$c: T^*M \rightarrow \operatorname{End}(E),$	$c: T^*M \rightarrow \operatorname{End}(E),$	$c: T^*M \rightarrow \operatorname{End}(E),$
cardona-qft-methods_ EQ0193	094	■ 1.000 ● N +0	$c: T_x^*M \rightarrow \operatorname{End}(E_x)$	$c: T_x^*M \rightarrow \operatorname{End}(E_x)$	$c: T_x^*M \rightarrow \operatorname{End}(E_x)$
cardona-qft-methods_ EQ0194	094	■ 1.000 ● K +0	$c(v) \omega = v \wedge \omega - \iota_v \omega.$	$c(v) \omega = v \wedge \omega - \iota_v \omega.$	$c(v) \omega = v \wedge \omega - \iota_v \omega.$
cardona-qft-methods_ EQ0195 (6)	094	■ 1.000 ● K +0	$\nabla_u(c(v)s) = c(\nabla_u^{LC} v)s + c(v) \nabla_u s,$	$\nabla_u(c(v)s) = c(\nabla_u^{LC} v)s + c(v) \nabla_u s,$	$\nabla_u(c(v)s) = c(\nabla_u^{LC} v)s + c(v) \nabla_u s,$
cardona-qft-methods_ EQ0196 (7)	095	■ 1.000 ● K +0	$\mathcal{D} := \sum_{j=1}^n c(e_j) \nabla_{e_j}: C^\infty(M, E) \rightarrow C^\infty(M, E),$	$\mathcal{D} := \sum_{j=1}^n c(e_j) \nabla_{e_j}: C^\infty(M, E) \rightarrow C^\infty(M, E),$	$\mathcal{D} := \sum_{j=1}^n c(e_j) \nabla_{e_j}: C^\infty(M, E) \rightarrow C^\infty(M, E),$
cardona-qft-methods_ EQ0197 (8)	095	■ 1.000 ● N +0	$\sigma_L(\mathcal{D})(x, \xi) = i c(\xi).$	$\sigma_L(\mathcal{D})(x, \xi) = i c(\xi).$	$\sigma_L(\mathcal{D})(x, \xi) = i c(\xi).$
cardona-qft-methods_ EQ0198	095	■ 1.000 ● N +1	$\nabla_{\partial/\partial x_i} = \frac{\partial}{\partial x_i}.$	$\nabla_{\partial/\partial x_i} = \frac{\partial}{\partial x_i}.$	$\nabla_{\partial/\partial x_i} = \frac{\partial}{\partial x_i}.$
cardona-qft-methods_ EQ0200	095	■ 1.000 ● K +0	$d^*: \Omega^*(M, E) \rightarrow \Omega^{*-1}(M, E)$	$d^*: \Omega^*(M, E) \rightarrow \Omega^{*-1}(M, E)$	$d^*: \Omega^*(M, E) \rightarrow \Omega^{*-1}(M, E)$
cardona-qft-methods_ EQ0203 (11)	096	■ 1.000 ● N +0	$\operatorname{Ker}(\mathcal{D}) \simeq H^*(M).$	$\operatorname{Ker}(\mathcal{D}) \simeq H^*(M).$	$\operatorname{Ker}(\mathcal{D}) \simeq H^*(M).$
cardona-qft-methods_ EQ0204 (12)	096	■ 1.000 ● K +0	$E = E^+ \oplus E^-,$	$E = E^+ \oplus E^-,$	$E = E^+ \oplus E^-,$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0205	096	■ 1.000 ● K +0	$c(v) \colon E^{\{ \text{ \pmb{ \}} \}} \longrightarrow E^{\{ \text{ \pmb{ \}} \}} \text{ .}$	$c(v) \colon E^{\pm} \longrightarrow E^{\mp}.$	$c(v) \colon E^{\pm} \longrightarrow E^{\mp}.$
cardona-qft-methods_ EQ0206	096	■ 1.000 ● K +0	$\nabla_{\text{v}} \colon C^{\{ \infty \}} \left(M, E^{\{ \text{ \pmb{ \}} \}} \right) \longrightarrow C^{\{ \infty \}} \left(M, E^{\{ \text{ \pmb{ \}} \}} \right) \text{ .}$	$\nabla_v \colon C^{\infty} \left(M, E^{\pm} \right) \longrightarrow C^{\infty} \left(M, E^{\pm} \right).$	$\nabla_v \colon C^{\infty} (M, E^{\pm}) \longrightarrow C^{\infty} (M, E^{\pm}).$
cardona-qft-methods_ EQ0207	096	■ 1.000 ● K +0	$\mathcal{D} \colon C^{\{ \infty \}} \left(M, E^{\{ \text{ \pmb{ \}} \}} \right) \longrightarrow C^{\{ \infty \}} \left(M, E^{\{ \text{ \pmb{ \}} \}} \right) \text{ .}$	$\mathcal{D} \colon C^{\infty} \left(M, E^{\pm} \right) \longrightarrow C^{\infty} \left(M, E^{\mp} \right).$	$\mathcal{D} \colon C^{\infty} (M, E^{\pm}) \longrightarrow C^{\infty} (M, E^{\mp}).$
cardona-qft-methods_ EQ0208 (13)	097	■ 1.000 ●	$\mathcal{D} = \left(\begin{array}{cc} 0 & \mathcal{D}^- \\ \mathcal{D}^+ & 0 \end{array} \right) \text{ .}$	$\mathcal{D} = \left(\begin{array}{cc} 0 & \mathcal{D}^- \\ \mathcal{D}^+ & 0 \end{array} \right).$	$\mathcal{D} = \left(\begin{array}{cc} 0 & \mathcal{D}^- \\ \mathcal{D}^+ & 0 \end{array} \right).$
cardona-qft-methods_ EQ0210	097	■ 1.000 ● N +1	$\operatorname{Ind} \left(\mathcal{D}^{\{ + \}} \right) = \sum_{j=0}^n (-1)^{\{ n \}} \operatorname{dim} H^{\{ j \}} (M)$	$\operatorname{Ind} \left(\mathcal{D}^+ \right) = \sum_{j=0}^n (-1)^n \dim H^j (M)$	$\operatorname{Ind} (\mathcal{D}^+) = \sum_{j=0}^n (-1)^n \dim H^j (M).$
cardona-qft-methods_ EQ0211	097	■ 1.000 ● N +0	$\Gamma \omega := i^{\{ p(p-1) + l \}} * \omega, \quad \omega \in \Lambda^p (M) \otimes \mathcal{C}.$	$\Gamma \omega := i^{p(p-1)+l} * \omega, \quad \omega \in \Lambda^p (M) \otimes \mathcal{C}.$	$\Gamma \omega := i^{p(p-1)+l} * \omega, \quad \omega \in \Lambda^p (M) \otimes \mathcal{C}.$
cardona-qft-methods_ EQ0212	098	■ 1.000 ● K +0	$\operatorname{Ind}_G \left(\mathcal{D}^{\{ + \}} \right) \in R(G) \text{ .}$	$\operatorname{Ind}_G \left(\mathcal{D}^+ \right) \in R(G).$	$\operatorname{Ind}_G (\mathcal{D}^+) \in R(G).$
cardona-qft-methods_ EQ0213 (1)	098	■ 1.000 ● N +0	$\mathbf{v} \colon M \rightarrow \mathfrak{g} = \operatorname{Lie} G \text{ .}$	$\mathbf{v} \colon M \rightarrow \mathfrak{g} = \operatorname{Lie} G.$	$\mathbf{v} \colon M \rightarrow \mathfrak{g} = \operatorname{Lie} G.$
cardona-qft-methods_ EQ0215	099	■ 1.000 ● K +0	$G = S^1 = \{ z \in \mathcal{C} \colon z = 1 \}$	$G = S^1 = \{ z \in \mathcal{C} \colon z = 1 \}$	$G = S^1 = \{ z \in \mathcal{C} \colon z = 1 \}$
cardona-qft-methods_ EQ0216 (3)	099	■ 1.000 ● N -1	$v(x) \neq 0, \quad \text{for all } x \notin K \text{ .}$	$v(x) \neq 0, \quad \text{for all } x \notin K.$	$v(x) \neq 0, \quad \text{for all } x \notin K.$
cardona-qft-methods_ EQ0218	100	■ 1.000 ● N +0	$c(\xi + v) \colon \pi^{\{ + \}} E^{\{ + \}} \rightarrow \pi^{\{ + \}} E^{\{ - \}} \text{ .}$	$c(\xi + v) \colon \pi^* E^+ \rightarrow \pi^* E^-.$	$c(\xi + v) \colon \pi^* E^+ \rightarrow \pi^* E^-.$
cardona-qft-methods_ EQ0219	100	■ 1.000 ● N +0	$\operatorname{t} - \operatorname{Ind}_G (c(\xi + v)) \in \widehat{R}(G) \text{ .}$	$\operatorname{t} - \operatorname{Ind}_G (c(\xi + v)) \in \widehat{R}(G).$	$\operatorname{t} - \operatorname{Ind}_G (c(\xi + v)) \in \widehat{R}(G).$
cardona-qft-methods_ EQ0220 (5)	100	■ 1.000 ● N +0	$\operatorname{t} - \operatorname{Ind}_G (E, \mathbf{v}) := \operatorname{t} - \operatorname{Ind}_G (c(\xi + v)) \text{ .}$	$\operatorname{t} - \operatorname{Ind}_G (E, \mathbf{v}) := \operatorname{t} - \operatorname{Ind}_G (c(\xi + v)).$	$\operatorname{t} - \operatorname{Ind}_G (E, \mathbf{v}) := \operatorname{t} - \operatorname{Ind}_G (c(\xi + v)).$
cardona-qft-methods_ EQ0221 (6)	101	■ 1.000 ● K +0	$\mathcal{D}_{fv} = \mathcal{D} + ic(fv),$	$\mathcal{D}_{fv} = \mathcal{D} + ic(fv),$	$\mathcal{D}_{fv} = \mathcal{D} + ic(fv),$
cardona-qft-methods_ EQ0222 (7)	101	■ 1.000 ● N +0	$\operatorname{Ker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^+ \cdot V, \quad \operatorname{Coker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^- \cdot V,$	$\operatorname{Ker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^+ \cdot V, \quad \operatorname{Coker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^- \cdot V,$	$\operatorname{Ker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^+ \cdot V, \quad \operatorname{Coker} \mathcal{D}_{fv}^+ = \sum_{V \in \operatorname{Irr}(G)} m_V^- \cdot V,$
cardona-qft-methods_ EQ0223 (8)	101	■ 1.000 ● N +0	$\operatorname{Ind}_G \left(\mathcal{D}_{fv}^+ \right) = \sum_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) \cdot V \in \widehat{R}(G).$	$\operatorname{Ind}_G \left(\mathcal{D}_{fv}^+ \right) = \sum_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) \cdot V \in \widehat{R}(G).$	$\operatorname{Ind}_G (\mathcal{D}_{fv}^+) = \sum_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) \cdot V \in \widehat{R}(G).$
cardona-qft-methods_ EQ0224 (9)	101	■ 1.000 ● K +0	$\operatorname{Ind}_G (\mathcal{D}, \mathbf{v}) := \operatorname{Ind}_G \left(\mathcal{D}_{fv}^+ \right),$	$\operatorname{Ind}_G (\mathcal{D}, \mathbf{v}) := \operatorname{Ind}_G \left(\mathcal{D}_{fv}^+ \right),$	$\operatorname{Ind}_G (\mathcal{D}, \mathbf{v}) := \operatorname{Ind}_G (\mathcal{D}_{fv}^+),$
cardona-qft-methods_ EQ0225	102	■ 1.000 ● N +0	$E^{\{ \text{ \pmb{ \}} \}} = M \times \mathbb{R}$	$E^{\pm} = M \times \mathbb{R}$	$E^{\pm} = M \times \mathbb{R}$
cardona-qft-methods_ EQ0226	102	■ 1.000 ● N +0	$c(\xi) := \frac{1}{i} \left(\xi_1 \sigma_1 + \xi_2 \sigma_2 \right), \quad \xi = (\xi_1, \xi_2) \in T^* M \simeq \mathbb{R}^2,$	$c(\xi) := \frac{1}{i} \left(\xi_1 \sigma_1 + \xi_2 \sigma_2 \right), \quad \xi = (\xi_1, \xi_2) \in T^* M \simeq \mathbb{R}^2,$	$c(\xi) := \frac{1}{i} \left(\xi_1 \sigma_1 + \xi_2 \sigma_2 \right), \quad \xi = (\xi_1, \xi_2) \in T^* M \simeq \mathbb{R}^2,$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0228 (10)	102	■ 1.000 ● K +0	$\operatorname{Ind}(\mathcal{D}, \mathbf{v}) = \mathrm{t} - \operatorname{Ind}(E, \mathbf{v})$.	$\operatorname{Ind}(\mathcal{D}, \mathbf{v}) = \mathrm{t} - \operatorname{Ind}(E, \mathbf{v})$.	$\operatorname{Ind}(\mathcal{D}, \mathbf{v}) = \mathrm{t} - \operatorname{Ind}(E, \mathbf{v})$.
cardona-qft-methods_ EQ0229 (11)	102	■ 1.000 ● K +0	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D})$.	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D})$.	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D})$.
cardona-qft-methods_ EQ0230	103	■ 1.000 ● N +0	$M \backslash \Sigma = M_1 \sqcup M_2$.	$M \backslash \Sigma = M_1 \sqcup M_2$.	$M \backslash \Sigma = M_1 \sqcup M_2$.
cardona-qft-methods_ EQ0231 (12)	103	■ 1.000 ● K +0	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D}_1, \mathbf{v}_1) + \operatorname{Ind}_G(\mathcal{D}_2, \mathbf{v}_2)$.	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D}_1, \mathbf{v}_1) + \operatorname{Ind}_G(\mathcal{D}_2, \mathbf{v}_2)$.	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v}) = \operatorname{Ind}_G(\mathcal{D}_1, \mathbf{v}_1) + \operatorname{Ind}_G(\mathcal{D}_2, \mathbf{v}_2)$.
cardona-qft-methods_ EQ0232	109	■ 1.000 ●	$\chi(X) = s_0 - s_1 + s_2 - \cdots$	$\chi(X) = s_0 - s_1 + s_2 - \cdots$	$\chi(X) = s_0 - s_1 + s_2 - \cdots$
cardona-qft-methods_ EQ0235	112	■ 1.000 ● K +0	$\left[\mathbb{P}^{n-1} \right] = 1 + \mathbb{L} + \cdots + \mathbb{L}^n = \frac{\mathbb{L}^n - 1}{\mathbb{L} - 1}$	$\left[\mathbb{P}^{n-1} \right] = 1 + \mathbb{L} + \cdots + \mathbb{L}^n = \frac{\mathbb{L}^n - 1}{\mathbb{L} - 1}$	$\left[\mathbb{P}^{n-1} \right] = 1 + \mathbb{L} + \cdots + \mathbb{L}^n = \frac{\mathbb{L}^n - 1}{\mathbb{L} - 1}$
cardona-qft-methods_ EQ0239 (4)	114	■ 1.000 ● N +1	$\int c(TX) \cap [X] = \chi(X)$	$\int c(TX) \cap [X] = \chi(X)$	$\int c(TX) \cap [X] = \chi(X)$
cardona-qft-methods_ EQ0240	115	■ 1.000 ● N +0	$f_*(\mathbb{1}_Z)(p) = \chi(f^{-1}(p) \cap Z)$	$f_*(\mathbb{1}_Z)(p) = \chi(f^{-1}(p) \cap Z)$	$f_*(\mathbb{1}_Z)(p) = \chi(f^{-1}(p) \cap Z)$
cardona-qft-methods_ EQ0242 (5)	116	■ 1.000 ● N +0	$\int c_{\mathrm{SM}}(X) = \kappa_* c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\kappa_* \mathbb{1}_X) = \chi(X) c_{\mathrm{SM}}(\mathbb{1}_{\mathrm{pt}}) = \chi(X)$	$\int c_{\mathrm{SM}}(X) = \kappa_* c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\kappa_* \mathbb{1}_X) = \chi(X) c_{\mathrm{SM}}(\mathbb{1}_{\mathrm{pt}}) = \chi(X)$	$\int c_{\mathrm{SM}}(X) = \kappa_* c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\kappa_* \mathbb{1}_X) = \chi(X) c_{\mathrm{SM}}(\mathbb{1}_{\mathrm{pt}}) = \chi(X)$
cardona-qft-methods_ EQ0243	116	■ 1.000 ● K +0	$\begin{aligned} c_{\mathrm{SM}}(X) &= c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\mathbb{1}_Z + \mathbb{1}_{X \setminus Z}) \\ &= c_{\mathrm{SM}}(\mathbb{1}_Z) + c_{\mathrm{SM}}(\mathbb{1}_{X \setminus Z}) = c_{\mathrm{SM}}(Z) + c_{\mathrm{SM}}(X \setminus Z) \end{aligned}$	$\begin{aligned} c_{\mathrm{SM}}(X) &= c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\mathbb{1}_Z + \mathbb{1}_{X \setminus Z}) \\ &= c_{\mathrm{SM}}(\mathbb{1}_Z) + c_{\mathrm{SM}}(\mathbb{1}_{X \setminus Z}) = c_{\mathrm{SM}}(Z) + c_{\mathrm{SM}}(X \setminus Z) \end{aligned}$	$\begin{aligned} c_{\mathrm{SM}}(X) &= c_{\mathrm{SM}}(\mathbb{1}_X) = c_{\mathrm{SM}}(\mathbb{1}_Z + \mathbb{1}_{X \setminus Z}) \\ &= c_{\mathrm{SM}}(\mathbb{1}_Z) + c_{\mathrm{SM}}(\mathbb{1}_{X \setminus Z}) = c_{\mathrm{SM}}(Z) + c_{\mathrm{SM}}(X \setminus Z) \end{aligned}$
cardona-qft-methods_ EQ0244	116	■ 1.000 ● S +0	$c_{\mathrm{SM}}(\mathbb{P}^1) = c(T\mathbb{P}^1) \cap [\mathbb{P}^1] = (1 + c_1(T\mathbb{P}^1)) \cap [\mathbb{P}^1] = [\mathbb{P}^1] + \chi(\mathbb{P}^1)[\mathbb{P}^0] = [\mathbb{P}^1] + 2[\mathbb{P}^0]$.	$c_{\mathrm{SM}}(\mathbb{P}^1) = c(T\mathbb{P}^1) \cap [\mathbb{P}^1] = (1 + c_1(T\mathbb{P}^1)) \cap [\mathbb{P}^1] = [\mathbb{P}^1] + \chi(\mathbb{P}^1)[\mathbb{P}^0] = [\mathbb{P}^1] + 2[\mathbb{P}^0]$.	$c_{\mathrm{SM}}(\mathbb{P}^1) = c(T\mathbb{P}^1) \cap [\mathbb{P}^1] = (1 + c_1(T\mathbb{P}^1)) \cap [\mathbb{P}^1] = [\mathbb{P}^1] + \chi(\mathbb{P}^1)[\mathbb{P}^0] = [\mathbb{P}^1] + 2[\mathbb{P}^0]$.
cardona-qft-methods_ EQ0245 (6)	116	■ 1.000 ● N +1	$c_{\mathrm{SM}}(\mathbb{P}^n) = [\mathbb{P}^n] + \binom{n+1}{1} [\mathbb{P}^{n-1}] + \binom{n+1}{2} [\mathbb{P}^{n-2}] + \cdots + \binom{n+1}{n} [\mathbb{P}^0]$.	$c_{\mathrm{SM}}(\mathbb{P}^n) = [\mathbb{P}^n] + \binom{n+1}{1} [\mathbb{P}^{n-1}] + \binom{n+1}{2} [\mathbb{P}^{n-2}] + \cdots + \binom{n+1}{n} [\mathbb{P}^0]$.	$c_{\mathrm{SM}}(\mathbb{P}^n) = [\mathbb{P}^n] + \binom{n+1}{1} [\mathbb{P}^{n-1}] + \binom{n+1}{2} [\mathbb{P}^{n-2}] + \cdots + \binom{n+1}{n} [\mathbb{P}^0]$.
cardona-qft-methods_ EQ0246	116	■ 1.000 ● N +0	$c_{\mathrm{SM}}(\mathbb{P}^n) = (1 + H)^{n+1} \cap [\mathbb{P}^n]$	$c_{\mathrm{SM}}(\mathbb{P}^n) = (1 + H)^{n+1} \cap [\mathbb{P}^n]$	$c_{\mathrm{SM}}(\mathbb{P}^n) = (1 + H)^{n+1} \cap [\mathbb{P}^n]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0247	117	■ 1.000 ● N +0	$c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}(\mathbb{1}_C)=c(TC)\cap[C]=(1+c_1(TC))\cap[C]=2\left[\mathbb{P}^1\right]+2\left[\mathbb{P}^0\right].$ $\cap[C]=2\left[\mathbb{P}^1\right]+2\left[\mathbb{P}^0\right].$	$c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}(\mathbb{1}_C)=c(TC)\cap[C]=(1+c_1(TC))\cap[C]=2\left[\mathbb{P}^1\right]+2\left[\mathbb{P}^0\right].$	$c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}(\mathbb{1}_C)=c(TC)\cap[C]=(1+c_1(TC))\cap[C]=2\left[\mathbb{P}^1\right]+2\left[\mathbb{P}^0\right].$
cardona-qft-methods_ EQ0248	117	■ 1.000 ● N +0	$c_{\mathrm{SM}}(\mathrm{L}_1)+c_{\mathrm{SM}}(\mathrm{L}_2)-c_{\mathrm{SM}}(\mathrm{L}_1\cap\mathrm{L}_2)=2c_{\mathrm{SM}}(\mathbb{P}^1)-c_{\mathrm{SM}}(\mathbb{P}^0)=2\left[\mathbb{P}^1\right]+3\left[\mathbb{P}^0\right].$	$c_{\mathrm{SM}}(L_1)+c_{\mathrm{SM}}(L_2)-c_{\mathrm{SM}}(L_1\cap L_2)=2c_{\mathrm{SM}}(\mathbb{P}^1)-c_{\mathrm{SM}}(\mathbb{P}^0)=2\left[\mathbb{P}^1\right]+3\left[\mathbb{P}^0\right].$	$c_{\mathrm{SM}}(L_1)+c_{\mathrm{SM}}(L_2)-c_{\mathrm{SM}}(L_1\cap L_2)=2c_{\mathrm{SM}}(\mathbb{P}^1)-c_{\mathrm{SM}}(\mathbb{P}^0)=2\left[\mathbb{P}^1\right]+3\left[\mathbb{P}^0\right].$
cardona-qft-methods_ EQ0249	118	■ 1.000 ● N +0	$S(\phi)=\int L(\phi)\,d^Dx$	$S(\phi)=\int L(\phi)d^Dx$	$S(\phi)=\int L(\phi)d^Dx$
cardona-qft-methods_ EQ0251	118	■ 1.000 ●	$G(\phi)=\frac{1}{N!}\int_{\sum_i p_i=0}\hat{\phi}(p_1)\cdots\hat{\phi}(p_N)U(G(p_1,\dots,p_N))dp_1\cdots dp_N$	$G(\phi)=\frac{1}{N!}\int_{\sum_i p_i=0}\hat{\phi}(p_1)\cdots\hat{\phi}(p_N)U(G(p_1,\dots,p_N))dp_1\cdots dp_N$	$G(\phi)=\frac{1}{N!}\int_{\sum_i p_i=0}\hat{\phi}(p_1)\cdots\hat{\phi}(p_N)U(G(p_1,\dots,p_N))dp_1\cdots dp_N$
cardona-qft-methods_ EQ0252	118	■ 1.000 ● N +1	$U(G(p_1,\dots,p_N))=\int I_G(k_1,\dots,k_\ell,p_1,\dots,p_N)d^Dk_1\cdots d^Dk_\ell$	$U(G(p_1,\dots,p_N))=\int I_G(k_1,\dots,k_\ell,p_1,\dots,p_N)d^Dk_1\cdots d^Dk_\ell$	$U(G(p_1,\dots,p_N))=\int I_G(k_1,\dots,k_\ell,p_1,\dots,p_N)d^Dk_1\cdots d^Dk_\ell$
cardona-qft-methods_ EQ0253 (2)	119	■ 1.000 ● N +0	$U(G)=\frac{\Gamma(n-D/2)}{(4\pi)^{D/2}}\int_{\sigma_n}\frac{P_G(\underline{t},p)^{-n+D/2}\omega_n}{\Psi_G(\underline{t})^{-n+D(\ell+1)/2}}$	$U(G)=\frac{\Gamma(n-D\ell/2)}{(4\pi)^{\ell D/2}}\int_{\sigma_n}\frac{P_G(\underline{t},p)^{-n+D\ell/2}\omega_n}{\Psi_G(\underline{t})^{-n+D(\ell+1)/2}}$	$U(G)=\frac{\Gamma(n-D\ell/2)}{(4\pi)^{\ell D/2}}\int_{\sigma_n}\frac{P_G(\underline{t},p)^{-n+D\ell/2}\omega_n}{\Psi_G(\underline{t})^{-n+D(\ell+1)/2}}$
cardona-qft-methods_ EQ0255	119	■ 1.000 ● W +0	$U(G)=\frac{\Gamma(2-D/2)}{(4\pi)^{D/2}}\int_{\sigma_2}\frac{(p^2t_1t_2)^{-2+D/2}\omega_2}{(t_1+t_2)^{-2+D}}$	$U(G)=\frac{\Gamma(2-D/2)}{(4\pi)^{D/2}}\int_{\sigma_2}\frac{(p^2t_1t_2)^{-2+D/2}\omega_2}{(t_1+t_2)^{-2+D}}$	$U(G)=\frac{\Gamma(2-D/2)}{(4\pi)^{D/2}}\int_{\sigma_2}\frac{(p^2t_1t_2)^{-2+D/2}\omega_2}{(t_1+t_2)^{-2+D}}$
cardona-qft-methods_ EQ0256	120	■ 1.000 ● N +0	$\int_{\text{cycle } \sigma} \frac{\text{diff. form}}{\Psi_G(t)^M}$	$\int_{\text{cycle } \sigma} \frac{\text{diff. form}}{\Psi_G(t)^M}$	$\int_{\text{cycle } \sigma} \frac{\text{diff. form}}{\Psi_G(t)^M}$
cardona-qft-methods_ EQ0257	120	■ 1.000 ● N +0	$\zeta(n_1,\dots,n_r)=\sum_{0<k_1<\dots<k_r}\frac{1}{k_1^{n_1}\dots k_r^{n_r}}$	$\zeta(n_1,\dots,n_r)=\sum_{0<k_1<\dots<k_r}\frac{1}{k_1^{n_1}\dots k_r^{n_r}}$	$\zeta(n_1,\dots,n_r)=\sum_{0<k_1<\dots<k_r}\frac{1}{k_1^{n_1}\dots k_r^{n_r}}$
cardona-qft-methods_ EQ0258	121	■ 1.000 ● N +0	$\Psi_G(t_1,\dots,t_n):=\sum_T\prod_{e\notin T}t_e$	$\Psi_G(t_1,\dots,t_n):=\sum_T\prod_{e\notin T}t_e$	$\Psi_G(t_1,\dots,t_n):=\sum_T\prod_{e\notin T}t_e$
cardona-qft-methods_ EQ0259	122	■ 1.000 ● K +0	$\Psi_G(\underline{t})=t_2t_3+t_1t_3+t_1t_2=t_1t_2t_3\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}\right)$	$\Psi_G(\underline{t})=t_2t_3+t_1t_3+t_1t_2=t_1t_2t_3\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}\right)$	$\Psi_G(\underline{t})=t_2t_3+t_1t_3+t_1t_2=t_1t_2t_3\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}\right)$
cardona-qft-methods_ EQ0260	122	■ 1.000 ● N +0	$\Psi_G(\underline{t})=t_1\cdots t_n\left(\frac{1}{t_1}+\cdots+\frac{1}{t_n}\right)$	$\Psi_G(\underline{t})=t_1\cdots t_n\left(\frac{1}{t_1}+\cdots+\frac{1}{t_n}\right)$	$\Psi_G(\underline{t})=t_1\cdots t_n\left(\frac{1}{t_1}+\cdots+\frac{1}{t_n}\right)$
cardona-qft-methods_ EQ0266	127	■ 1.000 ● N +1	$\Psi_G(t_1,\dots,t_{n_1},u_1,\dots,u_{n_2})\neq 0$	$\Psi_G(t_1,\dots,t_{n_1},u_1,\dots,u_{n_2})\neq 0$	$\Psi_G(t_1,\dots,t_{n_1},u_1,\dots,u_{n_2})\neq 0$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0269	129	■ 1.000 ● N +0	$c_{\mathrm{SM}}(\mathbb{L}_{\hat{X}})\left(\mathbb{H}_{\hat{X}}\right)=a_{\{0\}}\left[\mathbb{P}^0\right]+a_{\{1\}}\left[\mathbb{P}^1\right]+\cdots+a_n\left[\mathbb{P}^n\right]$	$c_{\mathrm{SM}}\left(\mathbb{L}_{\hat{X}}\right)=a_0\left[\mathbb{P}^0\right]+a_1\left[\mathbb{P}^1\right]+\cdots+a_n\left[\mathbb{P}^n\right].$	$c_{\mathrm{SM}}(\mathbb{L}_{\hat{X}})=a_0[\mathbb{P}^0]+a_1[\mathbb{P}^1]+\cdots+a_n[\mathbb{P}^n] \quad .$
cardona-qft-methods_ EQ0270	130	■ 1.000 ● N +0	$\begin{aligned} c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{A}^n}) &= c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{P}^n} - \mathbb{L}_{\mathbb{P}^{n-1}}) \\ &= ((1+H)^{n+1} - (1+H)^n H) \cap [\mathbb{P}^n] = (1+H)^n \cap [\mathbb{P}^n] \\ &= \sum_{i=0}^n \binom{n}{i} [\mathbb{P}^i] \rightsquigarrow F_{\mathbb{A}^n}(T) = \sum_{i=0}^n \binom{n}{i} T^i \end{aligned}$	$c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{A}^n}) = c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{P}^n} - \mathbb{L}_{\mathbb{P}^{n-1}}) \\ = ((1+H)^{n+1} - (1+H)^n H) \cap [\mathbb{P}^n] = (1+H)^n \cap [\mathbb{P}^n] \\ = \sum_{i=0}^n \binom{n}{i} [\mathbb{P}^i] \rightsquigarrow F_{\mathbb{A}^n}(T) = \sum_{i=0}^n \binom{n}{i} T^i \quad .$	$c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{A}^n}) = c_{\mathrm{SM}}(\mathbb{L}_{\mathbb{P}^n} - \mathbb{L}_{\mathbb{P}^{n-1}}) \\ = ((1+H)^{n+1} - (1+H)^n H) \cap [\mathbb{P}^n] = (1+H)^n \cap [\mathbb{P}^n] \\ = \sum_{i=0}^n \binom{n}{i} [\mathbb{P}^i] \rightsquigarrow F_{\mathbb{A}^n}(T) = \sum_{i=0}^n \binom{n}{i} T^i \quad .$
cardona-qft-methods_ EQ0271	130	■ 1.000 ●	$\begin{aligned} C_G(T) &= F_{\mathbb{A}^n \setminus \mathbb{A}^{n-1}}(T) = (1+T)^n - (1+T)^{n-1} \\ &= (1+T-1)(1+T)^{n-1} = T(1+T)^{n-1} \end{aligned}$	$C_G(T) = F_{\mathbb{A}^n \setminus \mathbb{A}^{n-1}}(T) = (1+T)^n - (1+T)^{n-1} \\ = (1+T-1)(1+T)^{n-1} = T(1+T)^{n-1}$	$C_G(T) = F_{\mathbb{A}^n \setminus \mathbb{A}^{n-1}}(T) = (1+T)^n - (1+T)^{n-1} \\ = (1+T-1)(1+T)^{n-1} = T(1+T)^{n-1}$
cardona-qft-methods_ EQ0273	132	■ 1.000 ● N +0	$T_G(x, y) = T_{G/e}(x, y) + T_{G \setminus e}(x, y)$	$T_G(x, y) = T_{G/e}(x, y) + T_{G \setminus e}(x, y)$	$T_G(x, y) = T_{G/e}(x, y) + T_{G \setminus e}(x, y)$
cardona-qft-methods_ EQ0274	133	■ 1.000 ● N +0	$\gamma^{b_0(G)} \alpha^{\#V(G)-b_0(G)} \beta^{b_1(G)} T_G \left(\frac{\gamma x}{\alpha}, \frac{y}{\beta} \right)$	$\gamma^{b_0(G)} \alpha^{\#V(G)-b_0(G)} \beta^{b_1(G)} T_G \left(\frac{\gamma x}{\alpha}, \frac{y}{\beta} \right)$	$\gamma^{b_0(G)} \alpha^{\#V(G)-b_0(G)} \beta^{b_1(G)} T_G \left(\frac{\gamma x}{\alpha}, \frac{y}{\beta} \right)$
cardona-qft-methods_ EQ0275	134	■ 1.000 ● N +0	$T_{G_1 \cup G_2}(x, y) = T_{G_1}(x, y) \cdot T_{G_2}(x, y)$	$T_{G_1 \cup G_2}(x, y) = T_{G_1}(x, y) \cdot T_{G_2}(x, y)$	$T_{G_1 \cup G_2}(x, y) = T_{G_1}(x, y) \cdot T_{G_2}(x, y)$
cardona-qft-methods_ EQ0276	134	■ 1.000 ● W +0	$\begin{aligned} T_{G_{2e}}(x, y) &= T_G(x, y) + y T_{G/e}(x, y) \\ &= T_{G \setminus e}(x, y) + (y+1) T_{G/e}(x, y) \end{aligned}$	$T_{G_{2e}}(x, y) = T_G(x, y) + y T_{G/e}(x, y) \\ = T_{G \setminus e}(x, y) + (y+1) T_{G/e}(x, y)$	$T_{G_{2e}}(x, y) = T_G(x, y) + y T_{G/e}(x, y) \\ = T_{G \setminus e}(x, y) + (y+1) T_{G/e}(x, y)$
cardona-qft-methods_ EQ0277	134	■ 1.000 ● N +1	$\sum_{m \geq 0} T_{G_{me}}(x, y) \frac{s^m}{m!} = e^s \left(T_{G \setminus e}(x, y) + \frac{e^{(y-1)s} - 1}{y-1} T_{G/e}(x, y) \right)$	$\sum_{m \geq 0} T_{G_{me}}(x, y) \frac{s^m}{m!} = e^s \left(T_{G \setminus e}(x, y) + \frac{e^{(y-1)s} - 1}{y-1} T_{G/e}(x, y) \right)$	$\sum_{m \geq 0} T_{G_{me}}(x, y) \frac{s^m}{m!} = e^s \left(T_{G \setminus e}(x, y) + \frac{e^{(y-1)s} - 1}{y-1} T_{G/e}(x, y) \right) \quad ,$
cardona-qft-methods_ EQ0280	136	■ 1.000 ● K +0	$\chi(\mathbb{L}_{G_m}) = (-1)^{m-1} (\chi(Y_G) - \chi(Y_{G/e}))$	$\chi(Y_{G_{me}}) = (-1)^{m-1} (\chi(Y_G) - \chi(Y_{G/e}))$	$\chi(Y_{G_{me}}) = (-1)^{m-1} (\chi(Y_G) - \chi(Y_{G/e})) \quad ,$
cardona-qft-methods_ EQ0281 (1)	137	■ 1.000 ● N +1	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e}$	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e}$	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e} \quad .$
cardona-qft-methods_ EQ0285	137	■ 1.000 ● N +1	$C_G(T) = C_{G \setminus e, G/e}(T) + (T-1) C_{G \setminus e}(T)$	$C_G(T) = C_{G \setminus e, G/e}(T) + (T-1) C_{G \setminus e}(T)$	$C_G(T) = C_{G \setminus e, G/e}(T) + (T-1) C_{G \setminus e}(T) \quad .$
cardona-qft-methods_ EQ0288	138	■ 1.000 ● K +0	$C_G(T) = (2T-1) T(1+T)^2 - T(T-1)(1+T)^2 + T(1+T) = T(1+T)(1+T+T^2)$	$C_G(T) = (2T-1) T(1+T)^2 - T(T-1)(1+T)^2 + T(1+T) = T(1+T)(1+T+T^2)$	$C_G(T) = (2T-1) T(1+T)^2 - T(T-1)(1+T)^2 + T(1+T) = T(1+T)(1+T+T^2)$
cardona-qft-methods_ EQ0291	139	■ 1.000 ● N +0	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e}$	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e}$	$\Psi_G = t_e \Psi_{G \setminus e} + \Psi_{G/e}$
cardona-qft-methods_ EQ0293	142	■ 1.000 ● N +0	$\mathbb{U}(G) \equiv -\mathbb{U}(G \setminus e) \pmod{\mathbb{L}}$	$\mathbb{U}(G) \equiv -\mathbb{U}(G \setminus e) \pmod{\mathbb{L}}$	$\mathbb{U}(G) \equiv -\mathbb{U}(G \setminus e) \pmod{\mathbb{L}}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0294	142	■ 1.000 ● N +0	$\mathbb{U}(G)=(\mathbb{L}-1)\mathbb{U}(G\backslash e)\equiv\mathbb{U}(G\backslash e)\pmod{\mathbb{L}};$	$\mathbb{U}(G)=(\mathbb{L}-1)\mathbb{U}(G\backslash e)\equiv-\mathbb{U}(G\backslash e)\pmod{\mathbb{L}};$	$\mathbb{U}(G)=(\mathbb{L}-1)\mathbb{U}(G\searrow e)\equiv-\mathbb{U}(G\searrow e)\pmod{\mathbb{L}};$
cardona-qft-methods_ EQ0295	142	■ 1.000 ● N +0	$\mathbb{U}(G)=\mathbb{L}\mathbb{U}(G\backslash e)\equiv 0\pmod{\mathbb{L}}$	$\mathbb{U}(G)=\mathbb{L}\mathbb{U}(G\backslash e)\equiv 0\pmod{\mathbb{L}}$	$\mathbb{U}(G)=\mathbb{L}\mathbb{U}(G\searrow e)\equiv 0\pmod{\mathbb{L}}$
cardona-qft-methods_ EQ0297 (1)	150	■ 1.000 ● K +0	$\mathbf{A}_{\mu}(x)\rightarrow\mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)[\mathbf{A}_{\mu}(x)+\partial_{\mu}g(x),$ $\mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)\mathbf{A}_{\mu}(x)+\partial_{\mu}g(x),$	$\mathbf{A}_{\mu}(x)\rightarrow\mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)[\mathbf{A}_{\mu}(x)+\partial_{\mu}]g(x),$	$\mathbf{A}_{\mu}(x)\rightarrow\mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)[\mathbf{A}_{\mu}(x)+\partial_{\mu}]g(x),$
cardona-qft-methods_ EQ0298 (2)	150	■ 1.000 ● K +0	$\mathcal{C}(\mathbf{A}^{\prime})=\mathcal{C}(\mathbf{A})+d\Omega,$	$\mathcal{C}(\mathbf{A}^{\prime})=\mathcal{C}(\mathbf{A})+d\Omega,$	$\mathcal{C}(\mathbf{A}^{\prime})=\mathcal{C}(\mathbf{A})+d\Omega,$
cardona-qft-methods_ EQ0299 (3)	150	■ 1.000 ● N +1	$L(q,\dot{q})\rightarrow L(q^{\prime},\dot{q}^{\prime})=L(q,\dot{q})+d\Omega(q,t).$	$L(q,\dot{q})dt\rightarrow L(q^{\prime},\dot{q}^{\prime})dt^{\prime}=L(q,\dot{q})dt+d\Omega(q,t)$	$L(q,\dot{q})dt\rightarrow L(q^{\prime},\dot{q}^{\prime})dt^{\prime}=L(q,\dot{q})dt+d\Omega(q,t).$
cardona-qft-methods_ EQ0300 (4)	150	■ 1.000 ● N +1	$\int\mathbf{A}_{\mu}(x)j^{\mu}(x)d^nx$	$\int\mathbf{A}_{\mu}(x)j^{\mu}(x)d^nx$	$\int\mathbf{A}_{\mu}(x)j^{\mu}(x)d^nx.$
cardona-qft-methods_ EQ0304 (8)	152	■ 1.000 ● N +1	$\mathbf{A}(x)\xrightarrow{g}\mathbf{A}^{\prime}(x)=g^{-1}(x)\mathbf{A}(x)g(x)+g^{-1}(x)dg(x).$	$\mathbf{A}(x)\xrightarrow{g}\mathbf{A}^{\prime}(x)=g^{-1}(x)\mathbf{A}(x)g(x)+g^{-1}(x)dg(x).$	$\mathbf{A}(x)\xrightarrow{g}\mathbf{A}^{\prime}(x)=g^{-1}(x)\mathbf{A}(x)g(x)+g^{-1}(x)dg(x).$
cardona-qft-methods_ EQ0305 (9)	152	■ 1.000 ●	$\mathbf{F}\xrightarrow{g}\mathbf{F}^{\prime}=g\mathbf{F}g^{-1}.$	$\mathbf{F}\xrightarrow{g}\mathbf{F}^{\prime}=g\mathbf{F}g^{-1}.$	$\mathbf{F}\xrightarrow{g}\mathbf{F}^{\prime}=g\mathbf{F}g^{-1}.$
cardona-qft-methods_ EQ0306 (10)	152	■ 1.000 ● K +0	$\mathrm{Tr}(\mathbf{F}^k),$	$\mathrm{Tr}(\mathbf{F}^k),$	$Tr(\mathbf{F}^k),$
cardona-qft-methods_ EQ0307 (11)	154	■ 1.000 ● N +0	$\delta_{\mathrm{gauge}}P_{2k}=d\left(\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}\right),$	$\delta_{\mathrm{gauge}}P_{2k}=d\left(\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}\right),$	$\delta_{gauge}P_{2k}=d(\delta_{gauge}\mathcal{C}_{2k-1}),$
cardona-qft-methods_ EQ0309 (13)	154	■ 1.000 ● N +2	$\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}=d\Omega$	$\delta_{\mathrm{gauge}}\mathcal{C}_{2k-1}=d\Omega$	$\delta_{gauge}\mathcal{C}_{2k-1}=d\Omega.$ ■
cardona-qft-methods_ EQ0310 (14)	154	■ 1.000 ● N +1	$\delta_{\mathrm{gauge}}I[A]=\int_{M^{2n-1}}\delta_{\mathrm{gauge}}\mathcal{C}_{2n-1}=\int_{M^{2n-1}}d\Omega$	$\delta_{\mathrm{gauge}}I[A]=\int_{M^{2n-1}}\delta_{\mathrm{gauge}}\mathcal{C}_{2n-1}=\int_{M^{2n-1}}d\Omega$	$\delta_{gauge}I[A]=\int_{M^{2n-1}}\delta_{gauge}\mathcal{C}_{2n-1}=\int_{M^{2n-1}}d\Omega,$
cardona-qft-methods_ EQ0311 (15)	155	■ 1.000 ● N +0	$x^{\mu}\rightarrow x^{\prime\mu}=x^{\mu}+\xi^{\mu}(x).$	$x^{\mu}\rightarrow x^{\prime\mu}=x^{\mu}+\xi^{\mu}(x).$	$x^{\mu}\rightarrow x^{\prime\mu}=x^{\mu}+\xi^{\mu}(x).$
cardona-qft-methods_ EQ0312 (16)	155	■ 1.000 ● N +0	$v^{\prime\mu}(x^{\prime})=L_{\nu}^{\mu}(x)v^{\nu}(x),$ where $L_{\nu}^{\mu}(x)=\frac{\partial x^{\prime\mu}}{\partial x^{\nu}}=\delta_{\nu}^{\mu}+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}.$	$v^{\prime\mu}(x^{\prime})=L_{\nu}^{\mu}(x)v^{\nu}(x),$ where $L_{\nu}^{\mu}(x)=\frac{\partial x^{\prime\mu}}{\partial x^{\nu}}=\delta_{\nu}^{\mu}+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}.$	$v^{\prime\mu}(x^{\prime})=L_{\nu}^{\mu}(x)v^{\nu}(x),$ where $L_{\nu}^{\mu}(x)=\frac{\partial x^{\prime\mu}}{\partial x^{\nu}}=\delta_{\nu}^{\mu}+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}.$
cardona-qft-methods_ EQ0313 (17)	155	■ 1.000 ● W +0	$\begin{aligned} v^{\prime\mu}(x)&=L_{\nu}^{\mu}(x)v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \\ &=v^{\mu}(x)+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \end{aligned}$	$\begin{aligned} v^{\prime\mu}(x)&=L_{\nu}^{\mu}(x)v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \\ &=v^{\mu}(x)+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \end{aligned}$	$\begin{aligned} v^{\prime\mu}(x)&=L_{\nu}^{\mu}(x)v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \\ &=v^{\mu}(x)+\frac{\partial\xi^{\mu}}{\partial x^{\nu}}v^{\nu}(x)-\xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \end{aligned}$
cardona-qft-methods_ EQ0317 (3)	159	■ 1.000 ● N +0	$g_{\mu\nu}(x)\equiv\eta_{ab}e_{\mu}^a(x)e_{\nu}^b(x),$	$g_{\mu\nu}(x)\equiv\eta_{ab}e_{\mu}^a(x)e_{\nu}^b(x),$	$g_{\mu\nu}(x)\equiv\eta_{ab}e_{\mu}^a(x)e_{\nu}^b(x),$
cardona-qft-methods_ EQ0318 (4)	160	■ 1.000 ● N +0	$e_{\mu}^a(x)\longrightarrow e_{\mu}^{\prime a}(x)=\Lambda_b^a(x)e_{\mu}^b(x),$	$e_{\mu}^a(x)\longrightarrow e_{\mu}^{\prime a}(x)=\Lambda_b^a(x)e_{\mu}^b(x),$	$e_{\mu}^a(x)\longrightarrow e_{\mu}^{\prime a}(x)=\Lambda_b^a(x)e_{\mu}^b(x),$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0319 (5)	160	1.000 ● K +0	$\Lambda_{\mathrm{c}}^{\mathrm{a}}(x)\Lambda_{\mathrm{d}}^{\mathrm{b}}(x)\eta_{\mathrm{a}\mathrm{b}}=\eta_{\mathrm{cd}},$	$\Lambda_c^a(x)\Lambda_d^b(x)\eta_{ab} = \eta_{cd},$	$\Lambda_c^a(x)\Lambda_d^b(x)\eta_{ab} = \eta_{cd},$
cardona-qft-methods_ EQ0322 (8)	161	1.000 ● N +0	$\omega_{\mathrm{b}\mathrm{\mu}}^{\mathrm{a}}(x)\xrightarrow{\Lambda}\omega_{\mathrm{b}\mathrm{\mu}}^{\mathrm{\prime a}}(x)=\Lambda_{\mathrm{c}}^{\mathrm{a}}(x)\Lambda_{\mathrm{b}}^{\mathrm{d}}(x)\omega_{\mathrm{d}\mathrm{\mu}}^{\mathrm{c}}(x)+\Lambda_{\mathrm{c}}^{\mathrm{a}}(x)\partial_{\mathrm{\mu}}\Lambda_{\mathrm{b}}^{\mathrm{c}}(x)\omega_{\mathrm{d}\mathrm{\mu}}^{\mathrm{c}}(x)+\Lambda_{\mathrm{c}}^{\mathrm{a}}(x)\Lambda_{\mathrm{b}}^{\mathrm{d}}(x)\omega_{\mathrm{d}\mathrm{\mu}}^{\mathrm{c}}(x)+\Lambda_{\mathrm{c}}^{\mathrm{a}}(x)\partial_{\mathrm{\mu}}\Lambda_{\mathrm{b}}^{\mathrm{c}}(x).$	$\omega_{b\mu}^a(x)\xrightarrow{\Lambda}\omega_{b\mu}^{\prime a}(x)=\Lambda_c^a(x)\Lambda_b^d(x)\omega_{d\mu}^c(x)+\Lambda_c^a(x)\partial_\mu\Lambda_b^c(x).$	$\omega_{b\mu}^a(x)\xrightarrow{\Lambda}\omega_{b\mu}^{\prime a}(x)=\Lambda_c^a(x)\Lambda_b^d(x)\omega_{d\mu}^c(x)+\Lambda_c^a(x)\partial_\mu\Lambda_b^c(x).$
cardona-qft-methods_ EQ0327 (14)	162	1.000 ● K +0	$\mathrm{d}\left(\mathrm{d}\alpha_{\mathrm{p}}\right)=\mathrm{d}^2\alpha_{\mathrm{p}}=0.$	$d\left(d\alpha_p\right)=d^2\alpha_p=0.$	$d(d\alpha_p)=d^2\alpha_p=0.$
cardona-qft-methods_ EQ0328 (15)	162	1.000 ● W +0	$\begin{aligned} D^2\phi^a&=D\left[d\phi^a+\omega_b^a\phi^b\right] \\ &=d\left[d\phi^a+\omega_b^a\phi^b\right]+\omega_b^a\left[d\phi^b+\omega_c^b\phi^c\right] \\ &=d\omega_b^a+\omega_c^a\wedge\omega_b^c\phi^b. \end{aligned}$	$D^2\phi^a=D\left[d\phi^a+\omega_b^a\phi^b\right]=d\left[d\phi^a+\omega_b^a\phi^b\right]+\omega_b^a\left[d\phi^b+\omega_c^b\phi^c\right]=[d\omega_b^a+\omega_c^a\wedge\omega_b^c]\phi^b.$	$D^2\phi^a=D[d\phi^a+\omega_b^a\phi^b]=d[d\phi^a+\omega_b^a\phi^b]+\omega_b^a[d\phi^b+\omega_c^b\phi^c]=[d\omega_b^a+\omega_c^a\wedge\omega_b^c]\phi^b.$
cardona-qft-methods_ EQ0331 (18)	163	1.000 ● N +1	$T^{\mathrm{a}}=\mathrm{d}\,e^{\mathrm{a}}+\omega_{\mathrm{b}}^{\mathrm{a}}\wedge e^{\mathrm{b}}$	$T^a=de^a+\omega_b^a\wedge e^b$	$T^a=de^a+\omega_b^a\wedge e^b,$
cardona-qft-methods_ EQ0340 (8)	166	1.000 ● N +0	$\mathbf{P}_D=\mathbf{P}_{4k_1}\mathbf{P}_{4k_2}\cdots\mathbf{P}_{4k_r},$	$\mathbf{P}_D=\mathbf{P}_{4k_1}\mathbf{P}_{4k_2}\cdots\mathbf{P}_{4k_r},$	$\mathbf{P}_D=\mathbf{P}_{4k_1}\mathbf{P}_{4k_2}\cdots\mathbf{P}_{4k_r},$
cardona-qft-methods_ EQ0341 (9)	166	1.000 ●	$I_{\mathrm{D}}[e,\omega]=\kappa\int_M\sum_{p=0}^{[\mathrm{D}/2]}\alpha_pL_p^{\mathrm{D}}$	$I_D[e,\omega]=\kappa\int_M\sum_{p=0}^{[D/2]}\alpha_pL_p^D$	$I_D[e,\omega]=\kappa\int_M\sum_{p=0}^{[D/2]}\alpha_pL_p^D$
cardona-qft-methods_ EQ0342 (10)	166	1.000 ● N +2	$L_{\mathrm{p}}^{\mathrm{D}}=\epsilon_{\mathrm{a}_1\cdots\mathrm{a}_\mathrm{D}}R^{\mathrm{a}_1\mathrm{a}_2}\cdots R^{\mathrm{a}_{2p-1}\mathrm{a}_{2p}}e^{\mathrm{a}_{2p+1}}\cdots e^{\mathrm{a}_\mathrm{D}}$	$L_p^D=\epsilon_{a_1\cdots a_D}R^{a_1a_2}\cdots R^{a_{2p-1}a_{2p}}e^{a_{2p+1}}\cdots e^{a_D}$	$L_p^D=\epsilon_{a_1\cdots a_D}R^{a_1a_2}\cdots R^{a_{2p-1}a_{2p}}e^{a_{2p+1}}\cdots e^{a_D}.$ ■
cardona-qft-methods_ EQ0344 (12)	167	1.000 ● C +5	$\begin{aligned} I_4&=\kappa\int_M\epsilon_{abcd}\left[\alpha_2R^{ab}R^{cd}+\alpha_1R^{ab}e^ce^d+\alpha_0e^ae^be^ce^d\right] \\ &=-\kappa\int_M\sqrt{ g }\left[\alpha_2\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+2\alpha_1R+24\alpha_0\right]d^4x \\ &=-\kappa\alpha_2\cdot\mathbf{E}_4-2\alpha_1\int_M\sqrt{ g }Rd^4x-24\kappa\alpha_0\cdot V_4. \end{aligned}$	$I_4=\kappa\int_M\epsilon_{abcd}\left[\alpha_2R^{ab}R^{cd}+\alpha_1R^{ab}e^ce^d+\alpha_0e^ae^be^ce^d\right]=-\kappa\int_M\sqrt{ g }\left[\alpha_2\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+2\alpha_1R+24\alpha_0\right]d^4x=-\kappa\alpha_2\cdot\mathbf{E}_4-2\alpha_1\int_M\sqrt{ g }Rd^4x-24\kappa\alpha_0\cdot V_4$	$I_4=\kappa\int_M\epsilon_{abcd}\left[\alpha_2R^{ab}R^{cd}+\alpha_1R^{ab}e^ce^d+\alpha_0e^ae^be^ce^d\right]=-\kappa\int_M\sqrt{ g }\left[\alpha_2\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+2\alpha_1R+24\alpha_0\right]d^4x=-\kappa\alpha_2\cdot\mathbf{E}_4-2\alpha_1\int_M\sqrt{ g }Rd^4x-24\kappa\alpha_0\cdot V_4.$ (12)
cardona-qft-methods_ EQ0345 (13)	167	1.000 ● N +0	$\epsilon_{\mathrm{abcd}}R^{\mathrm{ab}}R^{\mathrm{cd}}e^e=\sqrt{ g }\left[R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right]d^5x.$	$\epsilon_{abcde}R^{ab}R^{cd}e^e=\sqrt{ g }\left[R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right]d^5x.$	$\epsilon_{abcde}R^{ab}R^{cd}e^e=\sqrt{ g }\left[R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right]d^5x.$
cardona-qft-methods_ EQ0346 (14)	168	1.000 ● N +1	$\delta I_D=\int\left[\delta e^a\mathcal{E}_a+\delta\omega^{ab}\mathcal{E}_{ab}\right]=0,$	$\delta I_D=\int\left[\delta e^a\mathcal{E}_a+\delta\omega^{ab}\mathcal{E}_{ab}\right]=0,$	$\delta I_D=\int\left[\delta e^a\mathcal{E}_a+\delta\omega^{ab}\mathcal{E}_{ab}\right]=0,$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08347 (15)	168	■ 1.000 ● N +1	$\mathcal{E}_{\{a\}}=\sum_{p=0}^{\left\lfloor\frac{D-1}{2}\right\rfloor}\alpha_{\{p\}}(D-2p)\mathcal{E}_{\{a\}}^{\{p\}}=0$	$\mathcal{E}_a=\sum_{p=0}^{\left[\frac{D-1}{2}\right]}\alpha_p(D-2p)\mathcal{E}_a^{(p)}=0$	$\mathcal{E}_a=\sum_{p=0}^{\left[\frac{D-1}{2}\right]}\alpha_p(D-2p)\mathcal{E}_a^{(p)}=0,$
cardona-qft-methods_ E08348 (16)	168	■ 1.000 ● N +1	$\mathcal{E}_{\{a\ b\}}=\sum_{p=1}^{\left\lfloor\frac{D-1}{2}\right\rfloor}\alpha_{\{p\}}p(D-2p)\mathcal{E}_{\{a\ b\}}^{\{p\}}=0$	$\mathcal{E}_{ab}=\sum_{p=1}^{\left[\frac{D-1}{2}\right]}\alpha_pp(D-2p)\mathcal{E}_{ab}^{(p)}=0$	$\mathcal{E}_{ab}=\sum_{p=1}^{\left[\frac{D-1}{2}\right]}\alpha_pp(D-2p)\mathcal{E}_{ab}^{(p)}=0,$
cardona-qft-methods_ E08350 (19)	168	■ 1.000 ● N +0	$T^{\{a\}}=d\ e^{\{a\}}+\omega^{\{a\}}_{\{ \}_b}e^{\{b\}}=0,$	$T^a=de^a+\omega^a{}_be^b=0,$	$T^a=de^a+\omega^a{}_be^b=0,$
cardona-qft-methods_ E08352	169	■ 1.000 ● N +1	$I_{\{D\}}[g]=\int_M d^Dx\sqrt{g}\left[\alpha_0'+\alpha_1'R+\alpha_2'\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+\cdots\right]$	$I_D[g]=\int_M d^Dx\sqrt{g}\left[\alpha_0'+\alpha_1'R+\alpha_2'\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+\cdots\right]$	$I_D[g]=\int_M d^Dx\sqrt{g}\left[\alpha_0'+\alpha_1'R+\alpha_2'\left(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2\right)+\cdots\right].$
cardona-qft-methods_ E08353 (21)	170	■ 1.000 ● N +0	$\zeta_{\{0\}}=e^{\{a\}}\ T_{\{a\}},$	$\zeta_0=e^aT_a,$	$\zeta_0=e^aT_a,$
cardona-qft-methods_ E08361 (1)	173	■ 1.000 ● N +0	$L_{\{3\}}=\epsilon_{abc}R^{\{a\ b\ c\}}\ R^{\{a\ b\}}\ e^{\{c\}}\ .$	$L_3=\epsilon_{abc}R^{ab}e^c.$	$L_3=\epsilon_{abc}R^{ab}e^c.$
cardona-qft-methods_ E08365 (4)	174	■ 1.000 ● K +0	$\delta\omega^{\{a\ b\}}=0,$	$\delta\omega^{ab}=0,$	$\delta\omega^{ab}=0,$
cardona-qft-methods_ E08366 (5)	174	■ 1.000 ● K +0	$\delta L_{\{3\}}=d\left[\epsilon_{abc}R^{\{a\ b\ c\}}\ R^{\{a\ b\}}\ \lambda^{\{c\}}\right],$	$\delta L_3=d\left[\epsilon_{abc}R^{ab}\lambda^c\right],$	$\delta L_3=d[\epsilon_{abc}R^{ab}\lambda^c],$
cardona-qft-methods_ E08367 (6)	175	■ 1.000 ●	$L_{\{3\}}^{\{A\ d\ S\}}=\epsilon_{abc}\left[R^{\{a\ b\ c\}}\left(R^{\{a\ b\}}\ e^{\{c\}}\ \pm\ \frac{1}{3l^2}e^ae^be^c\right)\right]$	$L_3^{AdS}=\epsilon_{abc}\left(R^{ab}e^c\pm\frac{1}{3l^2}e^ae^be^c\right)$	$L_3^{AdS}=\epsilon_{abc}(R^{ab}e^c\pm\frac{1}{3l^2}e^ae^be^c),$
cardona-qft-methods_ E08369 (9)	175	■ 1.000 ● N +0	$\begin{aligned} &\delta\left[\begin{array}{cc} \omega^{\{a\ b\}} & e^{\{a\ b\}} \\ -e^{\{a\ b\}} & 0 \end{array}\right] = d\left[\begin{array}{cc} \lambda^{\{a\ b\}} & \lambda^{\{a\ b\}} \\ -\lambda^{\{a\ b\}} & 0 \end{array}\right] \\ &\quad + \left[\begin{array}{cc} \omega_c^{\{a\ b\}} & e^{\{a\ b\}} \\ -e_c^{\{a\ b\}} & 0 \end{array}\right] \left[\begin{array}{cc} \lambda^{cb} & \lambda^{cb} \\ \pm\lambda^{cb} & 0 \end{array}\right] \\ &\quad - \left[\begin{array}{cc} \lambda^{ac} & \lambda^{ac} \\ -\lambda^{ac} & 0 \end{array}\right] \left[\begin{array}{cc} \omega_c^{\{a\ b\}} & e^{\{a\ b\}} \\ \pm e^{\{a\ b\}} & 0 \end{array}\right]. \end{aligned}$	$\delta\left[\begin{array}{cc} \omega^{\{a\ b\}} & e^{\{a\ b\}} \\ -e^{\{a\ b\}} & 0 \end{array}\right] = d\left[\begin{array}{cc} \lambda^{\{a\ b\}} & \lambda^{\{a\ b\}} \\ -\lambda^{\{a\ b\}} & 0 \end{array}\right] \\ \quad + \left[\begin{array}{cc} \omega_c^{\{a\ b\}} & e^{\{a\ b\}} \\ -e_c^{\{a\ b\}} & 0 \end{array}\right] \left[\begin{array}{cc} \lambda^{cb} & \lambda^{cb} \\ \pm\lambda^{cb} & 0 \end{array}\right] \\ \quad - \left[\begin{array}{cc} \lambda^{ac} & \lambda^{ac} \\ -\lambda^{ac} & 0 \end{array}\right] \left[\begin{array}{cc} \omega_c^{\{a\ b\}} & e^{\{a\ b\}} \\ \pm e^{\{a\ b\}} & 0 \end{array}\right].$	
cardona-qft-methods_ E08370 (11)	175	■ 1.000 ● N +0	$\delta W^{\{A\ B\}}=d\ \Lambda^{\{A\ B\}}+W_{\{C\}}^{\{A\}}\ \Lambda^{\{B\}C}-\Lambda^{AC}W_C^{\{B\}}.$	$\delta W^{AB}=d\Lambda^{AB}+W_C^A\Lambda^{CB}-\Lambda^{AC}W_C^B,$	$\delta W^{AB}=d\Lambda^{AB}+W_C^A\Lambda^{CB}-\Lambda^{AC}W_C^B,$

Conf.

MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

—

no value

LaTeX source

MathPix’s reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0371 (10)	175	■ 1.000 ● N +0	$\begin{aligned} W^{\{A\ B\}} &= \left[\begin{array}{cc} \omega^{\{a\ b\}} & e^{\{a\} \ l^{-1}} \\ -e^{\{b\} \ l^{-1}} & 0 \end{array} \right] \\ \Lambda^{\{A\ B\}} &= \left[\begin{array}{cc} \lambda^{\{a\ b\}} & \lambda^{\{a\} \ l^{-1}} \\ -\lambda^{\{b\} \ l^{-1}} & 0 \end{array} \right], \end{aligned}$	$W^{AB} = \begin{bmatrix} \omega^{ab} & e^a l^{-1} \\ -e^b l^{-1} & 0 \end{bmatrix}$ $\Lambda^{AB} = \begin{bmatrix} \lambda^{ab} & \lambda^a l^{-1} \\ -\lambda^b l^{-1} & 0 \end{bmatrix},$	$W^{AB} = \begin{bmatrix} \omega^{ab} & e^a l^{-1} \\ -e^b l^{-1} & 0 \end{bmatrix}$ $\Lambda^{AB} = \begin{bmatrix} \lambda^{ab} & \lambda^a l^{-1} \\ -\lambda^b l^{-1} & 0 \end{bmatrix},$
cardona-qft-methods_ EQ0373 (13)	176	■ 1.000 ● K +0	$D_{\{W\}} \Upsilon^{\{A\ B\}} = d \Upsilon^{\{A\ B\}} + W_{\{C\}}^A \Upsilon^{CB} + W_C^B \Upsilon^{AC} = 0.$	$D_W \Upsilon^{AB} = d \Upsilon^{AB} + W_C^A \Upsilon^{CB} + W_C^B \Upsilon^{AC} = 0.$	$D_W \Upsilon^{AB} = d \Upsilon^{AB} + W_C^A \Upsilon^{CB} + W_C^B \Upsilon^{AC} = 0.$
cardona-qft-methods_ EQ0376 (17)	177	■ 1.000 ● K +0	$L_{\{3\}^{\{E\ H\}}} = \epsilon_{abc} R^{\{a\ b\} \ c} R^{\{a\ b\}} e^{\{c\}},$	$L_3^{EH} = \epsilon_{abc} R^{ab} e^c,$	$L_3^{EH} = \epsilon_{abc} R^{ab} e^c,$
cardona-qft-methods_ EQ0381 (22)	178	■ 1.000 ● K +0	$L_{\{2\ n+1\}} = \epsilon_{a_1 \dots a_{2n+1}} R^{a_1 a_2} \dots R^{a_{2n-1} a_{2n}} e^{a_{2n+1}},$	$L_{2n+1} = \epsilon_{a_1 \dots a_{2n+1}} R^{a_1 a_2} \dots R^{a_{2n-1} a_{2n}} e^{a_{2n+1}},$	$L_{2n+1} = \epsilon_{a_1 \dots a_{2n+1}} R^{a_1 a_2} \dots R^{a_{2n-1} a_{2n}} e^{a_{2n+1}},$
cardona-qft-methods_ EQ0382	178	■ 1.000 ● K +0	$F^{\{A\ B\}} = d W^{\{A\ B\}} + W_{\{C\}}^A W^{CB},$	$F^{AB} = d W^{AB} + W_C^A W^{CB},$	$F^{AB} = d W^{AB} + W_C^A W^{CB},$
cardona-qft-methods_ EQ0383 (23)	178	■ 1.000 ● N +0	$F^{\{A\ B\}} = \left[\begin{array}{cc} R^{\{a\ b\}} \pm l^{-2} e^a e^b & l^{-1} T^a \\ -l^{-1} T^b & 0 \end{array} \right].$	$F^{AB} = \begin{bmatrix} R^{ab} \pm l^{-2} e^a e^b & l^{-1} T^a \\ -l^{-1} T^b & 0 \end{bmatrix}.$	$F^{AB} = \begin{bmatrix} R^{ab} \pm l^{-2} e^a e^b & l^{-1} T^a \\ -l^{-1} T^b & 0 \end{bmatrix}.$
cardona-qft-methods_ EQ0384 (24)	178	■ 1.000 ● K +0	$\mathbf{E}_{\{4\}} = \epsilon_{ABCD} F^{\{A\ B\}} F^{\{C\ D\}}.$	$\mathbf{E}_4 = \epsilon_{ABCD} F^{AB} F^{CD}.$	$\mathbf{E}_4 = \epsilon_{ABCD} F^{AB} F^{CD}.$
cardona-qft-methods_ EQ0385 (25)	179	■ 1.000 ● W +2	$\begin{aligned} \mathbf{E}_{\{4\}} &= 4 \epsilon_{abc} (R^{\{a\ b\}} \pm l^{-2} e^a e^b) l^{-1} T^a \\ &= \frac{4}{l} d \left[\epsilon_{abc} \left(R^{ab} \pm \frac{1}{3 l^2} e^a e^b \right) e^c \right] \end{aligned}$	$\mathbf{E}_4 = 4 \epsilon_{abc} (R^{ab} \pm l^{-2} e^a e^b) l^{-1} T^a$ $= \frac{4}{l} d \left[\epsilon_{abc} \left(R^{ab} \pm \frac{1}{3 l^2} e^a e^b \right) e^c \right],$	$\mathbf{E}_4 = 4 \epsilon_{abc} (R^{ab} \pm l^{-2} e^a e^b) l^{-1} T^a$ $= \frac{4}{l} d \left[\epsilon_{abc} \left(R^{ab} \pm \frac{1}{3 l^2} e^a e^b \right) e^c \right],$
cardona-qft-methods_ EQ0387	179	■ 1.000 ● K +0	$\delta \left(d L_{\{3\}^{\{A\ d\ S\}}} \right) = 0.$	$\delta \left(d L_3^{AdS} \right) = 0.$	$\delta \left(d L_3^{AdS} \right) = 0.$
cardona-qft-methods_ EQ0390 (28)	180	■ 1.000 ●	$L_{\{2\ n-1\}^{\{A\ d\ S\}}} = \sum_{p=0}^{n-1} \bar{\alpha}_p L_p^{2n-1}$	$L_{2n-1}^{(A)dS} = \sum_{p=0}^{n-1} \bar{\alpha}_p L_p^{2n-1}$	$L_{2n-1}^{(A)dS} = \sum_{p=0}^{n-1} \bar{\alpha}_p L_p^{2n-1},$
cardona-qft-methods_ EQ0391 (29)	180	■ 1.000 ● N +0	$\bar{\alpha}_{\{p\}} = \kappa \cdot \frac{(\pm 1)^{p+1} l^{2p-D}}{(D-2p)} \binom{n-1}{p}, p = 1, 2, \dots, n-1 = \frac{D-1}{2},$	$\bar{\alpha}_p = \kappa \cdot \frac{(\pm 1)^{p+1} l^{2p-D}}{(D-2p)} \binom{n-1}{p}, p = 1, 2, \dots, n-1 = \frac{D-1}{2},$	$\bar{\alpha}_p = \kappa \cdot \frac{(\pm 1)^{p+1} l^{2p-D}}{(D-2p)} \binom{n-1}{p}, p = 1, 2, \dots, n-1 = \frac{D-1}{2},$
cardona-qft-methods_ EQ0393 (31)	180	■ 1.000 ● N +0	$L_{\{5\}^{\{A\ d\ S\}}} = \frac{\kappa}{l} \epsilon_{abcde} \left[e^a R^{bc} R^{de} \pm \frac{2}{3 l^2} e^a e^b e^c R^{de} + \frac{1}{5 l^4} e^a e^b e^c e^d e^e \right].$	$L_5^{(A)dS} = \frac{\kappa}{l} \epsilon_{abcde} \left[e^a R^{bc} R^{de} \pm \frac{2}{3 l^2} e^a e^b e^c R^{de} + \frac{1}{5 l^4} e^a e^b e^c e^d e^e \right].$	$L_5^{(A)dS} = \frac{\kappa}{l} \epsilon_{abcde} \left[e^a R^{bc} R^{de} \pm \frac{2}{3 l^2} e^a e^b e^c R^{de} + \frac{1}{5 l^4} e^a e^b e^c e^d e^e \right].$
cardona-qft-methods_ EQ0394	180	■ 1.000 ● N +0	$\tilde{e}^{\{a\}} = l^{-1} e^{\{a\}},$	$\tilde{e}^a = l^{-1} e^a,$	$\tilde{e}^a = l^{-1} e^a,$
cardona-qft-methods_ EQ0396	181	■ 1.000 ● S -1	$L_{\{3\}^{\{A\ d\ S\}}} = \epsilon_{abc} \left(R^{\{a\ b\}} \pm \frac{e^a e^b}{3 l^2} \right) e^{\{c\}}$	$L_3^{(A)dS} = \epsilon_{abc} \left(R^{ab} \pm \frac{e^a e^b}{3 l^2} \right) e^c$	$L_3^{(A)dS} = \epsilon_{abc} (R^{ab} \pm \frac{e^a e^b}{3 l^2}) e^c$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 ■ no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00400 (33)	183	■ 1.000 ● N +0	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}-\int_{\Omega'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}-\int_{\Omega'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}-\int_{\Omega'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$
cardona-qft-methods_ E00401 (34)	183	■ 1.000 ● K +0	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}+\int_{\widetilde{\Omega}'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}+\int_{\widetilde{\Omega}'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$	$\kappa\left[\int_{\Omega}\mathbf{E}_{2n}+\int_{\widetilde{\Omega}'}\mathbf{E}_{2n}\right]=2n\pi\hbar.$
cardona-qft-methods_ E00402	183	■ 1.000 ● K +0	$\kappa=n\hbar,$	$\kappa=n\hbar,$	$\kappa=n\hbar,$
cardona-qft-methods_ E00403 (35)	183	■ 1.000 ● N +0	$\epsilon_{ab_1b_2\cdots b_{2n}}\left[R^{b_1b_2}+\frac{1}{l^2}e^{b_1}e^{b_2}\right]\cdots\left[R^{b_{2n-1}b_{2n}}+\frac{1}{l^2}e^{b_{2n-1}}e^{b_{2n}}\right]=0$	$\epsilon_{ab_1b_2\cdots b_{2n}}\left[R^{b_1b_2}+\frac{1}{l^2}e^{b_1}e^{b_2}\right]\cdots\left[R^{b_{2n-1}b_{2n}}+\frac{1}{l^2}e^{b_{2n-1}}e^{b_{2n}}\right]=0$	$\epsilon_{ab_1b_2\cdots b_{2n}}\left[R^{b_1b_2}+\frac{1}{l^2}e^{b_1}e^{b_2}\right]\cdots\left[R^{b_{2n-1}b_{2n}}+\frac{1}{l^2}e^{b_{2n-1}}e^{b_{2n}}\right]=0$
cardona-qft-methods_ E00404 (36)	184	■ 1.000 ● N +1	$\epsilon_{abb_2\cdots b_{2n}}\left[R^{b_2b_3}+\frac{1}{l^2}e^{b_2}e^{b_3}\right]\cdots\left[R^{b_{2n-2}b_{2n-1}}+\frac{1}{l^2}e^{b_{2n-2}}e^{b_{2n-1}}\right]T^{b_{2n}}=0$	$\epsilon_{abb_2\cdots b_{2n}}\left[R^{b_2b_3}+\frac{1}{l^2}e^{b_2}e^{b_3}\right]\cdots\left[R^{b_{2n-2}b_{2n-1}}+\frac{1}{l^2}e^{b_{2n-2}}e^{b_{2n-1}}\right]T^{b_{2n}}=0$	$\epsilon_{abb_2\cdots b_{2n}}\left[R^{b_2b_3}+\frac{1}{l^2}e^{b_2}e^{b_3}\right]\cdots\left[R^{b_{2n-2}b_{2n-1}}+\frac{1}{l^2}e^{b_{2n-2}}e^{b_{2n-1}}\right]T^{b_{2n}}=0$
cardona-qft-methods_ E00406	185	■ 1.000 ● K +0	$\operatorname{Pfaff}[M]=\epsilon_{a_1b_1a_2b_2\cdots a_nb_n}M^{a_1b_1}M^{a_2b_2}\cdots M^{a_nb_n}=\sqrt{\det[M]}.$	$\operatorname{Pfaff}[M]=\epsilon_{a_1b_1a_2b_2\cdots a_nb_n}M^{a_1b_1}\cdots M^{a_nb_n}=\sqrt{\det[M]}.$	$\operatorname{Pfaff}[M]=\epsilon_{a_1b_1a_2b_2\cdots a_nb_n}M^{a_1b_1}\cdots M^{a_nb_n}=\sqrt{\det[M]}.$
cardona-qft-methods_ E00407 (38)	186	■ 1.000 ● N +1	$\delta I[\phi]=\int_M\delta\phi^r\frac{\delta L}{\delta\phi^r}+\int_{\partial M}\Theta(\phi,\delta\phi).$	$\delta I[\phi]=\int_M\delta\phi^r\frac{\delta L}{\delta\phi^r}+\int_{\partial M}\Theta(\phi,\delta\phi).$	$\delta I[\phi]=\int_M\delta\phi^r\frac{\delta L}{\delta\phi^r}+\int_{\partial M}\Theta(\phi,\delta\phi).$
cardona-qft-methods_ E00410 (40)	187	■ 1.000 ● N +1	$I_{2n-1}=\int_M L_{2n-1}^{(A)dS}+\int_{\partial M}B_{2n}$	$I_{2n-1}=\int_M L_{2n-1}^{(A)dS}+\int_{\partial M}B_{2n}$	$I_{2n-1}=\int_M L_{2n-1}^{(A)dS}+\int_{\partial M}B_{2n},$
cardona-qft-methods_ E00416 (1)	190	■ 1.000 ● N +1	$I_{\mathrm{Int}}=\int_{\Gamma}A_{\mu}j^{\mu}d^4x$	$I_{\mathrm{Int}}=\int_{\Gamma}A_{\mu}j^{\mu}d^4x$	$I_{\mathrm{Int}}=\int_{\Gamma}A_{\mu}j^{\mu}d^4x,$
cardona-qft-methods_ E00417 (2)	191	■ 1.000 ● N +1	$I_{\mathrm{Int}}=q\int_{\Gamma}A_{\mu}(z)dz^{\mu}=q\int_{\Gamma}A$	$I_{\mathrm{Int}}=q\int_{\Gamma}A_{\mu}(z)dz^{\mu}=q\int_{\Gamma}A$	$I_{\mathrm{Int}}=q\int_{\Gamma}A_{\mu}(z)dz^{\mu}=q\int_{\Gamma}A,$
cardona-qft-methods_ E00418 (3)	192	■ 1.000 ●	$I[\mathbf{A};\mathbf{j}]=\int\langle\mathbf{j}_{2p}\wedge\mathfrak{C}_{2p+1}(\mathbf{A})\rangle,$	$I[\mathbf{A};\mathbf{j}]=\int\langle\mathbf{j}_{2p}\wedge\mathfrak{C}_{2p+1}(\mathbf{A})\rangle,$	$I[\mathbf{A};\mathbf{j}]=\int\langle\mathbf{j}_{2p}\wedge\mathfrak{C}_{2p+1}(\mathbf{A})\rangle,$
cardona-qft-methods_ E00419 (4)	192	■ 1.000 ● N +1	$\mathbf{j}_{2p}=qj^{a_1a_2\cdots a_s}\mathbf{K}_{a_1}\mathbf{K}_{a_2}\cdots\mathbf{K}_{a_s}\delta(\Gamma)dx^{\alpha_1}\wedge dx^{\alpha_2}\wedge\cdots dx^{\alpha_{D-2p-1}},$	$\mathbf{j}_{2p}=qj^{a_1a_2\cdots a_s}\mathbf{K}_{a_1}\mathbf{K}_{a_2}\cdots\mathbf{K}_{a_s}\delta(\Gamma)dx^{\alpha_1}\wedge dx^{\alpha_2}\wedge\cdots dx^{\alpha_{D-2p-1}},$	$\mathbf{j}_{2p}=qj^{a_1a_2\cdots a_s}\mathbf{K}_{a_1}\mathbf{K}_{a_2}\cdots\mathbf{K}_{a_s}\delta(\Gamma)dx^{\alpha_1}\wedge dx^{\alpha_2}\wedge\cdots dx^{\alpha_{D-2p-1}},$
cardona-qft-methods_ E00420 (5)	192	■ 1.000 ● K +0	$D\mathbf{j}_{2p}=d\mathbf{j}_{2p}+[\mathbf{A},\mathbf{j}_{2p}]=0,$	$D\mathbf{j}_{2p}=d\mathbf{j}_{2p}+[\mathbf{A},\mathbf{j}_{2p}]=0,$	$D\mathbf{j}_{2p}=d\mathbf{j}_{2p}+[\mathbf{A},\mathbf{j}_{2p}]=0,$
cardona-qft-methods_ E00421 (6)	193	■ 1.000 ● N +0	$I[A]=\frac{1}{2}\int_M(\mathrm{Tr}[F\wedge*F]-\Theta(x)\mathrm{Tr}[F\wedge F]),$	$I[A]=\frac{1}{2}\int_M(\mathrm{Tr}[F\wedge*F]-\Theta(x)\mathrm{Tr}[F\wedge F]),$	$I[A]=\frac{1}{2}\int_M(\mathrm{Tr}[F\wedge*F]-\Theta(x)\mathrm{Tr}[F\wedge F]),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0423	202	■ 1.000 ● N +1	$S_{\{E\,H\}}\!\left(g_{\{\mu\,\nu\}}\right)\!=\!\frac{1}{16\,\pi\,G}\,\int_M R\,\sqrt{g}\,d^4\,x$	$S_{EH}(g_{\mu\nu})=\frac{1}{16\pi G}\int_MR\sqrt{g}\,d^4x$	$S_{EH}(g_{\mu\nu})=\frac{1}{16\pi G}\int_MR\sqrt{g}\,d^4x,$
cardona-qft-methods_ EQ0425	203	■ 1.000 ● N +0	$\int_M\!\!\left(\frac{1}{2}\!\left D_\mu\mathbf{H}\right ^2\!-\!\mu_0^2\!\left \mathbf{H}\right ^2\!-\!\xi_0R\!\left \mathbf{H}\right ^2\!+\!\lambda_0\!\left \mathbf{H}\right ^4\right)\!\sqrt{g}\,d^4x.$	$\int_M\left(\frac{1}{2} D_\mu\mathbf{H} ^2-\mu_0^2 \mathbf{H} ^2-\xi_0R \mathbf{H} ^2+\lambda_0 \mathbf{H} ^4\right)\sqrt{g}d^4x.$	$\int_M\left(\frac{1}{2} D_\mu\mathbf{H} ^2-\mu_0^2 \mathbf{H} ^2-\xi_0R \mathbf{H} ^2+\lambda_0 \mathbf{H} ^4\right)\sqrt{g}\,d^4x.$
cardona-qft-methods_ EQ0426	204	■ 1.000 ● K +0	$[\gamma,a]=0,\,\forall a\in\mathcal{A},\quad\text{and}\quad D\gamma=-\gamma D.$	$[\gamma,a]=0,\forall a\in\mathcal{A},\quad\text{and}\quad D\gamma=-\gamma D.$	$[\gamma,a]=0,\,\forall a\in\mathcal{A},\quad\text{and}\quad D\gamma=-\gamma D.$
cardona-qft-methods_ EQ0428	205	■ 1.000 ● N +0	$\left[[D,a],b^0\right]=0\quad\text{for all }a,b\in\mathcal{A}.$	$[[D,a],b^0]=0\quad\forall a,b\in\mathcal{A}.$	$[[D,a],b^0]=0\quad\forall a,b\in\mathcal{A}.$
cardona-qft-methods_ EQ0430	207	■ 1.000 ● N +0	$\mathcal{A}_{LR}=\mathbb{C}\oplus\mathbb{H}_L\oplus\mathbb{H}_R\oplus M_3(\mathbb{C}).$	$\mathcal{A}_{LR}=\mathbb{C}\oplus\mathbb{H}_L\oplus\mathbb{H}_R\oplus M_3(\mathbb{C}).$	$\mathcal{A}_{LR}=\mathbb{C}\oplus\mathbb{H}_L\oplus\mathbb{H}_R\oplus M_3(\mathbb{C}).$
cardona-qft-methods_ EQ0431	208	■ 1.000 ● N +0	$\operatorname{Ad}(u)\xi=u\xi u^*\quad\text{for all }\xi\in\mathcal{M}.$	$\operatorname{Ad}(u)\xi=u\xi u^*\quad\forall\xi\in\mathcal{M}.$	$\operatorname{Ad}(u)\xi=u\xi u^*\quad\forall\xi\in\mathcal{M}.$
cardona-qft-methods_ EQ0433	208	■ 1.000 ● N +0	$\mathcal{M}^0=\{\bar{\xi};\xi\in\mathcal{M}\},\quad a\bar{\xi}b=\overline{b^*\xi a^*}$	$\mathcal{M}^0=\{\bar{\xi};\xi\in\mathcal{M}\},\quad a\bar{\xi}b=\overline{b^*\xi a^*}$	$\mathcal{M}^0=\{\bar{\xi};\xi\in\mathcal{M}\},\quad a\bar{\xi}b=\overline{b^*\xi a^*}$
cardona-qft-methods_ EQ0440	210	■ 1.000 ● K +0	$\mathrm{SU}\!\left(\mathcal{A}_F\right)\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$	$\mathrm{SU}(\mathcal{A}_F)\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$	$\mathrm{SU}(\mathcal{A}_F)\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$
cardona-qft-methods_ EQ0442	211	■ 1.000 ● N +0	$\mathbb{C}_F\subset\mathcal{A}_F,\quad\mathbb{C}_F=\{(\lambda,\lambda,0),\lambda\in\mathbb{C}\}.$	$\mathbb{C}_F\subset\mathcal{A}_F,\quad\mathbb{C}_F=\{(\lambda,\lambda,0),\lambda\in\mathbb{C}\}.$	$\mathbb{C}_F\subset\mathcal{A}_F,\quad\mathbb{C}_F=\{(\lambda,\lambda,0),\lambda\in\mathbb{C}\}.$
cardona-qft-methods_ EQ0446	211	■ 1.000 ● K +0	$T:E_R=\uparrow_R\otimes\mathbf{1}^0\rightarrow J_F E_R.$	$T:E_R=\uparrow_R\otimes\mathbf{1}^0\rightarrow J_F E_R.$	$T:E_R=\uparrow_R\otimes\mathbf{1}^0\rightarrow J_F E_R.$
cardona-qft-methods_ EQ0447	211	■ 1.000 ● N +0	$Y_{\{\downarrow 3\}}^{\{\prime\}}=W_1Y_{(\downarrow 3)}W_3^*,\quad Y_{(\uparrow 3)}^{\prime}=W_2Y_{(\uparrow 3)}W_3^*$	$Y_{(\downarrow 3)}^{\prime}=W_1Y_{(\downarrow 3)}W_3^*,\quad Y_{(\uparrow 3)}^{\prime}=W_2Y_{(\uparrow 3)}W_3^*$	$Y_{(\downarrow 3)}^{\prime}=W_1Y_{(\downarrow 3)}W_3^*,\quad Y_{(\uparrow 3)}^{\prime}=W_2Y_{(\uparrow 3)}W_3^*$
cardona-qft-methods_ EQ0450	212	■ 1.000 ● W +0	$\begin{gathered} Y_{(\uparrow 3)}=\delta_{(\uparrow 3)}\quad Y_{(\downarrow 3)}=U_{CKM}\delta_{(\downarrow 3)}U_{CKM}^*\\ Y_{(\uparrow 1)}=U_{PMNS}^*\delta_{(\uparrow 1)}U_{PMNS}\quad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)} \end{gathered}$	$Y_{(\uparrow 3)}=\delta_{(\uparrow 3)}\quad Y_{(\downarrow 3)}=U_{CKM}\delta_{(\downarrow 3)}U_{CKM}^*\\ Y_{(\uparrow 1)}=U_{PMNS}^*\delta_{(\uparrow 1)}U_{PMNS}\quad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)}$	$Y_{(\uparrow 3)}=\delta_{(\uparrow 3)}\quad Y_{(\downarrow 3)}=U_{CKM}\delta_{(\downarrow 3)}U_{CKM}^*\\ Y_{(\uparrow 1)}=U_{PMNS}^*\delta_{(\uparrow 1)}U_{PMNS}\quad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)}$
cardona-qft-methods_ EQ0451	212	■ 1.000 ●	$U=\left(\begin{array}{ccc} c_1&&-s_1c_3&&-s_1s_3\\ s_1c_2&c_1c_2c_3-s_2s_3e_\delta&c_1c_2s_3+s_2c_3e_\delta\\ s_1s_2&c_1s_2c_3+c_2s_3e_\delta&c_1s_2s_3-c_2c_3e_\delta \end{array}\right)$	$U=\left(\begin{array}{ccc} c_1&&-s_1c_3&&-s_1s_3\\ s_1c_2&c_1c_2c_3-s_2s_3e_\delta&c_1c_2s_3+s_2c_3e_\delta\\ s_1s_2&c_1s_2c_3+c_2s_3e_\delta&c_1s_2s_3-c_2c_3e_\delta \end{array}\right)$	$U=\left(\begin{array}{ccc} c_1&&-s_1c_3&&-s_1s_3\\ s_1c_2&c_1c_2c_3-s_2s_3e_\delta&c_1c_2s_3+s_2c_3e_\delta\\ s_1s_2&c_1s_2c_3+c_2s_3e_\delta&c_1s_2s_3-c_2c_3e_\delta \end{array}\right)$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0453	214	■ 1.000 ● W +0	$\begin{gathered} \mathcal{A} = C^{\infty}(M) \otimes \mathcal{A}_F = C^{\infty}(M, \mathcal{A}_F) \\ \mathcal{H} = L^2(M, S) \otimes \mathcal{H}_F = L^2(M, S \otimes \mathcal{H}_F) \\ D = \not{D}_M \otimes 1 + \gamma_5 \otimes D_F \end{gathered}$	$\mathcal{A} = C^{\infty}(M) \otimes \mathcal{A}_F = C^{\infty}(M, \mathcal{A}_F)$ $\mathcal{H} = L^2(M, S) \otimes \mathcal{H}_F = L^2(M, S \otimes \mathcal{H}_F)$ $D = \not{D}_M \otimes 1 + \gamma_5 \otimes D_F$	$\mathcal{A} = C^{\infty}(M) \otimes \mathcal{A}_F = C^{\infty}(M, \mathcal{A}_F)$ $\mathcal{H} = L^2(M, S) \otimes \mathcal{H}_F = L^2(M, S \otimes \mathcal{H}_F)$ $D = \not{D}_M \otimes 1 + \gamma_5 \otimes D_F$
cardona-qft-methods_ EQ0455	215	■ 1.000 ● N +0	$\xi b = b^0 \xi, \quad \xi \in \mathcal{H}, \quad b \in \mathcal{A}$	$\xi b = b^0 \xi, \quad \xi \in \mathcal{H}, \quad b \in \mathcal{A}$	$\xi b = b^0 \xi, \quad \xi \in \mathcal{H}, \quad b \in \mathcal{A}$
cardona-qft-methods_ EQ0456	215	■ 1.000 ● N +0	$\xi \in \mathcal{H} \rightarrow \text{Ad}(u)\xi = u\xi u^* \quad \xi \in \mathcal{H}$	$\xi \in \mathcal{H} \rightarrow \text{Ad}(u)\xi = u\xi u^* \quad \xi \in \mathcal{H}$	$\xi \in \mathcal{H} \rightarrow \text{Ad}(u)\xi = u\xi u^* \quad \xi \in \mathcal{H}$
cardona-qft-methods_ EQ0457	215	■ 1.000 ● K +0	$D \rightarrow D_A = D + A + \varepsilon' J A J^{-1}$	$D \rightarrow D_A = D + A + \varepsilon' J A J^{-1}$	$D \rightarrow D_A = D + A + \varepsilon' J A J^{-1}$
cardona-qft-methods_ EQ0458	215	■ 1.000 ● N +0	$A = \sum a_j [D, b_j], \quad a_j, b_j \in \mathcal{A}.$	$A = \sum a_j [D, b_j], \quad a_j, b_j \in \mathcal{A}.$	$A = \sum a_j [D, b_j], \quad a_j, b_j \in \mathcal{A}.$
cardona-qft-methods_ EQ0459	216	■ 1.000 ● K +0	$\text{SU}(\mathcal{A}_F) = \{u \in \text{U}(\mathcal{A}_F) \mid \det(u) = 1\},$	$\text{SU}(\mathcal{A}_F) = \{u \in \text{U}(\mathcal{A}_F) \mid \det(u) = 1\},$	$\text{SU}(\mathcal{A}_F) = \{u \in \text{U}(\mathcal{A}_F) \mid \det(u) = 1\},$
cardona-qft-methods_ EQ0460	216	■ 1.000 ● K +0	$\text{SU}(\mathcal{A}_F) \sim \text{U}(1) \times \text{SU}(2) \times \text{SU}(3),$	$\text{SU}(\mathcal{A}_F) \sim \text{U}(1) \times \text{SU}(2) \times \text{SU}(3),$	$\text{SU}(\mathcal{A}_F) \sim \text{U}(1) \times \text{SU}(2) \times \text{SU}(3),$
cardona-qft-methods_ EQ0462	216	■ 1.000 ● N +0	$\left(\begin{array}{cccc} \frac{4}{3}\Lambda + V & 0 & 0 & 0 \\ 0 & -\frac{2}{3}\Lambda + V & 0 & 0 \\ 0 & 0 & Q_{11} + \frac{1}{3}\Lambda + V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} + \frac{1}{3}\Lambda + V \end{array} \right)$	$\left(\begin{array}{cccc} \frac{4}{3}\Lambda + V & 0 & 0 & 0 \\ 0 & -\frac{2}{3}\Lambda + V & 0 & 0 \\ 0 & 0 & Q_{11} + \frac{1}{3}\Lambda + V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} + \frac{1}{3}\Lambda + V \end{array} \right)$	$\left(\begin{array}{cccc} \frac{4}{3}\Lambda + V & 0 & 0 & 0 \\ 0 & -\frac{2}{3}\Lambda + V & 0 & 0 \\ 0 & 0 & Q_{11} + \frac{1}{3}\Lambda + V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} + \frac{1}{3}\Lambda + V \end{array} \right)$
cardona-qft-methods_ EQ0463	217	■ 1.000 ● K +0	$\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2\Lambda & 0 & 0 \\ 0 & 0 & Q_{11} - \Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} - \Lambda \end{array} \right)$	$\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2\Lambda & 0 & 0 \\ 0 & 0 & Q_{11} - \Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} - \Lambda \end{array} \right)$	$\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2\Lambda & 0 & 0 \\ 0 & 0 & Q_{11} - \Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22} - \Lambda \end{array} \right)$
cardona-qft-methods_ EQ0464	217	■ 1.000 ● N +0	$\sum_i a_i [\gamma_5 \otimes D_F, a'_i](x) \Big _{\mathcal{H}_f} = \gamma_5 \otimes (A_q^{(0,1)} + A_\ell^{(0,1)})$	$\sum_i a_i [\gamma_5 \otimes D_F, a'_i](x) \Big _{\mathcal{H}_f} = \gamma_5 \otimes (A_q^{(0,1)} + A_\ell^{(0,1)})$	$\sum_i a_i [\gamma_5 \otimes D_F, a'_i](x) \Big _{\mathcal{H}_f} = \gamma_5 \otimes (A_q^{(0,1)} + A_\ell^{(0,1)})$
cardona-qft-methods_ EQ0466	217	■ 1.000 ● N +0	$q = \left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array} \right).$	$q = \left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array} \right).$	$q = \left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array} \right).$
cardona-qft-methods_ EQ0468	217	■ 1.000 ●	$-E = \frac{1}{4} s \otimes \text{id} + \sum_{\mu < \nu} \gamma^\mu \gamma^\nu \otimes \mathbb{F}_{\mu\nu} - i \gamma_5 \gamma^\mu \otimes \mathbb{M}(D_\mu \varphi) + 1_4 \otimes (D^{0,1})^2$	$-E = \frac{1}{4} s \otimes \text{id} + \sum_{\mu < \nu} \gamma^\mu \gamma^\nu \otimes \mathbb{F}_{\mu\nu} - i \gamma_5 \gamma^\mu \otimes \mathbb{M}(D_\mu \varphi) + 1_4 \otimes (D^{0,1})^2$	$-E = \frac{1}{4} s \otimes \text{id} + \sum_{\mu < \nu} \gamma^\mu \gamma^\nu \otimes \mathbb{F}_{\mu\nu} - i \gamma_5 \gamma^\mu \otimes \mathbb{M}(D_\mu \varphi) + 1_4 \otimes (D^{0,1})^2,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0469	218	1.000 N +0	$D_{\{\mu\}} \varphi = \partial_{\mu} \varphi + \frac{i}{2} g_2 W_{\mu}^{\alpha} \varphi \sigma^{\alpha} - \frac{i}{2} g_1 B_{\mu} \varphi$	$D_{\mu} \varphi = \partial_{\mu} \varphi + \frac{i}{2} g_2 W_{\mu}^{\alpha} \varphi \sigma^{\alpha} - \frac{i}{2} g_1 B_{\mu} \varphi$	$D_{\mu} \varphi = \partial_{\mu} \varphi + \frac{i}{2} g_2 W_{\mu}^{\alpha} \varphi \sigma^{\alpha} - \frac{i}{2} g_1 B_{\mu} \varphi$
cardona-qft-methods_ EQ0470	218	1.000 N -1	$\operatorname{Tr}(f(D/\Lambda)),$	$\operatorname{Tr}(f(D/\Lambda)),$	$\operatorname{Tr}(f(D/\Lambda)),$
cardona-qft-methods_ EQ0472	218	1.000 N -1	$\operatorname{Tr}\!\left(e^{-t\,\Delta}\right)\sim\sum a_{\alpha}\,t^{\alpha}\quad(t\rightarrow 0)$	$\operatorname{Tr}\left(e^{-t\Delta}\right)\sim\sum a_{\alpha}t^{\alpha}\quad(t\rightarrow 0)$	$\operatorname{Tr}(e^{-t\Delta})\sim\sum a_{\alpha}t^{\alpha}\quad(t\rightarrow 0)$
cardona-qft-methods_ EQ0473	218	1.000 N -1	$\zeta_D(s)=\operatorname{Tr}\!\left(\Delta^{-s/2}\right).$	$\zeta_D(s)=\operatorname{Tr}\left(\Delta^{-s/2}\right).$	$\zeta_D(s)=\operatorname{Tr}(\Delta^{-s/2}).$
cardona-qft-methods_ EQ0474	218	1.000 K +0	$\operatorname{Res}_{s=-2\alpha}\zeta_D(s)=\frac{2\,a_{\alpha}}{\Gamma(-\alpha)}.$	$\operatorname{Res}_{s=-2\alpha}\zeta_D(s)=\frac{2a_{\alpha}}{\Gamma(-\alpha)}.$	$\operatorname{Res}_{s=-2\alpha}\zeta_D(s)=\frac{2\,a_{\alpha}}{\Gamma(-\alpha)}.$
cardona-qft-methods_ EQ0475	218	1.000 N +0	$ D ^{-s}=\Delta^{-s/2}=\frac{1}{\Gamma\left(\frac{s}{2}\right)}\int_0^\infty e^{-t\Delta}t^{s/2-1}dt$	$ D ^{-s}=\Delta^{-s/2}=\frac{1}{\Gamma\left(\frac{s}{2}\right)}\int_0^\infty e^{-t\Delta}t^{s/2-1}dt$	$ D ^{-s}=\Delta^{-s/2}=\frac{1}{\Gamma\left(\frac{s}{2}\right)}\int_0^\infty e^{-t\Delta}t^{s/2-1}dt$
cardona-qft-methods_ EQ0476	219	1.000 K +0	$\frac{1}{\Gamma\left(\frac{s}{2}\right)}\sim\frac{s}{2}\quad\text{as}\quad s\rightarrow 0$	$\frac{1}{\Gamma\left(\frac{s}{2}\right)}\sim\frac{s}{2}\quad\text{as}\quad s\rightarrow 0$	$\frac{1}{\Gamma\left(\frac{s}{2}\right)}\sim\frac{s}{2}\quad\text{as}\quad s\rightarrow 0.$
cardona-qft-methods_ EQ0477	219	1.000 N -1	$\int_0^\infty\operatorname{Tr}\!\left(e^{-t\,\Delta}\right)t^{s/2-1}dt$	$\int_0^\infty\operatorname{Tr}\left(e^{-t\Delta}\right)t^{s/2-1}dt$	$\int_0^\infty\operatorname{Tr}(e^{-t\Delta})t^{s/2-1}dt$
cardona-qft-methods_ EQ0479	219	1.000 N +0	$\frac{1}{2}\left\langle J\tilde{\xi},D_A\tilde{\xi}\right\rangle$	$\frac{1}{2}\left\langle J\tilde{\xi},D_A\tilde{\xi}\right\rangle$	$\frac{1}{2}\langle J\tilde{\xi},D_A\tilde{\xi}\rangle$
cardona-qft-methods_ EQ0480	219	1.000 N +0	$\mathcal{H}_{cl}^+=\{\xi\in\mathcal{H}_{cl}\mid\gamma\xi=\xi\}$	$\mathcal{H}_{cl}^+=\{\xi\in\mathcal{H}_{cl}\mid\gamma\xi=\xi\}$	$\mathcal{H}_{cl}^+=\{\xi\in\mathcal{H}_{cl}\mid\gamma\xi=\xi\}$
cardona-qft-methods_ EQ0482	220	1.000 N +0	$\frac{1}{2}\left\langle J\tilde{\xi},D_A\tilde{\xi}\right\rangle$	$\frac{1}{2}\left\langle J\tilde{\xi},D_A\tilde{\xi}\right\rangle$	$\frac{1}{2}\langle J\tilde{\xi},D_A\tilde{\xi}\rangle$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0483	220	■ 1.000	$\begin{aligned} S &= \frac{1}{\pi^2} \left(48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d} \right) \int \sqrt{g} d^4 x \\ &+ \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \int R \sqrt{g} d^4 x \\ &+ \frac{f_0}{10 \pi^2} \int \left(\frac{11}{6} R^* R^* - 3 C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} \right) \sqrt{g} d^4 x \\ &+ \frac{(-2 \mathfrak{a} f_2 \Lambda^2 + \mathfrak{c} f_0)}{\pi^2} \int \varphi ^2 \sqrt{g} d^4 x \\ &+ \frac{f_0 \mathfrak{a}}{2 \pi^2} \int D_\mu \varphi ^2 \sqrt{g} d^4 x \\ &- \frac{f_0 \mathfrak{a}}{12 \pi^2} \int R \varphi ^2 \sqrt{g} d^4 x \\ &+ \frac{f_0 \mathfrak{b}}{2 \pi^2} \int \varphi ^4 \sqrt{g} d^4 x \\ &+ \frac{f_0}{2 \pi^2} \int \left(g_3^2 G_{\mu\nu}^i G^{\mu\nu i} + g_2^2 F_{\mu\nu}^\alpha F^{\mu\nu\alpha} + \frac{5}{3} g_1^2 B_{\mu\nu} B^{\mu\nu} \right) \sqrt{g} d^4 x, \end{aligned}$	$S = \frac{1}{\pi^2} \left(48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d} \right) \int \sqrt{g} d^4 x$ $+ \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \int R \sqrt{g} d^4 x$ $+ \frac{f_0}{10 \pi^2} \int \left(\frac{11}{6} R^* R^* - 3 C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} \right) \sqrt{g} d^4 x$ $+ \frac{(-2 \mathfrak{a} f_2 \Lambda^2 + \mathfrak{c} f_0)}{\pi^2} \int \varphi ^2 \sqrt{g} d^4 x$ $+ \frac{f_0 \mathfrak{a}}{2 \pi^2} \int D_\mu \varphi ^2 \sqrt{g} d^4 x$ $- \frac{f_0 \mathfrak{a}}{12 \pi^2} \int R \varphi ^2 \sqrt{g} d^4 x$ $+ \frac{f_0 \mathfrak{b}}{2 \pi^2} \int \varphi ^4 \sqrt{g} d^4 x$ $+ \frac{f_0}{2 \pi^2} \int \left(g_3^2 G_{\mu\nu}^i G^{\mu\nu i} + g_2^2 F_{\mu\nu}^\alpha F^{\mu\nu\alpha} + \frac{5}{3} g_1^2 B_{\mu\nu} B^{\mu\nu} \right) \sqrt{g} d^4 x,$	$S = \frac{1}{\pi^2} (48 f_4 \Lambda^4 - f_2 \Lambda^2 \mathfrak{c} + \frac{f_0}{4} \mathfrak{d}) \int \sqrt{g} d^4 x$ $+ \frac{96 f_2 \Lambda^2 - f_0 \mathfrak{c}}{24 \pi^2} \int R \sqrt{g} d^4 x$ $+ \frac{f_0}{10 \pi^2} \int (\frac{11}{6} R^* R^* - 3 C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}) \sqrt{g} d^4 x$ $+ \frac{(-2 \mathfrak{a} f_2 \Lambda^2 + \mathfrak{c} f_0)}{\pi^2} \int \varphi ^2 \sqrt{g} d^4 x$ $+ \frac{f_0 \mathfrak{a}}{2 \pi^2} \int D_\mu \varphi ^2 \sqrt{g} d^4 x$ $- \frac{f_0 \mathfrak{a}}{12 \pi^2} \int R \varphi ^2 \sqrt{g} d^4 x$ $+ \frac{f_0 \mathfrak{b}}{2 \pi^2} \int \varphi ^4 \sqrt{g} d^4 x$ $+ \frac{f_0}{2 \pi^2} \int (g_3^2 G_{\mu\nu}^i G^{\mu\nu i} + g_2^2 F_{\mu\nu}^\alpha F^{\mu\nu\alpha} + \frac{5}{3} g_1^2 B_{\mu\nu} B^{\mu\nu}) \sqrt{g} d^4 x,$
cardona-qft-methods_ EQ0484	220	■ 1.000 ● N +1	$f_{\{k\}} = \int_0^\infty f(v) v^{k-1} d v$	$f_k = \int_0^\infty f(v) v^{k-1} dv$	$f_k = \int_0^\infty f(v) v^{k-1} dv.$
cardona-qft-methods_ EQ0486	221	■ 1.000 ● N -1	$\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{\frac{n-m}{2}} \int_M a_{\{n\}}(x, P) d v(x)$	$\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{\frac{n-m}{2}} \int_M a_n(x, P) dv(x)$	$\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{\frac{n-m}{2}} \int_M a_n(x, P) dv(x)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0487	221	■1.000 ●W +1	$\begin{gathered} \nabla_{\mu}=\partial_{\mu}+\omega_{\mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2} g_{\mu \nu}\left(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id}\right) \\ E=B-g^{\mu \nu}\left(\partial_{\mu} \omega_{\nu}^{\prime}+\omega_{\mu}^{\prime} \omega_{\nu}^{\prime}-\Gamma_{\mu \nu}^{\rho} \omega_{\rho}^{\prime}\right) \\ \Omega_{\mu \nu}=\partial_{\mu} \omega_{\nu}^{\prime}-\partial_{\nu} \omega_{\mu}^{\prime}+\left[\omega_{\mu}^{\prime}, \omega_{\nu}^{\prime}\right] \end{gathered}$	$\nabla_{\mu}=\partial_{\mu}+\omega_{\mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2} g_{\mu \nu}\left(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id}\right) \\ E=B-g^{\mu \nu}\left(\partial_{\mu} \omega_{\nu}^{\prime}+\omega_{\mu}^{\prime} \omega_{\nu}^{\prime}-\Gamma_{\mu \nu}^{\rho} \omega_{\rho}^{\prime}\right) \\ \Omega_{\mu \nu}=\partial_{\mu} \omega_{\nu}^{\prime}-\partial_{\nu} \omega_{\mu}^{\prime}+\left[\omega_{\mu}^{\prime}, \omega_{\nu}^{\prime}\right]$	$\nabla_{\mu}=\partial_{\mu}+\omega_{\mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2} g_{\mu \nu}\left(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id}\right) \\ E=B-g^{\mu \nu}\left(\partial_{\mu} \omega_{\nu}^{\prime}+\omega_{\mu}^{\prime} \omega_{\nu}^{\prime}-\Gamma_{\mu \nu}^{\rho} \omega_{\rho}^{\prime}\right) \\ \Omega_{\mu \nu}=\partial_{\mu} \omega_{\nu}^{\prime}-\partial_{\nu} \omega_{\mu}^{\prime}+\left[\omega_{\mu}^{\prime}, \omega_{\nu}^{\prime}\right].$
cardona-qft-methods_ EQ0491	222	■1.000 ●	$\begin{aligned} S &= \frac{1}{2} \kappa_0^2 \int R \sqrt{g} d^4 x + \gamma_0 \int \sqrt{g} d^4 x \\ &+ \alpha_0 \int C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma} \sqrt{g} d^4 x + \tau_0 \int R^* R^* \sqrt{g} d^4 x \\ &+ \frac{1}{2} \int D H ^2 \sqrt{g} d^4 x - \mu_0^2 \int H ^2 \sqrt{g} d^4 x \\ &- \xi_0 \int R H ^2 \sqrt{g} d^4 x + \lambda_0 \int H ^4 \sqrt{g} d^4 x \\ &+ \frac{1}{4} \int \left(G_{\mu \nu}^i G^{\mu \nu i} + F_{\mu \nu}^{\alpha} F^{\mu \nu \alpha} + B_{\mu \nu} B^{\mu \nu} \right) \sqrt{g} d^4 x \end{aligned}$	$S = \frac{1}{2\kappa_0^2} \int R \sqrt{g} d^4 x + \gamma_0 \int \sqrt{g} d^4 x \\ + \alpha_0 \int C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} \sqrt{g} d^4 x + \tau_0 \int R^* R^* \sqrt{g} d^4 x \\ + \frac{1}{2} \int DH ^2 \sqrt{g} d^4 x - \mu_0^2 \int H ^2 \sqrt{g} d^4 x \\ - \xi_0 \int R H ^2 \sqrt{g} d^4 x + \lambda_0 \int H ^4 \sqrt{g} d^4 x \\ + \frac{1}{4} \int \left(G_{\mu\nu}^i G^{\mu\nu i} + F_{\mu\nu}^{\alpha} F^{\mu\nu\alpha} + B_{\mu\nu} B^{\mu\nu} \right) \sqrt{g} d^4 x$	$S = \frac{1}{2\kappa_0^2} \int R \sqrt{g} d^4 x + \gamma_0 \int \sqrt{g} d^4 x \\ + \alpha_0 \int C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} \sqrt{g} d^4 x + \tau_0 \int R^* R^* \sqrt{g} d^4 x \\ + \frac{1}{2} \int DH ^2 \sqrt{g} d^4 x - \mu_0^2 \int H ^2 \sqrt{g} d^4 x \\ - \xi_0 \int R H ^2 \sqrt{g} d^4 x + \lambda_0 \int H ^4 \sqrt{g} d^4 x \\ + \frac{1}{4} \int \left(G_{\mu\nu}^i G^{\mu\nu i} + F_{\mu\nu}^{\alpha} F^{\mu\nu\alpha} + B_{\mu\nu} B^{\mu\nu} \right) \sqrt{g} d^4 x$
cardona-qft-methods_ EQ0493	222	■1.000 ●K +0	$\frac{g^2 f_0}{2 \pi^2} = \frac{1}{4}.$	$\frac{g^2 f_0}{2 \pi^2} = \frac{1}{4}.$	$\frac{g^2 f_0}{2 \pi^2} = \frac{1}{4}.$
cardona-qft-methods_ EQ0495	225	■1.000 ●N +0	$\partial_t x_i(t) = \beta_{x_i}(x(t)),$	$\partial_t x_i(t) = \beta_{x_i}(x(t)),$	$\partial_t x_i(t) = \beta_{x_i}(x(t)),$
cardona-qft-methods_ EQ0496 (1)	225	■1.000 ●N +0	$\beta_1 = \frac{41}{96 \pi^2} g_1^3, \quad \beta_2 = -\frac{19}{96 \pi^2} g_2^3, \quad \beta_3 = -\frac{7}{16 \pi^2} g_3^3.$	$\beta_1 = \frac{41}{96 \pi^2} g_1^3, \quad \beta_2 = -\frac{19}{96 \pi^2} g_2^3, \quad \beta_3 = -\frac{7}{16 \pi^2} g_3^3.$	$\beta_1 = \frac{41}{96 \pi^2} g_1^3, \quad \beta_2 = -\frac{19}{96 \pi^2} g_2^3, \quad \beta_3 = -\frac{7}{16 \pi^2} g_3^3.$
cardona-qft-methods_ EQ0497 (2)	225	■1.000 ●N +0	$g_1(0)=0.3575, \quad g_2(0)=0.6514, \quad g_3(0)=1.221$	$g_1(0)=0.3575, \quad g_2(0)=0.6514, \quad g_3(0)=1.221$	$g_1(0)=0.3575, \quad g_2(0)=0.6514, \quad g_3(0)=1.221$
cardona-qft-methods_ EQ0499	226	■1.000 ●N +0	$\frac{\sqrt{\mathfrak{a} f_0}}{\pi} = \frac{2 M_W}{g}$	$\frac{\sqrt{\mathfrak{a} f_0}}{\pi} = \frac{2 M_W}{g}$	$\frac{\sqrt{\mathfrak{a} f_0}}{\pi} = \frac{2 M_W}{g}$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0503 (4)	227	■ 1.000 ● N +0	$\begin{gathered} \delta_{(\uparrow 1)}=\frac{1}{246} \\ \left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right) \\ Y_{\nu}=U_{PMNS}^{\dagger} \delta_{(\uparrow 1)} U_{PMNS} \end{gathered}$	$\delta_{(\uparrow 1)}=\frac{1}{246} \left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right)$ $Y_{\nu}=U_{PMNS}^{\dagger} \delta_{(\uparrow 1)} U_{PMNS}$	$\delta_{(\uparrow 1)}=\frac{1}{246} \left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right)$ $Y_{\nu}=U_{PMNS}^{\dagger} \delta_{(\uparrow 1)} U_{PMNS}$
cardona-qft-methods_ EQ0504 (3)	227	■ 1.000 ●	$\begin{gathered} \beta_y=\frac{1}{16 \pi^2} \left(\frac{9}{2} y^3-8 g_{\{3\}^2} y-\frac{9}{4} g_2^2 y-\frac{17}{12} g_1^2 y-\frac{17}{12} g_{\{1\}^2} y\right) . \\ \beta_{\lambda}=\frac{1}{16 \pi^2} \left(24 \lambda^2+12 \lambda y^2-9 \lambda\left(g_2^2+\frac{1}{3} g_1^2\right)-6 y^4+\frac{9}{8} g_2^4+\frac{3}{8} g_1^4+\frac{3}{4} g_2^2 g_1^2\right) , \end{gathered}$	$\beta_y=\frac{1}{16 \pi^2} \left(\frac{9}{2} y^3-8 g_{\{3\}^2} y-\frac{9}{4} g_2^2 y-\frac{17}{12} g_1^2 y-\frac{17}{12} g_{\{1\}^2} y\right) .$ $\beta_{\lambda}=\frac{1}{16 \pi^2} \left(24 \lambda^2+12 \lambda y^2-9 \lambda\left(g_2^2+\frac{1}{3} g_1^2\right)-6 y^4+\frac{9}{8} g_2^4+\frac{3}{8} g_1^4+\frac{3}{4} g_2^2 g_1^2\right) ,$	$\beta_y=\frac{1}{16 \pi^2} \left(\frac{9}{2} y^3-8 g_{\{3\}^2} y-\frac{9}{4} g_2^2 y-\frac{17}{12} g_1^2 y-\frac{17}{12} g_{\{1\}^2} y\right) . \quad (3)$ $\beta_{\lambda}=\frac{1}{16 \pi^2} \left(24 \lambda^2+12 \lambda y^2-9 \lambda\left(g_2^2+\frac{1}{3} g_1^2\right)-6 y^4+\frac{9}{8} g_2^4+\frac{3}{8} g_1^4+\frac{3}{4} g_2^2 g_1^2\right) ,$
cardona-qft-methods_ EQ0505	228	■ 1.000 ● N +0	$\frac{f_0}{2 \pi^2} \int b \varphi ^4 \sqrt{g} d^4 x=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}}{\mathfrak{a}^2} \int \mathbf{H} ^4 \sqrt{g} d^4 x$	$\frac{f_0}{2 \pi^2} \int b \varphi ^4 \sqrt{g} d^4 x=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}}{\mathfrak{a}^2} \int \mathbf{H} ^4 \sqrt{g} d^4 x$	$\frac{f_0}{2 \pi^2} \int b \varphi ^4 \sqrt{g} d^4 x=\frac{\pi^2}{2 f_0} \frac{\mathfrak{b}}{\mathfrak{a}^2} \int \mathbf{H} ^4 \sqrt{g} d^4 x$
cardona-qft-methods_ EQ0506	228	■ 1.000 ● K +0	$\tilde{\lambda}(\Lambda)=g_3^2 \frac{\mathfrak{b}}{\mathfrak{a}^2} .$	$\tilde{\lambda}(\Lambda)=g_3^2 \frac{\mathfrak{b}}{\mathfrak{a}^2} .$	$\tilde{\lambda}(\Lambda)=g_3^2 \frac{\mathfrak{b}}{\mathfrak{a}^2} .$
cardona-qft-methods_ EQ0507	228	■ 1.000 ● N +0	$\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^2}\left(12 \lambda^2+B\right)$	$\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^2}\left(12 \lambda^2+B\right)$	$\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^2}\left(12 \lambda^2+B\right)$
cardona-qft-methods_ EQ0508	228	■ 1.000 ● N +0	$\gamma=\frac{1}{16 \pi^2}\left(12 y_t^2-9 g_2^2-3 g_1^2\right) \quad B=\frac{3}{16}\left(3 g_2^4+2 g_1^2 g_2^2+g_1^4\right)-3 y_t^4$	$\gamma=\frac{1}{16 \pi^2}\left(12 y_t^2-9 g_2^2-3 g_1^2\right) \quad B=\frac{3}{16}\left(3 g_2^4+2 g_1^2 g_2^2+g_1^4\right)-3 y_t^4$	$\gamma=\frac{1}{16 \pi^2}\left(12 y_t^2-9 g_2^2-3 g_1^2\right) \quad B=\frac{3}{16}\left(3 g_2^4+2 g_1^2 g_2^2+g_1^4\right)-3 y_t^4$
cardona-qft-methods_ EQ0509	228	■ 1.000 ● N +0	$m_{\mathrm{H}}^2=8 \lambda \frac{M^2}{g^2}, \quad m_H=\sqrt{2 \lambda} \frac{2 M}{g} .$	$m_{\mathrm{H}}^2=8 \lambda \frac{M^2}{g^2}, \quad m_H=\sqrt{2 \lambda} \frac{2 M}{g} .$	$m_{\mathrm{H}}^2=8 \lambda \frac{M^2}{g^2}, \quad m_H=\sqrt{2 \lambda} \frac{2 M}{g} .$
cardona-qft-methods_ EQ0510	229	■ 1.000 ● N +1	$\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^2+\frac{\theta}{\eta} R^* R^*\right) \sqrt{g} d^4 x$	$\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^2+\frac{\theta}{\eta} R^* R^*\right) \sqrt{g} d^4 x$	$\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^2+\frac{\theta}{\eta} R^* R^*\right) \sqrt{g} d^4 x$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08511	229	■1.000 ●W +0	$\begin{aligned} & \& \beta_{\eta}=-\frac{1}{(4\pi)^2}\frac{133}{10}\eta^2 \\ & \& \frac{133}{10}\eta^2 \\ & \& \& \beta_{\omega}=-\frac{1}{(4\pi)^2}\frac{25+1098\omega+200\omega^2}{60}\eta \\ & \& \& \& \beta_{\theta}=\frac{1}{(4\pi)^2}\frac{7(56-171\theta)}{90}\eta. \end{aligned}$	$\beta_{\eta}=-\frac{1}{(4\pi)^2}\frac{133}{10}\eta^2$ $\beta_{\omega}=-\frac{1}{(4\pi)^2}\frac{25+1098\omega+200\omega^2}{60}\eta$ $\beta_{\theta}=\frac{1}{(4\pi)^2}\frac{7(56-171\theta)}{90}\eta.$	$\beta_{\eta}=-\frac{1}{(4\pi)^2}\frac{133}{10}\eta^2$ $\beta_{\omega}=-\frac{1}{(4\pi)^2}\frac{25+1098\omega+200\omega^2}{60}\eta$ $\beta_{\theta}=\frac{1}{(4\pi)^2}\frac{7(56-171\theta)}{90}\eta.$
cardona-qft-methods_ E08516 (2)	236	■1.000 ●N +0	$\Delta x\,\Delta p\geq\frac{\hbar}{2},$	$\Delta x\Delta p\geq\frac{\hbar}{2},$	$\Delta x\Delta p\geq\frac{\hbar}{2},$
cardona-qft-methods_ E08517 (3)	236	■1.000 ●N +0	$x\,y-y\,x=i\,\theta,$	$xy-yx=i\theta,$	$x\,y-y\,x=i\theta,$
cardona-qft-methods_ E08518 (4)	236	■1.000 ●N +0	$\Delta x\,\Delta y\geq\frac{\theta}{2},$	$\Delta x\Delta y\geq\frac{\theta}{2},$	$\Delta x\,\Delta y\geq\frac{\theta}{2},$
cardona-qft-methods_ E08521 (1)	239	■1.000 ●N +0	$U(g)=\bigoplus_NU^{\{N\}}(g),$	$U(g)=\bigoplus_NU^{(N)}(g),$	$U(g)=\bigoplus_NU^{(N)}(g),$
cardona-qft-methods_ E08522 (2)	239	■1.000 ●	$U^{\{N\}}(g)\left \rho_1\right\rangle\left \rho_2\right\rangle\otimes\cdots\otimes\left \rho_N\right\rangle=U^{\{1\}}(g)\left \rho_1\right\rangle\left \rho_2\right\rangle\otimes\cdots\otimes U^{\{1\}}(g)\left \rho_N\right\rangle.$	$U^{(N)}(g)\left \rho_1\right\rangle\otimes\left \rho_2\right\rangle\otimes\cdots\otimes\left \rho_N\right\rangle=U^{(1)}(g)\left \rho_1\right\rangle\otimes U^{(1)}(g)\left \rho_2\right\rangle\otimes\cdots\otimes U^{(1)}(g)\left \rho_N\right\rangle.$	$U^{(N)}(g)\left \rho_1\right\rangle\otimes\left \rho_2\right\rangle\otimes\cdots\otimes\left \rho_N\right\rangle=U^{(1)}(g)\left \rho_1\right\rangle\otimes U^{(1)}(g)\left \rho_2\right\rangle\otimes\cdots\otimes U^{(1)}(g)\left \rho_N\right\rangle.$
cardona-qft-methods_ E08523 (3)	240	■1.000 ●N +0	$\alpha^*=\sum_{g\in G}\alpha(g)\,g,$	$\alpha^*=\sum_{g\in G}\alpha(g)g,$	$\alpha^*=\sum_{g\in G}\alpha(g)g,$
cardona-qft-methods_ E08524 (4)	240	■1.000 ●N +0	$\alpha^*\beta^*:=\sum_{g\in G}(\alpha*\beta)(g)\,g,$	$\alpha^*\beta^*:=\sum_{g\in G}(\alpha*\beta)(g)g,$	$\alpha^*\beta^*:=\sum_{g\in G}(\alpha*\beta)(g)\,g,$
cardona-qft-methods_ E08525 (5)	240	■1.000 ●N +0	$(\alpha*\beta)(g)=\sum_{h\in G}\alpha\left(gh^{-1}\right)\beta(h)\,.$	$(\alpha*\beta)(g)=\sum_{h\in G}\alpha\left(gh^{-1}\right)\beta(h).$	$(\alpha*\beta)(g)=\sum_{h\in G}\alpha(gh^{-1})\beta(h).$
cardona-qft-methods_ E08526 (6)	240	■1.000 ●N +0	$\alpha^*=\int d\mu(g)\,\alpha(g)\,g\,.$	$\alpha^*=\int d\mu(g)\alpha(g)g.$	$\alpha^*=\int d\mu(g)\alpha(g)g.$
cardona-qft-methods_ E08527 (7)	240	■1.000 ●N +2	$\begin{aligned} & U^{\{N\}}\left(\left(\alpha^*\right)\right)=\int d\mu(g)\,\alpha(g)\,U^{\{N\}}(g) \\ & U\left(\alpha^*\right)=\int d\mu(g)\,\alpha(g)\,U(g) \end{aligned}$	$U^{(N)}\left(\alpha^*\right)=\int d\mu(g)\alpha(g)U^{(N)}(g)$ $U\left(\alpha^*\right)=\int d\mu(g)\alpha(g)U(g)$	$U^{(N)}(\alpha^*)=\int d\mu(g)\alpha(g)U^{(N)}(g),$ $U(\alpha^*)=\int d\mu(g)\alpha(g)U(g).$
cardona-qft-methods_ E08528 (8)	241	■1.000 ●N +0	$U^{\{1\}}(g)\left \rho\right\rangle=\left \sigma\right\rangle D_{\sigma\rho}(g),$	$U^{(1)}(g) \rho\rangle= \sigma\rangle D_{\sigma\rho}(g),$	$U^{(1)}(g) \rho\rangle= \sigma\rangle D_{\sigma\rho}(g),$
cardona-qft-methods_ E08529 (9)	241	■1.000 ●K +0	$\Delta(g)=g\otimes g\,.$	$\Delta(g)=g\otimes g.$	$\Delta(g)=g\otimes g.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08531	241	■ 1.000 ● N +0	$\begin{aligned} & \text{\&U\^{(3)}(g)=\left(U\^{(2)}\right)\otimes U\^{(1)}\right)\Delta(g),} \\ & \text{\&U\^{(4)}(g)=\left(U\^{(3)}\otimes U\^{(1)}\right)\Delta(g),} \end{aligned}$	$U^{(3)}(g) = \left(U^{(2)} \otimes U^{(1)} \right) \Delta(g),$ $U^{(4)}(g) = \left(U^{(3)} \otimes U^{(1)} \right) \Delta(g),$	$U^{(3)}(g) = \left(U^{(2)} \otimes U^{(1)} \right) \Delta(g),$ $U^{(4)}(g) = \left(U^{(3)} \otimes U^{(1)} \right) \Delta(g),$
cardona-qft-methods_ E08532 (11)	241	■ 1.000 ● N +0	$U^{(N)}(g)=U^{(1)}(g)\otimes U^{(1)}(g)\otimes\cdots\otimes U^{(1)}(g).$	$U^{(N)}(g) = U^{(1)}(g) \otimes U^{(1)}(g) \otimes \cdots \otimes U^{(1)}(g).$	$U^{(N)}(g) = U^{(1)}(g) \otimes U^{(1)}(g) \otimes \cdots \otimes U^{(1)}(g).$
cardona-qft-methods_ E08533 (12)	241	■ 1.000 ● K +0	$(\Delta\otimes\mathrm{Id})\Delta=(\mathrm{Id}\otimes\Delta)\Delta.$	$(\Delta \otimes \text{Id})\Delta = (\text{Id} \otimes \Delta)\Delta.$	$(\Delta \otimes \text{Id})\Delta = (\text{Id} \otimes \Delta)\Delta.$
cardona-qft-methods_ E08534 (13)	241	■ 1.000 ● W +0	$\mathbb{C}G\rightarrow\mathbb{C}G\otimes\mathbb{C}G\otimes\cdots\mathbb{C}G.$	$\mathbb{C}G \rightarrow \underbrace{\mathbb{C}G \otimes \mathbb{C}G \otimes \cdots \mathbb{C}G}_{N \text{ factors}}.$	$\mathbb{C}G \rightarrow \underbrace{\mathbb{C}G \otimes \mathbb{C}G \otimes \cdots \mathbb{C}G}_{N \text{ factors}}.$
cardona-qft-methods_ E08535 (14)	241	■ 1.000 ● W +2	$\mathbb{C}G\rightarrow\mathbb{C}G\otimes\mathbb{C}G\otimes\cdots\mathbb{C}G.$	$\mathbb{C}G \rightarrow \underbrace{\mathbb{C}G \otimes \mathbb{C}G \otimes \cdots \mathbb{C}G}_{(N+1) \text{ factors}}$	$\mathbb{C}G \rightarrow \underbrace{\mathbb{C}G \otimes \mathbb{C}G \otimes \cdots \mathbb{C}G}_{(N+1) \text{ factors}},$
cardona-qft-methods_ E08536 (15)	242	■ 1.000 ● N +0	$\Delta_\theta(g)=F_\theta^{-1}(g\otimes g)F_\theta,$	$\Delta_\theta(g) = F_\theta^{-1}(g \otimes g)F_\theta,$	$\Delta_\theta(g) = F_\theta^{-1}(g \otimes g)F_\theta,$
cardona-qft-methods_ E08537	242	■ 1.000 ● N +0	$F_\theta=e^{\frac{1}{2}P_\mu\theta^{\mu\nu}\otimes P_\nu},$	$F_\theta = e^{\frac{1}{2}P_\mu\theta^{\mu\nu}\otimes P_\nu},$	$F_\theta = e^{\frac{1}{2}P_\mu\theta^{\mu\nu}\otimes P_\nu},$
cardona-qft-methods_ E08538	242	■ 1.000 ● N +0	$\epsilon:G\rightarrow\mathbb{C},\quad\epsilon(g)=1.$	$\epsilon:G\rightarrow\mathbb{C},\quad\epsilon(g)=1.$	$\epsilon:G\rightarrow\mathbb{C},\quad\epsilon(g)=1.$
cardona-qft-methods_ E08539	242	■ 1.000 ●	$\epsilon:\int d\mu(g)\alpha(g)g\rightarrow\int d\mu(g)\alpha(g)$	$\epsilon:\int d\mu(g)\alpha(g)g\rightarrow\int d\mu(g)\alpha(g)$	$\epsilon:\int d\mu(g)\alpha(g)g\rightarrow\int d\mu(g)\alpha(g)$
cardona-qft-methods_ E08540	243	■ 1.000 ● N +1	$S:\hat{\alpha}=\int d\mu(g)\alpha(g)g\rightarrow S(\hat{\alpha})=\int d\mu(g)\alpha(g)S(g)$	$S:\hat{\alpha}=\int d\mu(g)\alpha(g)g\rightarrow S(\hat{\alpha})=\int d\mu(g)\alpha(g)S(g)$	$S:\hat{\alpha}=\int d\mu(g)\alpha(g)g\rightarrow S(\hat{\alpha})=\int d\mu(g)\alpha(g)S(g).$
cardona-qft-methods_ E08541	243	■ 1.000 ● N +1	$m(g\otimes h)=gh,\quad;g,h\in G.$	$m(g \otimes h) = gh,\quad;g,h \in G.$	$m(g \otimes h) = gh,\quad;g,h \in G.$
cardona-qft-methods_ E08544	243	■ 1.000 ● K +0	$\left(m_{13}\otimes m_{24}\right)\left(a_1\otimes a_2\otimes a_3\otimes a_4\right)=a_1a_3\otimes a_2a_4.$	$(m_{13} \otimes m_{24})(a_1 \otimes a_2 \otimes a_3 \otimes a_4) = a_1a_3 \otimes a_2a_4.$	$(m_{13} \otimes m_{24})(a_1 \otimes a_2 \otimes a_3 \otimes a_4) = a_1a_3 \otimes a_2a_4.$
cardona-qft-methods_ E08545	243	■ 1.000 ● N +1	$\left(\int d\mu(g)\alpha(g)g\right)^*=\int d\mu(g)\bar{\alpha}(g)g^{-1}$	$\left(\int d\mu(g)\alpha(g)g\right)^*=\int d\mu(g)\bar{\alpha}(g)g^{-1}$	$\left(\int d\mu(g)\alpha(g)g\right)^*=\int d\mu(g)\bar{\alpha}(g)g^{-1}.$
cardona-qft-methods_ E08546 (16)	244	■ 1.000 ● K +0	$\Delta:H\rightarrow H\otimes H$	$\Delta:H\rightarrow H\otimes H$	$\Delta:H\rightarrow H\otimes H$
cardona-qft-methods_ E08550	244	■ 1.000 ● N +0	$m(\alpha\otimes\beta)=\alpha\beta.$	$m(\alpha \otimes \beta) = \alpha\beta.$	$m(\alpha \otimes \beta) = \alpha\beta.$
cardona-qft-methods_ E08551	244	■ 1.000 ● N +1	$\left(m_{13}\otimes m_{24}\right)\left(\alpha_1\otimes\alpha_2\otimes\alpha_3\otimes\alpha_4\right)=\alpha_1\alpha_3\otimes\alpha_2\alpha_4.$	$(m_{13} \otimes m_{24})(\alpha_1 \otimes \alpha_2 \otimes \alpha_3 \otimes \alpha_4) = \alpha_1\alpha_3 \otimes \alpha_2\alpha_4.$	$(m_{13} \otimes m_{24})(\alpha_1 \otimes \alpha_2 \otimes \alpha_3 \otimes \alpha_4) = \alpha_1\alpha_3 \otimes \alpha_2\alpha_4.$
cardona-qft-methods_ E08552 (19)	244	■ 1.000 ● N +0	$\Delta m(\alpha\otimes\beta)=(m_{13}\otimes m_{24})(\Delta\otimes\Delta)(\alpha\otimes\beta).$	$\Delta m(\alpha \otimes \beta) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(\alpha \otimes \beta).$	$\Delta m(\alpha \otimes \beta) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(\alpha \otimes \beta).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0553	245	■ 1.000 ● N -2	$(F \otimes Id)(\Delta \otimes Id)F = (Id \otimes F)(Id \otimes \Delta)F, \quad (\epsilon \otimes Id)F = (Id \otimes \epsilon)F = 1.$	$(F \otimes Id)(\Delta \otimes Id)F = (Id \otimes F)(Id \otimes \Delta)F, \quad (\epsilon \otimes Id)F = (Id \otimes \epsilon)F = 1.$	$(F \otimes Id)(\Delta \otimes Id)F = (Id \otimes F)(Id \otimes \Delta)F, \quad (\epsilon \otimes Id)F = (Id \otimes \epsilon)F = 1.$
cardona-qft-methods_ EQ0554	245	■ 1.000 ● N +0	$\Delta(g) = g \otimes g, \quad g \in G \subset GCG,$	$\Delta(g) = g \otimes g, \quad g \in G \subset GCG,$	$\Delta(g) = g \otimes g, \quad g \in G \subset GCG,$
cardona-qft-methods_ EQ0555	245	■ 1.000 ● N +0	$\xi \otimes \eta \rightarrow (\rho \otimes \rho) \Delta(g) \xi \otimes \eta = \rho(g) \xi \otimes \rho(g) \eta.$	$\xi \otimes \eta \rightarrow (\rho \otimes \rho) \Delta(g) \xi \otimes \eta = \rho(g) \xi \otimes \rho(g) \eta.$	$\xi \otimes \eta \rightarrow (\rho \otimes \rho) \Delta(g) \xi \otimes \eta = \rho(g) \xi \otimes \rho(g) \eta.$
cardona-qft-methods_ EQ0556	245	■ 1.000 ● N +0	$\tau(\xi \otimes \eta) = \eta \otimes \xi.$	$\tau(\xi \otimes \eta) = \eta \otimes \xi.$	$\tau(\xi \otimes \eta) = \eta \otimes \xi.$
cardona-qft-methods_ EQ0557	246	■ 1.000 ● N +0	$\tau_{i,i+1} = \underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes \tau \underbrace{1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}.$	$\tau_{i,i+1} = \underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes \tau \underbrace{1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}.$	$\tau_{i,i+1} = \underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes \tau \underbrace{1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}.$
cardona-qft-methods_ EQ0559	246	■ 1.000 ● N +1	$\Delta^{\text{op}}(\eta) = \eta^{(2)} \otimes \eta^{(2)}$	$\Delta^{\text{op}}(\eta) = \eta^{(2)} \otimes \eta^{(2)}$	$\Delta^{\text{op}}(\eta) = \eta^{(2)} \otimes \eta^{(2)}.$
cardona-qft-methods_ EQ0560	247	■ 1.000 ● N +0	$R_{i,i+1} = \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}$	$R_{i,i+1} = \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}$	$R_{i,i+1} = \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text{ factors}} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text{ factors}}$
cardona-qft-methods_ EQ0561	247	■ 1.000 ● K +0	$\mathcal{R}_{i,i+1} = \tau_{i,i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i,i+1}$	$\mathcal{R}_{i,i+1} = \tau_{i,i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i,i+1}$	$\mathcal{R}_{i,i+1} = \tau_{i,i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i,i+1}$
cardona-qft-methods_ EQ0562	247	■ 1.000 ● K +0	$R_{i+1,i+2} R_{i,i+2} R_{i,i+1} = R_{i,i+1} R_{i,i+2} R_{i+1,i+2}.$	$R_{i+1,i+2} R_{i,i+2} R_{i,i+1} = R_{i,i+1} R_{i,i+2} R_{i+1,i+2}.$	$R_{i+1,i+2} R_{i,i+2} R_{i,i+1} = R_{i,i+1} R_{i,i+2} R_{i+1,i+2}.$
cardona-qft-methods_ EQ0563	247	■ 1.000 ● N +0	$\Delta_{\theta}(g) = F_{\theta}^{-1}(g \otimes g) F_{\theta}$	$\Delta_{\theta}(g) = F_{\theta}^{-1}(g \otimes g) F_{\theta}$	$\Delta_{\theta}(g) = F_{\theta}^{-1}(g \otimes g) F_{\theta}$
cardona-qft-methods_ EQ0564	247	■ 1.000 ●	$\tau_{\theta} = (\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}$	$\tau_{\theta} = (\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}$	$\tau_{\theta} = (\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}$
cardona-qft-methods_ EQ0565	247	■ 1.000 ● K +0	$\tau_{\theta} = \tau(\rho \otimes \rho) F_{\theta}^2,$	$\tau_{\theta} = \tau(\rho \otimes \rho) F_{\theta}^2,$	$\tau_{\theta} = \tau(\rho \otimes \rho) F_{\theta}^2,$
cardona-qft-methods_ EQ0566	247	■ 1.000 ● K +0	$R_{\theta} = F_{\theta}^2.$	$R_{\theta} = F_{\theta}^2.$	$R_{\theta} = F_{\theta}^2.$
cardona-qft-methods_ EQ0567 (1)	248	■ 1.000 ● K +0	$R(t) = 0 \quad t < 0$	$R(t) = 0 \quad t < 0$	$R(t) = 0 \quad t < 0$
cardona-qft-methods_ EQ0569 (3)	248	■ 1.000 ● K +0	$\Im \omega > 0.$	$\Im \omega > 0.$	$\Im \omega > 0.$
cardona-qft-methods_ EQ0570 (4)	248	■ 1.000 ● W -2	$\begin{gathered} \{[\rho(x), \eta(y)] = 0 \quad \text{if} \\ \{ \text{if} \} \setminus \left(x^0 - y^0 \right)^2 - (\vec{x} - \vec{y})^2 < 0 \} \end{gathered}$	$[\rho(x), \eta(y)] = 0 \quad \text{if} \\ (x^0 - y^0)^2 - (\vec{x} - \vec{y})^2 < 0$	$[\rho(x), \eta(y)] = 0 \quad \text{if} \\ (x^0 - y^0)^2 - (\vec{x} - \vec{y})^2 < 0$
cardona-qft-methods_ EQ0571 (6)	249	■ 1.000 ● N +0	$[\varphi(x), \chi(y)]_- = 0 \quad x \sim y$	$[\varphi(x), \chi(y)]_- = 0 \quad x \sim y$	$[\varphi(x), \chi(y)]_- = 0 \quad x \sim y$
cardona-qft-methods_ EQ0572 (7)	249	■ 1.000 ● N +0	$\left. \left. \left[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y) \right] \right _+ = 0 \quad x \sim y$	$\left[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y) \right]_+ = 0 \quad x \sim y$	$[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y)]_+ = 0 \quad x \sim y$
cardona-qft-methods_ EQ0575 (10)	249	■ 1.000 ● N +0	$g \rightarrow [U_1 \otimes U_2](g \otimes g)$	$g \rightarrow [U_1 \otimes U_2](g \otimes g)$	$g \rightarrow [U_1 \otimes U_2](g \otimes g)$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08576 (11)	250	1.000 N +0	$\begin{aligned} \Delta: G &\rightarrow G \otimes G, \\ g &\rightarrow \Delta(g) := g \otimes g, \end{aligned}$	$\Delta: G \rightarrow G \otimes G, \\ g \rightarrow \Delta(g) := g \otimes g,$	$\Delta: G \rightarrow G \otimes G, \\ g \rightarrow \Delta(g) := g \otimes g,$
cardona-qft-methods_ E08577 (12)	250	1.000 N +1	$F_{\{\theta\}} = e^{\frac{i}{2} \partial_\mu \otimes \theta^{\mu\nu} \partial_\nu} \text{ "Twist element" }$	$F_\theta = e^{\frac{i}{2} \partial_\mu \otimes \theta^{\mu\nu} \partial_\nu} = \text{ "Twist element" }$	$F_\theta = e^{\frac{i}{2} \partial_\mu \otimes \theta^{\mu\nu} \partial_\nu} = \text{ "Twist element" }$
cardona-qft-methods_ E08578 (13)	250	1.000 N +0	$f \star g = m_{\{0\}} \left[F_{\{\theta\}} f \otimes g \right],$	$f \star g = m_0 [F_\theta f \otimes g],$	$f \star g = m_0 [F_\theta f \otimes g],$
cardona-qft-methods_ E08579 (14)	250	1.000 N +0	$m_{\{0\}}(\alpha \otimes \beta)(x) = \alpha(x) \beta(x)$	$m_0(\alpha \otimes \beta)(x) = \alpha(x) \beta(x)$	$m_0(\alpha \otimes \beta)(x) = \alpha(x) \beta(x)$
cardona-qft-methods_ E08581	250	1.000 N +0	$\Lambda: \alpha \rightarrow \Lambda^* \alpha, \quad (\Lambda^* \alpha)(x) = \alpha[\Lambda^{-1}(x)].$	$\Lambda: \alpha \rightarrow \Lambda^* \alpha, \quad (\Lambda^* \alpha)(x) = \alpha[\Lambda^{-1}(x)].$	$\Lambda: \alpha \rightarrow \Lambda^* \alpha, \quad (\Lambda^* \alpha)(x) = \alpha[\Lambda^{-1}(x)].$
cardona-qft-methods_ E08582 (15)	250	1.000 K +0	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta}.$	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta}.$	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta}.$
cardona-qft-methods_ E08583 (16)	250	1.000 N +0	$\phi \otimes_{S,A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$	$\phi \otimes_{S,A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$	$\phi \otimes_{S,A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$
cardona-qft-methods_ E08584 (17)	251	1.000 N +0	$\tau_{\{0\}}(\phi \otimes \chi) = \chi \otimes \phi$	$\tau_0(\phi \otimes \chi) = \chi \otimes \phi$	$\tau_0(\phi \otimes \chi) = \chi \otimes \phi$
cardona-qft-methods_ E08585 (18)	251	1.000 K +0	$\tau_{\{0\}} \Delta_{\{0\}}(\Lambda) = \Delta_{\{0\}}(\Lambda) \tau_{\{0\}}$	$\tau_0 \Delta_0(\Lambda) = \Delta_0(\Lambda) \tau_0$	$\tau_0 \Delta_0(\Lambda) = \Delta_0(\Lambda) \tau_0$
cardona-qft-methods_ E08586 (19)	251	1.000 N +0	$\left(\frac{1 \pm \tau_0}{2} \right) (\phi \otimes \chi).$	$\left(\frac{1 \pm \tau_0}{2} \right) (\phi \otimes \chi).$	$\left(\frac{1 \pm \tau_0}{2} \right) (\phi \otimes \chi).$
cardona-qft-methods_ E08587 (20)	251	1.000 N +0	$\tau_{\{0\}} F_{\{\theta\}} = F_{\{\theta\}}^{-1} \tau_{\{0\}},$	$\tau_0 F_\theta = F_\theta^{-1} \tau_0,$	$\tau_0 F_\theta = F_\theta^{-1} \tau_0,$
cardona-qft-methods_ E08589 (22)	251	1.000 N +0	$\tau_{\theta} \equiv F_{\theta}^{-1} \tau_0 F_{\theta}, \quad \tau_{\theta}^2 = 1 \otimes 1,$	$\tau_{\theta} \equiv F_{\theta}^{-1} \tau_0 F_{\theta}, \quad \tau_{\theta}^2 = 1 \otimes 1,$	$\tau_{\theta} \equiv F_{\theta}^{-1} \tau_0 F_{\theta}, \quad \tau_{\theta}^2 = 1 \otimes 1,$
cardona-qft-methods_ E08590 (23)	251	1.000 K +0	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1} \Lambda \otimes \Lambda F_{\theta}.$	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1} \Lambda \otimes \Lambda F_{\theta}.$	$\Delta_{\theta}(\Lambda) = F_{\theta}^{-1} \Lambda \otimes \Lambda F_{\theta}.$
cardona-qft-methods_ E08593 (2)	252	1.000	$U: (a, \Lambda) \rightarrow U(a, \Lambda)$	$U: (a, \Lambda) \rightarrow U(a, \Lambda)$	$U: (a, \Lambda) \rightarrow U(a, \Lambda)$
cardona-qft-methods_ E08594 (3)	252	1.000 K +0	$U(a, \Lambda) \varphi(x) U(a, \Lambda)^{-1} = \varphi((a, \Delta)x)$	$U(a, \Lambda) \varphi(x) U(a, \Lambda)^{-1} = \varphi((a, \Delta)x)$	$U(a, \Lambda) \varphi(x) U(a, \Lambda)^{-1} = \varphi((a, \Delta)x)$
cardona-qft-methods_ E08595 (4)	252	1.000 K +0	$\operatorname{Ad} U(a, \Lambda) \varphi = U(a, \Lambda) \varphi U(a, \Lambda)^{-1}$	$\operatorname{Ad} U(a, \Lambda) \varphi = U(a, \Lambda) \varphi U(a, \Lambda)^{-1}$	$\operatorname{Ad} U(a, \Lambda) \varphi = U(a, \Lambda) \varphi U(a, \Lambda)^{-1}$
cardona-qft-methods_ E08596 (5)	252	1.000 K +0	$(a, \Delta): \alpha \rightarrow (a, \Lambda) \triangleright \alpha, \quad [(a, \Lambda) \alpha](x) \quad \alpha \in S(M^{d+1})$	$(a, \Delta): \alpha \rightarrow (a, \Lambda) \triangleright \alpha, \quad [(a, \Lambda) \alpha](x) \quad \alpha \in S(M^{d+1})$	$(a, \Delta): \alpha \rightarrow (a, \Lambda) \triangleright \alpha, \quad [(a, \Lambda) \alpha](x) \quad \alpha \in S(M^{d+1})$
cardona-qft-methods_ E08597 (6)	252	1.000 K +0	$\Delta_0((a, \Lambda)) = (a, \Lambda) \otimes (a, \Lambda)$	$\Delta_0((a, \Lambda)) = (a, \Lambda) \otimes (a, \Lambda)$	$\Delta_0((a, \Lambda)) = (a, \Lambda) \otimes (a, \Lambda)$
cardona-qft-methods_ E08599 (8)	252	1.000 K +0	$\varphi(x_1) \varphi(x_2) \dots \varphi(x_N).$	$\varphi(x_1) \varphi(x_2) \dots \varphi(x_N).$	$\varphi(x_1) \varphi(x_2) \dots \varphi(x_N).$
cardona-qft-methods_ E08600 (9)	253	1.000 K +0	$(\varphi \otimes \dots \otimes \varphi)(x_1, x_2, \dots, x_N) = \varphi(x_1) \varphi(x_2) \dots \varphi(x_N)$	$(\varphi \otimes \dots \otimes \varphi)(x_1, x_2, \dots, x_N) = \varphi(x_1) \varphi(x_2) \dots \varphi(x_N)$	$(\varphi \otimes \dots \otimes \varphi)(x_1, x_2, \dots, x_N) = \varphi(x_1) \varphi(x_2) \dots \varphi(x_N)$
Conf. MathPix confidence: ≥ 0.9 $0.5-0.9$ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_E08601 (10)	253	■1.000 ● S -1	$\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I}}_{N-1} \otimes \underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I}}_{N-2} \otimes \Delta_0 \dots \Delta_0$	$\underbrace{(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I})}_{N-1} \underbrace{(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I})}_{N-2} \dots \Delta_0$	$\underbrace{(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I})}_{N-1} \underbrace{(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I})}_{N-2} \dots \Delta_0$
cardona-qft-methods_E08602 (11)	253	■1.000 ● K +0	$\Delta_{\{0\}}^N(a, \Lambda) = (a, \Lambda) \otimes (a, \Lambda) \otimes \dots \otimes (a, \Lambda) \otimes (a, \Lambda)$	$\Delta_0^N(a, \Lambda) = (a, \Lambda) \otimes (a, \Lambda) \otimes \dots \otimes (a, \Lambda).$	$\Delta_0^N(a, \Lambda) = (a, \Lambda) \otimes (a, \Lambda) \otimes \dots \otimes (a, \Lambda).$
cardona-qft-methods_E08603 (12)	253	■1.000 ● K +0	$U(a, \Lambda) \left(\varphi^{\otimes N} \left((a, \Lambda)^{-1} x_1, \dots, (a, \Lambda)^{-1} x_N \right) \right) U(a, \Lambda)^{-1} = \varphi^{\otimes N} (x_1, x_2, \dots x_N)$	$U(a, \Lambda) \left(\varphi^{\otimes N} \left((a, \Lambda)^{-1} x_1, \dots, (a, \Lambda)^{-1} x_N \right) \right) U(a, \Lambda)^{-1} = \varphi^{\otimes N} (x_1, x_2, \dots x_N)$	$U(a, \Lambda) \left(\varphi^{\otimes N} \left((a, \Lambda)^{-1} x_1, \dots, (a, \Lambda)^{-1} x_N \right) \right) U(a, \Lambda)^{-1} = \varphi^{\otimes N} (x_1, x_2, \dots x_N)$
cardona-qft-methods_E08604 (13)	253	■1.000 ● K +0	$\varphi = \int d\mu(p) \left(c_p^\dagger e_p + c_p e_{-p} \right) = \varphi^{(-)} + \varphi^{(+)}$	$\varphi = \int d\mu(p) \left(c_p^\dagger e_p + c_p e_{-p} \right) = \varphi^{(-)} + \varphi^{(+)}$	$\varphi = \int d\mu(p) \left(c_p^\dagger e_p + c_p e_{-p} \right) = \varphi^{(-)} + \varphi^{(+)}$
cardona-qft-methods_E08606	253	■1.000 ● S +2	$\begin{array}{r} \varphi^{(-)}(x_1) \varphi^{(-)}(x_2) \dots \varphi^{(-)}(x_N) 0\rangle = \\ \int \prod_i d\mu(p_i) c_{p_1}^\dagger(x_1) c_{p_2}^\dagger(x_2) \dots c_{p_N}^\dagger(x_N) 0\rangle e_{p_1}(x_1) e_{p_2}(x_2) \dots e_{p_N}(x_N) \end{array}$	$\varphi^{(-)}(x_1) \varphi^{(-)}(x_2) \dots \varphi^{(-)}(x_N) 0\rangle = \int \prod_i d\mu(p_i) c_{p_1}^\dagger(x_1) c_{p_2}^\dagger(x_2) \dots c_{p_N}^\dagger(x_N) 0\rangle e_{p_1}(x_1) e_{p_2}(x_2) \dots e_{p_N}(x_N)$	$\varphi^{(-)}(x_1) \varphi^{(-)}(x_2) \dots \varphi^{(-)}(x_N) 0\rangle = \int \prod_i d\mu(p_i) c_{p_1}^\dagger(x_1) c_{p_2}^\dagger(x_2) \dots c_{p_N}^\dagger(x_N) 0\rangle e_{p_1}(x_1) e_{p_2}(x_2) \dots e_{p_N}(x_N)$
cardona-qft-methods_E08607 (15)	253	■1.000 ● K +0	$[P_\mu, c_p^\dagger] = P_\mu c_p^\dagger, \quad P_\mu 0\rangle = 0$	$[P_\mu, c_p^\dagger] = P_\mu c_p^\dagger, \quad P_\mu 0\rangle = 0$	$[P_\mu, c_p^\dagger] = P_\mu c_p^\dagger, \quad P_\mu 0\rangle = 0$
cardona-qft-methods_E08608 (16)	254	■1.000 ● N +0	$\mathcal{P}_\mu e_p = -p_\mu e_p$	$\mathcal{P}_\mu e_p = -p_\mu e_p$	$\mathcal{P}_\mu e_p = -p_\mu e_p$
cardona-qft-methods_E08609 (17)	254	■1.000 ● N +0	$\Delta_0(\mathcal{P}_\mu) = \mathbb{I} \otimes \mathcal{P}_\mu + \mathcal{P}_\mu \otimes \mathbb{I}$	$\Delta_0(\mathcal{P}_\mu) = \mathbb{I} \otimes \mathcal{P}_\mu + \mathcal{P}_\mu \otimes \mathbb{I}$	$\Delta_0(\mathcal{P}_\mu) = \mathbb{I} \otimes \mathcal{P}_\mu + \mathcal{P}_\mu \otimes \mathbb{I}$
cardona-qft-methods_E08610 (18)	254	■1.000 ●	$\left(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I} \right) \Delta_0(\mathcal{P}_\mu) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = - \sum_i p_{ij} e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N}.$	$(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I}) \Delta_0(\mathcal{P}_\mu) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = - \sum_i p_{ij} e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N}.$	$(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \dots \mathbb{I} \otimes \mathbb{I}) \Delta_0(\mathcal{P}_\mu) e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N} = - \sum_i p_{ij} e_{p_1} \otimes e_{p_2} \otimes \dots e_{p_N}.$
cardona-qft-methods_E08614 (22)	254	■1.000 ● K +0	$d\mu(\Lambda^{-1} p_i) = d\mu(p_i)$	$d\mu(\Lambda^{-1} p_i) = d\mu(p_i)$	$d\mu(\Lambda^{-1} p_i) = d\mu(p_i)$
cardona-qft-methods_E08616 (24)	255	■1.000 ● N +0	$(\alpha^* f)(p) = f(\alpha p)$	$(\alpha^* f)(p) = f(\alpha p)$	$(\alpha^* f)(p) = f(\alpha p)$
cardona-qft-methods_E08619 (27)	256	■1.000 ● K +0	$U(a, \Lambda) \varphi^2((a, \Lambda)^{-1} x) U(a, \Lambda)^{-1} = \varphi^2(x).$	$U(a, \Lambda) \varphi^2((a, \Lambda)^{-1} x) U(a, \Lambda)^{-1} = \varphi^2(x).$	$U(a, \Lambda) \varphi^2((a, \Lambda)^{-1} x) U(a, \Lambda)^{-1} = \varphi^2(x).$
cardona-qft-methods_E08620 (28)	256	■1.000 ● N +0	$U(a, \Delta) \Psi((a, \Delta)^{-1} x) U(a, \Delta)^{-1} = \Psi$	$U(a, \Delta) \Psi((a, \Delta)^{-1} x) U(a, \Delta)^{-1} = \Psi$	$U(a, \Delta) \Psi((a, \Delta)^{-1} x) U(a, \Delta)^{-1} = \Psi$
cardona-qft-methods_E08622 (30)	257	■1.000 ● K +0	$m_\theta(\gamma \otimes \delta) = \gamma(x) \delta(x), \quad \gamma, \delta \in \mathcal{A}_0(M^{(d+1)}).$	$m_\theta(\gamma \otimes \delta) = \gamma(x) \delta(x), \quad \gamma, \delta \in \mathcal{A}_0(M^{(d+1)}).$	$m_\theta(\gamma \otimes \delta) = \gamma(x) \delta(x), \quad \gamma, \delta \in \mathcal{A}_0(M^{(d+1)}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00625 (33)	257	■ 1.000 ● K +0	$U_{\{\theta\}}(a, \Lambda) \varphi_{\{\theta\}} \left((a, \Lambda)^{-1} x \right) U_{\{\theta\}}(a, \Lambda)^{-1} = \varphi_{\{\theta\}}(x)$	$U_{\theta}(a, \Lambda) \varphi_{\theta} \left((a, \Lambda)^{-1} x \right) U_{\theta}(a, \Lambda)^{-1} = \varphi_{\theta}(x)$	$U_{\theta}(a, \Lambda) \varphi_{\theta}((a, \Lambda)^{-1} x) U_{\theta}(a, \Lambda)^{-1} = \varphi_{\theta}(x)$
cardona-qft-methods_ E00626 (34)	257	■ 1.000 ● N -1	$\varphi = \int d\mu(p) \left[a_p^{\dagger} e_p + a_p e_{-p} \right] = \varphi_{\theta}^{(-)} + \varphi_{\theta}^{(+)}, \quad d\mu(p) = \frac{d^d p}{2 p_0 }.$	$\varphi = \int d\mu(p) \left[a_p^{\dagger} e_p + a_p e_{-p} \right] = \varphi_{\theta}^{(-)} + \varphi_{\theta}^{(+)}, \quad d\mu(p) = \frac{d^d p}{2 p_0 }.$	$\varphi = \int d\mu(p) [a_p^{\dagger} e_p + a_p e_{-p}] = \varphi_{\theta}^{(-)} + \varphi_{\theta}^{(+)}, \quad d\mu(p) = \frac{d^d p}{2 p_0 }.$
cardona-qft-methods_ E00627 (35)	257	■ 1.000 ● K +0	$a_{\{p\}} 0\rangle = 0, \forall p.$	$a_p 0\rangle = 0, \forall p.$	$a_p 0\rangle = 0, \forall p.$
cardona-qft-methods_ E00628 (36)	257	■ 1.000 ● K +0	$U_{\{\theta\}}(a, \Lambda) a_{\{p\}}^{\dagger} U_{\{\theta\}}(a, \Lambda)^{-1} = a_{\Lambda p}^{\dagger}, \quad U_{\{\theta\}}(a, \Lambda) a_p U_{\{\theta\}}(a, \Lambda)^{-1} = a_{\Delta p}$	$U_{\theta}(a, \Lambda) a_p^{\dagger} U_{\theta}(a, \Lambda)^{-1} = a_{\Lambda p}^{\dagger}, \quad U_{\theta}(a, \Lambda) a_p U_{\theta}(a, \Lambda)^{-1} = a_{\Delta p}$	$U_{\theta}(a, \Lambda) a_p^{\dagger} U_{\theta}(a, \Lambda)^{-1} = a_{\Lambda p}^{\dagger}, \quad U_{\theta}(a, \Lambda) a_p U_{\theta}(a, \Lambda)^{-1} = a_{\Delta p}$
cardona-qft-methods_ E00629 (37)	257	■ 1.000 ● K +0	$U_{\{\theta\}}(a, \Lambda) 0\rangle = a_{\Lambda p}^{\dagger} 0\rangle.$	$U_{\theta}(a, \Lambda) 0\rangle = a_{\Lambda p}^{\dagger} 0\rangle.$	$U_{\theta}(a, \Lambda) 0\rangle = a_{\Lambda p}^{\dagger} 0\rangle.$
cardona-qft-methods_ E00630 (38)	258	■ 1.000 ● N +0	$\int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2}$	$\int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2}$	$\int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes e_{p_2}$
cardona-qft-methods_ E00631 (39)	258	■ 1.000 ● K +0	$\Delta_{\theta}(P_{\mu}) e_{p_1} \otimes e_{p_2} = (\mathbb{I} \otimes P_{\mu} + P_{\mu} \otimes \mathbb{I}) e_{p_1} \otimes e_{p_2} = - \left(\sum_i p_{i\mu} e_{p_1} \otimes e_{p_2} \right).$	$\Delta_{\theta}(P_{\mu}) e_{p_1} \otimes e_{p_2} = (\mathbb{I} \otimes P_{\mu} + P_{\mu} \otimes \mathbb{I}) e_{p_1} \otimes e_{p_2} = - \left(\sum_i p_{i\mu} e_{p_1} \otimes e_{p_2} \right).$	$\Delta_{\theta}(P_{\mu}) e_{p_1} \otimes e_{p_2} = (\mathbb{I} \otimes P_{\mu} + P_{\mu} \otimes \mathbb{I}) e_{p_1} \otimes e_{p_2} = - \left(\sum_i p_{i\mu} e_{p_1} \otimes e_{p_2} \right).$
cardona-qft-methods_ E00632 (40)	258	■ 1.000 ● N +1	$\left[P_{\mu}^{\theta}, a_p^{\dagger} \right] = P_{\mu} a_p^{\dagger}$	$\left[P_{\mu}^{\theta}, a_p^{\dagger} \right] = P_{\mu} a_p^{\dagger}$	$\left[P_{\mu}^{\theta}, a_p^{\dagger} \right] = P_{\mu} a_p^{\dagger},$
cardona-qft-methods_ E00633 (41)	258	■ 1.000 ● K +0	$\Delta_{\theta}(\Lambda) \triangleright e_{p_1} \otimes e_{p_2} = \mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_1} \otimes e_{p_2}.$	$\Delta_{\theta}(\Lambda) \triangleright e_{p_1} \otimes e_{p_2} = \mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_1} \otimes e_{p_2}.$	$\Delta_{\theta}(\Lambda) \triangleright e_{p_1} \otimes e_{p_2} = \mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_1} \otimes e_{p_2}.$
cardona-qft-methods_ E00635 (42)	258	■ 1.000 ● N +0	$a_{\{p\}}^{\dagger} = c_p^{\dagger} e^{\frac{i}{2} p \wedge P}$	$a_p^{\dagger} = c_p^{\dagger} e^{\frac{i}{2} p \wedge P}$	$a_p^{\dagger} = c_p^{\dagger} e^{\frac{i}{2} p \wedge P}$
cardona-qft-methods_ E00636 (43)	258	■ 1.000 ● K +0	$U_{\{\theta\}}(a, \Lambda) = U_0(a, \Lambda) = U(a, \Lambda).$	$U_{\theta}(a, \Lambda) = U_0(a, \Lambda) = U(a, \Lambda).$	$U_{\theta}(a, \Lambda) = U_0(a, \Lambda) = U(a, \Lambda).$
cardona-qft-methods_ E00637 (44)	258	■ 1.000 ●	$a_p = e^{-\frac{i}{2} p \wedge P} c_p = c_p e^{-\frac{i}{2} p \wedge P}$	$a_p = e^{-\frac{i}{2} p \wedge P} c_p = c_p e^{-\frac{i}{2} p \wedge P},$	$a_p = e^{-\frac{i}{2} p \wedge P} c_p = c_p e^{-\frac{i}{2} p \wedge P},$
cardona-qft-methods_ E00639 (46)	259	■ 1.000 ● W +0	$\begin{gathered} U(a, \Lambda) \varphi_{\{\theta\}} \left((a, \Lambda)^{-1} x_1 \right) \varphi_{\{\theta\}} \left((a, \Lambda)^{-1} x_2 \right) \dots \varphi_{\{\theta\}} \left((a, \Lambda)^{-1} x_N \right) U(a, \Lambda)^{-1} 0\rangle = \\ \varphi_{\theta}((a, \Lambda)(x_1)) \varphi_{\theta}((a, \Lambda)(x_2)) \dots \varphi_{\theta}((a, \Lambda)(x_N)) 0\rangle \end{gathered}$	$U(a, \Lambda) \varphi_{\theta}(x_1) \varphi_{\theta}(x_2) \dots \varphi_{\theta}(x_N) U(a, \Lambda)^{-1} 0\rangle = \\ \varphi_{\theta}((a, \Lambda)(x_1)) \varphi_{\theta}((a, \Lambda)(x_2)) \dots \varphi_{\theta}((a, \Lambda)(x_N)) 0\rangle$	$U(a, \Lambda) \varphi_{\theta}(x_1) \varphi_{\theta}(x_2) \dots \varphi_{\theta}(x_N) U(a, \Lambda)^{-1} 0\rangle = \\ \varphi_{\theta}((a, \Lambda)(x_1)) \varphi_{\theta}((a, \Lambda)(x_2)) \dots \varphi_{\theta}((a, \Lambda)(x_N)) 0\rangle$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0640 (47)	259	■ 1.000 ● N +0	$\begin{gathered} \tau_{\{\theta\}}=\mathcal{F}_{\{\theta\}}^{-1} \\ \tau_{\{\emptyset\}} \mathcal{F}, \quad \tau_{\{\emptyset\}} \alpha \otimes \beta := \beta \otimes \alpha, \\ \beta \otimes \alpha, \end{gathered}$	$\tau_{\theta} = \mathcal{F}_{\theta}^{-1} \tau_0 \mathcal{F},$ $\tau_0 \alpha \otimes \beta := \beta \otimes \alpha,$	$\tau_{\theta} = \mathcal{F}_{\theta}^{-1} \tau_0 \mathcal{F},$ $\tau_0 \alpha \otimes \beta := \beta \otimes \alpha,$
cardona-qft-methods_ EQ0641 (49)	259	■ 1.000 ● K +0	$\tau_{\{\emptyset\}}^2=\mathbb{I} \quad \tau_{\{\theta\}}^2=\mathbb{I}.$	$\tau_0^2 = \mathbb{I} \quad \tau_{\theta}^2 = \mathbb{I}.$	$\tau_0^2 = \mathbb{I} \quad \tau_{\theta}^2 = \mathbb{I}.$
cardona-qft-methods_ EQ0642 (50)	259	■ 1.000 ● N +0	$e_{\{p_1\}} \otimes S_{\{\theta\}} e_{\{p_2\}}=\frac{\mathbb{I}}{2} e_{p_1} \otimes e_{p_2}$	$e_{p_1} \otimes S_{\theta} e_{p_2} = \frac{\mathbb{I} \pm \tau_{\theta}}{2} e_{p_1} \otimes e_{p_2}$	$e_{p_1} \otimes S_{\theta} e_{p_2} = \frac{\mathbb{I} \pm \tau_{\theta}}{2} e_{p_1} \otimes e_{p_2}$
cardona-qft-methods_ EQ0644 (48)	259	■ 1.000 ● N +0	$\begin{gathered} \int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes S_{\theta} e_{\Lambda p_2} \\ = \int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e^{ip_2 \wedge p_1} e_{\Lambda p_2} \otimes S_{\theta} e_{\Lambda p_1} \end{gathered}$	$\int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes S_{\theta} e_{\Lambda p_2}$ $= \int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e^{ip_2 \wedge p_1} e_{\Lambda p_2} \otimes S_{\theta} e_{\Lambda p_1}$	$\int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e_{p_1} \otimes S_{\theta} e_{\Lambda p_2}$ $= \int \prod_i d\mu(p_i) a_{p_1}^{\dagger} a_{p_2}^{\dagger} 0\rangle e^{ip_2 \wedge p_1} e_{\Lambda p_2} \otimes S_{\theta} e_{\Lambda p_1}$
cardona-qft-methods_ EQ0645 (52)	260	■ 1.000 ● N +0	$=\int \prod_i d\mu(p_i) \left(\exp^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger} \right) 0\rangle e^{ip_1 \wedge p_2} e_{\Lambda p_1} \otimes S_{\theta} e_{\Lambda p_2}.$	$= \int \prod_i d\mu(p_i) \left(\exp^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger} \right) 0\rangle e^{ip_1 \wedge p_2} e_{\Lambda p_1} \otimes S_{\theta} e_{\Lambda p_2}.$	$= \int \prod_i d\mu(p_i) (\exp^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger}) 0\rangle e^{ip_1 \wedge p_2} e_{\Lambda p_1} \otimes S_{\theta} e_{\Lambda p_2}.$
cardona-qft-methods_ EQ0647 (54)	260	■ 1.000 ● N +0	$\tau_{\{\theta\}}^{ij} = \mathcal{F}_{\theta}^{-1} \tau_0^{ij} \mathcal{F} = \mathcal{F}_{\theta}^{-2} \tau_0^{ij},$	$\tau_{\theta}^{ij} = \mathcal{F}_{\theta}^{-1} \tau_0^{ij} \mathcal{F} = \mathcal{F}_{\theta}^{-2} \tau_0^{ij},$	$\tau_{\theta}^{ij} = \mathcal{F}_{\theta}^{-1} \tau_0^{ij} \mathcal{F} = \mathcal{F}_{\theta}^{-2} \tau_0^{ij},$
cardona-qft-methods_ EQ0648 (55)	260	■ 1.000 ● K +0	$\left(\tau_{\{\emptyset\}}^{ij}\right)^2=\mathbb{I}$	$\left(\tau_0^{ij}\right)^2 = \mathbb{I}$	$(\tau_0^{ij})^2 = \mathbb{I}$
cardona-qft-methods_ EQ0649 (56)	260	■ 1.000 ● N +0	$\tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} = \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2}$	$\tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} = \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2}$	$\tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} = \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2}$
cardona-qft-methods_ EQ0651 (58)	260	■ 1.000 ● N +1	$\varphi_{\theta}^* = \varphi_0^* e - \frac{i}{2} \overleftarrow{\partial} \wedge P$	$\varphi_{\theta}^* = \varphi_0^* e - \frac{i}{2} \overleftarrow{\partial} \wedge P$	$\varphi_{\theta}^* = \varphi_0^* e - \frac{i}{2} \overleftarrow{\partial} \wedge P.$
cardona-qft-methods_ EQ0653 (3)	261	■ 1.000 ● K +0	$\tau_{\theta} x_1, x_2\rangle = - x_1, x_2\rangle.$	$\tau_{\theta} x_1, x_2\rangle = - x_1, x_2\rangle.$	$\tau_{\theta} x_1, x_2\rangle = - x_1, x_2\rangle.$
cardona-qft-methods_ EQ0656 (6)	262	■ 1.000 ● N +0	$\langle [0 \mid \rho_{\theta}(x), \eta_{\theta}(y)] \mid 0 \rangle = 0.$	$\langle [0 \mid \rho_{\theta}(x), \eta_{\theta}(y)] \mid 0 \rangle = 0.$	$\langle [0 \mid \rho_{\theta}(x), \eta_{\theta}(y)] \mid 0 \rangle = 0.$
cardona-qft-methods_ EQ0657 (7)	262	■ 1.000 ●	$\omega_{\beta}(\rho_{\theta}(x)\eta_{\theta}(y)) = \frac{\mathrm{Tr}\left[e^{-\beta H}\rho_{\theta}(x)\eta_{\theta}(y)\right]}{\mathrm{Tr}\,e^{-\beta H}},$	$\omega_{\beta}(\rho_{\theta}(x)\eta_{\theta}(y)) = \frac{\mathrm{Tr}\left[e^{-\beta H}\rho_{\theta}(x)\eta_{\theta}(y)\right]}{\mathrm{Tr}\,e^{-\beta H}},$	$\omega_{\beta}(\rho_{\theta}(x)\eta_{\theta}(y)) = \frac{\mathrm{Tr}\left[e^{-\beta H}\rho_{\theta}(x)\eta_{\theta}(y)\right]}{\mathrm{Tr}\,e^{-\beta H}},$
cardona-qft-methods_ EQ0658 (8)	262	■ 1.000 ● K +0	$ds^2 = dt^2 - a(t)^2 \left[dr^2 + r^2 d\Omega^2 \right]$	$ds^2 = dt^2 - a(t)^2 \left[dr^2 + r^2 d\Omega^2 \right]$	$ds^2 = dt^2 - a(t)^2 \left[dr^2 + r^2 d\Omega^2 \right]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0662 (13)	263	■ 1.000 ● N +2	$U(\backslash\Lambda\mathrm{bda})\ S^{\wedge\{2\}}\ U^{\wedge\{-1\}}(\backslash\Lambda\mathrm{bda})=\frac{\mathrm{frac}\{-i^{\wedge\{2\}}\}{2!}}\ \int\mathrm{d}_{\mathrm{x}}^{\wedge\{4\}}\ \mathrm{d}^{\wedge\{4\}}\ y\left(\backslash\mathrm{theta}\backslash\mathrm{left}(\mathrm{x}_{\{0\}}-\mathrm{y}_{\{0\}}\backslash\mathrm{right})\right.\backslash\mathrm{left}[\mathrm{H}_{\{I\}}(\mathrm{x}),\ \mathrm{H}_{\{I\}}(\mathrm{y})\backslash\mathrm{right}]+\mathrm{H}_{\{I\}}(\mathrm{y})\ \mathrm{H}_{\{I\}}(\mathrm{x})\backslash\mathrm{right})$	$U(\Lambda)S^{(2)}U^{-1}(\Lambda)=\frac{-i^2}{2!}\int d_x^4d^4y\left(\theta\left(x_0-y_0\right)\left[H_I(x),H_I(y)\right]+H_I(y)H_I(x)\right)$	$U(\Lambda)S^{(2)}U^{-1}(\Lambda)=\frac{-i^2}{2!}\int d_x^4d^4y\left(\theta(x_0-y_0)[H_I(x),H_I(y)]+H_I(y)H_I(x)\right)$
cardona-qft-methods_ EQ0663 (14)	263	■ 1.000 ● K +0	$\backslash\mathrm{theta}\backslash\mathrm{left}(\mathrm{x}_{\{0\}}-\mathrm{y}_{\{0\}}\backslash\mathrm{right})\backslash\mathrm{left}[\mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{x}),\ \mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{y})\backslash\mathrm{right}]=\backslash\mathrm{theta}\backslash\mathrm{left}((\backslash\Lambda\mathrm{bda}\ \mathrm{x})_{\{0\}}-(\backslash\Lambda\mathrm{bda}\ \mathrm{y})_{\{0\}}\backslash\mathrm{right})\backslash\mathrm{left}[\mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{x}),\ \mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{y})\backslash\mathrm{right}]$	$\theta\left(x_0-y_0\right)\left[H_I(\Lambda x),H_I(\Lambda y)\right]=\theta\left((\Lambda x)_0-(\Lambda y)_0\right)\left[H_I(\Lambda x),H_I(\Lambda y)\right]$	$\theta(x_0-y_0)[H_I(\Lambda x),H_I(\Lambda y)]=\theta((\Lambda x)_0-(\Lambda y)_0)[H_I(\Lambda x),H_I(\Lambda y)]$
cardona-qft-methods_ EQ0665 (16)	263	■ 1.000 ● N +0	$\backslash\mathrm{left}[\mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{x}),\ \mathrm{H}_{\{I\}}(\backslash\Lambda\mathrm{bda}\ \mathrm{y})\backslash\mathrm{right}]=0,\ \backslash\mathrm{quad}\ \mathrm{x}\ \backslash\mathrm{sim}\ \mathrm{y},$	$[H_I(\Lambda x),H_I(\Lambda y)]=0,\quad x\sim y,$	$[H_I(\Lambda x),H_I(\Lambda y)]=0,\quad x\sim y,$
cardona-qft-methods_ EQ0666 (17)	263	■ 1.000 ● K +0	$\backslash\mathrm{theta}_{\{0\ i\}}\backslash\mathrm{left}(\backslash\mathrm{vec}\{\mathrm{P}_{\{i\ n\ c\}}\}\backslash\mathrm{right})_{\{i\}}$	$\theta_{0i}\left(\vec{P}_{inc}\right)_i$	$\theta_{0i}\left(\vec{P}_{inc}\right)_i$
cardona-qft-methods_ EQ0667 (18)	263	■ 1.000 ● K +0	$\backslash\mathrm{theta}_{\{0\ i\}}\backslash\mathrm{left}(\backslash\mathrm{vec}\{\mathrm{P}_{\{i\ n\ c\}}\}\backslash\mathrm{right})_{\{i\}}=0$	$\theta_{0i}\left(\vec{P}_{inc}\right)_i=0$	$\theta_{0i}\left(\vec{P}_{inc}\right)_i=0$
cardona-qft-methods_ EQ0668 (19)	264	■ 1.000 ● K +0	$\frac{\Delta T(\widehat{n})}{T}=\sum_{lm}a_{lm}Y_{lm}(\widehat{n})\ .$	$\frac{\Delta T(\widehat{n})}{T}=\sum_{lm}a_{lm}Y_{lm}(\widehat{n}).$	$\frac{\Delta T(\widehat{n})}{T}=\sum_{lm}a_{lm}Y_{lm}(\widehat{n}).$
cardona-qft-methods_ EQ0669 (20)	264	■ 1.000 ● K +0	$\mathrm{a}_{\{l\ m\}}=4\ \pi\mathrm{i}^{\wedge\{l\}}\ \int\mathrm{d}^{\{3\}}\ \mathrm{k}\frac{\{2\ \pi\}^{\{3\}}}{\backslash\Phi(\mathrm{k})\ \backslash\Delta_{\{l\}^{\wedge\{T\}}(\mathrm{k})\ Y_{\{l\ m\}}^{\wedge\{*\}}(\backslash\widehat{\mathrm{k}})}$	$a_{lm}=4\pi(-i)^l\int\frac{d^3k}{(2\pi)^3}\Phi(k)\Delta_l^T(k)Y_{lm}^*(\widehat{k})$	$a_{lm}=4\pi(-i)^l\int\frac{d^3k}{(2\pi)^3}\Phi(k)\Delta_l^T(k)Y_{lm}^*(\widehat{k})$
cardona-qft-methods_ EQ0671 (21)	264	■ 1.000 ● K +0	$\mathrm{P}_{\{\Phi\}}=\backslash\mathrm{left}.\frac{\{8\ \pi\ G\ H^{\{2\}}\}}{9\varepsilon\mathrm{k}^3}\bigg _{aH=\mathrm{k}}$	$P_\Phi=\left.\frac{8\pi GH^2}{9\varepsilon k^3}\right _{aH=k}$	$P_\Phi=\left.\frac{8\pi GH^2}{9\varepsilon k^3}\right _{aH=k}$
cardona-qft-methods_ EQ0672 (22)	264	■ 1.000 ● N +1	$\backslash\mathrm{varphi}(\backslash\alpha,\ \mathrm{x})=\backslash\mathrm{left}(\frac{\{\alpha\}}{\{3\}}\backslash\mathrm{right})^{\{ \frac{\{\alpha\}}{\{3\}}\}}\ \int\mathrm{d}^{\{3\}}\ \mathrm{x}^{\{\prime\}}\ \backslash\mathrm{varphi}_{\{\backslash\mathrm{theta}\}}(\mathrm{x})\ \backslash\mathrm{exp}\ ^{\{-\backslash\alpha\}\backslash\mathrm{left}(\mathrm{x}^{\{\prime\}}\backslash\mathrm{right})-\mathrm{x}\backslash\mathrm{right})^{\{2\}}}$	$\varphi(\alpha,x)=\left(\frac{\alpha}{3}\right)^{\frac{3}{2}}\int d^3x'\varphi_\theta(x)\exp^{-\alpha(x'-x)^2}$	$\varphi(\alpha,x)=\left(\frac{\alpha}{3}\right)^{\frac{3}{2}}\int d^3x'\varphi_\theta(x)\exp^{-\alpha(x'-x)^2},$
cardona-qft-methods_ EQ0673 (23)	264	■ 1.000 ● K +0	$\backslash\Delta\ \backslash\mathrm{varphi}_{\{\backslash\mathrm{theta}\}}\backslash\mathrm{left}(\backslash\alpha,\ \mathrm{x}_{\{1\}}\backslash\mathrm{right})\ \backslash\Delta\ \backslash\mathrm{varphi}_{\{\backslash\mathrm{theta}\}}\backslash\mathrm{left}(\backslash\alpha,\ \mathrm{x}_{\{2\}}\backslash\mathrm{right})\ \backslash\mathrm{geq}\backslash\mathrm{angle}\ 0 \ \backslash\mathrm{varphi}_{\{\backslash\mathrm{theta}\}}\backslash\mathrm{left}(\backslash\alpha,\ \mathrm{x}_{\{1\}}\backslash\mathrm{right}),\ \backslash\mathrm{varphi}_{\{\backslash\mathrm{theta}\}}\backslash\mathrm{left}(\backslash\alpha,\ \mathrm{x}_{\{2\}}\backslash\mathrm{right}) 0\backslash\mathrm{angle}$	$\Delta\varphi_\theta\left(\alpha,x_1\right)\Delta\varphi_\theta\left(\alpha,x_2\right)\geq\left\langle 0 \varphi_\theta\left(\alpha,x_1\right),\varphi_\theta\left(\alpha,x_2\right) 0\right\rangle$	$\Delta\varphi_\theta(\alpha,x_1)\Delta\varphi_\theta(\alpha,x_2)\geq\left\langle 0 \varphi_\theta(\alpha,x_1),\varphi_\theta(\alpha,x_2) 0\right\rangle$
cardona-qft-methods_ EQ0675 (1)	268	■ 1.000 ● K +0	$\backslash\mathrm{vec}\{S\}=\frac{\{1\}}{\{2\}}\ \backslash\mathrm{vec}\{\backslash\sigma\},$	$\vec{S}=\frac{1}{2}\vec{\sigma},$	$\vec{S}=\frac{1}{2}\vec{\sigma},$
cardona-qft-methods_ EQ0676 (2)	268	■ 1.000 ● K +0	$\backslash\sigma^{\wedge\{1\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ 0\ \&\ +1\ \backslash\backslash\ +1\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ \backslash\sigma^{\wedge\{2\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ 0\ \&\ -i\ \backslash\backslash\ +i\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ \backslash\sigma^{\wedge\{3\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ +1\ \&\ 0\ \backslash\backslash\ 0\ \&\ -1\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})\ .$	$\sigma^1=\left(\begin{array}{cc}0&+1\\+1&0\end{array}\right),\quad\sigma^2=\left(\begin{array}{cc}0&-i\\+i&0\end{array}\right),\quad\sigma^3=\left(\begin{array}{cc}+1&0\\0&-1\end{array}\right).$	$\sigma^1=\left(\begin{array}{cc}0&+1\\+1&0\end{array}\right),\quad\sigma^2=\left(\begin{array}{cc}0&-i\\+i&0\end{array}\right),\quad\sigma^3=\left(\begin{array}{cc}+1&0\\0&-1\end{array}\right).$
cardona-qft-methods_ EQ0677	268	■ 1.000 ● N +1	$\backslash\mathrm{left}[\backslash\sigma^{\wedge\{i\}},\ \backslash\sigma^{\wedge\{j\}}\backslash\mathrm{right}]=2\ i\ \epsilon^{\{i\ j\ k\}}\ \backslash\sigma^{\wedge\{k\}}$	$[\sigma^i,\sigma^j]=2i\epsilon^{ijk}\sigma^k$	$[\sigma^i,\sigma^j]=2i\epsilon^{ijk}\sigma^k,$
cardona-qft-methods_ EQ0679 (4)	269	■ 1.000 ● N +1	$s=0,\ \frac{\{1\}}{\{2\}},\ 1,\ \frac{\{3\}}{\{2\}},\ \backslash\mathrm{ldots},$	$s=0,\frac{1}{2},1,\frac{3}{2},\ldots,$	$s=0,\frac{1}{2},1,\frac{3}{2},\ldots,$
cardona-qft-methods_ EQ0680 (5)	269	■ 1.000 ●	$\backslash\mathrm{vec}\{S\}\ \backslash\mathrm{cdot}\ \backslash\mathrm{mid}\ \backslash\mathrm{text}\ \{\ \mathrm{state}\ \}\backslash\mathrm{rangle}=0,$	$\vec{S}\cdot\ \ \mathrm{state}\ \rangle=0,$	$\vec{S}\cdot \mathrm{state}\rangle=0,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0682 (6)	269	■ 1.000 ● N +1	$S^{\wedge\{ \mathrm{pm}\}}=S^{\wedge\{1\}} \mathrm{pm} \mathrm{~i} \mathrm{~} S^{\wedge\{2\}}$	$S^{\pm} = S^1 \pm iS^2$	$S^{\pm} = S^1 \pm iS^2 \, .$
cardona-qft-methods_ EQ0683	269	■ 1.000 ● N +1	$\left[S^{\wedge\{3\}}, S^{\wedge\{ \mathrm{pm}\}}\right]= \mathrm{pm} \mathrm{~} S^{\wedge\{ \mathrm{pm}\}}, \quad \left[S^{\wedge\{+\}}, S^{\wedge\{-}\}}\right]=2 \mathrm{~} S^{\wedge\{3\}}$	$[S^3, S^{\pm}] = \pm S^{\pm}, \quad [S^+, S^-] = 2S^3$	$[S^3, S^{\pm}] = \pm S^{\pm}, \quad [S^+, S^-] = 2S^3 \, .$
cardona-qft-methods_ EQ0685	269	■ 1.000 ● K +0	$ \uparrow\rangle\!\rangle=\mathrm{binom}\{1\}{0}, \quad \downarrow\rangle\!\rangle=\mathrm{binom}\{0\}{1} \mathrm{~} .$	$ \uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$	$ \uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \, .$
cardona-qft-methods_ EQ0686	269	■ 1.000 ● N +0	$S^{\wedge\{3\}} \uparrow\rangle\!\rangle=+\frac{1}{2} \uparrow\rangle\!\rangle, \quad S^{\wedge\{3\}} \downarrow\rangle\!\rangle=-\frac{1}{2} \downarrow\rangle\!\rangle,$	$S^3 \uparrow\rangle = +\frac{1}{2} \uparrow\rangle, \quad S^3 \downarrow\rangle = -\frac{1}{2} \downarrow\rangle,$	$S^3 \uparrow\rangle = +\frac{1}{2} \uparrow\rangle, \quad S^3 \downarrow\rangle = -\frac{1}{2} \downarrow\rangle,$
cardona-qft-methods_ EQ0688 (7)	270	■ 1.000 ● N +0	$\mathrm{mathcal}\{H\}=\mathrm{mathbb}\{C\}^{\wedge\{2\}} \otimes \cdots \otimes \mathrm{mathbb}\{C\}^{\wedge\{2\}}=\mathrm{left}(\mathrm{mathbb}\{C\}^{\wedge\{2\}}\mathrm{right})^{\wedge\{ \otimes L\}}$	$\mathcal{H} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{\otimes L}$	$\mathcal{H} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{\otimes L}$
cardona-qft-methods_ EQ0689	270	■ 1.000 ● N -1	$ \uparrow\rangle\!\rangle\cdots \downarrow\rangle\!\rangle=\mathrm{binom}\{1\}{0} \otimes \mathrm{binom}\{1\}{0} \otimes \cdots \otimes \mathrm{binom}\{0\}{1} \otimes \mathrm{binom}\{0\}{1} \otimes \mathrm{binom}\{1\}{0} \mathrm{~} .$	$ \uparrow\uparrow\cdots\downarrow\downarrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \cdots \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$	$ \uparrow\uparrow\cdots\downarrow\downarrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \cdots \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \, .$
cardona-qft-methods_ EQ0690	270	■ 1.000 ● N +0	$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0 \mathrm{~} .$	$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0.$	$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0 \, .$
cardona-qft-methods_ EQ0691	270	■ 1.000 ● N +0	$\mathrm{left}\{\! \uparrow\rangle\!\rangle\uparrow\rangle\!\rangle, \frac{1}{\sqrt{2}}\left(\uparrow\downarrow\rangle+ \downarrow\uparrow\rangle, \downarrow\downarrow\rangle, \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle- \downarrow\uparrow\rangle)\right\}.$	$\left\{ \uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}\left(\uparrow\downarrow\rangle+ \downarrow\uparrow\rangle, \downarrow\downarrow\rangle, \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle- \downarrow\uparrow\rangle)\right\}.$	$\{ \uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle+ \downarrow\uparrow\rangle, \downarrow\downarrow\rangle, \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle- \downarrow\uparrow\rangle)\} \, .$
cardona-qft-methods_ EQ0692 (8)	270	■ 1.000 ● N +1	$\hat{H}=4 \sum_{l=1}^L\mathrm{left}(\frac{1}{4}-\vec{S}_{-l} \cdot \vec{S}_{l+1} \mathrm{~} \cdot \vec{S}_{-l+1}\mathrm{right}),$	$\hat{H} = 4 \sum_{l=1}^L \left(\frac{1}{4} - \vec{S}_l \cdot \vec{S}_{l+1} \right),$	$\hat{H} = 4 \sum_{l=1}^L \left(\frac{1}{4} - \vec{S}_l \cdot \vec{S}_{l+1} \right) \, ,$
cardona-qft-methods_ EQ0693	270	■ 1.000 ● K +0	$\vec{S}_{L+1}=\vec{S}_1 \mathrm{~} .$	$\vec{S}_{L+1} = \vec{S}_1.$	$\vec{S}_{L+1} = \vec{S}_1 \, .$
cardona-qft-methods_ EQ0694 (9)	271	■ 1.000 ● N +0	$\hat{H} \psi\rangle=E \psi\rangle, \quad \psi\rangle\in\mathrm{left}(\mathrm{mathbb}\{C\}^{\wedge\{2\}}\mathrm{right})^{\wedge\{ \otimes L\}},$	$\hat{H} \psi\rangle = E \psi\rangle, \quad \psi\rangle \in (\mathbb{C}^2)^{\otimes L},$	$\hat{H} \psi\rangle = E \psi\rangle, \quad \psi\rangle \in (\mathbb{C}^2)^{\otimes L},$
cardona-qft-methods_ EQ0696	271	■ 1.000 ● K +0	$\mathrm{mathbb}\{P\}_{-l, \mathrm{~} l+1}=\mathrm{left}(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\mathrm{right}),$	$\mathbb{P}_{l, l+1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$	$\mathbb{P}_{l, l+1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$
cardona-qft-methods_ EQ0700	272	■ 1.000 ● K +0	$\hat{U}=\mathrm{mathbb}\{P\}_{\{1,2\}} \cdots \mathrm{mathbb}\{P\}_{\{L-1, \mathrm{~} L\}} \mathrm{~} .$	$\hat{U} = \mathbb{P}_{1,2} \cdots \mathbb{P}_{L-1,L}.$	$\hat{U} = \mathbb{P}_{1,2} \cdots \mathbb{P}_{L-1,L} \, .$
cardona-qft-methods_ EQ0701 (11)	272	■ 1.000 ● K +0	$\hat{U}^L=\mathrm{mathbb}\{I\} \mathrm{~} .$	$\hat{U}^L = \mathbb{I}.$	$\hat{U}^L = \mathbb{I} \, .$
cardona-qft-methods_ EQ0702 (12)	272	■ 1.000 ● K +0	$U=\mathrm{e}^{\{2 \mathrm{~} \pi \mathrm{~} \mathrm{i} \mathrm{~} \mathrm{n} \mathrm{~} / \mathrm{~} L\}},$	$U = e^{2\pi i n/L},$	$U = e^{2\pi i n/L},$
cardona-qft-methods_ EQ0703 (13)	272	■ 1.000 ● N +2	$\mathrm{left}(\frac{u_{\mathrm{~} k}+\frac{i}{2}}{u_{\mathrm{~} k}-\frac{i}{2}}\mathrm{right})^{\wedge\{L\}}=\prod_{j\neq k}^M\frac{u_{\mathrm{~} k}-u_{\mathrm{~} j}+i}{u_{\mathrm{~} k}-u_{\mathrm{~} j}-i}, \quad k=1, \mathrm{~} \ldots, \mathrm{~} M$	$\left(\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\right)^L = \prod_{j\neq k}^M \frac{u_k-u_j+i}{u_k-u_j-i}, \quad k=1, \dots, M$	$\left(\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\right)^L = \prod_{j\neq k}^M \frac{u_k-u_j+i}{u_k-u_j-i}, \quad k=1, \dots, M \, .$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0704	272	■ 1.000	$E=2\sum_{k=1}^M\frac{1}{u_k^2+\frac{1}{4}},\quad U=\prod_{k=1}^M\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}},$	$E=2\sum_{k=1}^M\frac{1}{u_k^2+\frac{1}{4}},\quad U=\prod_{k=1}^M\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}.$	$E=2\sum_{k=1}^M\frac{1}{u_k^2+\frac{1}{4}},\quad U=\prod_{k=1}^M\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}.$
cardona-qft-methods_ EQ0705 (1)	272	■ 1.000 ● K +0	$ \psi\rangle=\sum_{1\leq l_1<\dots<l_M\leq L}\psi(l_1,\dots,l_M)S_{l_1}^-\cdots S_{l_M}^- 0\rangle,$	$ \psi\rangle=\sum_{1\leq l_1<\dots<l_M\leq L}\psi(l_1,\dots,l_M)S_{l_1}^-\cdots S_{l_M}^- 0\rangle,$	$ \psi\rangle=\sum_{1\leq l_1<\dots<l_M\leq L}\psi(l_1,\dots,l_M)S_{l_1}^-\cdots S_{l_M}^- 0\rangle,$
cardona-qft-methods_ EQ0706 (2)	273	■ 1.000 ● N +1	$\psi(l_1,\dots,l_M)=\sum_{\tau\in S_n}A(\tau)e^{ip_{\tau_1}l_1+\dots+ip_{\tau_M}l_M}$	$\psi(l_1,\dots,l_M)=\sum_{\tau\in S_n}A(\tau)e^{ip_{\tau_1}l_1+\dots+ip_{\tau_M}l_M}$	$\psi(l_1,\dots,l_M)=\sum_{\tau\in S_n}A(\tau)e^{ip_{\tau_1}l_1+\dots+ip_{\tau_M}l_M},$
cardona-qft-methods_ EQ0707	273	■ 1.000 ● N +0	$A(\tau)=\operatorname{sgn}(\tau)\prod_{j<k}(e^{ip_k+ip_j}-2e^{ip_k}+1),$	$A(\tau)=\operatorname{sgn}(\tau)\prod_{j<k}(e^{ip_k+ip_j}-2e^{ip_k}+1),$	$A(\tau)=\operatorname{sgn}(\tau)\prod_{j<k}(e^{ip_k+ip_j}-2e^{ip_k}+1),$
cardona-qft-methods_ EQ0708	273	■ 1.000 ● K +0	$E=\sum_{k=1}^M8\sin^2\left(\frac{p_k}{2}\right).$	$E=\sum_{k=1}^M8\sin^2\left(\frac{p_k}{2}\right).$	$E=\sum_{k=1}^M8\sin^2\left(\frac{p_k}{2}\right).$
cardona-qft-methods_ EQ0710	273	■ 1.000 ● N +0	$S(p_k,p_j)=-\frac{e^{ip_k+ip_j}-2e^{ip_k}+1}{e^{ip_k+ip_j}-2e^{ip_j}+1}.$	$S(p_k,p_j)=-\frac{e^{ip_k+ip_j}-2e^{ip_k}+1}{e^{ip_k+ip_j}-2e^{ip_j}+1}.$	$S(p_k,p_j)=-\frac{e^{ip_k+ip_j}-2e^{ip_k}+1}{e^{ip_k+ip_j}-2e^{ip_j}+1}.$
cardona-qft-methods_ EQ0711	273	■ 1.000 ● W +0	$e^{ip_k}=\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\iff\frac{1}{2}\cot\left(\frac{p_k}{2}\right)=u_k,$	$e^{ip_k}=\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\iff\frac{1}{2}\cot\left(\frac{p_k}{2}\right)=u_k,$	$e^{ip_k}=\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}}\iff\frac{1}{2}\cot\left(\frac{p_k}{2}\right)=u_k,$
cardona-qft-methods_ EQ0713	274	■ 1.000 ● N +2	$R_{a,b}(u-u')R_{a,c}(u)R_{b,c}(u')=R_{b,c}(u')R_{a,c}(u)R_{a,b}(u-u')$	$R_{a,b}(u-u')R_{a,c}(u)R_{b,c}(u')=R_{b,c}(u')R_{a,c}(u)R_{a,b}(u-u')$	$R_{a,b}(u-u')R_{a,c}(u)R_{b,c}(u')=R_{b,c}(u')R_{a,c}(u)R_{a,b}(u-u')$
cardona-qft-methods_ EQ0714	274	■ 1.000 ● N +0	$R_{a,b}(u-u')=$	$R_{a,b}(u-u')=$	$R_{a,b}(u-u')=$
cardona-qft-methods_ EQ0715	275	■ 1.000 ● K +0	$R_{a,b}(u-u')L_{a,l}(u)L_{b,l}(u')=L_{b,l}(u')L_{a,l}(u)R_{a,b}(u-u').$	$R_{a,b}(u-u')L_{a,l}(u)L_{b,l}(u')=L_{b,l}(u')L_{a,l}(u)R_{a,b}(u-u').$	$R_{a,b}(u-u')L_{a,l}(u)L_{b,l}(u')=L_{b,l}(u')L_{a,l}(u)R_{a,b}(u-u').$
cardona-qft-methods_ EQ0719	276	■ 1.000 ● K +0	$\hat{T}(u)\hat{T}(u')=\hat{T}(u')\hat{T}(u)$	$\hat{T}(u)\hat{T}(u')=\hat{T}(u')\hat{T}(u)$	$\hat{T}(u)\hat{T}(u')=\hat{T}(u')\hat{T}(u)$
cardona-qft-methods_ EQ0720	276	■ 1.000 ● K +0	$[\hat{T}(u),\hat{T}(u')]=0.$	$[\hat{T}(u),\hat{T}(u')]=0.$	$[\hat{T}(u),\hat{T}(u')]=0.$
cardona-qft-methods_ EQ0721	276	■ 1.000 ● N +0	$\frac{1}{i^L}\hat{T}\left(\frac{i}{2}\right)=\operatorname{tr}_a(P_{a,L}\cdots P_{a,L-1}\cdots P_{a,1})=P_{12}P_{23}\cdots P_{L-1,L}=\hat{U}=e^{i\hat{P}}$	$\frac{1}{i^L}\hat{T}\left(\frac{i}{2}\right)=\operatorname{tr}_a(P_{a,L}\cdots P_{a,L-1}\cdots P_{a,1})=P_{12}P_{23}\cdots P_{L-1,L}=\hat{U}=e^{i\hat{P}}$	$\frac{1}{i^L}\hat{T}\left(\frac{i}{2}\right)=\operatorname{tr}_a(P_{a,L}\cdots P_{a,L-1}\cdots P_{a,1})=P_{12}P_{23}\cdots P_{L-1,L}=\hat{U}=e^{i\hat{P}},$
cardona-qft-methods_ EQ0722	276	■ 1.000 ● N +0	$i\hat{T}(u)^{-1}\frac{\mathrm{d}}{\mathrm{d}u}\hat{T}(u)\Big _{u=\frac{i}{2}}=\sum_{l=1}^LP_{l-1,l}.$	$i\hat{T}(u)^{-1}\frac{\mathrm{d}}{\mathrm{d}u}\hat{T}(u)\Big _{u=\frac{i}{2}}=\sum_{l=1}^LP_{l-1,l}.$	$i\hat{T}(u)^{-1}\frac{\mathrm{d}}{\mathrm{d}u}\hat{T}(u)\Big _{u=\frac{i}{2}}=\sum_{l=1}^LP_{l-1,l}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0723	276	1.000 N +0	$\hat{H}=2\sum_{l=1}^L\left(1_{l,l+1}-P_{l,l+1}\right)=2L-2i\frac{\mathrm{d}}{\mathrm{d}u}\log\hat{T}(u)\Big _{u=\frac{i}{2}}.$	$\hat{H}=2\sum_{l=1}^L(1_{l,l+1}-P_{l,l+1})=2L-2i\frac{\mathrm{d}}{\mathrm{d}u}\log\hat{T}(u)\Big _{u=\frac{i}{2}}.$	$\hat{H}=2\sum_{l=1}^L(1_{l,l+1}-P_{l,l+1})=2L-2i\frac{\mathrm{d}}{\mathrm{d}u}\log\hat{T}(u)\Big _{u=\frac{i}{2}}.$
cardona-qft-methods_ EQ0724	276	1.000	$\log\hat{T}(u)=\hat{Q}_1+\left(u-\frac{i}{2}\right)\hat{Q}_2+\left(u-\frac{i}{2}\right)^2\hat{Q}_3+\left(u-\frac{i}{2}\right)^3\hat{Q}_4+\ldots,$	$\log\hat{T}(u)=\hat{Q}_1+\left(u-\frac{i}{2}\right)\hat{Q}_2+\left(u-\frac{i}{2}\right)^2\hat{Q}_3+\left(u-\frac{i}{2}\right)^3\hat{Q}_4+\ldots,$	$\log\hat{T}(u)=\hat{Q}_1+\left(u-\frac{i}{2}\right)\hat{Q}_2+\left(u-\frac{i}{2}\right)^2\hat{Q}_3+\left(u-\frac{i}{2}\right)^3\hat{Q}_4+\ldots,$
cardona-qft-methods_ EQ0725	276	1.000 K +0	$\hat{Q}_1\propto\hat{P},\quad\hat{Q}_2\propto\hat{H}-2L.$	$\hat{Q}_1\propto\hat{P},\quad\hat{Q}_2\propto\hat{H}-2L.$	$\hat{Q}_1\propto\hat{P},\quad\hat{Q}_2\propto\hat{H}-2L.$
cardona-qft-methods_ EQ0726	276	1.000 N +1	$\left[\hat{Q}_n,\hat{Q}_m\right]=0$	$\left[\hat{Q}_n,\hat{Q}_m\right]=0$	$[\hat{Q}_n,\hat{Q}_m]=0.$
cardona-qft-methods_ EQ0727	277	1.000 K +0	$\hat{T}(u) \psi\rangle=T(u) \psi\rangle.$	$\hat{T}(u) \psi\rangle=T(u) \psi\rangle.$	$\hat{T}(u) \psi\rangle=T(u) \psi\rangle.$
cardona-qft-methods_ EQ0729	277	1.000 N +1	$ \psi\rangle=\hat{B}(u_1)\cdots\hat{B}(u_M) 0\rangle$	$ \psi\rangle=\hat{B}(u_1)\cdots\hat{B}(u_M) 0\rangle$	$ \psi\rangle=\hat{B}(u_1)\cdots\hat{B}(u_M) 0\rangle$
cardona-qft-methods_ EQ0731	277	1.000 N +0	$T(u)=\left(u+\frac{i}{2}\right)^L\prod_{j=1}^M\frac{u-u_j-i}{u-u_j}+\left(u-\frac{i}{2}\right)^L\prod_{j=1}^M\frac{u-u_j+i}{u-u_j}.$	$T(u)=\left(u+\frac{i}{2}\right)^L\prod_{j=1}^M\frac{u-u_j-i}{u-u_j}+\left(u-\frac{i}{2}\right)^L\prod_{j=1}^M\frac{u-u_j+i}{u-u_j}.$	$T(u)=(u+\frac{i}{2})^L\prod_{j=1}^M\frac{u-u_j-i}{u-u_j}+(u-\frac{i}{2})^L\prod_{j=1}^M\frac{u-u_j+i}{u-u_j}.$
cardona-qft-methods_ EQ0732	278	1.000 N +1	$Q(u)=\prod_{j=1}^M(u-u_j).$	$Q(u)=\prod_{j=1}^M(u-u_j).$	$Q(u)=\prod_{j=1}^M(u-u_j).$
cardona-qft-methods_ EQ0733	278	1.000 K +0	$T(u)Q(u)=\left(u+\frac{i}{2}\right)^LQ(u-i)+\left(u-\frac{i}{2}\right)^LQ(u+i).$	$T(u)Q(u)=\left(u+\frac{i}{2}\right)^LQ(u-i)+\left(u-\frac{i}{2}\right)^LQ(u+i).$	$T(u)Q(u)=(u+\frac{i}{2})^LQ(u-i)+(u-\frac{i}{2})^LQ(u+i).$
cardona-qft-methods_ EQ0737 (7)	281	1.000 N +0	$[\mathcal{D},P_\mu]=iP_\mu,\quad[\mathcal{D},K_\mu]=-iK_\mu$	$[\mathcal{D},P_\mu]=iP_\mu,\quad[\mathcal{D},K_\mu]=-iK_\mu$	$[\mathcal{D},P_\mu]=iP_\mu,\quad[\mathcal{D},K_\mu]=-iK_\mu$
cardona-qft-methods_ EQ0738	281	1.000 N +1	$L=\begin{bmatrix} S^3 & S^+ \\ S^- & -S^3 \end{bmatrix}.$	$L=\begin{bmatrix} S^3 & S^+ \\ S^- & -S^3 \end{bmatrix}.$	$L=\begin{bmatrix} S^3 & S^+ \\ S^- & -S^3 \end{bmatrix}.$
cardona-qft-methods_ EQ0739 (9)	282	1.000 N +1	$\begin{aligned} \left\{Q_{a\alpha},\bar{Q}_{\dot{\alpha}}^b\right\} &= (\sigma^\mu)_{\alpha\dot{\alpha}}P_\mu\delta_a^b \\ [\mathcal{D},Q_{a\alpha}] &= \frac{i}{2}Q_{a\alpha} \end{aligned}$	$\begin{aligned} \{Q_{a\alpha},\bar{Q}_{\dot{\alpha}}^b\} &= (\sigma^\mu)_{\alpha\dot{\alpha}}P_\mu\delta_a^b \\ [\mathcal{D},Q_{a\alpha}] &= \frac{i}{2}Q_{a\alpha} \end{aligned}$	$\begin{aligned} \{Q_{a\alpha},\bar{Q}_{\dot{\alpha}}^b\} &= (\sigma^\mu)_{\alpha\dot{\alpha}}P_\mu\delta_a^b \\ [\mathcal{D},Q_{a\alpha}] &= \frac{i}{2}Q_{a\alpha} \end{aligned}$
cardona-qft-methods_ EQ0741	282	1.000 N -1	$\mathcal{O}\equiv\mathcal{O}_{(x)}=\mathrm{Tr}\left(ZX\bar{Z}(DY)\left(D^5Y\right)\psi\left(D^3\psi\right)\bar{\psi}F\left(D^7\bar{F}\right)\dots\right)_{(x)}.$	$\mathcal{O}\equiv\mathcal{O}_{(x)}=\mathrm{Tr}\left(ZX\bar{Z}(DY)\left(D^5Y\right)\psi\left(D^3\psi\right)\bar{\psi}F\left(D^7\bar{F}\right)\dots\right)_{(x)}.$	$\mathcal{O}\equiv\mathcal{O}_{(x)}=\mathrm{Tr}\left(ZX\bar{Z}(DY)(D^5Y)\psi(D^3\psi)\bar{\psi}F(D^7\bar{F})\dots\right)_{(x)}.$
cardona-qft-methods_ EQ0742	282	1.000 K +0	$[\mathcal{D},\mathcal{O}]=i\Delta\mathcal{O},$	$[\mathcal{D},\mathcal{O}]=i\Delta\mathcal{O},$	$[\mathcal{D},\mathcal{O}]=i\Delta\mathcal{O},$
cardona-qft-methods_ EQ0743	282	1.000 N +0	$\gamma=g_{YM}^2\,\underbrace{\gamma_1}_{1\text{-loop}}+g_{YM}^4\,\underbrace{\gamma_2}_{2\text{-loop}}+\dots$	$\gamma=g_{YM}^2\,\underbrace{\gamma_1}_{1\text{-loop}}+g_{YM}^4\,\underbrace{\gamma_2}_{2\text{-loop}}+\dots$	$\gamma=g_{YM}^2\,\underbrace{\gamma_1}_{1\text{-loop}}+g_{YM}^4\,\underbrace{\gamma_2}_{2\text{-loop}}+\dots$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured
[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00746	283	■1.000 ● N +0	$\left\langle \left(\bar{Z}^j \right) (x) \left(Z^j \right) (y) \right\rangle = \frac{1}{(x-y)^{2j}}$		
cardona-qft-methods_ E00750	284	■1.000 ● W +0	$\frac{p}{s} \bigg _{(2,2)} = \begin{array}{ c c } \hline c c \\ \hline \frac{u}{(2,2)} & \text{supercharges} \\ \hline \text{supercharges} & \frac{u}{(4)} \\ \hline \end{array}$	$\mathfrak{psu}(2,2 4) = \begin{array}{ c c } \hline \mathfrak{u}(2,2) & \text{supercharges} \\ \hline \text{supercharges} & \mathfrak{u}(4) \\ \hline \end{array}$	$\mathfrak{psu}(2,2 4) = \begin{array}{ c c } \hline \mathfrak{u}(2,2) & \text{supercharges} \\ \hline \text{supercharges} & \mathfrak{u}(4) \\ \hline \end{array}$
cardona-qft-methods_ E00752 (1)	284	■1.000 ●	$S = \frac{R^2}{\alpha'} \int d\tau \frac{d\sigma}{2\pi} \left(\frac{1}{2} \partial_a Z^M \partial^a Z_M + \frac{1}{2} \partial_a Y^N \partial^a Y_N \right) + \text{fermions}$	$S = \frac{R^2}{\alpha'} \int d\tau \frac{d\sigma}{2\pi} \left(\frac{1}{2} \partial_a Z^M \partial^a Z_M + \frac{1}{2} \partial_a Y^N \partial^a Y_N \right) + \text{fermions}$	$S = \frac{R^2}{\alpha'} \int d\tau \frac{d\sigma}{2\pi} \left(\frac{1}{2} \partial_a Z^M \partial^a Z_M + \frac{1}{2} \partial_a Y^N \partial^a Y_N \right) + \text{fermions}$
cardona-qft-methods_ E00753	285	■1.000 ● N +0	$\sqrt{\lambda} = \frac{R^2}{\alpha'} \quad , \quad 4\pi g_s = \frac{\lambda}{N}.$	$\sqrt{\lambda} = \frac{R^2}{\alpha'} \quad , \quad 4\pi g_s = \frac{\lambda}{N}.$	$\sqrt{\lambda} = \frac{R^2}{\alpha'} \quad , \quad 4\pi g_s = \frac{\lambda}{N}.$
cardona-qft-methods_ E00754	285	■1.000 ● N +0	$S_{SYM} = \int \frac{N}{\lambda} \frac{d^4 x}{4\pi^2} \text{Tr}\{\dots\} \longleftrightarrow S_{\sigma\text{-model}} = \sqrt{\lambda} \int d\tau \frac{d\sigma}{2\pi} \{\dots\},$	$S_{SYM} = \int \frac{N}{\lambda} \frac{d^4 x}{4\pi^2} \text{Tr}\{\dots\} \longleftrightarrow S_{\sigma\text{-model}} = \sqrt{\lambda} \int d\tau \frac{d\sigma}{2\pi} \{\dots\},$	$S_{SYM} = \int \frac{N}{\lambda} \frac{d^4 x}{4\pi^2} \text{Tr}\{\dots\} \longleftrightarrow S_{\sigma\text{-model}} = \sqrt{\lambda} \int d\tau \frac{d\sigma}{2\pi} \{\dots\},$
cardona-qft-methods_ E00755	285	■1.000 ● K +0	$Z_4 + Z_5 = \frac{R}{z}, \quad (Z_0, Z_1, Z_2, Z_3) = \frac{R}{z} x_\mu, \quad ds_{AdS_5}^2 = \frac{R^2}{z^2} (dx^\mu dx_\mu + dz^2).$	$Z_4 + Z_5 = \frac{R}{z}, \quad (Z_0, Z_1, Z_2, Z_3) = \frac{R}{z} x_\mu, \quad ds_{AdS_5}^2 = \frac{R^2}{z^2} (dx^\mu dx_\mu + dz^2).$	$Z_4 + Z_5 = \frac{R}{z}, \quad (Z_0, Z_1, Z_2, Z_3) = \frac{R}{z} x_\mu, \quad ds_{AdS_5}^2 = \frac{R^2}{z^2} (dx^\mu dx_\mu + dz^2).$
cardona-qft-methods_ E00756	286	■1.000 ● N +1	$Y_1 + iY_2 = r_1 e^{i\phi_1}, \quad Y_3 + iY_4 = r_2 e^{i\phi_2}, \quad Y_5 + iY_6 = r_3 e^{i\phi_3}$	$Y_1 + iY_2 = r_1 e^{i\phi_1}, \quad Y_3 + iY_4 = r_2 e^{i\phi_2}, \quad Y_5 + iY_6 = r_3 e^{i\phi_3}$	$Y_1 + iY_2 = r_1 e^{i\phi_1}, \quad Y_3 + iY_4 = r_2 e^{i\phi_2}, \quad Y_5 + iY_6 = r_3 e^{i\phi_3},$
cardona-qft-methods_ E00757	286	■1.000 ● N +1	$Z_0 - iZ_5 = \rho_3 e^{it}, \quad Z_1 + iZ_2 = \rho_1 e^{i\alpha_1}, \quad ZY_3 + iZ_4 = \rho_2 e^{i\alpha_2}$	$Z_0 - iZ_5 = \rho_3 e^{it}, \quad Z_1 + iZ_2 = \rho_1 e^{i\alpha_1}, \quad ZY_3 + iZ_4 = \rho_2 e^{i\alpha_2}$	$Z_0 - iZ_5 = \rho_3 e^{it}, \quad Z_1 + iZ_2 = \rho_1 e^{i\alpha_1}, \quad ZY_3 + iZ_4 = \rho_2 e^{i\alpha_2}.$
cardona-qft-methods_ E00760 (1)	292	■1.000 ● N +1	$S = -\frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} \eta_{MN} \partial_\alpha X^M \partial_\beta X^N$	$S = -\frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} \eta_{MN} \partial_\alpha X^M \partial_\beta X^N$	$S = -\frac{1}{4\pi\alpha'} \int d\sigma d\tau \sqrt{\gamma} \gamma^{\alpha\beta} \eta_{MN} \partial_\alpha X^M \partial_\beta X^N,$
cardona-qft-methods_ E00761 (2)	292	■1.000 ● N +0	$T = \frac{1}{2\pi\alpha'} = \frac{1}{2\pi l_s^2}.$	$T = \frac{1}{2\pi\alpha'} = \frac{1}{2\pi l_s^2}.$	$T = \frac{1}{2\pi\alpha'} = \frac{1}{2\pi l_s^2}.$
cardona-qft-methods_ E00762 (3)	293	■1.000 ● N +1	$\left(\frac{\partial^2}{\partial \sigma^2} - \frac{\partial^2}{\partial \tau^2} \right) X^M(\tau, \sigma) = 0$	$\left(\frac{\partial^2}{\partial \sigma^2} - \frac{\partial^2}{\partial \tau^2} \right) X^M(\tau, \sigma) = 0$	$\left(\frac{\partial^2}{\partial \sigma^2} - \frac{\partial^2}{\partial \tau^2} \right) X^M(\tau, \sigma) = 0.$
cardona-qft-methods_ E00763 (4)	293	■1.000 ● N +1	$\frac{\partial}{\partial \sigma^+} \frac{\partial}{\partial \sigma^-} X^M(\tau, \sigma) = 0$	$\frac{\partial}{\partial \sigma^+} \frac{\partial}{\partial \sigma^-} X^M(\tau, \sigma) = 0$	$\frac{\partial}{\partial \sigma^+} \frac{\partial}{\partial \sigma^-} X^M(\tau, \sigma) = 0,$
cardona-qft-methods_ E00764 (5)	293	■1.000 ● N +0	$X^M(\tau, \sigma) = X_R^M(\sigma^-) + X_L^M(\sigma^+).$	$X^M(\tau, \sigma) = X_R^M(\sigma^-) + X_L^M(\sigma^+).$	$X^M(\tau, \sigma) = X_R^M(\sigma^-) + X_L^M(\sigma^+).$

Conf.

MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

LaTeX source

MathPix's reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00769 (10)	294	1.000 N +0	$M^2=\frac{2}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}-2\right)$	$M^2=\frac{2}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}-2\right)$	$M^2=\frac{2}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}-2\right)$
cardona-qft-methods_ E00770 (11)	294	1.000 N +1	$\hat{L}_0=\frac{1}{2}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\right)$	$\hat{L}_0=\frac{1}{2}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\right)$	$\hat{L}_0=\frac{1}{2}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\right)$
cardona-qft-methods_ E00771 (12)	294	1.000 N +0	$\left \xi_{MN}\right\rangle\equiv\xi_M\tilde{\xi}_N\alpha_1^M\tilde{\alpha}_1^N 0\rangle$	$ \xi_{MN}\rangle\equiv\xi_M\tilde{\xi}_N\alpha_1^M\tilde{\alpha}_1^N 0\rangle$	$ \xi_{MN}\rangle\equiv\xi_M\tilde{\xi}_N\alpha_1^M\tilde{\alpha}_1^N 0\rangle$
cardona-qft-methods_ E00772 (13)	294	1.000 N +1	$\xi_{MN}=\xi_{MN}^s+\xi_t\eta_{MN}+\xi_{MN}^a$	$\xi_{MN}=\xi_{MN}^s+\xi_t\eta_{MN}+\xi_{MN}^a$	$\xi_{MN}=\xi_{MN}^s+\xi_t\eta_{MN}+\xi_{MN}^a$
cardona-qft-methods_ E00774 (15)	295	1.000	$S=-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\,\mathrm{d}\tau\sqrt{\gamma}\gamma^{\alpha\beta}G_{MN}(X)\partial_\alpha X^M\partial_\beta X^N$	$S=-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\,\mathrm{d}\tau\sqrt{\gamma}\gamma^{\alpha\beta}G_{MN}(X)\partial_\alpha X^M\partial_\beta X^N$	$S=-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\gamma^{\alpha\beta}G_{MN}(X)\partial_\alpha X^M\partial_\beta X^N$
cardona-qft-methods_ E00776 (17)	295	1.000 C +4	$\begin{aligned} S=&-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\,\mathrm{d}\tau\sqrt{\gamma}\left[\left(\gamma^{\alpha\beta}g_{MN}(X)+\mathrm{i}\epsilon^{\alpha\beta}B_{MN}(X)\right)\partial_\alpha X^M\partial_\beta X^N\right.\\&\left.+\alpha'\Phi R\right]\end{aligned}$	$S=-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\,\mathrm{d}\tau\sqrt{\gamma}\left[\left(\gamma^{\alpha\beta}g_{MN}(X)+\mathrm{i}\epsilon^{\alpha\beta}B_{MN}(X)\right)\partial_\alpha X^M\partial_\beta X^N+\alpha'\Phi R\right]$	$S=-\frac{1}{4\pi\alpha'}\int\mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\left[\left(\gamma^{\alpha\beta}g_{MN}(X)+\mathrm{i}\epsilon^{\alpha\beta}B_{MN}(X)\right)\partial_\alpha X^M\partial_\beta X^N+\alpha'\Phi R\right]$
cardona-qft-methods_ E00777 (18)	296	1.000 K +0	$\chi=\frac{1}{4\pi}\int d^2\sigma(-\gamma)^{1/2}R.$	$\chi=\frac{1}{4\pi}\int d^2\sigma(-\gamma)^{1/2}R.$	$\chi=\frac{1}{4\pi}\int d^2\sigma(-\gamma)^{1/2}R.$
cardona-qft-methods_ E00782 (23)	298	1.000 N +0	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-1\right).$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-1\right).$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-1\right).$
cardona-qft-methods_ E00783 (24)	298	1.000 N +0	$ \xi_M\rangle\equiv\xi_M\alpha_1^M 0\rangle$	$ \xi_M\rangle\equiv\xi_M\alpha_1^M 0\rangle$	$ \xi_M\rangle\equiv\xi_M\alpha_1^M 0\rangle$
cardona-qft-methods_ E00784 (25)	298	1.000 N +0	$\int_{\partial M}d\tau A_M\partial_\tau X^M$	$\int_{\partial M}d\tau A_M\partial_\tau X^M$	$\int_{\partial M}d\tau A_M\partial_\tau X^M$
cardona-qft-methods_ E00785 (26)	298	1.000 N +1	$S=-\frac{1}{4}\int\mathrm{d}^D X e^{-\Phi}F_{MN}F^{MN}+O\left(\alpha'\right)$	$S=-\frac{1}{4}\int\mathrm{d}^D X e^{-\Phi}F_{MN}F^{MN}+O\left(\alpha'\right)$	$S=-\frac{1}{4}\int\mathrm{d}^D X e^{-\Phi}F_{MN}F^{MN}+O\left(\alpha'\right)$
cardona-qft-methods_ E00786 (27)	299	1.000 N +0	$S=\frac{1}{4\pi}\int_M\mathrm{d}^2\sigma\eta_{MN}\left[\frac{1}{\alpha'}\partial X^M\bar{\partial}X^N+\Psi^M\bar{\partial}\Psi^N+\tilde{\Psi}^M\bar{\partial}\tilde{\Psi}^N\right]$	$S=\frac{1}{4\pi}\int_M\mathrm{d}^2\sigma\eta_{MN}\left[\frac{1}{\alpha'}\partial X^M\bar{\partial}X^N+\Psi^M\bar{\partial}\Psi^N+\tilde{\Psi}^M\bar{\partial}\tilde{\Psi}^N\right]$	$S=\frac{1}{4\pi}\int_M\mathrm{d}^2\sigma\eta_{MN}\left[\frac{1}{\alpha'}\partial X^M\bar{\partial}X^N+\Psi^M\bar{\partial}\Psi^N+\tilde{\Psi}^M\bar{\partial}\tilde{\Psi}^N\right]$
cardona-qft-methods_ E00788 (29)	299	1.000 N +0	$\left\{\psi_r^M,\psi_s^N\right\}=\left\{\tilde{\psi}_r^M,\tilde{\psi}_s^N\right\}=\eta^{MN}\delta_{r+s}.$	$\left\{\psi_r^M,\psi_s^N\right\}=\left\{\tilde{\psi}_r^M,\tilde{\psi}_s^N\right\}=\eta^{MN}\delta_{r+s}.$	$\left\{\psi_r^M,\psi_s^N\right\}=\left\{\tilde{\psi}_r^M,\tilde{\psi}_s^N\right\}=\eta^{MN}\delta_{r+s}.$

Conf. MathPix confidence: ≥ 0.9 $0.5\text{--}0.9$ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean — not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E00789 (30)	300	■ 1.000 ● N +0	$M^{\{2\}}=\frac{1}{\alpha^{\prime}}\{\frac{1}{\alpha^{\prime}}\}^{\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a+\text{ same with tilde }\right)},$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a+\text{ same with tilde }\right),$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a+\text{ same with tilde }\right),$
cardona-qft-methods_ E00790 (31)	300	■ 1.000 ● N +0	$\tilde{L}_0=\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a-\text{ same with tilde }\right).$	$\tilde{L}_0=\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a-\text{ same with tilde }\right).$	$\tilde{L}_0=\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a-\text{ same with tilde }\right).$
cardona-qft-methods_ E00792 (33)	300	■ 1.000 ● N +0	$R:\psi_0^M 0\rangle\rightarrow s\rangle=\left \pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2}\right\rangle$	$R:\psi_0^M 0\rangle\rightarrow s\rangle=\left \pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2}\right\rangle$	$R:\psi_0^M 0\rangle\rightarrow s\rangle=\left \pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2}\right\rangle$
cardona-qft-methods_ E00793 (34)	301	■ 1.000 ● K +0	$NS:\psi_{1/2}^M 0\rangle,$	$NS:\psi_{1/2}^M 0\rangle,$	$NS:\psi_{1/2}^M 0\rangle,$
cardona-qft-methods_ E00797 (38)	302	■ 1.000 ● N +0	$M^{\{2\}}=\frac{n^2}{R^2}+\frac{m^2R^2}{\alpha'^2}+\frac{2}{\alpha'}(N+\tilde{N}-2)$	$M^2=\frac{n^2}{R^2}+\frac{m^2R^2}{\alpha'^2}+\frac{2}{\alpha'}(N+\tilde{N}-2)$	$M^2=\frac{n^2}{R^2}+\frac{m^2R^2}{\alpha'^2}+\frac{2}{\alpha'}(N+\tilde{N}-2)$
cardona-qft-methods_ E00798 (39)	304	■ 1.000 ●	$S_{DBI}=-T_p\int d^{p+1}\xi e^{-\Phi}\sqrt{\det(G_{ab}+B_{ab}+2\pi\alpha'F_{ab})}$	$S_{DBI}=-T_p\int d^{p+1}\xi e^{-\Phi}\sqrt{\det(G_{ab}+B_{ab}+2\pi\alpha'F_{ab})}$	$S_{DBI}=-T_p\int d^{p+1}\xi e^{-\Phi}\sqrt{\det(G_{ab}+B_{ab}+2\pi\alpha'F_{ab})}$
cardona-qft-methods_ E00799 (40)	304	■ 1.000 ● K +0	$S_{CS}=\mathrm{i}T_p\int_{p+1}e^{2\pi\alpha'F+B}\wedge C$	$S_{CS}=\mathrm{i}T_p\int_{p+1}e^{2\pi\alpha'F+B}\wedge C$	$S_{CS}=\mathrm{i}T_p\int_{p+1}e^{2\pi\alpha'F+B}\wedge C$
cardona-qft-methods_ E00800 (41)	304	■ 1.000 ● N +1	$G_{\{a\}b}(\xi)=\frac{\partial X^M}{\partial \xi^a}\frac{\partial X^N}{\partial \xi^b}G_{MN}(X)$	$G_{ab}(\xi)=\frac{\partial X^M}{\partial \xi^a}\frac{\partial X^N}{\partial \xi^b}G_{MN}(X)$	$G_{ab}(\xi)=\frac{\partial X^M}{\partial \xi^a}\frac{\partial X^N}{\partial \xi^b}G_{MN}(X)$
cardona-qft-methods_ E00801 (1)	305	■ 1.000 ● N +0	$\mathcal{G}=\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right).$	$\mathcal{G}=\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right).$	$\mathcal{G}=\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right).$
cardona-qft-methods_ E00802 (2)	306	■ 1.000 ● N +0	$\mathcal{B}=\left(\begin{array}{cc} 0 & 0 \\ B & 0 \end{array}\right),$	$\mathcal{B}=\left(\begin{array}{cc} 0 & 0 \\ B & 0 \end{array}\right),$	$\mathcal{B}=\left(\begin{array}{cc} 0 & 0 \\ B & 0 \end{array}\right),$
cardona-qft-methods_ E00805 (5)	306	■ 1.000 ● N +0	$\Lambda_{(\alpha\beta)}+\Lambda_{(\beta\gamma)}+\Lambda_{(\gamma\alpha)}=g_{(\alpha\beta\gamma)}\mathrm{d}g_{(\alpha\beta\gamma)}$	$\Lambda_{(\alpha\beta)}+\Lambda_{(\beta\gamma)}+\Lambda_{(\gamma\alpha)}=g_{(\alpha\beta\gamma)}\mathrm{d}g_{(\alpha\beta\gamma)}$	$\Lambda_{(\alpha\beta)}+\Lambda_{(\beta\gamma)}+\Lambda_{(\gamma\alpha)}=g_{(\alpha\beta\gamma)}\mathrm{d}g_{(\alpha\beta\gamma)}$
cardona-qft-methods_ E00806 (6)	306	■ 1.000 ● N +1	$\mathcal{J}_1\equiv\left(\begin{array}{cc} I & 0 \\ 0 & -I^t \end{array}\right)$	$\mathcal{J}_1\equiv\left(\begin{array}{cc} I & 0 \\ 0 & -I^t \end{array}\right)$	$\mathcal{J}_1\equiv\left(\begin{array}{cc} I & 0 \\ 0 & -I^t \end{array}\right),$
cardona-qft-methods_ E00807 (7)	307	■ 1.000 ● N +1	$\mathcal{J}_2\equiv\left(\begin{array}{cc} 0 & -J^{-1} \\ J & 0 \end{array}\right).$	$\mathcal{J}_2\equiv\left(\begin{array}{cc} 0 & -J^{-1} \\ J & 0 \end{array}\right).$	$\mathcal{J}_2\equiv\left(\begin{array}{cc} 0 & -J^{-1} \\ J & 0 \end{array}\right).$
cardona-qft-methods_ E00810 (10)	308	■ 1.000 ● K +0	$\mathcal{H}=\mathcal{G}\mathcal{J}_1\mathcal{J}_2.$	$\mathcal{H}=\mathcal{G}\mathcal{J}_1\mathcal{J}_2.$	$\mathcal{H}=\mathcal{G}\mathcal{J}_1\mathcal{J}_2.$
cardona-qft-methods_ E00812 (12)	308	■ 1.000 ● N +0	$\eta_+^1\otimes\eta_{\pm}^{2\dagger}=\frac{1}{4}\sum_{k=0}^6\frac{1}{k!}\eta_{\pm}^{2\dagger}\gamma_{i_1\dots i_k}\eta_+^1\gamma^{i_k\dots i_1}$	$\eta_+^1\otimes\eta_{\pm}^{2\dagger}=\frac{1}{4}\sum_{k=0}^6\frac{1}{k!}\eta_{\pm}^{2\dagger}\gamma_{i_1\dots i_k}\eta_+^1\gamma^{i_k\dots i_1}$	$\eta_+^1\otimes\eta_{\pm}^{2\dagger}=\frac{1}{4}\sum_{k=0}^6\frac{1}{k!}\eta_{\pm}^{2\dagger}\gamma_{i_1\dots i_k}\eta_+^1\gamma^{i_k\dots i_1}$

Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08813 (13)	308	■ 1.000 ● W +1	$\sum_{k} \frac{1}{k!} C_{i_1 \dots i_k} \gamma_{\alpha\beta}^{i_i \dots i_k} \longleftrightarrow \sum_k \frac{1}{k!} C_{i_1 \dots i_k}^{(k)} \mathrm{d} x^{i_i} \wedge \dots \wedge \mathrm{d} x^{i_k}$	$\sum_k \frac{1}{k!} C_{i_1 \dots i_k}^{(k)} \gamma_{\alpha\beta}^{i_i \dots i_k} \longleftrightarrow \sum_k \frac{1}{k!} C_{i_1 \dots i_k}^{(k)} \mathrm{d} x^{i_i} \wedge \dots \wedge \mathrm{d} x^{i_k}$	$\sum_k \frac{1}{k!} C_{i_1 \dots i_k}^{(k)} \gamma_{\alpha\beta}^{i_i \dots i_k} \longleftrightarrow \sum_k \frac{1}{k!} C_{i_1 \dots i_k}^{(k)} \mathrm{d} x^{i_i} \wedge \dots \wedge \mathrm{d} x^{i_k}$
cardona-qft-methods_ E08814 (14)	309	■ 1.000 ● N +0	$\Phi_{\pm} \rightarrow e^{-B} \Phi_{\pm} \equiv \Phi_{\pm}^D$	$\Phi_{\pm} \rightarrow e^{-B} \Phi_{\pm} \equiv \Phi_{\pm}^D$	$\Phi_{\pm} \rightarrow e^{-B} \Phi_{\pm} \equiv \Phi_{\pm}^D$
cardona-qft-methods_ E08815 (15)	309	■ 1.000 ● K +0	$\Phi_{+}=e^{-\mathrm{i}J}, \quad \Phi_{-}=-\mathrm{i}\Omega,$	$\Phi_{+}=e^{-\mathrm{i}J}, \quad \Phi_{-}=-\mathrm{i}\Omega,$	$\Phi_{+}=e^{-\mathrm{i}J}, \quad \Phi_{-}=-\mathrm{i}\Omega,$
cardona-qft-methods_ E08817 (17)	309	■ 1.000 ● N +0	$J \wedge \Omega = 0, \quad J \wedge J \wedge J = \frac{3\mathrm{i}}{4} \Omega \wedge \bar{\Omega},$	$J \wedge \Omega = 0, \quad J \wedge J \wedge J = \frac{3\mathrm{i}}{4} \Omega \wedge \bar{\Omega},$	$J \wedge \Omega = 0, \quad J \wedge J \wedge J = \frac{3\mathrm{i}}{4} \Omega \wedge \bar{\Omega},$
cardona-qft-methods_ E08821 (21)	311	■ 1.000 ● N +0	$[X+\zeta,Y+\eta]_H=[X+\zeta,Y+\eta]_C+\iota_X\iota_YH,$	$[X+\zeta,Y+\eta]_H=[X+\zeta,Y+\eta]_C+\iota_X\iota_YH,$	$[X+\zeta,Y+\eta]_H=[X+\zeta,Y+\eta]_C+\iota_X\iota_YH,$
cardona-qft-methods_ E08822	311	■ 1.000 ● K +0	$(\mathrm{d}-H\wedge)\Phi=0$	$(\mathrm{d}-H\wedge)\Phi=0$	$(\mathrm{d}-H\wedge)\Phi=0$
cardona-qft-methods_ E08825 (4)	313	■ 1.000 ● K +0	$F^{(10)}=\mathrm{d}C-H\wedge C+m\,e^B=\hat{F}-H\wedge C$	$F^{(10)}=\mathrm{d}C-H\wedge C+m\,e^B=\hat{F}-H\wedge C$	$F^{(10)}=\mathrm{d}C-H\wedge C+m\,e^B=\hat{F}-H\wedge C$
cardona-qft-methods_ E08826 (5)	313	■ 1.000 ● N +1	$F_n^{(10)}=(-1)^{\mathrm{Int}[n/2]}\star F_{10-n}^{(10)},$	$F_n^{(10)}=(-1)^{\mathrm{Int}[n/2]}\star F_{10-n}^{(10)},$	$F_n^{(10)}=(-1)^{\mathrm{Int}[n/2]}\star F_{10-n}^{(10)},$
cardona-qft-methods_ E08827 (6)	314	■ 1.000 ●	$d\,H=0,\quad d\,F^{(10)}-H\wedge F^{(10)}=0.$	$dH=0,\quad dF^{(10)}-H\wedge F^{(10)}=0.$	$dH=0,\quad dF^{(10)}-H\wedge F^{(10)}=0.$
cardona-qft-methods_ E08828 (7)	314	■ 1.000 ● N +0	$\tilde{\nabla}_\mu\epsilon+\frac{e^{-A}}{2}(\gamma_\mu\gamma_5\otimes\not{N}A)\epsilon=0$	$\tilde{\nabla}_\mu\epsilon+\frac{e^{-A}}{2}(\gamma_\mu\gamma_5\otimes\not{N}A)\epsilon=0$	$\tilde{\nabla}_\mu\epsilon+\frac{e^{-A}}{2}(\gamma_\mu\gamma_5\otimes\nabla A)\epsilon=0$
cardona-qft-methods_ E08829 (8)	314	■ 1.000 ● K +0	$\left[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu\right]\epsilon=-\frac{1}{2}\left(\nabla_mA\right)\left(\nabla^mA\right)\gamma_{\mu\nu}\epsilon$	$\left[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu\right]\epsilon=-\frac{1}{2}\left(\nabla_mA\right)\left(\nabla^mA\right)\gamma_{\mu\nu}\epsilon$	$[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu]\epsilon=-\frac{1}{2}(\nabla_mA)(\nabla^mA)\gamma_{\mu\nu}\epsilon$
cardona-qft-methods_ E08830 (9)	314	■ 1.000 ● K +0	$\left[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu\right]\epsilon=\frac{1}{4}\tilde{R}_{\mu\nu\lambda\rho}\gamma^{\lambda\rho}\epsilon=0$	$\left[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu\right]\epsilon=\frac{1}{4}\tilde{R}_{\mu\nu\lambda\rho}\gamma^{\lambda\rho}\epsilon=0$	$[\tilde{\nabla}_\mu,\tilde{\nabla}_\nu]\epsilon=\frac{1}{4}\tilde{R}_{\mu\nu\lambda\rho}\gamma^{\lambda\rho}\epsilon=0$
cardona-qft-methods_ E08832 (11)	314	■ 1.000 ● N +0	$\epsilon_{\mathrm{IIB}}^A=\xi_+^A\otimes\eta_+^A+\xi_-^A\otimes\eta_-^A,\quad A=1,2.$	$\epsilon_{\mathrm{IIB}}^A=\xi_+^A\otimes\eta_+^A+\xi_-^A\otimes\eta_-^A,\quad A=1,2.$	$\epsilon_{\mathrm{IIB}}^A=\xi_+^A\otimes\eta_+^A+\xi_-^A\otimes\eta_-^A,\quad A=1,2.$
cardona-qft-methods_ E08833 (12)	314	■ 1.000 ● K +0	$\nabla_m\eta_{\pm}^A=0.$	$\nabla_m\eta_{\pm}^A=0.$	$\nabla_m\eta_{\pm}^A=0.$
cardona-qft-methods_ E08836 (15)	317	■ 1.000 ● K +0	$ \Phi_+ = \Phi_- =e^A$	$ \Phi_+ = \Phi_- =e^A$	$ \Phi_+ = \Phi_- =e^A$
cardona-qft-methods_ E08839 (18)	317	■ 1.000 ● K +0	$\lambda(A_n)=(-1)^{\mathrm{Int}[n/2]}A_n.$	$\lambda(A_n)=(-1)^{\mathrm{Int}[n/2]}A_n.$	$\lambda(A_n)=(-1)^{\mathrm{Int}[n/2]}A_n.$
cardona-qft-methods_ E08840	328	■ 1.000 ● K +0	$\{f,g\}=\Pi(df,dg)$	$\{f,g\}=\Pi(df,dg)$	$\{f,g\}=\Pi(df,dg)$
cardona-qft-methods_ E08841	328	■ 1.000 ● N +0	$\{f,g\,h\}=g\{f,h\}+h\{f,g\},\,\forall f,g,h\in\mathcal{C}^\infty(M).$	$\{f,gh\}=g\{f,h\}+h\{f,g\},\,\forall f,g,h\in\mathcal{C}^\infty(M).$	$\{f,gh\}=g\{f,h\}+h\{f,g\},\,\forall f,g,h\in\mathcal{C}^\infty(M).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0843	329	■ 1.000 ● N +1	$S(X, \eta) := \int_{\Sigma} \langle \eta, dX \rangle + \frac{1}{2} \langle \eta, (\Pi^{\#} \circ X) \eta \rangle$	$S(X, \eta) := \int_{\Sigma} \langle \eta, dX \rangle + \frac{1}{2} \langle \eta, (\Pi^{\#} \circ X) \eta \rangle$	$S(X, \eta) := \int_{\Sigma} \langle \eta, dX \rangle + \frac{1}{2} \langle \eta, (\Pi^{\#} \circ X) \eta \rangle,$
cardona-qft-methods_ EQ0844	329	■ 1.000 ● N +0	$\mathrm{EL} = \{ \text{ Solutions of the Euler-Lagrange equations } \} \subset \Phi,$	$\mathrm{EL} = \{ \text{ Solutions of the Euler-Lagrange equations } \} \subset \Phi,$	$\mathrm{EL} = \{ \text{Solutions of the Euler-Lagrange equations} \} \subset \Phi,$
cardona-qft-methods_ EQ0847	330	■ 1.000 ● K +0	$\Phi_{\partial} := \{ \text{ vector bundle morphisms between } T(\partial \Sigma) \text{ and } T^* M \}.$	$\Phi_{\partial} := \{ \text{ vector bundle morphisms between } T(\partial \Sigma) \text{ and } T^* M \}.$	$\Phi_{\partial} := \{ \text{vector bundle morphisms between } T(\partial \Sigma) \text{ and } T^* M \}.$
cardona-qft-methods_ EQ0848	330	■ 1.000 ● N +0	$L_{\Sigma} := p(EL).$	$L_{\Sigma} := p(EL).$	$L_{\Sigma} := p(EL).$
cardona-qft-methods_ EQ0849	330	■ 1.000 ● N +0	$C_{\Pi} := \{ (X, \eta) \mid dX = \pi^{\#}(X) \eta, X : \partial \Sigma \rightarrow M, \eta \in \Gamma(T^* I \otimes X^*(T^* M)) \}.$	$C_{\Pi} := \{ (X, \eta) \mid dX = \pi^{\#}(X) \eta, X : \partial \Sigma \rightarrow M, \eta \in \Gamma(T^* I \otimes X^*(T^* M)) \}.$	$C_{\Pi} := \{ (X, \eta) \mid dX = \pi^{\#}(X) \eta, X : \partial \Sigma \rightarrow M, \eta \in \Gamma(T^* I \otimes X^*(T^* M)) \}.$
cardona-qft-methods_ EQ0850	330	■ 1.000 ● N +0	$[X, fY]_A = f[X, Y] + \rho_*(X)(f)Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^{\infty}(M).$	$[X, fY]_A = f[X, Y] + \rho_*(X)(f)Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^{\infty}(M).$	$[X, fY]_A = f[X, Y] + \rho_*(X)(f)Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^{\infty}(M).$
cardona-qft-methods_ EQ0852	331	■ 1.000 ● N +0	$\delta_A f := \rho^* df,$	$\delta_A f := \rho^* df,$	$\delta_A f := \rho^* df,$
cardona-qft-methods_ EQ0853	331	■ 1.000 ● N +0	$\langle \delta_A \alpha, X \wedge Y \rangle := - \langle \alpha, [X, Y]_A \rangle + \langle \delta \langle \alpha, X \rangle, Y \rangle - \langle \delta \langle \alpha, Y \rangle, X \rangle,$	$\langle \delta_A \alpha, X \wedge Y \rangle := - \langle \alpha, [X, Y]_A \rangle + \langle \delta \langle \alpha, X \rangle, Y \rangle - \langle \delta \langle \alpha, Y \rangle, X \rangle,$	$\langle \delta_A \alpha, X \wedge Y \rangle := - \langle \alpha, [X, Y]_A \rangle + \langle \delta \langle \alpha, X \rangle, Y \rangle - \langle \delta \langle \alpha, Y \rangle, X \rangle,$
cardona-qft-methods_ EQ0854	331	■ 1.000 ● N +0	$\delta_A \varphi^* = \varphi^* \delta_B.$	$\delta_A \varphi^* = \varphi^* \delta_B.$	$\delta_A \varphi^* = \varphi^* \delta_B.$
cardona-qft-methods_ EQ0856	332	■ 1.000 ● K +0	$Gr_{\mu} := \{ (a, b, c) \in G^3 \mid c = \mu(a, b) \}$	$Gr_{\mu} := \{ (a, b, c) \in G^3 \mid c = \mu(a, b) \}$	$Gr_{\mu} := \{ (a, b, c) \in G^3 \mid c = \mu(a, b) \}$
cardona-qft-methods_ EQ0857	333	■ 1.000 ●	$\overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \mathcal{G} \text{ and } \mathcal{G} \rightharpoonup \overline{\mathcal{G}}$	$\overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \mathcal{G} \text{ and } \mathcal{G} \rightharpoonup \overline{\mathcal{G}}$	$\overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \mathcal{G} \text{ and } \mathcal{G} \rightharpoonup \overline{\mathcal{G}}$
cardona-qft-methods_ EQ0858	333	■ 1.000 ● W +0	$T : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G} \times \mathcal{G}$ $(x, y) \mapsto (y, x)$	$T : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G} \times \mathcal{G}$ $(x, y) \mapsto (y, x)$	$T : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G} \times \mathcal{G}$ $(x, y) \mapsto (y, x)$
cardona-qft-methods_ EQ0859	334	■ 1.000 ● W +0	$T_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G} \times \mathcal{G} \text{ and } \overline{T_{rel}} : \overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \overline{\mathcal{G}} \times \overline{\mathcal{G}}.$	$T_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G} \times \mathcal{G} \text{ and } \overline{T_{rel}} : \overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \overline{\mathcal{G}} \times \overline{\mathcal{G}}.$	$T_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G} \times \mathcal{G} \text{ and } \overline{T_{rel}} : \overline{\mathcal{G}} \times \overline{\mathcal{G}} \rightharpoonup \overline{\mathcal{G}} \times \overline{\mathcal{G}}.$
cardona-qft-methods_ EQ0860	334	■ 1.000 ● N +0	$L_3 := I_{rel} \circ L_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$	$L_3 := I_{rel} \circ L_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$	$L_3 := I_{rel} \circ L_{rel} : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$
cardona-qft-methods_ EQ0861	334	■ 1.000 ● K +0	$\overline{I_{rel}} \circ L_1 = \overline{L_1},$	$\overline{I_{rel}} \circ L_1 = \overline{L_1},$	$\overline{I_{rel}} \circ L_1 = \overline{L_1},$
cardona-qft-methods_ EQ0862	334	■ 1.000 ● K +0	$I(L_1) = \overline{L_1},$	$I(L_1) = \overline{L_1},$	$I(L_1) = \overline{L_1},$
cardona-qft-methods_ EQ0864	335	■ 1.000 ● N +0	$L_2 := L_3 \circ (L_1 \times Id) : \mathcal{G} \rightharpoonup \mathcal{G}.$	$L_2 := L_3 \circ (L_1 \times Id) : \mathcal{G} \rightharpoonup \mathcal{G}.$	$L_2 := L_3 \circ (L_1 \times Id) : \mathcal{G} \rightharpoonup \mathcal{G}.$
cardona-qft-methods_ EQ0865	335	■ 1.000 ● K +0	$L_2 = L_3 \circ (Id \times L_1).$	$L_2 = L_3 \circ (Id \times L_1).$	$L_2 = L_3 \circ (Id \times L_1).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0866	335	■ 1.000 ● K +0	$\begin{aligned} & L_2 \circ L_1 = L_1 \\ & L_2 \circ L_2 = L_2 \\ & L_2 \circ L_3 = L_3. \end{aligned}$	$\begin{aligned} L_2 \circ L_1 &= L_1 \\ L_2 \circ L_2 &= L_2 \\ L_2 \circ L_3 &= L_3. \end{aligned}$	$\begin{aligned} L_2 \circ L_1 &= L_1 \\ L_2 \circ L_2 &= L_2 \\ L_2 \circ L_3 &= L_3. \end{aligned}$
cardona-qft-methods_ EQ0867	335	■ 1.000 ● K +0	$\overline{I_{rel}} \circ L_2 = \overline{L_2} \circ \overline{I_{rel}} \text{ and } L_2^\dagger = L_2.$	$\overline{I_{rel}} \circ L_2 = \overline{L_2} \circ \overline{I_{rel}} \text{ and } L_2^\dagger = L_2.$	$\overline{I_{rel}} \circ L_2 = \overline{L_2} \circ \overline{I_{rel}} \text{ and } L_2^\dagger = L_2.$
cardona-qft-methods_ EQ0868	335	■ 1.000 ● K +0	$C^* = \mathcal{G}^* \circ L_2$	$C^* = \mathcal{G}^* \circ L_2$	$C^* = \mathcal{G}^* \circ L_2$
cardona-qft-methods_ EQ0869	336	■ 1.000 ● N +0	$T := \left\{ (c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (c, l, g) \in L_3 \right\}$	$T := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (c, l, g) \in L_3\}$	$T := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (c, l, g) \in L_3\}$
cardona-qft-methods_ EQ0871	337	■ 1.000 ● W +0	$\begin{aligned} & \mathcal{G} = G. \\ & L = \mathcal{L} \times \mathcal{L} \times \mathcal{L}. \\ & I = \{\text{identity of } G\}. \end{aligned}$	$\begin{aligned} \mathcal{G} &= G. \\ L &= \mathcal{L} \times \mathcal{L} \times \mathcal{L}. \\ I &= \{\text{identity of } G\}. \end{aligned}$	$\begin{aligned} \mathcal{G} &= G. \\ L &= \mathcal{L} \times \mathcal{L} \times \mathcal{L}. \\ I &= \{\text{identity of } G\}. \end{aligned}$
cardona-qft-methods_ EQ0872	338	■ 1.000 ● N +0	$P := \left\{ (x, \alpha, \beta) \mid x \in \underline{G_{(n)}}, [\alpha] = [\beta] = x \right\},$	$P := \left\{ (x, \alpha, \beta) \mid x \in \underline{G_{(n)}}, [\alpha] = [\beta] = x \right\},$	$P := \{(x, \alpha, \beta) \mid x \in \underline{G_{(n)}}, [\alpha] = [\beta] = x\},$
cardona-qft-methods_ EQ0873	338	■ 1.000 ● K +0	$P^\dagger \circ P = Gr(Id_G), P \circ P^\dagger = \{(g, h) \in G^n \mid [g] = [h]\}.$	$P^\dagger \circ P = Gr(Id_G), P \circ P^\dagger = \{(g, h) \in G^n \mid [g] = [h]\}.$	$P^\dagger \circ P = Gr(Id_G), P \circ P^\dagger = \{(g, h) \in G^n \mid [g] = [h]\}.$
cardona-qft-methods_ EQ0875	338	■ 1.000 ● N +0	$\begin{aligned} & \phi : [0, 1] \rightarrow [0, 1] \\ & t \mapsto 1 - t \end{aligned}$	$\begin{aligned} \phi : [0, 1] &\rightarrow [0, 1] \\ t &\mapsto 1 - t \end{aligned}$	$\begin{aligned} \phi : [0, 1] &\rightarrow [0, 1] \\ t &\mapsto 1 - t \end{aligned}$
cardona-qft-methods_ EQ0876	338	■ 1.000 ● N +0	$\overline{L_1} := \cup_{x_0 \in M} T_{(X, \eta)}^* PM \cap L_1,$	$\overline{L_1} := \cup_{x_0 \in M} T_{(X, \eta)}^* PM \cap L_1,$	$\overline{L_1} := \cup_{x_0 \in M} T_{(X, \eta)}^* PM \cap L_1,$
cardona-qft-methods_ EQ0877	339	■ 1.000 ● K +0	$m : H \otimes H \rightarrow H.$	$m : H \otimes H \rightarrow H.$	$m : H \otimes H \rightarrow H.$
cardona-qft-methods_ EQ0878	339	■ 1.000 ●	$\langle a, b \rangle_H := \langle \bar{a}, b \rangle,$	$\langle a, b \rangle_H := \langle \bar{a}, b \rangle,$	$\langle a, b \rangle_H := \langle \bar{a}, b \rangle,$
cardona-qft-methods_ EQ0879	339	■ 1.000 ● K +0	$\langle m(a, b), c \rangle_H = \langle a, m(b, c) \rangle_H$	$\langle m(a, b), c \rangle_H = \langle a, m(b, c) \rangle_H$	$\langle m(a, b), c \rangle_H = \langle a, m(b, c) \rangle_H$
cardona-qft-methods_ EQ0880	339	■ 1.000 ● N +0	$e := \sum_i m(e_i, \overline{e_i})$	$e := \sum_i m(e_i, \overline{e_i})$	$e := \sum_i m(e_i, \overline{e_i})$
cardona-qft-methods_ EQ0881	339	■ 1.000 ● K +0	$\begin{aligned} & P : H \rightarrow H \\ & a \mapsto m(a, e) \end{aligned}$	$\begin{aligned} P : H &\rightarrow H \\ a &\mapsto m(a, e) \end{aligned}$	$\begin{aligned} P : H &\rightarrow H \\ a &\mapsto m(a, e) \end{aligned}$
cardona-qft-methods_ EQ0882	342	■ 1.000 ● N +0	$H_p(LM) \otimes H_q(LM) \rightarrow H_{p+q-n}(LM),$	$H_p(LM) \otimes H_q(LM) \rightarrow H_{p+q-n}(LM),$	$H_p(LM) \otimes H_q(LM) \rightarrow H_{p+q-n}(LM),$
cardona-qft-methods_ EQ0885	344	■ 1.000 ● N +0	$\tau_\Delta : M \times M \rightarrow (M \times M) / \nu_\Delta^c$	$\tau_\Delta : M \times M \rightarrow (M \times M) / \nu_\Delta^c$	$\tau_\Delta : M \times M \rightarrow (M \times M) / \nu_\Delta^c$
cardona-qft-methods_ EQ0886	344	■ 1.000 ● N +0	$\tau_\Delta : M \times M \rightarrow M^{TM}$	$\tau_\Delta : M \times M \rightarrow M^{TM}$	$\tau_\Delta : M \times M \rightarrow M^{TM}$
cardona-qft-methods_ EQ0891	345	■ 1.000 ● N +0	$LM^{-TM} \wedge LM^{-TM} \rightarrow LM^{-TM},$	$LM^{-TM} \wedge LM^{-TM} \rightarrow LM^{-TM},$	$LM^{-TM} \wedge LM^{-TM} \rightarrow LM^{-TM},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E08892	346	■ 1.000 ● N +0	$H_{\{p\}\backslash\left(M^{\wedge\{-T\}M}\right)}\cong H_{\{p+n\}}(M)$	$H_p\left(M^{-TM}\right)\cong H_{p+n}(M)$	$H_p(M^{-TM})\cong H_{p+n}(M)$
cardona-qft-methods_ E08893	346	■ 1.000 ● N +0	$M^{\wedge\{-T\}M}:=\Sigma^{\wedge\{-k\}}M^{\nu(e)}$	$M^{-TM}:=\Sigma^{-k}M^{\nu(e)}$	$M^{-TM}:=\Sigma^{-k}M^{\nu(e)}$
cardona-qft-methods_ E08894	346	■ 1.000 ● N +0	$\Delta^{\{*\}}:M^{\wedge\{-T\}M}\wedge M^{\wedge\{-T\}M}=(M\times M)^{\wedge\{-2\}T\}M\rightarrowtail M^{\wedge\{T\}M-2\}T\}M=M^{\wedge\{-T\}M}$	$\Delta^*:M^{-TM}\wedge M^{-TM}=(M\times M)^{-2TM}\rightarrow M^{TM-2TM}=M^{-TM}$	$\Delta^*:M^{-TM}\wedge M^{-TM}=(M\times M)^{-2TM}\rightarrow M^{TM-2TM}=M^{-TM}$
cardona-qft-methods_ E09901	347	■ 1.000 ● N +0	$H_{\{*\}+n}(L\,M)\cong H^{\{*\}}\left(C^{\{*\}}(M),\,C^{\{*\}}(M)\right)$.	$H_{*+n}(LM)\cong HH^*(C^*(M),C^*(M))$.	$H_{*+n}(LM)\cong HH^*(C^*(M),C^*(M))$.
cardona-qft-methods_ E09902	348	■ 1.000 ● N +0	$G_{\{x\}}=\{g\in G\mid g\,\overline{x}=\overline{x}\}$	$G_x=\{g\in G\mid g\bar{x}=\bar{x}\}$	$G_x=\{g\in G\mid g\bar{x}=\bar{x}\}$
cardona-qft-methods_ E09903	349	■ 1.000 ● N +0	$P_{\{G\}}M\times_{\{G\}}EG\stackrel{\sim}{=}L\left(M\times_{\{G\}}EG\right)$.	$P_GM\times_GEG\simeq L\left(M\times_GEG\right)$.	$P_GM\times_GEG\simeq L(M\times_GEG)$.
cardona-qft-methods_ E09904	349	■ 1.000 ● N +1	$H_{\{*\}}\backslash\left(\mathcal{L}\backslash\left(M\times_{\{G\}}EG\right);\mathbb{Q}\right)\cong H_{\{*\}}\backslash\left(P_{\{G\}}M;\mathbb{Q}\right)^G$	$H_{*}\left(\mathcal{L}\left(M\times_GEG\right);\mathbb{Q}\right)\cong H_{*}\left(P_GM;\mathbb{Q}\right)^G$	$H_{*}(\mathcal{L}(M\times_GEG);\mathbb{Q})\cong H_{*}(P_GM;\mathbb{Q})^G$
cardona-qft-methods_ E09905	349	■ 1.000 ● N +0	$P_{\{G\}}X:=\bigsqcup_{\{g\in G\}}P_{\{g\}}X\times\{g\}\quad\text{with}\quad P_{\{g\}}X:=\{f:[0,1]\rightarrow X\mid f(0)g=f(1)\}$	$P_GX:=\bigsqcup_{g\in G}P_gX\times\{g\}\quad\text{with}\quad P_gX:=\{f:[0,1]\rightarrow X\mid f(0)g=f(1)\}$	$P_GX:=\bigsqcup_{g\in G}P_gX\times\{g\}\quad\text{with}\quad P_gX:=\{f:[0,1]\rightarrow X\mid f(0)g=f(1)\}$
cardona-qft-methods_ E09907	350	■ 1.000 ● N +0	$P_{\{g\}}M:=\{_{\{g\in G\}}P_{\{h\}}M\mid\phi(1)=\psi(0)\}$	$P_gM:{}_1\times_0P_hM=\{(\phi,\psi)\in P_gM\times P_hM\mid\phi(1)=\psi(0)\}$	$P_gM:{}_1\times_0P_hM=\{(\phi,\psi)\in P_gM\times P_hM\mid\phi(1)=\psi(0)\}$
cardona-qft-methods_ E09909	350	■ 1.000 ● K +0	$P_{\{G\}}M^{\wedge\{-T\}M}:=\bigsqcup_{\{g\in G\}}P_{\{g\}}M^{\wedge\{-T\}M}$	$P_GM^{-TM}:=\bigsqcup_{g\in G}P_gM^{-TM}$	$P_GM^{-TM}:=\bigsqcup_{g\in G}P_gM^{-TM}$
cardona-qft-methods_ E09910	350	■ 1.000 ● K +0	$\mu:P_{\{G\}}M^{\wedge\{-T\}M}\wedge P_{\{G\}}M^{\wedge\{-T\}M}\rightarrowtail P_{\{G\}}M^{\wedge\{-T\}M}$.	$\mu:P_GM^{-TM}\wedge P_GM^{-TM}\rightarrow P_GM^{-TM}$.	$\mu:P_GM^{-TM}\wedge P_GM^{-TM}\rightarrow P_GM^{-TM}$.
cardona-qft-methods_ E09911	350	■ 1.000 ● N +0	$H_{\{*\}}\backslash\left(P_{\{G\}}M^{\wedge\{-T\}M};\mathbb{Q}\right)^G\cong H_{\{*\}+n}\left(L\left(M\times_{\{G\}}EG\right);\mathbb{Q}\right)$	$H_{*}\left(P_GM^{-TM};\mathbb{Q}\right)^G\cong H_{*+n}\left(L\left(M\times_GEG\right);\mathbb{Q}\right)$	$H_{*}(P_GM^{-TM};\mathbb{Q})^G\cong H_{*+n}(L(M\times_GEG);\mathbb{Q})$
cardona-qft-methods_ E09912	351	■ 1.000 ● N +0	$H^{\{*\}}\backslash\left(C^{\{*\}}(M;\mathbb{Q})\right)\#G,C^{\{*\}}(M;\mathbb{Q})\#G\right)\cong H_{\{*\}}\backslash\left(P_{\{G\}}M^{\wedge\{-T\}M};\mathbb{Q}\right)^G$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong H_{*}\left(P_GM^{-TM};\mathbb{Q}\right)^G$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong H_{*}(P_GM^{-TM};\mathbb{Q})^G$
cardona-qft-methods_ E09913	351	■ 1.000 ●	$C^{\{*\}}(M)\#G:=C^{\{*\}}(M)\otimes\mathbb{Q}G\quad(\alpha,g)\cdot(\beta,h):=(\alpha\cdot g^*\beta,gh)$.	$C^*(M)\#G:=C^*(M)\otimes\mathbb{Q}G\quad(\alpha,g)\cdot(\beta,h):=(\alpha\cdot g^*\beta,gh)$.	$C^*(M)\#G:=C^*(M)\otimes\mathbb{Q}G\quad(\alpha,g)\cdot(\beta,h):=(\alpha\cdot g^*\beta,gh)$.
cardona-qft-methods_ E09914	351	■ 1.000 ● N +0	$H^{\{*\}}\backslash\left(C^{\{*\}}(M;\mathbb{Q})\right)\#G,C^{\{*\}}(M;\mathbb{Q})\#G\right)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$
cardona-qft-methods_ E09918	351	■ 1.000 ● K +0	$C^{\{-*\}}(M)\rightarrowtail C_{\{*\}}\left(M^{\wedge\{-T\}M}\right)$	$C^{-*}(M)\rightarrow C_{*}\left(M^{-TM}\right)$	$C^{-*}(M)\rightarrow C_{*}(M^{-TM})$
cardona-qft-methods_ E09920	352	■ 1.000 ● N +0	$H^{\{*\}}\backslash\left(C^{\{*\}}(M;\mathbb{Q})\right)\#G,C^{\{*\}}(M;\mathbb{Q})\#G\right)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong\mathrm{Ext}^*_{\mathbb{Q}G}(\mathbb{Q},C_{*}(P_GM^{-TM};\mathbb{Q}))$
cardona-qft-methods_ E09921	352	■ 1.000 ● N +0	$H^{\{*\}}\backslash\left(C^{\{*\}}(M;\mathbb{Q})\right)\#G,C^{\{*\}}(M;\mathbb{Q})\#G\right)\cong H_{*}\left(P_GM^{-TM};\mathbb{Q}\right)^G$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong H_{*}\left(P_GM^{-TM};\mathbb{Q}\right)^G$	$HH^*(C^*(M;\mathbb{Q})\#G,C^*(M;\mathbb{Q})\#G)\cong H_{*}(P_GM^{-TM};\mathbb{Q})^G$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ E09922	352	■ 1.000 ● K +0	$H\,H^*\!\left(C^{\scriptscriptstyle\wedge}\!\left\{*\right\}(M)\,\#\,G,\,C^{\scriptscriptstyle\wedge}\!\left\{*\right\}(M)\,\#\,G\right)$	$HH^*(C^*(M)\#G,C^*(M)\#G)$	$HH^*(C^*(M)\#G,C^*(M)\#G)$
cardona-qft-methods_ E09923	352	■ 1.000 ● N +0	$\operatorname{Ext}_{\mathbb{Z}G}^*\!\left(\mathbb{Z},C_*\left(P_GM^{-TM}\right)\right)\cong H^*\left(\left(C^*(EG)\otimes C_*\left(P_GM^{-TM}\right)\right)^G\right)$ $C_*\!\left(P_GM^{-TM}\right)\right)\cong H^*\left(\left(C^*(EG)\otimes C_*\left(P_GM^{-TM}\right)\right)^G\right)$	$\operatorname{Ext}_{\mathbb{Z}G}^*\left(\mathbb{Z},C_*\left(P_GM^{-TM}\right)\right)\cong H^*\left(\left(C^*(EG)\otimes C_*\left(P_GM^{-TM}\right)\right)^G\right)$	$\operatorname{Ext}_{\mathbb{Z}G}^*\left(\mathbb{Z},C_*\left(P_GM^{-TM}\right)\right)\cong H^*\left(\left(C^*(EG)\otimes C_*\left(P_GM^{-TM}\right)\right)^G\right)$
cardona-qft-methods_ E09924	352	■ 1.000 ● N -1	$P_{\scriptscriptstyle G}M\times_GEG_n\rightarrowtail M\times_GEG_n$	$P_GM\times_GEG_n\stackrel{e_0}{\rightarrow}M\times_GEG_n$	$P_GM\times_GEG_n\stackrel{e_0}{\rightarrow}M\times_GEG_n$
cardona-qft-methods_ E09926	352	■ 1.000 ● N +0	$\left(P_{\scriptscriptstyle G}M\times_GEG_n\right)^{-e_0^*T(M_n)}\leftarrow\left(P_{\scriptscriptstyle G}M\times_GEG_{n+1}\right)^{-e_0^*T(M_{n+1})}\leftarrow\cdots$	$(P_GM\times_GEG_n)^{-e_0^*T(M_n)}\leftarrow(P_GM\times_GEG_{n+1})^{-e_0^*T(M_{n+1})}\leftarrow\dots$	$(P_GM\times_GEG_n)^{-e_0^*T(M_n)}\leftarrow(P_GM\times_GEG_{n+1})^{-e_0^*T(M_{n+1})}\leftarrow\dots$
cardona-qft-methods_ E09929	353	■ 1.000 ● N +0	$\left[G^{\scriptscriptstyle ad}/G\right]\cong\bigcup_{\scriptscriptstyle (g)}[* / C(g)]$	$[G^{ad}/G]\cong\bigcup_{\scriptscriptstyle (g)}[* / C(g)]$	$[G^{ad}/G]\cong\bigcup_{\scriptscriptstyle (g)}[* / C(g)]$
cardona-qft-methods_ E09930 (g)	353	■ 1.000 ● K +0	$\bigcup B C(g)_{\scriptscriptstyle n}^{-T B C(g)_{\scriptscriptstyle n}}$	$\bigcup BC(g)_n^{-TBC(g)_n}$	$\bigcup BC(g)_n^{-TBC(g)_n}$
cardona-qft-methods_ E09932	354	■ 1.000 ● N +0	$H_*\left(EG_n/C(g)\times EG_n/C(h)\right)\rightarrow H_{*-k_n}\left(EG_n/(C(g)\cap C(h))\right)$	$H_*(EG_n/C(g)\times EG_n/C(h))\rightarrow H_{*-k_n}(EG_n/(C(g)\cap C(h)))$	$H_*(EG_n/C(g)\times EG_n/C(h))\rightarrow H_{*-k_n}(EG_n/(C(g)\cap C(h)))$
cardona-qft-methods_ E09933	354	■ 1.000 ● N +0	$H_*\left(EG_n/(C(g)\cap C(h))\right)\rightarrow H_*\left(EG_n/C(gh)\right)$	$H_*(EG_n/(C(g)\cap C(h)))\rightarrow H_*(EG_n/C(gh))$	$H_*(EG_n/(C(g)\cap C(h)))\rightarrow H_*(EG_n/C(gh))$
cardona-qft-methods_ E09934	354	■ 1.000 ● N +0	$H^*\left(EG_n/(C(g)\cap C(h))\right)\rightarrow H^*\left(EG_n/C(gh)\right)$	$H^*(EG_n/(C(g)\cap C(h)))\rightarrow H^*(EG_n/C(gh))$	$H^*(EG_n/(C(g)\cap C(h)))\rightarrow H^*(EG_n/C(gh))$
cardona-qft-methods_ E09935	354	■ 1.000 ● N +0	$H^*(B(C(g)\cap C(h)))\rightarrow H^*(BC(gh)).$	$H^*(B(C(g)\cap C(h)))\rightarrow H^*(BC(gh)).$	$H^*(B(C(g)\cap C(h)))\rightarrow H^*(BC(gh)).$
cardona-qft-methods_ E09938	355	■ 1.000 ● N +0	$H_*^{pro}\left(LBG^{-TBG}\right)\cong H^*(LBG)$	$H_*^{pro}\left(LBG^{-TBG}\right)\cong H^*(LBG)$	$H_*^{pro}\left(LBG^{-TBG}\right)\cong H^*(LBG)$
cardona-qft-methods_ E09939	355	■ 1.000 ● N +0	$LBG\leftarrow\operatorname{Map}(\Theta,BG)\rightarrow LBG\times LBG$	$LBG\leftarrow\operatorname{Map}(\Theta,BG)\rightarrow LBG\times LBG$	$LBG\leftarrow\operatorname{Map}(\Theta,BG)\rightarrow LBG\times LBG$
cardona-qft-methods_ E09940	355	■ 1.000 ● N +0	$H^*(LBG)\otimes H^*(LBG)\rightarrow H^*(LBG)$	$H^*(LBG)\otimes H^*(LBG)\rightarrow H^*(LBG)$	$H^*(LBG)\otimes H^*(LBG)\rightarrow H^*(LBG)$
cardona-qft-methods_ E09941 (1)	358	■ 1.000 ● K +0	$[X]-[Y]=[X\backslash Y],$	$[X]-[Y]=[X\backslash Y],$	$[X]-[Y]=[X\backslash Y],$
cardona-qft-methods_ E09944 (4)	358	■ 1.000 ● N +0	$\left[\mathbb{A}_{\scriptscriptstyle k}^n\right]=\mathbb{L}^n.$	$[\mathbb{A}_k^n]=\mathbb{L}^n.$	$[\mathbb{A}_k^n]=\mathbb{L}^n.$
cardona-qft-methods_ E09945 (5)	359	■ 1.000 ●	$\mathbb{T}=\left[\mathbb{A}_{\scriptscriptstyle k}^1\right]-\left[\mathbb{A}_{\scriptscriptstyle k}^0\right]=\mathbb{L}-1,$	$\mathbb{T}=\left[\mathbb{A}_k^1\right]-\left[\mathbb{A}_k^0\right]=\mathbb{L}-1,$	$\mathbb{T}=\left[\mathbb{A}_k^1\right]-\left[\mathbb{A}_k^0\right]=\mathbb{L}-1,$
cardona-qft-methods_ E09946 (6)	359	■ 1.000 ● K +0	$[X\cup Y]=[X]+[Y]-[X\cap Y]$	$[X\cup Y]=[X]+[Y]-[X\cap Y]$	$[X\cup Y]=[X]+[Y]-[X\cap Y]$
cardona-qft-methods_ E09948 (8)	359	■ 1.000 ● N +0	$\left[\mathbb{P}_{\scriptscriptstyle k}^n\right]=\frac{\mathbb{L}^{n+1}-1}{\mathbb{L}-1},$	$[\mathbb{P}_k^n]=\frac{\mathbb{L}^{n+1}-1}{\mathbb{L}-1},$	$[\mathbb{P}_k^n]=\frac{\mathbb{L}^{n+1}-1}{\mathbb{L}-1},$
cardona-qft-methods_ E09949 (9)	359	■ 1.000 ● N +0	$\left[\mathbb{P}_{\scriptscriptstyle k}^n\right]=1+\mathbb{L}+\mathbb{L}^2+\cdots+\mathbb{L}^n,$	$[\mathbb{P}_k^n]=1+\mathbb{L}+\mathbb{L}^2+\cdots+\mathbb{L}^n,$	$[\mathbb{P}_k^n]=1+\mathbb{L}+\mathbb{L}^2+\cdots+\mathbb{L}^n,$
cardona-qft-methods_ E09950 (1)	359	■ 1.000 ● K +0	$\Psi_{\Gamma}(t)=\sum_{T\subset\Gamma}\prod_{e\notin E(T)}t_e,$	$\Psi_{\Gamma}(t)=\sum_{T\subset\Gamma}\prod_{e\notin E(T)}t_e,$	$\Psi_{\Gamma}(t)=\sum_{T\subset\Gamma}\prod_{e\notin E(T)}t_e,$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0951 (2)	360	■ 1.000 ● N +0	$X_{\{\Gamma\}}=\left\{t=\left(t_1:t_2:\cdots:t_n\right.\right.\left.\left.\right\}\in\mathbb{P}_{\{k\}}^{\{n-1\}}\mid\Psi_{\{\Gamma\}}(t)=0\right\}.$	$X_\Gamma=\left\{t=(t_1:t_2:\cdots:t_n)\in\mathbb{P}_k^{n-1}\mid\Psi_\Gamma(t)=0\right\}.$	$X_\Gamma=\left\{t=(t_1:t_2:\cdots:t_n)\in\mathbb{P}_k^{n-1}\mid\Psi_\Gamma(t)=0\right\}.$
cardona-qft-methods_ EQ0952 (4)	360	■ 1.000 ● N +0	$\begin{gathered}\mathcal{C}:\mathbb{P}_{\{k\}}^{\{n-1\}}\rightarrow\mathbb{P}_{\{k\}}^{\{n-1\}}\\\left(t_1:t_2:\cdots:t_n\right)\mapsto\left(\frac{1}{t_1}:\frac{1}{t_2}:\cdots:\frac{1}{t_n}\right)\end{gathered}$	$\mathcal{C}:\mathbb{P}_k^{n-1}\rightarrow\mathbb{P}_k^{n-1}$ $(t_1:t_2:\cdots:t_n)\mapsto\left(\frac{1}{t_1}:\frac{1}{t_2}:\cdots:\frac{1}{t_n}\right)$	$\mathcal{C}:\mathbb{P}_k^{n-1}\rightarrow\mathbb{P}_k^{n-1}$ $(t_1:t_2:\cdots:t_n)\mapsto\left(\frac{1}{t_1}:\frac{1}{t_2}:\cdots:\frac{1}{t_n}\right)$
cardona-qft-methods_ EQ0953 (5)	360	■ 1.000 ● N +0	$\Sigma_{\{n\}}=\left\{\left(t_1:t_2:\cdots:t_n\right)\in\mathbb{P}_{\{k\}}^{\{n-1\}}\mid\prod_{i=1}^nt_i=0\right\}.$	$\Sigma_n=\left\{(t_1:t_2:\cdots:t_n)\in\mathbb{P}_k^{n-1}\mid\prod_{i=1}^nt_i=0\right\}.$	$\Sigma_n=\left\{(t_1:t_2:\cdots:t_n)\in\mathbb{P}_k^{n-1}\mid\prod_{i=1}^nt_i=0\right\}.$
cardona-qft-methods_ EQ0954 (6)	361	■ 1.000 ● K +0	$\mathcal{C}\circ\pi_1=\pi_2.$	$\mathcal{C}\circ\pi_1=\pi_2.$	$\mathcal{C}\circ\pi_1=\pi_2.$
cardona-qft-methods_ EQ0955 (7)	361	■ 1.000 ● K +0	$t_{\{1\}}t_{\{n+1\}}=t_{\{2\}}t_{\{n+2\}}=\cdots=t_{\{n\}}t_{\{2n\}},$	$t_1t_{n+1}=t_2t_{n+2}=\cdots=t_nt_{2n},$	$t_1t_{n+1}=t_2t_{n+2}=\cdots=t_nt_{2n},$
cardona-qft-methods_ EQ0961 (1)	362	■ 1.000 ● K +0	$\Psi_{\{\Gamma\}}(t)=1,$	$\Psi_\Gamma(t)=1,$	$\Psi_\Gamma(t)=1,$
cardona-qft-methods_ EQ0962 (2)	362	■ 1.000 ● K +0	$X_{\{\Gamma\}}=\emptyset.$	$X_\Gamma=\emptyset.$	$X_\Gamma=\emptyset.$
cardona-qft-methods_ EQ0963 (3)	362	■ 1.000 ● K +0	$\Psi_{\{\Gamma^{\vee}\}}(t)=t_{\{1\}}t_{\{2\}}\cdots t_{\{n\}},$	$\Psi_{\Gamma^\vee}(t)=t_1t_2\cdots t_n,$	$\Psi_{\Gamma^\vee}(t)=t_1t_2\cdots t_n,$
cardona-qft-methods_ EQ0964 (4)	362	■ 1.000 ● N +0	$X_{\{\Gamma^{\vee}\}}=\left\{t\in\mathbb{P}_{\{k\}}^{\{n-1\}}\mid\Psi_{\{\Gamma^{\vee}\}}(t)=0\right\}=\Sigma_{\{n\}}.$	$X_{\Gamma^\vee}=\left\{t\in\mathbb{P}_k^{n-1}\mid\Psi_{\Gamma^\vee}(t)=0\right\}=\Sigma_n.$	$X_{\Gamma^\vee}=\left\{t\in\mathbb{P}_k^{n-1}\mid\Psi_{\Gamma^\vee}(t)=0\right\}=\Sigma_n.$
cardona-qft-methods_ EQ0966 (6)	362	■ 1.000 ● K +0	$\left[X_{\{\Gamma\}}\right]=0.$	$[X_\Gamma]=0.$	$[X_\Gamma]=0.$
cardona-qft-methods_ EQ0968 (8)	363	■ 1.000 ● K +0	$\Psi_{\{\Gamma\}}(t)=t_{\{1\}}+t_{\{2\}}+\cdots+t_{\{n\}},$	$\Psi_\Gamma(t)=t_1+t_2+\cdots+t_n,$	$\Psi_\Gamma(t)=t_1+t_2+\cdots+t_n,$
cardona-qft-methods_ EQ0970 (10)	363	■ 1.000 ● N +0	$\left[\mathcal{L}\right]=\left[X_{\{\Gamma\}}\right]=\left[\frac{\mathbb{P}_{\{k\}}^{\{n-2\}}}{\mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}}\right].$	$[\mathcal{L}]=[X_\Gamma]=[\mathbb{P}_k^{n-2}]=\frac{\mathbb{L}^{n-1}-1}{\mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}}.$	$[\mathcal{L}]=[X_\Gamma]=[\mathbb{P}_k^{n-2}]=\frac{\mathbb{L}^{n-1}-1}{\mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}}.$
cardona-qft-methods_ EQ0973 (13)	363	■ 1.000 ● N +0	$\mathcal{H}:\mathcal{G}(\mathbb{T})\mapsto\frac{\mathcal{G}(\mathbb{T})-\mathcal{G}(-1)}{\mathbb{T}+1}$	$\mathcal{H}:g(\mathbb{T})\mapsto\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}$	$\mathcal{H}:g(\mathbb{T})\mapsto\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}$
cardona-qft-methods_ EQ0974 (14)	364	■ 1.000 ●	$\frac{\mathcal{G}(\mathbb{T})-\mathcal{G}(-1)}{\mathbb{T}+1}=\frac{(\mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=\left[\mathbb{P}^{n-1}\right],$	$\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(\mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=\left[\mathbb{P}^{n-1}\right],$	$\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(\mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=\left[\mathbb{P}^{n-1}\right],$
cardona-qft-methods_ EQ0975 (15)	364	■ 1.000 ● W +0	$\left[\mathcal{L}\cap\Sigma_n\right]=\frac{\mathcal{G}(\mathbb{T})-\mathcal{G}(-1)}{\mathbb{T}+1}=\frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1},$	$[\mathcal{L}\cap\Sigma_n]=\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1},$	$[\mathcal{L}\cap\Sigma_n]=\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ9976 (16)	364	■ 1.000 ● W +0	$\backslash\left[X_{\backslash\Gamma}\backslash\Sigma_n\right]=\backslash\left[\mathcal{L}\backslash\Sigma_n\right]=\mathcal{L}-\backslash\left[\mathcal{L}\cap\Sigma_n\right]=\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1}$	$[X_{\Gamma^\vee}\backslash\Sigma_n] = [\mathcal{L}\backslash\Sigma_n] = [\mathcal{L}] - [\mathcal{L} \cap \Sigma_n] = \frac{\mathbb{T}^{n-1} - (-1)^{n-1}}{\mathbb{T} + 1}.$	$[X_{\Gamma^\vee}\backslash\Sigma_n] = [\mathcal{L}\backslash\Sigma_n] = [\mathcal{L}] - [\mathcal{L} \cap \Sigma_n] = \frac{\mathbb{T}^{n-1} - (-1)^{n-1}}{\mathbb{T} + 1}.$
cardona-qft-methods_ EQ9977 (17)	364	■ 1.000 ● K +0	$\mathcal{I}=\left(t_2t_3\cdots t_n,t_1t_3\cdots t_n,\ldots,t_1t_2\cdots\hat{t}_i\cdots t_n,\ldots,t_1t_2\cdots t_{n-1}\right).$	$\mathcal{I}=\left(t_2t_3\cdots t_n,t_1t_3\cdots t_n,\ldots,t_1t_2\cdots\hat{t}_i\cdots t_n,\ldots,t_1t_2\cdots t_{n-1}\right).$	$\mathcal{I}=\left(t_2t_3\cdots t_n,t_1t_3\cdots t_n,\ldots,t_1t_2\cdots\hat{t}_i\cdots t_n,\ldots,t_1t_2\cdots t_{n-1}\right).$
cardona-qft-methods_ EQ9980 (20)	365	■ 1.000 ● K +0	$\backslash\left[X_{\backslash\Gamma}\backslash\Sigma\right]=\backslash\left[X_{\backslash\Gamma}\backslash\Sigma\cap\Sigma_n\right]=\backslash\left[X_{\backslash\Gamma}\backslash\Sigma\right]+\left[\mathcal{S}_n\right],$	$[X_{\Gamma^\vee}] = [X_{\Gamma^\vee} \backslash \Sigma] + [X_{\Gamma^\vee} \cap \Sigma] = [X_{\Gamma^\vee} \backslash \Sigma] + [\mathcal{S}_n],$	$[X_{\Gamma^\vee}] = [X_{\Gamma^\vee} \backslash \Sigma] + [X_{\Gamma^\vee} \cap \Sigma] = [X_{\Gamma^\vee} \backslash \Sigma] + [\mathcal{S}_n],$
cardona-qft-methods_ EQ9981 (1)	366	■ 1.000 ● K +0	$\Psi_\Gamma(t)=p(t)q(t)$	$\Psi_\Gamma(t)=p(t)q(t)$	$\Psi_\Gamma(t)=p(t)q(t)$
cardona-qft-methods_ EQ9982 (2)	367	■ 1.000 ● N +0	$P=\left\{t_i\mid\text{such that }t_i\text{ appears in }p(t)\right\}$	$P=\left\{t_i\mid\text{such that }t_i\text{ appears in }p(t)\right\}$	$P=\left\{t_i\mid\text{such that }t_i\text{ appears in }p(t)\right\}$
cardona-qft-methods_ EQ9984	369	■ 1.000 ● K +0	$XY-YX-[X,Y],\quad X,Y\in\tilde{m}g.$	$XY-YX-[X,Y],\quad X,Y\in\tilde{m}g.$	$XY-YX-[X,Y],\quad X,Y\in\tilde{m}g.$
cardona-qft-methods_ EQ9985	369	■ 1.000 ● N +0	$gf(x)=f\left(g^{-1}x\right),\quad f\in C(M).$	$gf(x)=f\left(g^{-1}x\right),\quad f\in C(M).$	$gf(x)=f(g^{-1}x),\quad f\in C(M).$
cardona-qft-methods_ EQ9986	369	■ 1.000 ● N +0	$e_\xi:g\in G\mapsto\langle J\xi,g\nu\rangle\in\mathbb{C}.$	$e_\xi:g\in G\mapsto\langle J\xi,g\nu\rangle\in\mathbb{C}.$	$e_\xi:g\in G\mapsto\langle J\xi,g\nu\rangle\in\mathbb{C}.$
cardona-qft-methods_ EQ9990	371	■ 1.000 ● N +0	$\omega(A)=\langle\nu,A\nu\rangle,\quad A\in\mathcal{A}.$	$\omega(A)=\langle\nu,A\nu\rangle,\quad A\in\mathcal{A}.$	$\omega(A)=\langle\nu,A\nu\rangle,\quad A\in\mathcal{A}.$
cardona-qft-methods_ EQ9991	372	■ 1.000 ● K +0	$\langle[A],[B]\rangle=\omega\left(A^\dagger B\right),\quad A,B\in\mathcal{A},$	$\langle[A],[B]\rangle=\omega\left(A^\dagger B\right),\quad A,B\in\mathcal{A},$	$\langle[A],[B]\rangle=\omega(A^\dagger B),\quad A,B\in\mathcal{A},$
cardona-qft-methods_ EQ9993	373	■ 1.000 ● N +0	$X\in\tilde{m}g\mapsto1\otimes X+X\otimes1\in\mathcal{T}(\tilde{m}g)\otimes\mathcal{T}(\tilde{m}g)$	$X\in\tilde{m}g\mapsto1\otimes X+X\otimes1\in\mathcal{T}(\tilde{m}g)\otimes\mathcal{T}(\tilde{m}g)$	$X\in\tilde{m}g\mapsto1\otimes X+X\otimes1\in\mathcal{T}(\tilde{m}g)\otimes\mathcal{T}(\tilde{m}g)$
cardona-qft-methods_ EQ9994	373	■ 1.000 ● N -1	$\Delta\mathcal{I}\subseteq\mathcal{I}\otimes\mathcal{T}(\tilde{m}g)+\mathcal{T}(\tilde{m}g)\otimes\mathcal{I}.$	$\Delta\mathcal{I}\subseteq\mathcal{I}\otimes\mathcal{T}(\tilde{m}g)+\mathcal{T}(\tilde{m}g)\otimes\mathcal{I}.$	$\Delta\mathcal{I}\subseteq\mathcal{I}\otimes\mathcal{T}(\tilde{m}g)+\mathcal{T}(\tilde{m}g)\otimes\mathcal{I}.$
cardona-qft-methods_ EQ9996	374	■ 1.000 ● K +0	$ab(E)=a\otimes b(\Delta E),\quad a,b\in\mathcal{E}^\dagger,E\in\mathcal{E}.$	$ab(E)=a\otimes b(\Delta E),\quad a,b\in\mathcal{E}^\dagger,E\in\mathcal{E}.$	$ab(E)=a\otimes b(\Delta E),\quad a,b\in\mathcal{E}^\dagger,E\in\mathcal{E}.$
cardona-qft-methods_ EQ9997 (1)	374	■ 1.000 ● K +0	$Ea(F)=a\left(\overline{E}^\dagger F\right),\quad E,F\in\mathcal{E},a\in\mathcal{E}^\dagger.$	$Ea(F)=a\left(\overline{E}^\dagger F\right),\quad E,F\in\mathcal{E},a\in\mathcal{E}^\dagger.$	$Ea(F)=a(\overline{E}^\dagger F),\quad E,F\in\mathcal{E},a\in\mathcal{E}^\dagger.$
cardona-qft-methods_ EQ9998	374	■ 1.000 ● N +0	$\left(\forall a,b\in\mathcal{E}^\dagger\right)\delta(ab)=\delta(a)b+a\delta(b).$	$\left(\forall a,b\in\mathcal{E}^\dagger\right)\delta(ab)=\delta(a)b+a\delta(b).$	$\left(\forall a,b\in\mathcal{E}^\dagger\right)\delta(ab)=\delta(a)b+a\delta(b).$
cardona-qft-methods_ EQ9999	374	■ 1.000 ● N +0	$X(ab)=X((a\otimes b)\circ\Delta)=(a\otimes b)\circ\Delta\circ X^\dagger(\cdot).$	$X(ab)=X((a\otimes b)\circ\Delta)=(a\otimes b)\circ\Delta\circ X^\dagger(\cdot).$	$X(ab)=X((a\otimes b)\circ\Delta)=(a\otimes b)\circ\Delta\circ X^\dagger(\cdot).$
cardona-qft-methods_ EQ1002	375	■ 1.000 ● N +0	$a\in\mathcal{E}^\dagger\mapsto f_a\in\mathbb{C}[[x_1,\ldots x_n]],\quad f_a(x)=\sum_{\alpha\in\mathbb{N}^n}\frac{x^\alpha}{\alpha!}a(X^\alpha).$	$a\in\mathcal{E}^\dagger\mapsto f_a\in\mathbb{C}[[x_1,\ldots x_n]],\quad f_a(x)=\sum_{\alpha\in\mathbb{N}^n}\frac{x^\alpha}{\alpha!}a(X^\alpha).$	$a\in\mathcal{E}^\dagger\mapsto f_a\in\mathbb{C}[[x_1,\ldots x_n]],\quad f_a(x)=\sum_{\alpha\in\mathbb{N}^n}\frac{x^\alpha}{\alpha!}a(X^\alpha).$
cardona-qft-methods_ EQ1004	375	■ 1.000 ●	$ab(1)=a\otimes b(\Delta 1)=a(1)b(1).$	$ab(1)=a\otimes b(\Delta 1)=a(1)b(1).$	$ab(1)=a\otimes b(\Delta 1)=a(1)b(1).$
cardona-qft-methods_ EQ1005	375	■ 1.000 ● N +0	$a^\dagger(E+fF)=\overline{a(E)}+f\overline{a(F)},\quad E,F\in\mathcal{E}_\mathbb{R}.$	$a^\dagger(E+fF)=\overline{a(E)}+f\overline{a(F)},\quad E,F\in\mathcal{E}_\mathbb{R}.$	$a^\dagger(E+fF)=\overline{a(E)}+f\overline{a(F)},\quad E,F\in\mathcal{E}_\mathbb{R}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ1006	375	■ 1.000 ● N +0	$\chi_{\{0\}}\!\left(a^{\dagger}\!\right)\!=\!a^{\dagger}\!\left(1\!=\!\overline{\overline{a(1)}}\!=\!\overline{\overline{\chi_{\{0\}}(a)}}\right)$.	$\chi_0\left(a^\dagger\right)=a^\dagger(1)=\overline{a(1)}=\overline{\chi_0(a)}.$	$\chi_0(a^\dagger)=a^\dagger(1)=\overline{a(1)}=\overline{\chi_0(a)}.$
cardona-qft-methods_ EQ1007	376	■ 1.000 ● N +0	$\xi\in\mathcal{F}\mapsto e_\xi\in\mathcal{E}^\dagger,\quad e_\xi(E)=\langle J\xi,E\nu\rangle,$	$\xi\in\mathcal{F}\mapsto e_\xi\in\mathcal{E}^\dagger,\quad e_\xi(E)=\langle J\xi,E\nu\rangle,$	$\xi\in\mathcal{F}\mapsto e_\xi\in\mathcal{E}^\dagger,\quad e_\xi(E)=\langle J\xi,E\nu\rangle,$
cardona-qft-methods_ EQ1008	376	■ 1.000 ● N +0	$M\!=\!\left\{\chi\in\mathcal{A}^{\dagger}\mid\chi(a\,b)\!=\!\chi(a)\,\chi(b),\,\chi\!\left(a^{\dagger}\right)\!=\!\overline{\overline{\chi(a)}}\right\}.$	$M=\left\{\chi\in\mathcal{A}^\dagger\mid\chi(ab)=\chi(a)\chi(b),\chi\left(a^\dagger\right)=\overline{\overline{\chi(a)}}\right\}.$	$M=\left\{\chi\in\mathcal{A}^\dagger\mid\chi(ab)=\chi(a)\chi(b),\chi(a^\dagger)=\overline{\overline{\chi(a)}}\right\}.$
cardona-qft-methods_ EQ1009	376	■ 1.000 ● N +0	$E\,e_{\xi}(F)\!=\!\left\langle J\xi,\bar{E}^{\dagger}F\nu\right\rangle\!=\!\left\langle \bar{E}J\xi,F\nu\right\rangle\!=\!\left\langle JE\xi,F\nu\right\rangle\!=\!e_{E\xi}(F).$	$Ee_\xi(F)=\left\langle J\xi,\bar{E}^\dagger F\nu\right\rangle=\left\langle \bar{E}J\xi,F\nu\right\rangle=\left\langle JE\xi,F\nu\right\rangle=e_{E\xi}(F).$	$Ee_\xi(F)=\left\langle J\xi,\bar{E}^\dagger F\nu\right\rangle=\left\langle \bar{E}J\xi,F\nu\right\rangle=\left\langle JE\xi,F\nu\right\rangle=e_{E\xi}(F).$
cardona-qft-methods_ EQ1010	376	■ 1.000 ● N +0	$\omega\in\mathcal{E}^\dagger,\quad\omega(E)=e_\nu(E)=\langle\nu,E\nu\rangle$	$\omega\in\mathcal{E}^\dagger,\quad\omega(E)=e_\nu(E)=\langle\nu,E\nu\rangle$	$\omega\in\mathcal{E}^\dagger,\quad\omega(E)=e_\nu(E)=\langle\nu,E\nu\rangle$
cardona-qft-methods_ EQ1011	377	■ 1.000 ● N +0	$\tilde{m}g=\mathbb{R}X,\quad\mathcal{F}=\mathcal{H}=\mathbb{C}^2,\quad X=\begin{pmatrix}0&1\\-1&0\end{pmatrix},\quad\nu=\begin{pmatrix}1\\0\end{pmatrix}.$	$\tilde{m}g=\mathbb{R}X,\quad\mathcal{F}=\mathcal{H}=\mathbb{C}^2,\quad X=\begin{pmatrix}0&1\\-1&0\end{pmatrix},\quad\nu=\begin{pmatrix}1\\0\end{pmatrix}.$	$\tilde{m}g=\mathbb{R}X,\quad\mathcal{F}=\mathcal{H}=\mathbb{C}^2,\quad X=\begin{pmatrix}0&1\\-1&0\end{pmatrix},\quad\nu=\begin{pmatrix}1\\0\end{pmatrix}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																								
cardona-qft-methods_TAB_007-1	643	1.000	$\begin{tabular}[t]{l l l l l l l l l l}\hline \backslash(d\backslash) & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline \backslash(\epsilon\backslash) & 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 \\ \hline \backslash(\epsilon^{\prime}) & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ \hline \backslash(\epsilon^{\prime}\backslash) & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ \hline \backslash(\epsilon^{\prime}\backslash)\backslash(\epsilon^{\prime}\backslash)\backslash(\epsilon^{\prime}\backslash) & 1 & & -1 & & 1 & & -1 & \\ \hline \end{tabular}$	<table><tr><td>d</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ϵ</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ′</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ″</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	d	0	1	2	3	4	5	6	7	ϵ	1	-1	1	1	1	-1	1	1	ϵ′	1	1	-1	-1	-1	-1	1	1	ϵ″	1		-1		1		-1		<table><tr><td>d</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ϵ</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ′</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ″</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	d	0	1	2	3	4	5	6	7	ϵ	1	-1	1	1	1	-1	1	1	ϵ′	1	1	-1	-1	-1	-1	1	1	ϵ″	1		-1		1		-1	
d	0	1	2	3	4	5	6	7																																																																					
ϵ	1	-1	1	1	1	-1	1	1																																																																					
ϵ′	1	1	-1	-1	-1	-1	1	1																																																																					
ϵ″	1		-1		1		-1																																																																						
d	0	1	2	3	4	5	6	7																																																																					
ϵ	1	-1	1	1	1	-1	1	1																																																																					
ϵ′	1	1	-1	-1	-1	-1	1	1																																																																					
ϵ″	1		-1		1		-1																																																																						
cardona-qft-methods_TAB_007-2	181	0.546	$\begin{tabular}[t]{l l l l l}\hline \backslash(D=3\backslash) CS Lagrangians & Characteristic classes & Groups \\ \hline \backslash[L_3^{(A)} d S=\epsilon_{a b c}\backslashleft(R^{a} b\right) \pm \frac{e^{a} e^{b}}{3 l^2}\backslashright) e^{c} \\ \backslash] & \backslash(\mathbf{E}_4=\epsilon_{a b c}\backslashleft(R^{a} b\right) \pm \frac{e^{a} e^{b}}{3 l^2}\backslashright) T^c \\ \backslash] & \backslash(\mathbf{T}_4=\epsilon_{a b c}\backslashleft(T^{a} b\right) \pm \frac{e^{a} e^{b}}{3 l^2}\backslashright) R^c \\ \backslash] & \backslash(S O(4)\backslash), \backslash(S O(3,1)\backslash) \\ \backslash] or \backslash(S O(2,2)\backslash) \backslashend{tabular} \\ \backslash] \backslash hline \backslash(L_3)^{\text {Lor }}=\omega^{a}{ }_b d \omega^{b}{ }_a+\frac{2}{3} \omega^{a}{ }_b \omega^{b}{ }_c \omega^{c}{ }_a \\ \backslash] \backslash \omega^{a}{ }_b \omega^{b}{ }_c \omega^{c}{ }_a \\ \backslash] & \backslash(\mathbf{P}_4)^{\text {Lor }}=R^{a}{ }_b R^{b}{ }_a \\ \backslash] & \backslash(S O(2,1)\backslash) \backslash hline \backslash(L_3)^{\text {T o r }}=e^{a}{ }_b T_a \\ \backslash] & \backslash(\mathbf{N}_4=T^{a}{ }_b T_a-e^{a}{ }_b e^{b}{ }_a R_{a b} \\ \backslash] & \backslash(S O(2,1)\backslash) \backslash hline \backslash(L_3)^{\text {U(1)}}=A d A \\ \backslash] & \backslash(\mathbf{F}_4)^{\text {U(1)}}=F F \\ \backslash] & \backslash(U(1)\backslash) \backslash hline \backslash(L_3)^{\text {SU(N)}}=\operatorname{Tr}\backslashleft[\mathbf{A} d \mathbf{A}+\frac{2}{3} \mathbf{A A A}\right] \\ \backslash] & \backslash(\mathbf{P}_4)^{\text {SU(4)}}=\operatorname{Tr}\backslashleft[\mathbf{P}_4\right] \\ \backslash] & \backslash(S U(N)\backslash) \backslash hline \backslashend{tabular}$	(not rendered)	<table><tr><td>D = 3 CS Lagrangians</td><td>Characteristic classes</td><td>Groups</td></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$</td><td>$\mathbf{P}_4^{Lor} = R^a_b R^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$\mathbf{P}_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\mathbf{A}d\mathbf{A} + \frac{2}{3}\mathbf{A} \mathbf{A} \mathbf{A}]$</td><td>$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{F} \mathbf{F}]$</td><td>$SU(N)$</td></tr></table>	D = 3 CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\mathbf{A}d\mathbf{A} + \frac{2}{3}\mathbf{A} \mathbf{A} \mathbf{A}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{F} \mathbf{F}]$	$SU(N)$																																																						
D = 3 CS Lagrangians	Characteristic classes	Groups																																																																											
$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$																																																																											
$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$																																																																											
$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$																																																																											
$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$																																																																											
$L_3^{SU(N)} = Tr[\mathbf{A}d\mathbf{A} + \frac{2}{3}\mathbf{A} \mathbf{A} \mathbf{A}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{F} \mathbf{F}]$	$SU(N)$																																																																											

Conf.

MathPix confidence:

≥0.9

0.5-0.9

<0.5

— no value

LaTeX source

MathPix's reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: basis]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																								
cardona-qft-methods_TAB_048	182	■0.562	$\begin{tabular}{t}{ l l } \hline \mathrm{(D=7 \mathrm{CS} \text{) Lagrangians \& Characteristic classes }} \hline (L_7^{\text{Lor}}=\omega(d \omega)^3+\frac{8}{9}\omega^3(d \omega)^2+\cdots+\frac{4}{7}\omega^7 \text{) \& } (\mathbf{P}_8^{\text{Lor}}=R^a{}_bR^b{}_cR^c{}_dR^d{}_a \text{) } \hline (L_7^I=(\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_bR^b{}_a \text{) } \hline (L_7^{II}=(e^aT_a)(R^a{}_bR^b{}_a) \text{) } \hline \end{tabular}$	(not rendered)	<table><tr><td>$D = 7$</td><td>CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{9}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td colspan="2">$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td colspan="2">$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td colspan="2">$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$	CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{9}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$		$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$		$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$																																																													
$D = 7$	CS Lagrangians	Characteristic classes																																																																											
$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{9}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$																																																																												
$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$																																																																												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$																																																																												
cardona-qft-methods_TAB_049	205	■0.993	$\begin{tabular}{t}{ l l l l l l l l l } \hline n \text{ \& } \theta \text{ \& } 1 \text{ \& } 2 \text{ \& } 3 \text{ \& } 4 \text{ \& } 5 \text{ \& } 6 \text{ \& } 7 \text{ \\ \hline } \backslash (\backslash \mathrm{varepsilon} \backslash) \text{ \& } 1 \text{ \& } 1 \text{ \& } -1 \text{ \& } -1 \text{ \& } -1 \text{ \& } 1 \text{ \& } 1 \text{ \& } 1 \text{ \\ \hline } \backslash (\backslash \mathrm{varepsilon}^{\prime} \backslash) \text{ \& } 1 \text{ \& } -1 \text{ \& } 1 \text{ \& } 1 \text{ \& } 1 \text{ \& } -1 \text{ \& } 1 \text{ \& } 1 \text{ \\ \hline } \backslash (\backslash \mathrm{varepsilon}^{\prime\prime} \backslash) \text{ \& } 1 \text{ \& } -1 \text{ \& } 1 \text{ \& } -1 \text{ \& } \backslash \text{ \\ \hline } \end{tabular}$	<table><tr><td>n</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1		<table><tr><td>n</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1	
n	0	1	2	3	4	5	6	7																																																																					
ε	1	1	-1	-1	-1	-1	1	1																																																																					
ε'	1	-1	1	1	1	-1	1	1																																																																					
ε''	1		-1		1		-1																																																																						
n	0	1	2	3	4	5	6	7																																																																					
ε	1	1	-1	-1	-1	-1	1	1																																																																					
ε'	1	-1	1	1	1	-1	1	1																																																																					
ε''	1		-1		1		-1																																																																						
cardona-qft-methods_TAB_050	281	■0.574	$\begin{tabular}{t}{ l l } \hline (L \backslash) \text{ \& } (p \text{ } p \text{ } p \text{ } p \backslash) \text{ \\ \hline } k \text{ } (k \text{ } k \backslash) \text{ } k \text{ \& } (\bar{L} \backslash) \text{ \\ \hline } \end{tabular}$	<table><tr><td>L</td><td>$pppp$</td></tr><tr><td>$k \text{ } k \text{ } k \text{ } k$</td><td>$\bar{L}$</td></tr></table>	L	$pppp$	$k \text{ } k \text{ } k \text{ } k$	$\bar{L}$	<table><tr><td>L</td><td>pp</td></tr><tr><td>$k \text{ } k$</td><td>$\bar{L}$</td></tr></table> ,	L	pp	$k \text{ } k$	$\bar{L}$																																																																
L	$pppp$																																																																												
$k \text{ } k \text{ } k \text{ } k$	$\bar{L}$																																																																												
L	pp																																																																												
$k \text{ } k$	$\bar{L}$																																																																												
cardona-qft-methods_TAB_051	281	■0.887	$\begin{tabular}{t}{ l l l } \hline (L \backslash) \text{ \& } (P \backslash) \text{ \& } (Q \backslash) \text{ \\ \hline } (K \backslash) \text{ \& } (L \backslash) \text{ \& } (S \backslash) \text{ \\ \hline } (\bar{Q} \backslash) \text{ \& } (R \backslash) \text{ \\ \hline } \end{tabular}$	<table><tr><td>L</td><td>P</td><td>Q</td></tr><tr><td>K</td><td>L</td><td>S</td></tr><tr><td>S</td><td>Q</td><td>R</td></tr></table>	L	P	Q	K	L	S	S	Q	R	<table><tr><td>L</td><td>P</td><td>Q</td></tr><tr><td>K</td><td>$\bar{L}$</td><td>$\bar{S}$</td></tr><tr><td>S</td><td>$\bar{Q}$</td><td>R</td></tr></table> .	L	P	Q	K	$\bar{L}$	$\bar{S}$	S	$\bar{Q}$	R																																																						
L	P	Q																																																																											
K	L	S																																																																											
S	Q	R																																																																											
L	P	Q																																																																											
K	$\bar{L}$	$\bar{S}$																																																																											
S	$\bar{Q}$	R																																																																											
cardona-qft-methods_TAB_052	286	■0.474	$\begin{tabular}{t}{ l l } \hline (L \backslash) \text{ \& } (p \text{ } p \text{ } p \text{ } p \backslash) \text{ \\ \hline } k \text{ } (k \text{ } k \text{ } k \backslash) \text{ \& } (\bar{L} \backslash) \text{ \\ \hline } \end{tabular}$	<table><tr><td>L</td><td>$pppp$</td></tr><tr><td>$k \text{ } k \text{ } k \text{ } k$</td><td>$\bar{L}$</td></tr></table>	L	$pppp$	$k \text{ } k \text{ } k \text{ } k$	$\bar{L}$	<table><tr><td>L</td><td>pp</td></tr><tr><td>$k \text{ } k$</td><td>$\bar{L}$</td></tr></table> .	L	pp	$k \text{ } k$	$\bar{L}$																																																																
L	$pppp$																																																																												
$k \text{ } k \text{ } k \text{ } k$	$\bar{L}$																																																																												
L	pp																																																																												
$k \text{ } k$	$\bar{L}$																																																																												

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

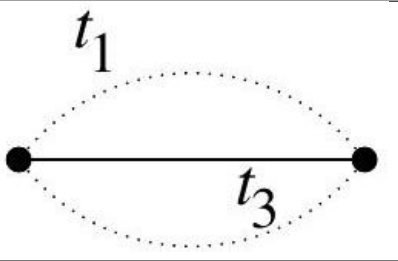
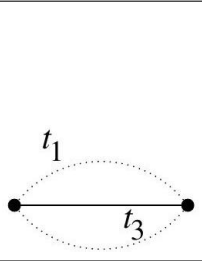
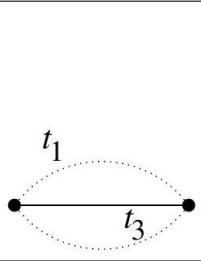
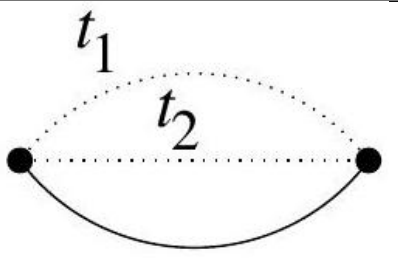
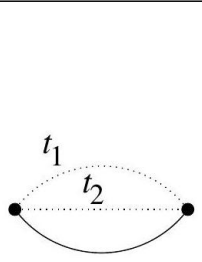
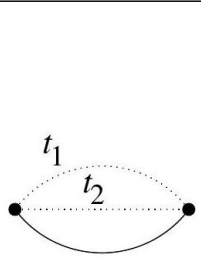
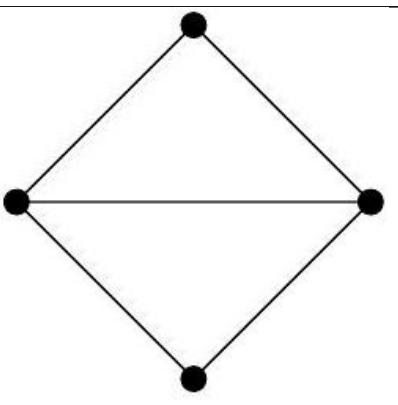
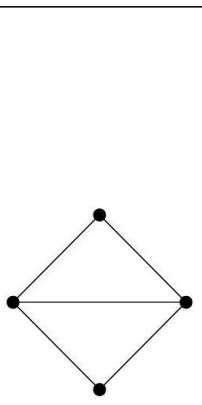
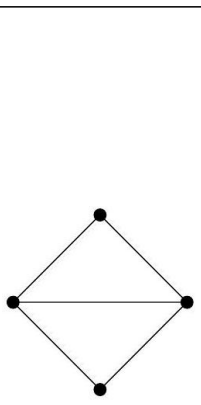
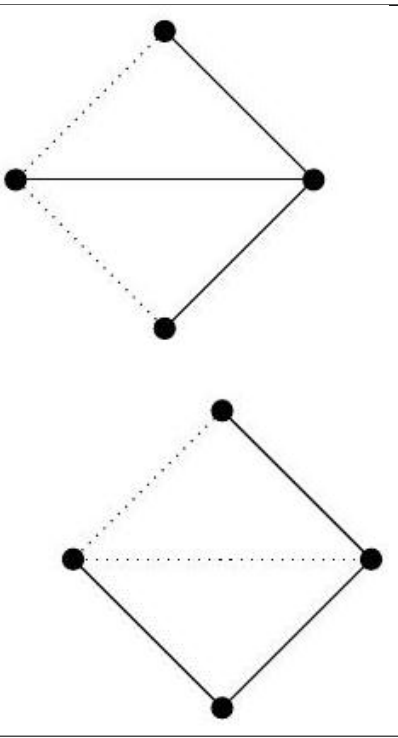
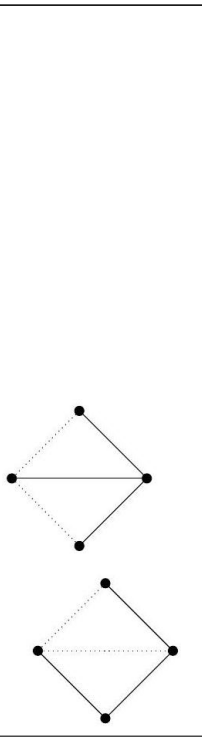
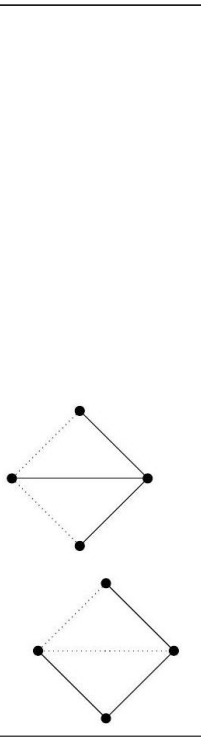
[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

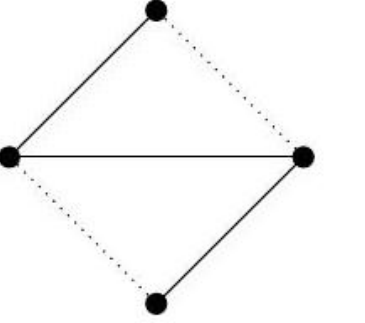
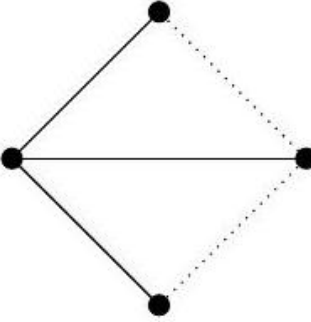
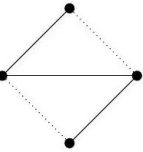
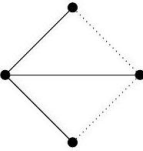
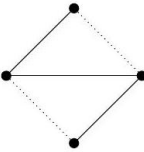
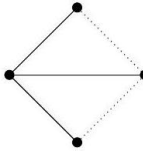
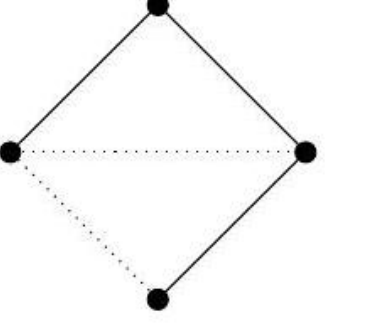
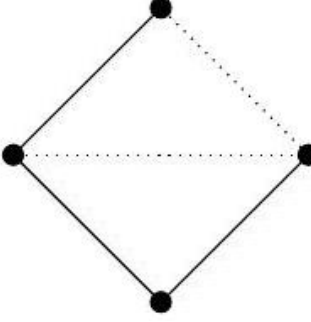
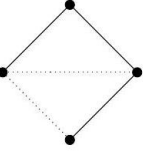
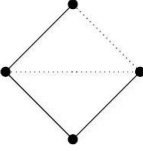
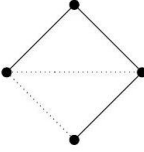
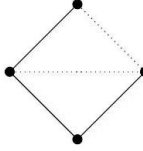
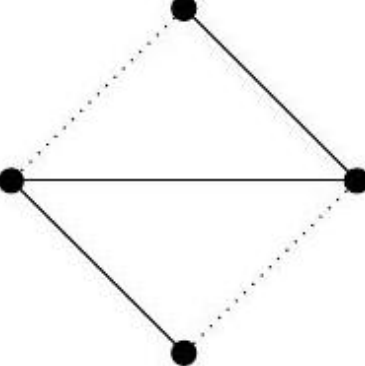
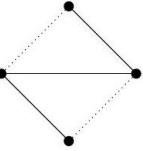
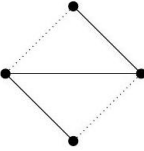
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image								
cardona-qft-methods_TAB_287	287	0.514	$\begin{tabular}{t}{ l l } \hline \frac{s}{\frac{u}{2 \mid 2}} \& \text{create } (8_B + 8_F) \backslash \text{magnons} \\ \hline \text{destroy } 8 + 8 \text{ magnons} \& \overline{\frac{s}{\frac{u}{2 \mid 2}}} \\ \hline \end{tabular}$	<table><tr><td>$\frac{s}{u}(2 \mid 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy $8 + 8$ magnons</td><td>$\overline{\frac{s}{u}(2 \mid 2)}$</td></tr></table>	$\frac{s}{u}(2 \mid 2)$	create $8_B + 8_F$ magnons	destroy $8 + 8$ magnons	$\overline{\frac{s}{u}(2 \mid 2)}$	<table><tr><td>$\frac{s}{u}(2 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy $8 + 8$ magnons</td><td>$\overline{\frac{s}{u}(2 2)}$</td></tr></table>	$\frac{s}{u}(2 2)$	create $8_B + 8_F$ magnons	destroy $8 + 8$ magnons	$\overline{\frac{s}{u}(2 2)}$
$\frac{s}{u}(2 \mid 2)$	create $8_B + 8_F$ magnons												
destroy $8 + 8$ magnons	$\overline{\frac{s}{u}(2 \mid 2)}$												
$\frac{s}{u}(2 2)$	create $8_B + 8_F$ magnons												
destroy $8 + 8$ magnons	$\overline{\frac{s}{u}(2 2)}$												
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.													

Image regions — rendered against scan

Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

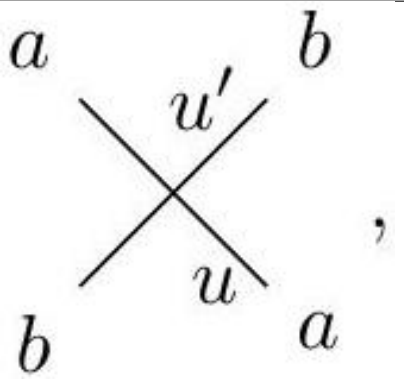
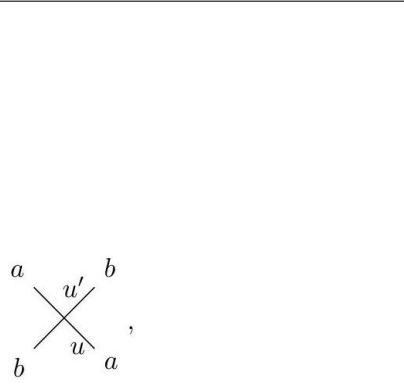
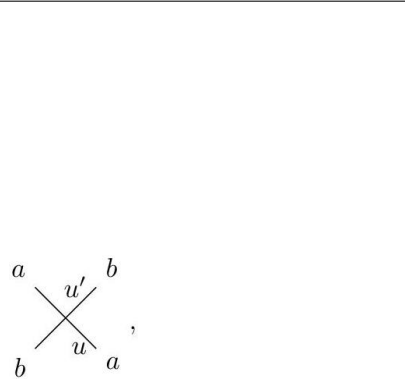
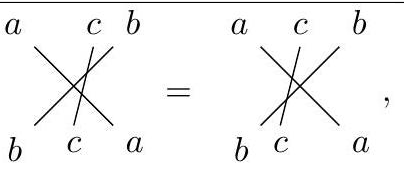
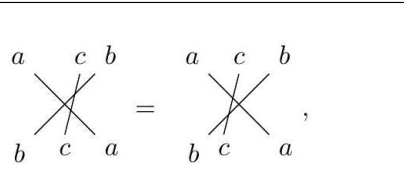
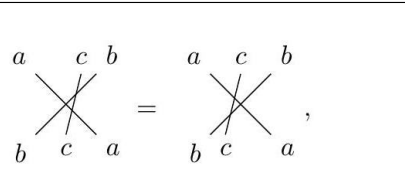
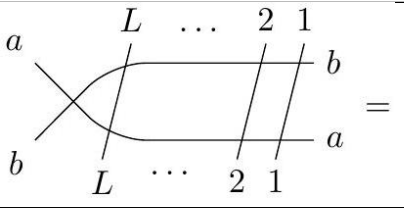
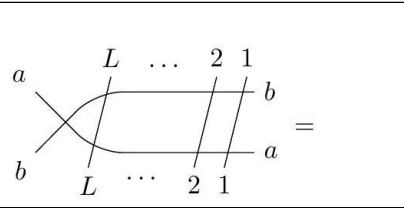
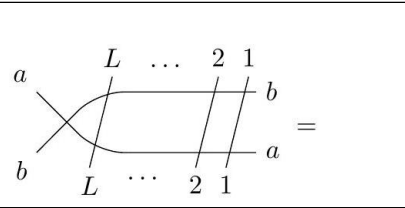
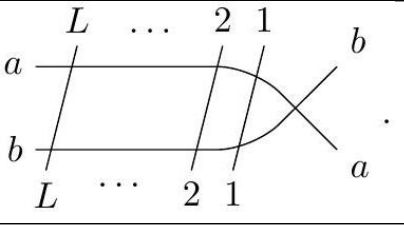
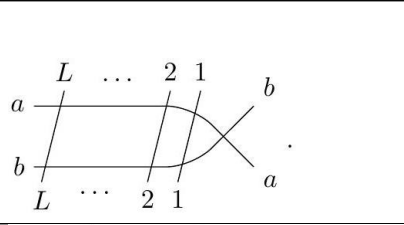
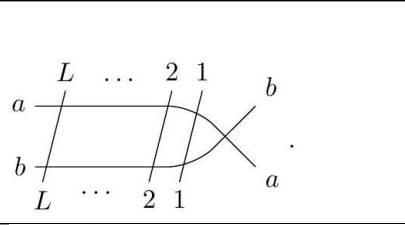
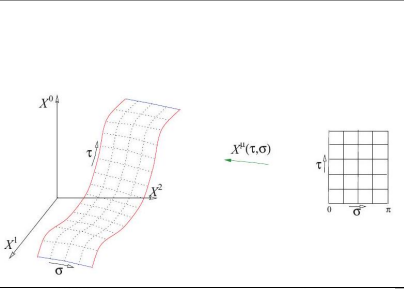
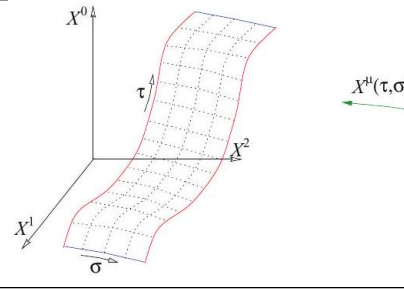
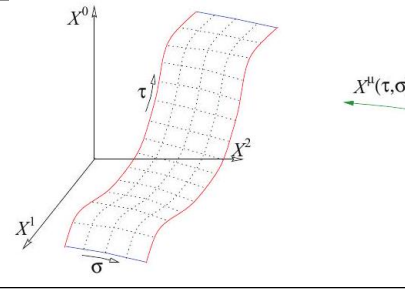
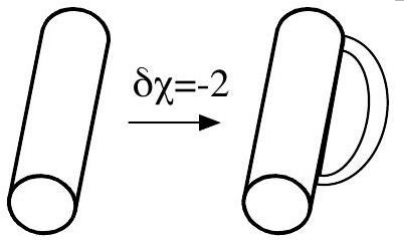
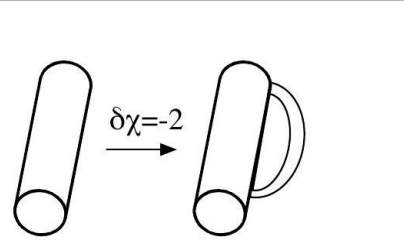
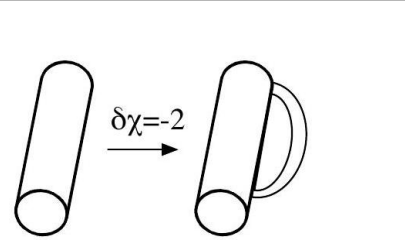
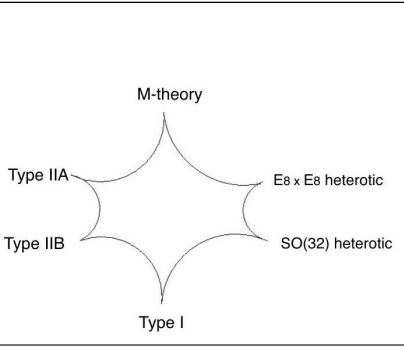
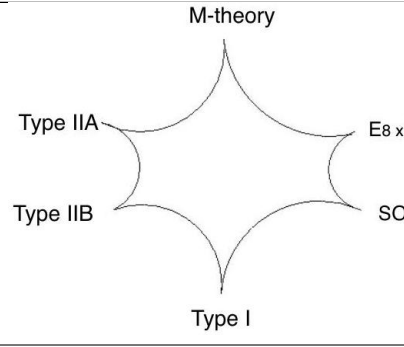
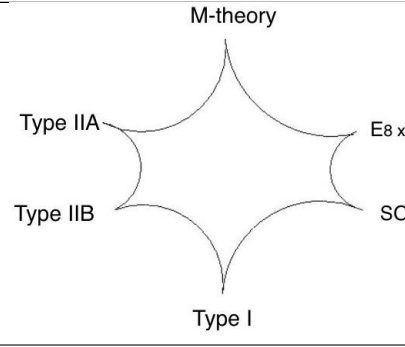
Identifier	Page	Class	Source	Author source	Rendered	Scan
cardona-qft-methods_PIC_0023	023	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-023_303_699_1682_342.jpg			
cardona-qft-methods_DIA_0034	034	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-034_145_712_1101_333.jpg	$\begin{array}{ccc} \text{Spin}(TM) \times \text{Spin}_d & \longrightarrow & \text{Spin}(TM) \\ \eta \times \xi \downarrow & & \eta \downarrow \\ \text{SO}(TM) \times \text{SO}_d & \longrightarrow & \text{SO}(TM) \end{array} \begin{array}{c} \xrightarrow{\pi} \\ \xrightarrow{\pi} \end{array} M$	$\begin{array}{ccc} \text{Spin}(TM) \times \text{Spin}_d & \longrightarrow & \text{Spin}(TM) \\ \eta \times \xi \downarrow & & \eta \downarrow \\ \text{SO}(TM) \times \text{SO}_d & \longrightarrow & \text{SO}(TM) \end{array} \begin{array}{c} \xrightarrow{\pi} \\ \xrightarrow{\pi} \end{array} M$	$\begin{array}{ccc} \text{Spin}(TM) \times \text{Spin}_d & \longrightarrow & \text{Spin}(TM) \\ \eta \times \xi \downarrow & & \eta \downarrow \\ \text{SO}(TM) \times \text{SO}_d & \longrightarrow & \text{SO}(TM) \end{array} \begin{array}{c} \xrightarrow{\pi} \\ \xrightarrow{\pi} \end{array} M$
cardona-qft-methods_DIA_0084	084	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-084_276_260_1585_505.jpg	$\begin{array}{ccc} & \sigma & \\ E & \longrightarrow & F \\ & \searrow \quad \swarrow & \\ & X & \end{array}$	$\begin{array}{ccc} & \sigma & \\ E & \longrightarrow & F \\ & \searrow \quad \swarrow & \\ & X & \end{array}$	$\begin{array}{ccc} & \sigma & \\ E & \longrightarrow & F \\ & \searrow \quad \swarrow & \\ & X & \end{array}$
cardona-qft-methods_DIA_0086	086	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-086_294_460_1735_404.jpg	$\begin{array}{ccc} K(T^*M) & \xrightarrow{j_!} & K(\mathbb{C}^n) \\ & \uparrow i_! & \\ & K(\{pt\}) & \end{array}$	$\begin{array}{ccc} K(T^*M) & \xrightarrow{j_!} & K(\mathbb{C}^n) \\ & \uparrow i_! & \\ & K(\{pt\}) & \end{array}$	$\begin{array}{ccc} K(T^*M) & \xrightarrow{j_!} & K(\mathbb{C}^n) \\ & \uparrow i_! & \\ & K(\{pt\}) & \end{array}$
cardona-qft-methods_DIA_0119	119	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-119_90_363_1455_526.jpg			
cardona-qft-methods_DIA_0122	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_181_280_312_568.jpg			
cardona-qft-methods_DIA_0122	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_120_253_623_212.jpg			

cardona-qft-methods_DIA_0102	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_163_257_581_584.jpg			
cardona-qft-methods_DIA_0102	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_166_255_581_949.jpg			
cardona-qft-methods_DIA_0102	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_250_251_1222_582.jpg			
cardona-qft-methods_DIA_0102	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_425_232_1579_198.jpg			

cardona-qft-methods_DIA_01122	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_423_235_1581_460.jpg	 	 	 
cardona-qft-methods_DIA_01122	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_421_234_1583_722.jpg	 	 	 
cardona-qft-methods_DIA_01122	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_200_200_1581_986.jpg			

cardona-qft-methods_DIA_0112	122	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_198_200_1806_1018.jpg			
cardona-qft-methods_DIA_0112	126	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-126_202_496_1818_460.jpg			
cardona-qft-methods_DIA_0112	127	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_127_280_460_197.jpg			
cardona-qft-methods_DIA_0112	127	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_124_462_522.jpg			
cardona-qft-methods_DIA_0112	127	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_181_462_673.jpg			
cardona-qft-methods_DIA_0112	127	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_324_462_898.jpg			
cardona-qft-methods_DIA_0112	132	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-132_110_225_253_597.jpg			

cardona-qft-methods_DIA_01233	133	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_230_701_995_370.jpg	$\text{line} + \text{loop} = x+y$ $\text{line} + \text{two-loops} = x+y+y^2$	$\text{line} + \text{loop} = x+y$ $\text{line} + \text{two-loops} = x+y+y^2$	$\text{line} + \text{loop} = x+y$ $\text{line} + \text{two-loops} = x+y+y^2$
cardona-qft-methods_DIA_01233	133	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_234_875_1328_276.jpg	$\text{loop} + \text{line} = x^2 + (1+y)(x+y)$ $\text{two-loops} + \text{line} = x^2 + (1+y)(x+y)$	$\text{loop} + \text{line} = x^2 + (1+y)(x+y)$ $\text{two-loops} + \text{line} = x^2 + (1+y)(x+y)$	$\text{loop} + \text{line} = x^2 + (1+y)(x+y)$ $\text{two-loops} + \text{line} = x^2 + (1+y)(x+y)$
cardona-qft-methods_DIA_01233	133	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_207_839_1664_294.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$
cardona-qft-methods_DIA_01235	135	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-135_191_876_1466_142.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$
cardona-qft-methods_DIA_01236	136	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-136_198_358_775_248.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$
cardona-qft-methods_DIA_01236	136	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-136_198_356_775_811.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$
cardona-qft-methods_DIA_01238	138	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-138_230_276_591_570.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$
cardona-qft-methods_DIA_01270	270	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-270_243_783_250_317.jpg	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$	$\text{loop} + \text{line} = x(x+y) + (x+y+y^2)$ $\text{two-loops} + \text{line} = x(x+y) + (x+y+y^2)$

cardona-qft-methods_DIA_00274	274	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-274_159_166_1108_754.jpg			
cardona-qft-methods_DIA_00274	274	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-274_163_395_1416_526.jpg			
cardona-qft-methods_DIA_00275	275	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-275_200_404_1696_319.jpg			
cardona-qft-methods_DIA_00275	275	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-275_202_374_1696_754.jpg			
cardona-qft-methods_DIA_00292	292	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-292_361_765_1261_326.jpg			
cardona-qft-methods_DIA_00296	296	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-296_239_393_1319_535.jpg			
cardona-qft-methods_DIA_00305	305	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-305_432_678_251_370.jpg			

cardona-qft-methods_DIA_00361	344	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-344_193_652_251_382.jpg	$ \begin{array}{ccc} H_p(M) \otimes H_q(M) & \xrightarrow{\quad \star \quad} & H_{p+q-n}(M) \\ \downarrow P.D \otimes P.D & & \uparrow P.D^{-1} \\ H^{n-p}(M) \otimes H^{n-q}(M) & \xrightarrow{\quad \sqcup \quad} & H^{2n-p-q}(M) \end{array} $	$ \begin{array}{ccc} H_p(M) \otimes H_q(M) & \xrightarrow{\quad \star \quad} & H_{p+q-n}(M) \\ \downarrow P.D \otimes P.D & & \uparrow P.D^{-1} \\ H^{n-p}(M) \otimes H^{n-q}(M) & \xrightarrow{\quad \sqcup \quad} & H^{2n-p-q}(M) \end{array} $	$ \begin{array}{ccc} H_p(M) \otimes H_q(M) & \xrightarrow{\quad \star \quad} & H_{p+q-n}(M) \\ \downarrow P.D \otimes P.D & & \uparrow P.D^{-1} \\ H^{n-p}(M) \otimes H^{n-q}(M) & \xrightarrow{\quad \sqcup \quad} & H^{2n-p-q}(M) \end{array} $
cardona-qft-methods_DIA_00373	373	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-373_170_248_269_584.jpg	$ \begin{array}{ccc} \tilde{\mathcal{E}} & \dashrightarrow & \mathcal{C}^*(\tilde{\mathcal{F}}) \\ \downarrow & & \downarrow \\ \mathcal{E} & \longrightarrow & \mathcal{L}^+(\mathcal{F}), \end{array} $	$ \begin{array}{ccc} \tilde{\mathcal{E}} & \dashrightarrow & \mathcal{C}^*(\tilde{\mathcal{F}}) \\ \downarrow & & \downarrow \\ \mathcal{E} & \longrightarrow & \mathcal{L}^+(\mathcal{F}), \end{array} $	$ \begin{array}{ccc} \tilde{\mathcal{E}} & \dashrightarrow & \mathcal{C}^*(\tilde{\mathcal{F}}) \\ \downarrow & & \downarrow \\ \mathcal{E} & \longrightarrow & \mathcal{L}^+(\mathcal{F}), \end{array} $
cardona-qft-methods_DIA_00373	373	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-373_191_354_1310_535.jpg	$ \begin{array}{ccc} \mathcal{E} & \xrightarrow{\Delta} & \mathcal{E} \otimes \mathcal{E} \\ \downarrow \Delta & & \downarrow \text{Id} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{E} & \xrightarrow{\Delta \otimes \text{Id}} & \mathcal{E} \otimes \mathcal{E} \otimes \mathcal{E}. \end{array} $	$ \begin{array}{ccc} \mathcal{E} & \xrightarrow{\Delta} & \mathcal{E} \otimes \mathcal{E} \\ \downarrow \Delta & & \downarrow \text{Id} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{E} & \xrightarrow{\Delta \otimes \text{Id}} & \mathcal{E} \otimes \mathcal{E} \otimes \mathcal{E}. \end{array} $	$ \begin{array}{ccc} \mathcal{E} & \xrightarrow{\Delta} & \mathcal{E} \otimes \mathcal{E} \\ \downarrow \Delta & & \downarrow \text{Id} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{E} & \xrightarrow{\Delta \otimes \text{Id}} & \mathcal{E} \otimes \mathcal{E} \otimes \mathcal{E}. \end{array} $
cardona-qft-methods_DIA_00374	374	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-374_181_244_937_586.jpg	$ \begin{array}{ccc} & & \mathcal{E} \otimes \mathcal{E} \\ & \nearrow \Delta & \downarrow \tau \\ \mathcal{E} & & \mathcal{E} \otimes \mathcal{E}, \\ & \searrow \Delta & \end{array} $	$ \begin{array}{ccc} & & \mathcal{E} \otimes \mathcal{E} \\ & \nearrow \Delta & \downarrow \tau \\ \mathcal{E} & & \mathcal{E} \otimes \mathcal{E}, \\ & \searrow \Delta & \end{array} $	$ \begin{array}{ccc} & & \mathcal{E} \otimes \mathcal{E} \\ & \nearrow \Delta & \downarrow \tau \\ \mathcal{E} & & \mathcal{E} \otimes \mathcal{E}, \\ & \searrow \Delta & \end{array} $